AK High-Dimensional Projection Structural Theory

Appendix BFA

Navier–Stokes Blow-up, Source-Faithful Reconstruction,
Adversarial Falsification, and Generic Extraction

Post-v21 Problem-Interface Research Supplement
BFA 1.0 — Construction Draft
Sections BFA.1–BFA.15

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Abstract

Appendix BFA is a post-v21 research supplement for adversarial analysis of an external Navier–Stokes finite-time blow-up proof candidate. It preserves the frozen AK-HPDST v21.0.0 source set and Claim Register and treats the September 2026 OpenAI release as an external benchmark family rather than an AK theorem. The global re-audit separates source authority from official-target authority, source-faithful reconstruction from benchmark-quarantined independent recovery, route nonclosure from route falsification, and route falsification from quantifier-aware target falsification. It also requires directional paper–formalization–target interpretation relations, frozen external-input identity, and an explicit promotion ladder from BFA-local candidate to Technical Extension candidate to successor-Core candidate. BFA.1 fixes the constitutional boundary for the full appendix. BFA.2 fixes the official Clay C/D targets at exact quantifier strength, includes the periodic-pressure erratum, separates the analytic paper, public announcement, formal reference, and formal proof source, and pins the OpenAI formal family to a content-addressed repository commit. BFA.3 reconstructs the analytic proof as a typed dependency DAG from leading profile and stress construction through oscillatory realization, residual-improvement cycles, stage-uniform derivative control, finite-to-infinite completion, compact forcing, energy, uniqueness, viscosity scaling, and periodization. BFA.4 projects every registered analytic node into AK ownership, source-specific Bridge role, source/audit defect sorts, finite–infinite and terminal boundaries, target-adapter semantics, and exact Return prerequisite packets while deliberately leaving Core-P and Core-DG not invoked. BFA.5 proves the conditional source-faithful reconstruction theorem: if the registered external analytic lane is true at its exact source strength and the typed interpretation edges are valid, then the BFA composition exports precisely the Clay-C/Clay-D targets and no stronger target, while preserving external theorem authority and a strict no-pass-leakage discipline. BFA.6 then audits that route against the frozen AK-v21 source-binding, finite-to-infinite, local-to-global, noncompensation, Core-sovereignty, Return, quarantine, converse, target-strength, formalization, and release-governance rules. It separates source-faithful use of the external proof from a future benchmark-quarantined independent-recovery lane and finds no identified rule-level contradiction, without verifying any external analytic lemma. BFA.7 then formalizes independent-recovery eligibility itself: it freezes the target/cutoff/precursor package, audits precursor-selection provenance and the accessible theorem-bearing knowledge state, distinguishes strict from bounded isolation, requires a seal-before-compare two-phase protocol, and fixes the present source-exposed BFA context as non-certifiable for strict independent recovery while preserving its value for post-hoc mathematics and protocol design. BFA.8 then defines the adversarial falsification semantics used by the remainder of the appendix: it separates suspicion from certified negative evidence, binding error from proof failure and theorem falsity, route nonclosure from route rejection, formal replay from semantic mismatch, source truth from target interpretation, and proof-route failure from quantifier-aware Clay-target falsification; it also fixes strength-indexed failure propagation, repair identity, and Navier–Stokes-specific adversarial obligations. BFA.9 instantiates those semantics as a critical-cut architecture, separating graph cuts, selected-route blockers, derivational dominators, and audit frontiers, while recognizing the shared C/D role of the viscosity-scaling node and fixing the completion and terminal audit order without turning priority into falsity. BFA.10 then stress-tests the finite-to-infinite completion lane against the actual source statements of Proposition 9.6, Lemmas 9.7–9.8, Lemma 5.4, and Proposition 9.9: it checks

the common-domain and stage-independent-loss contracts, the shared summation-theorem hypotheses, recursive shrinking-cutoff completion, nonlinear fixed-stage comparison, flat-remainder handling, divergence preservation, and the quantifier order producing infinite-order residual flatness. No certified adverse finding is obtained at that declared source-interface resolution, while complete independent re-proof remains explicitly unclaimed. BFA.11 independently audits the terminal localization, smooth forcing extension, energy class, comparison argument, viscosity transport, periodicization, and direct Clay-C/Clay-D Return edges, again finding no certified adverse witness at the declared source-semantic resolution while keeping mathematical verifier status uninvoked. BFA.12 then separates the analytic paper, pinned Lean Comparator statements, and corrected official Clay targets, replacing any claim of literal theorem equivalence by one-way witness-forgetting projections and an explicit cross-language semantic-bridge ledger; the global re-audit additionally records independent third-party formal-audit evidence without converting it into a BFA verifier token. BFA.13 begins the generic-extraction phase: it proves only two modest BFA-local filtered-residual lemmas, treats the adaptive-localizer RIC and monotone-defect-improvement patterns as research programs, and rebinds joint-obligation, negative-subject, and cross-artifact semantic-projection views to their existing v21 owners. BFA.14 separates internal owner admission, external novelty, and adoption value, and nominates no present Technical-Extension or successor-Core candidate. BFA.15 closes the construction by a six-axis typed verdict, an explicit open-obligation ledger, a non-claim register, immutable-history/reopen rules, and a global consistency re-audit. No AK-only solution of the Navier–Stokes Millennium Prize Problem and no successor-Core promotion are claimed here.

Contents

BFA.1 Scope, Authority, and Constitutional Boundary . . . 6
BFA.1.1 Role, release relation, and local authority . . . 6
BFA.1.2 Frozen v21 authority and noncompetition . . . 7
BFA.1.3 External source family, target sovereignty, and authority partition . . . 7
BFA.1.4 Eight distinct logical tasks . . . 9
BFA.1.5 Source-faithful reconstruction versus independent recovery . . . 9
BFA.1.6 Status sorts and evidence discipline . . . 9
BFA.1.7 Compatibility is weaker than proof . . . 10
BFA.1.8 Adversarial nonclosure, route falsification, and target falsification . . . 11
BFA.1.9 Use of existing failure-cut and target-cut calculus . . . 12
BFA.1.10 Core-P, Core-DG, and Bridge invocation discipline . . . 12
BFA.1.11 Analytic proof, formal proof, and official target . . . 12
BFA.1.12 Generic extraction and promotion quarantine . . . 13
BFA.1.13 Local identities, frozen-input discipline, and governance . . . 13
BFA.1.14 Explicit non-claims . . . 14
BFA.1.15 BFA dependency and section contract . . . 14
BFA.2 Exact External Target and Source Binding . . . 15
BFA.2.1 Purpose and binding discipline . . . 15
BFA.2.2 Official Clay target artifact and exact Fefferman predicates . . . 16
BFA.2.3 Analytic-paper source binding: whole-space theorem . . . 18
BFA.2.4 Analytic-paper source binding: periodic corollary . . . 18
BFA.2.5 Announcement binding and lifecycle separation . . . 19
BFA.2.6 Pinned Lean formal-source family . . . 19
BFA.2.7 Directional interpretation graph and semantic firewalls . . . 20
BFA.2.8 Maximal binding matrix and theorem-facing statuses . . . 21
BFA.2.9 BFA.2 closure and BFA.3 handoff . . . 22
BFA.3 Analytic Proof Architecture and Dependency DAG . . . 23
BFA.3.1 Purpose, authority boundary, and non-verification status . . . 23
BFA.3.2 Edge taxonomy for the source proof . . . 23
BFA.3.3 Source proof spine at viscosity one . . . 24
BFA.3.4 Dependency DAG and mandatory paths . . . 26
BFA.3.5 Leading profile, annular stress, and all-order base correction . . . 26
BFA.3.6 Oscillatory realization and exact defect bookkeeping . . . 27

BFA.3.7 The finite correction state and residual-improvement engine . . . 28
BFA.3.8 Common-domain and derivative-loss gate . . . 28
BFA.3.9 Finite-to-infinite completion and local-field realization . . . 29
BFA.3.10 Localization, terminal force extension, and solution-class closure . . . 29
BFA.3.11 Scaling and periodic transport . . . 30
BFA.3.12 Critical audit cut and priority map . . . 30
BFA.3.13 AK ownership map and deliberate Core non-invocation . . . 31
BFA.3.14 Architecture closure and handoff . . . 32
BFA.4 AK Typed Projection of the External Proof . . . 32
BFA.4.1 Purpose and projection discipline . . . 32
BFA.4.2 Authority, carrier, role, and deliberate Core non-invocation . . . 34
BFA.4.3 Proof-boundary taxonomy . . . 34
BFA.4.4 Node-by-node typed projection . . . 35
BFA.4.5 Defect/obligation vector and correction semantics . . . 36
BFA.4.6 Finite-to-infinite and endpoint boundaries as typed obligations . . . 37
BFA.4.7 Route-closure contracts and exact target witness packets . . . 37
BFA.4.8 Direct BFA target slices and relation to the canonical v21 schema . . . 38
BFA.4.9 Node-to-authority/role matrix . . . 39
BFA.4.10 What AK contributes at this stage . . . 40
BFA.4.11 BFA.4 closure and handoff . . . 40
BFA.5 Source-Faithful Reconstruction Theorem . . . 41
BFA.5.1 Purpose and theorem mode . . . 41
BFA.5.2 Reconstruction assumption packets . . . 41
BFA.5.3 Internal route reconstruction . . . 42
BFA.5.4 Typed export from route packets to exact target witnesses . . . 43
BFA.5.5 Main source-faithful reconstruction theorems . . . 43
BFA.5.6 Conservativity and theorem ceiling . . . 44
BFA.5.7 Status discipline after reconstruction . . . 45
BFA.5.8 BFA.5 closure and handoff . . . 46
BFA.6 AK–OpenAI Compatibility and No-Contradiction Audit . . . 46
BFA.6.1 Purpose, scope, and audit mode . . . 46
BFA.6.2 The frozen v21 compatibility basis . . . 47
BFA.6.3 Source binding, typing, and ownership audit . . . 48
BFA.6.4 Finite-to-infinite and local-to-global audit . . . 49
BFA.6.5 Non-compensation, Return, and target-strength audit . . . 50
BFA.6.6 Source-faithful use versus benchmark quarantine . . . 50
BFA.6.7 Compatibility with the frozen v21 Navier–Stokes package . . . 51
BFA.6.8 Formalization, infrastructure, and status-separation audit . . . 52
BFA.6.9 Compatibility matrix and aggregate verdict . . . 52
BFA.6.10 BFA.6 closure and handoff . . . 53
BFA.7 Independent-Recovery Eligibility and Quarantine Protocol . . . 54
BFA.7.1 Purpose and non-claim . . . 54
BFA.7.2 The independent-recovery experiment object . . . 54
BFA.7.3 Precursor-corpus admissibility and selection bias . . . 56
BFA.7.4 Contamination closure and accessible knowledge state . . . 56
BFA.7.5 Isolation evidence levels . . . 57
BFA.7.6 Two-phase sealed recovery protocol . . . 58
BFA.7.7 Execution-environment requirements . . . 59
BFA.7.8 Canaries, logs, and limits of negative evidence . . . 60
BFA.7.9 Mathematical verification after provenance classification . . . 60
BFA.7.10 The BFA independent-recovery certificate . . . 60
BFA.7.11 Current BFA status and permitted research within this context . . . 61
BFA.7.12 Protocol closure and handoff . . . 61

BFA.8 Adversarial Falsification Semantics . . . 62
BFA.8.1 Purpose, scope, and non-claim . . . 62
BFA.8.2 Negative-evidence lifecycle and typed subjects . . . 62
BFA.8.3 Source identity, proof invalidity, and source-theorem falsity . . . 64
BFA.8.4 Route instances, theorem truth, and strength-indexed propagation . . . 64
BFA.8.5 Shared dependencies, alternate routes, and repair identity . . . 65
BFA.8.6 Audit-expanded DAG for node, edge, and joint-composition obligations . . . 66
BFA.8.7 Formal-artifact and semantic-interpretation failures . . . 67
BFA.8.8 Source-to-target interpretation failure . . . 67
BFA.8.9 Quantifier-aware source and target falsification . . . 68
BFA.8.10 Navier–Stokes adversarial priority map . . . 68
BFA.8.11 Existing v21 blocker/cut calculus and BFA adversarial semantics . . . 69
BFA.8.12 Adversarial verdict record and no-scalar-collapse rule . . . 69
BFA.8.13 BFA.8 closure and handoff . . . 70
BFA.9 Critical-Cut Audit . . . 71
BFA.9.1 Purpose, scope, and audit ceiling . . . 71
BFA.9.2 Normative source graph and two target slices . . . 71
BFA.9.3 Four notions that must remain distinct . . . 72
BFA.9.4 BFA-local vector criticality profile . . . 73
BFA.9.5 Exact load-bearing regions of the source route . . . 74
BFA.9.6 Trivial terminal cuts versus informative interior cuts . . . 75
BFA.9.7 Target-specific interpretation instances . . . 75
BFA.9.8 Structural audit profile of the principal candidates . . . 76
BFA.9.9 Joint-composition obligations are first-class audit objects . . . 77
BFA.9.10 Prospective structural frontiers . . . 77
BFA.9.11 Whole-space and periodic frontier records . . . 78
BFA.9.12 Retrospective causal blockers from verifier states . . . 78
BFA.9.13 Adverse-result ceiling for critical objects . . . 79
BFA.9.14 Audit execution order exported to BFA.10–BFA.12 . . . 79
BFA.9.15 BFA.9 critical-cut verdict . . . 80
BFA.9.16 BFA.9 closure and handoff . . . 81
BFA.10 Finite-to-Infinite / Completion Stress Test . . . 82
BFA.10.1 Purpose, audit resolution, and non-promotion ceiling . . . 82
BFA.10.2 Exact source completion contract . . . 83
BFA.10.3 Stress Test 0: stage-zero initialization and consumer-domain closure . . . 84
BFA.10.4 Stress Test I: one-cycle residual improvement . . . 84
BFA.10.5 Stress Test II: one common domain for all finite stages . . . 85
BFA.10.6 Stress Test III: stage-independent derivative loss . . . 86
BFA.10.7 Stress Test IV: exact hypothesis crosswalk into the shared summation theorem . . . 86
BFA.10.8 Stress Test V: shrinking cutoffs, local finiteness, and summability . . . 87
BFA.10.9 Stress Test VI: nonlinear residual comparison and quantifier order . . . 87
BFA.10.10 Stress Test VII: the flat remainders are not illegally summed . . . 88
BFA.10.11 Stress Test VIII: divergence, representatives, support, and overlap . . . 88
BFA.10.12 Stress Test IX: preservation of the local output class . . . 89
BFA.10.13 Interchange audit . . . 89
BFA.10.14 Adversarial audit matrix . . . 90
BFA.10.15 Strongest current completion verdict . . . 90
BFA.10.16 Residual open obligations and falsification handles . . . 91
BFA.10.17 Generic extraction seeds, without promotion . . . 92
BFA.10.18 BFA.10 closure and BFA.11 handoff . . . 93
BFA.11 Terminal Return Stress Test . . . 94
BFA.11.1 Purpose, inherited input, and audit ceiling . . . 94
BFA.11.2 Exact source contract for the terminal region . . . 95

BFA.11.3 Stress test of O_{10}: localization without destroying the singular core . . . 95
BFA.11.4 Stress test of O_{11}: terminal force jets and smooth compact extension . . . 96
BFA.11.5 Stress test of O_{12}: energy and dissipation . . . 98
BFA.11.6 Stress test of O_{13}: comparison with an arbitrary smooth bounded-energy competitor . . . 98
BFA.11.7 Stress test of O_{14}: arbitrary positive viscosity . . . 100
BFA.11.8 Stress test of O_{15}: periodic transport and pressure periodicity . . . 100
BFA.11.9 Statement-level Clay target audit and lifecycle firewall . . . 102
BFA.11.10 Adversarial terminal audit matrix . . . 103
BFA.11.11 Open obligations retained after BFA.11 . . . 103
BFA.11.12 BFA.11 verdict and BFA.12 handoff . . . 103
BFA.12 Paper–Lean–Problem Triangular Audit . . . 105
BFA.12.1 Purpose, resolution, and nonpromotion ceiling . . . 105
BFA.12.2 Eight objects in the triangular audit . . . 105
BFA.12.3 Reference–submission signature audit . . . 106
BFA.12.4 The paper and the exposed formal theorem are not equivalent statements . . . 107
BFA.12.5 Cross-language semantic bridge ledger . . . 108
BFA.12.6 Formal whole-space target versus Clay-C . . . 109
BFA.12.7 Formal periodic target versus corrected Clay-D . . . 110
BFA.12.8 Paper-to-formal relation: projection, not equivalence . . . 110
BFA.12.9 Target-consequence commutativity of the triangle . . . 111
BFA.12.10 Formal bridge implementation and witness specificity . . . 112
BFA.12.11 Source identity, metadata, and replay separation . . . 112
BFA.12.12 Independent third-party formal-audit evidence . . . 113
BFA.12.13 Adversarial mismatch catalogue . . . 113
BFA.12.14 Triangular audit matrix . . . 114
BFA.12.15 BFA.12 verdict and exact claim ceiling . . . 115
BFA.13 Generic Mathematical Extraction . . . 116
BFA.13.1 Purpose, extraction ceiling, and reaudited classification . . . 116
BFA.13.2 Frozen-v21 and classical-overlap screen . . . 118
BFA.13.3 Extraction I: separated-filter residual limit closure . . . 118
BFA.13.4 Extraction II: fixed-stage filtered residual closure . . . 119
BFA.13.5 Extraction III: monotone defect improvement as a research schema . . . 120
BFA.13.6 Extraction IV: adaptive-localizer completion remains quarantined . . . 120
BFA.13.7 Extraction V: terminal jet realization is a contract, not a new theorem . . . 121
BFA.13.8 Extraction VI: joint-obligation vertices are a v21 normal form . . . 121
BFA.13.9 Extraction VII: cross-artifact semantic projection is a Search/Platform specialization . . . 122
BFA.13.10 Extraction VIII: negative-evidence subject typing is a Chapter 8 specialization . . . 122
BFA.13.11 Source-specific material that does not survive extraction . . . 122
BFA.13.12 Reaudited extraction matrix . . . 123
BFA.13.13 Research-program consequences after re-audit . . . 124
BFA.13.14 Status vector, closure, and handoff . . . 124
BFA.14 Promotion / Non-Promotion Matrix . . . 125
BFA.14.1 Purpose, constitutional boundary, and decision semantics . . . 125
BFA.14.2 Three gate families: owner admission, novelty, and adoption . . . 126
BFA.14.3 Mathematical-candidate decisions . . . 128
BFA.14.4 Audit, proof-semantic, and infrastructure decisions . . . 129
BFA.14.5 Source-specific and document-level decisions . . . 130
BFA.14.6 Canonical Promotion / Non-Promotion Matrix . . . 130
BFA.14.7 Promotion debt and successor-entry conditions . . . 131
BFA.14.8 BFA.14 status vector and final handoff . . . 132
BFA.15 Final Verdict, Open Obligations, and Explicit Non-Claims . . . 133
BFA.15.1 Purpose and closure discipline . . . 133
BFA.15.2 Final six-axis verdict . . . 133

BFA.15.3 What BFA has actually established . . . 136
BFA.15.4 Open obligations after closure . . . 136
BFA.15.5 Explicit non-claims . . . 138
BFA.15.6 Reopen triggers and immutable-history rule . . . 139
BFA.15.7 Global cross-section re-audit closure . . . 139
BFA.15.8 Final release record . . . 140

BFA.1 Scope, Authority, and Constitutional Boundary

BFA.1.1 Role, release relation, and local authority

Declaration BFA.1.1 (BFA document role)**.** Appendix BFA is a *post-v21 Problem-Interface research supplement* for AK High-Dimensional Projection Structural Theory (AK-HPDST). Its purpose is to audit an external Navier–Stokes finite-time blow-up proof candidate by means of source-faithful reconstruction, typed proof-DAG analysis, adversarial verification, falsification semantics, benchmark-quarantined independent recovery, and controlled extraction of possible domain-independent mathematics.

Appendix BFA is not part of the frozen AK-HPDST v21.0.0 canonical source set. It therefore creates no retroactive v21 theorem identity, no v21 Claim UID, no migration of a frozen v21 statement, and no modification of the final v21 Claim Register. Its baseline relation is

$$\mathsf{parent_baseline} = \mathsf{AK\text{-}HPDST\ v21.0.0}, \qquad \mathsf{release_relation} = \mathsf{external_research_supplement}.$$

Remark BFA.1.2 (Why the supplement is external to the frozen v21 release)**.** The v21 final Claim Register closes the canonical source set and separates registration, package release, publication, and mathematical proof [7]. Adding Appendix BFA directly to that already frozen source set would blur the release boundary. BFA therefore begins in a local research-supplement namespace. Any later integration into a successor release requires a fresh migration, ownership, dependency, source-binding, consumer, and Claim-Register audit.

Definition BFA.1.3 (BFA primary layer)**.** The primary layer of Appendix BFA is

$$\mathsf{ProblemInterfaceResearchSupplement}.$$

Its permitted local statement roles are:

(i) source-binding definition;

(ii) target-binding definition;

(iii) proof-architecture reconstruction;

(iv) source-faithful conditional theorem;

(v) compatibility theorem;

(vi) audit obligation;

(vii) route-rejection or route-falsification theorem;

(viii) counterexample or failure witness;

(ix) benchmark-quarantine rule;

(x) formalization-interpretation audit;

(xi) generic-theorem candidate;

(xii) promotion or non-promotion assessment;

(xiii) explicit non-claim.

No local BFA role is a stable-Core role merely by appearing in this appendix.

Definition BFA.1.4 (BFA statement record and axis separation)**.** Every theorem-facing BFA item is represented by a local record

$$\begin{aligned}\mathfrak{C}_{\mathrm{BFA}} = (&\mathsf{id}, \mathsf{role}, \mathsf{evidence}, \mathsf{invocation}, \mathsf{conformance}, \mathsf{verifier}, \mathsf{workflow},\\ &\mathsf{baseline_relation}, \mathsf{owner}, \mathsf{C_{src}}, \mathsf{C_{tgt}}, \mathsf{Hyp}, \mathsf{Out},\\ &\mathsf{Fail}, \mathsf{A}).\end{aligned}$$

The fields are independent. In particular, successful workflow, correct serialization, publication, repository availability, formal compilation, source recognition, or successor-candidate status does not change the mathematical evidence field. The local baseline_relation is one of

$$\mathrm{consumes_frozen_v21}, \qquad \mathrm{bfa_local}, \qquad \mathrm{successor_candidate},$$

and is neither a v21 migration status nor a promotion act.

Policy 1 (BFA-TYPE-01: no untyped theorem-facing consumption)**.** *A BFA statement may be consumed as theorem-facing evidence only when its source type, target type, exact hypotheses, exact output strength, evidence status, failure semantics, owner, and immutable artifact closure are all resolved. Missing required fields yield* undefined*; favorable fields elsewhere in the record do not compensate for the omission.*

BFA.1.2 Frozen v21 authority and noncompetition

Policy 2 (BFA-OWN-01: canonical-owner noncompetition)**.** *Appendix BFA consumes AK-HPDST v21.0.0 only through its frozen canonical owners [1, 3, 4, 5, 7]. In particular:*

(i) Main Chapters 1–10 retain canonical authority over v21 generic Core, Interface, verifier, Bridge/Return, Validation, and governance rules;

(ii) the v21 Problem Interface appendices retain authority over the existing Navier–Stokes exploration, soundness, and proof-first architecture;

(iii) Appendix UB, Search/Platform, and Companion retain authority over their registered causal-blocker, target-cut, route, replay, and execution semantics;

(iv) Appendix CR-v21 retains authority over the frozen v21 identity, admissible-use, and release-closure records.

A BFA restatement of a v21 rule is a consumer restatement or proof locator; it does not create a competing theorem authority.

Proposition BFA.1.5 (Conservative post-release attachment)**.** *Suppose every BFA statement uses a fresh local identity, consumes every v21 premise at no stronger than its frozen source-supported strength, and exports no conclusion back into the frozen v21 dependency DAG. Then adjoining BFA as a research supplement is conservative with respect to AK-HPDST v21.0.0.*

Proof. The frozen v21 theorem identities, source artifacts, hypotheses, target strengths, dependency edges, and Claim-Register states remain unchanged. Every BFA statement is downstream of those immutable records and carries a new local identity. Since no BFA conclusion is imported as a premise of a frozen v21 theorem, no truth condition or admissible-use condition of the v21 release is altered. Hence the attachment is conservative. □

BFA.1.3 External source family, target sovereignty, and authority partition

Definition BFA.1.6 (Provisional benchmark source family)**.** The initial benchmark family for Appendix BFA is the September 2026 OpenAI Navier–Stokes finite-time singularity release, including the public announcement, the analytic paper, and the associated Lean repository [11, 12, 13]. Before any of these artifacts is consumed

theorem-facing, BFA.2 must split the family into exact, immutable artifact bindings with separate locators and roles. The family label

$$\mathsf{SRC}_{\text{OAI-NS-2026}}$$

is therefore a provisional provenance label, not a theorem identity and not a claim that all member artifacts are already semantically equivalent.

Definition BFA.1.7 (Source-sovereignty record)**.** A theorem-facing BFA external-source record is

$$\begin{aligned}\mathfrak{S} = (&\mathsf{source_id}, \mathsf{source_authority}, \mathsf{artifact_id}, \mathsf{version}, \mathsf{locator},\\ &\mathsf{exact_statement}, \mathsf{Hyp}, \mathsf{Concl}, \mathsf{permitted_use},\\ &\mathsf{forbidden_stronger_reading}, \mathsf{evidence_status}, \mathsf{provenance},\\ &\mathsf{canonicalization}, \mathsf{hash}, \mathsf{retrieved_at}).\end{aligned}$$

A title, URL, DOI, repository name, hash, or institutional announcement without an exact theorem-facing statement and locator is provenance, not sufficient theorem evidence. Freshness is recorded separately from frozen artifact identity.

Definition BFA.1.8 (Official-target sovereignty record)**.** A BFA official-target record is

$$\begin{aligned}\mathfrak{T} = (&\mathsf{target_id}, \mathsf{target_authority}, \mathsf{artifact_id}, \mathsf{locator},\\ &\mathsf{exact_statement}, \mathsf{domain}, \mathsf{quantifiers}, \mathsf{Hyp}, \mathsf{Concl},\\ &\mathsf{solution_class}, \mathsf{strength}, \mathsf{hash}).\end{aligned}$$

For the Millennium Prize comparison, the official Clay/Fefferman formulation owns the benchmark target semantics; an announcement or analytic paper may claim to establish that target but does not rewrite it [14].

Policy 3 (BFA-SRC-01: field-relative source authority)**.** *BFA uses no single global "highest-authority" ordering across heterogeneous artifacts. Authority is field-relative:*

(i) the official problem formulation controls the official target;

(ii) the analytic paper controls the exact analytic theorem and proof text attributed to that paper;

(iii) the formal repository and checked environment control the exact formal theorem actually represented there;

(iv) an institutional announcement controls provenance and the claims made in that announcement, but not an unstated analytic lemma;

(v) secondary summaries are explanatory evidence only unless separately bound as an authoritative mathematical source.

If artifacts disagree within a consumed field, the weakest exact source-supported reading governs until the discrepancy is repaired. If two authority domains cannot yet be related by a proved interpretation edge, the cross-artifact claim is $\mathrm{undefined}$ *rather than silently resolved by precedence.*

Remark BFA.1.9 (Current external-recognition state is lifecycle metadata)**.** At the time of this re-audit, OpenAI publicly states that its analytic proof and Lean formalization establish alternatives C and D, whereas the Clay Mathematics Institute still lists Navier–Stokes among the unsolved Millennium problems. These are distinct, time-indexed institutional states; neither is copied into an atomic mathematical verifier field. BFA.2 freezes the exact artifacts and retrieval dates used by the audit.

Remark BFA.1.10 (The benchmark proof is not a premise of independent recovery)**.** The benchmark proof may be consumed by the *source-faithful reconstruction track*. It is quarantined from any track that advertises an independent AK recovery of the same target. Merely knowing the benchmark answer does not license importing its witnesses, constants, correction scheme, proof lemmas, or conclusion-derived search constraints into an independent-recovery derivation [2, 5].

BFA.1.4 Eight distinct logical tasks

Definition BFA.1.11 (BFA task separation)**.** Appendix BFA separates the following tasks:

$$\begin{aligned}
\mathsf{SourceBind} &: \text{freeze the exact external theorem sources,}\\
\mathsf{TargetBind} &: \text{freeze the exact official target semantics,}\\
\mathsf{Reconstruct} &: \text{rebuild the external proof as a typed AK-facing DAG,}\\
\mathsf{Interpret} &: \text{prove the relation from source theorem to official target,}\\
\mathsf{Verify} &: \text{independently discharge selected mathematical obligations,}\\
\mathsf{Falsify} &: \text{search for certified failures at the exact logical level,}\\
\mathsf{Extract} &: \text{isolate domain-independent theorem candidates,}\\
\mathsf{Promote} &: \text{assess successor-layer admissibility under a new governance act.}
\end{aligned}$$

No task implies another without a separately proved edge.

Proposition BFA.1.12 (Task noncollapse)**.** *None of the following implications is valid by default:*

$$\mathsf{SourceBind} \Rightarrow \mathsf{TargetBind}, \qquad \mathsf{Reconstruct} \Rightarrow \mathsf{Verify}, \qquad \mathsf{Verify} \Rightarrow \mathsf{Interpret},$$
$$\mathsf{Falsify} \Rightarrow \mathsf{TargetFalsity}, \qquad \mathsf{Extract} \Rightarrow \mathsf{Promote}.$$

Proof. Source identity, target identity, dependency reconstruction, mathematical truth, source-to-target interpretation, route failure, target negation, abstraction, and canonical promotion answer different predicates and may have different authorities. No conversion theorem is part of their definitions. □

BFA.1.5 Source-faithful reconstruction versus independent recovery

Definition BFA.1.13 (Source-faithful reconstruction track)**.** The source-faithful reconstruction track, denoted $\mathsf{SFR}_{\mathrm{BFA}}$, may consume the exact external analytic and formal proof artifacts. Its target is to determine whether the displayed external proof can be represented as a complete typed route

$$\mathsf{ExactExternalSource} \longrightarrow \mathsf{AnalyticProofDAG} \longrightarrow \mathsf{Completion} \longrightarrow \mathsf{SourceTheorem}$$
$$\longrightarrow \mathsf{TargetInterpretation} \longrightarrow \mathsf{OfficialTarget},$$

with every required source, target, hypothesis, strength, artifact, interpretation edge, and failure route made explicit.

Definition BFA.1.14 (Independent recovery track)**.** The independent recovery track, denoted $\mathsf{IR}_{\mathrm{BFA}}$, is a benchmark-quarantined program that asks whether a frozen precursor package, excluding the benchmark proof and every conclusion-derived witness or equivalent oracle, suffices to derive the same official target by an independently auditable route.

Policy 4 (BFA-KTR-01: reconstruction is not independent recovery)**.** *A successful* $\mathsf{SFR}_{\mathrm{BFA}}$ *run is never reported as an independent AK proof. Conversely, failure or incompleteness of* $\mathsf{IR}_{\mathrm{BFA}}$ *does not weaken a successful source-faithful reconstruction of a valid external theorem.*

Remark BFA.1.15 (Initial status of the independent track)**.** At BFA inception, the independent recovery track has

$$\mathsf{workflow} = \mathrm{proposed}, \qquad \mathsf{evidence} = \mathrm{spec}, \qquad \mathsf{recovery_relation} = \mathrm{not_yet_recovered}.$$

No Clay-level AK-only Navier–Stokes proof is asserted by these values.

BFA.1.6 Status sorts and evidence discipline

Definition BFA.1.16 (BFA audit-state vector)**.** For a fixed external target, BFA records the state vector

$$\mathbf{V}_{\mathrm{BFA}} = (s_{\mathrm{src}}, s_{\mathrm{target}}, s_{\mathrm{recon}}, s_{\mathrm{interp}}, s_{\mathrm{math}}, s_{\mathrm{formal}}, s_{\mathrm{external}}, s_{\mathrm{extract}}, s_{\mathrm{promotion}}).$$

The coordinates mean, respectively:

(i) source identity and source binding;

(ii) exact official-target binding;

(iii) source-faithful proof-DAG reconstruction;

(iv) source-theorem to official-target interpretation;

(v) independent mathematical audit of required nodes;

(vi) paper–formalization interpretation and formal verification;

(vii) external consensus, institutional recognition, or other lifecycle state;

(viii) generic-mathematics extraction;

(ix) successor-release promotion readiness.

The coordinates are independent typed fields. No favorable value is copied from one coordinate to another.

Policy 5 (BFA-STATUS-01: atomic verifier compatibility)**.** *Whenever a BFA node is consumed as a theorem-facing AK obligation, its atomic verifier status is exactly one of*

$$\text{pass}, \quad \text{reject}, \quad \text{undefined}, \quad \text{not_invoked}.$$

External publication, successful Lean compilation, repository availability, review attention, citation count, institutional endorsement, Clay lifecycle status, or BFA workflow completion is not an atomic mathematical pass *unless a declared verifier theorem converts that evidence into the consumed mathematical predicate.*

Remark BFA.1.17 (No scalar confidence substitution)**.** A probability, confidence score, agent vote, numerical reserve, or informal credibility estimate may be stored as advisory metadata. It does not replace any required atomic verifier status, exact interpretation edge, or target-relative result field.

BFA.1.7 Compatibility is weaker than proof

Definition BFA.1.18 (AK compatibility)**.** An external proof route is *AK-compatible* when its theorem-facing nodes can be typed without contradiction against the frozen AK-v21 constitutional rules, its dependencies can be represented by an admissible proof DAG or separately justified recursive semantics, every finite-to-infinite promotion is supported by an explicit theorem, every source-to-target change of semantics has an explicit interpretation edge, and the terminal external conclusion is reached only at its declared target strength.

Proposition BFA.1.19 (Compatibility nonpromotion)**.** *AK compatibility of an external proof does not establish the truth of that proof.*

Proof. Compatibility is a type and dependency property. A false but well-typed lemma can occur inside a perfectly well-formed proof DAG. Truth additionally requires each consumed theorem-facing obligation to pass its mathematical verifier. Hence compatibility is necessary for BFA composition but is not sufficient for mathematical correctness. □

Corollary BFA.1.20 (No-contradiction is not validation)**.** *Failure to find an AK-v21 structural contradiction with an external Navier–Stokes proof is not evidence that all analytic lemmas in that proof are correct. It establishes only that no audited AK structural prohibition has been triggered on the examined route.*

BFA.1.8 Adversarial nonclosure, route falsification, and target falsification

Definition BFA.1.21 (BFA negative-result classes)**.** BFA distinguishes at least the following negative-result classes:

$$\begin{aligned} F_{\text{src}} &: \text{certified source identity, statement, or binding failure,} \\ U_{\text{route}} &: \text{route nonclosure from missing or ill-typed required evidence,} \\ F_{\text{route}} &: \text{a well-typed required mathematical route obligation is false,} \\ F_{\text{formal}} &: \text{formal proof failure or paper–formalization semantic mismatch,} \\ F_{\text{interp}} &: \text{source theorem does not establish the claimed official target relation,} \\ F_{\text{target}} &: \text{a proof of the negation of the exact official target.} \end{aligned}$$

The symbol U_{route} is deliberately not called falsification: absence of evidence and evidence of falsity are distinct states.

Theorem BFA.1.22 (Route nonclosure and route falsification are distinct, and neither implies target falsification)**.** *Let $\mathcal{R}$ be one declared proof route to target T.*

(i) If a required node is missing, unresolved, or ill-typed, then $\mathcal{R}$ is nonclosing and the requested route result is undefined *at that node.*

(ii) If a required, well-typed mathematical obligation is certified false, then $\mathcal{R}$ is rejected at every strength requiring that obligation.

(iii) Neither (i) nor (ii) alone implies $\neg T$.

Proof. Part (i) is non-evaluability rather than falsity. Part (ii) removes the declared route from the admissible derivations of the requested target strength. In either case, another route may exist. Target falsity requires a valid proof of the exact logical negation $\neg T$, not merely failure of one attempted derivation. □

Definition BFA.1.23 (Quantifier-aware target falsification)**.** A BFA claim of F_{target} is admissible only when the evidence proves the logical negation of the exact target record $\mathfrak{T}$. Consequently, a single counterexample is sufficient only when the quantified form of the target makes it sufficient. In particular, for an existential target of the form

$$T \equiv \exists x \in \mathcal{X}\ P(x),$$

invalidating one proposed witness proves only failure of that witness or route; target falsification requires a proof of

$$\forall x \in \mathcal{X}\ \neg P(x).$$

The same rule applies to the exact quantifier structure of Clay alternatives C and D once bound in BFA.2.

Policy 6 (BFA-FALS-01: weakest exact negative conclusion)**.** *Every negative BFA result is reported at the weakest exact logical level established by the evidence. In particular:*

(i) missing evidence yields undefined*, not falsity;*

(ii) a false required route lemma yields route rejection at the strengths depending on that lemma, not automatic target falsity;

(iii) a certified nonzero obstruction is reported as the obstruction supplied by its governing theorem and is not renamed missing evidence;

(iv) a paper–Lean mismatch invalidates the claimed correspondence at the affected strength, not automatically the analytic theorem;

(v) failure of the source-to-target interpretation edge blocks the claimed target Return without rewriting the source theorem;

(vi) only a proof of the exact target negation may support F_{target}.

BFA.1.9 Use of existing failure-cut and target-cut calculus

Policy 7 (BFA-CUT-01: no duplicate cut theory)**.** *Appendix BFA does not introduce a competing generic theory of minimal causal failure cuts. Where a finite BFA target slice satisfies the hypotheses of the registered v21 causal-blocker or target-cut calculus, BFA consumes that calculus at its frozen strength [4, 6]. BFA distinguishes:*

(i) a target cut*, which meets every complete source-to-target route;*

(ii) a causal blocker antichain*, which contains the earliest required nodes currently carrying* reject *or* undefined*;*

(iii) a BFA adversarial audit set*, which is a selected set of high-leverage obligations chosen for verification or falsification and need not be minimal or an antichain.*

BFA-local novelty, if any, must lie in the mathematical interpretation of adversarial failures for external PDE proofs, not in renaming an existing graph-theoretic cut result.

Remark BFA.1.24 (Critical nodes versus new mathematics)**.** Identifying completion, residual flatness, forcing regularity, energy control, uniqueness, or final problem interpretation as critical nodes is an audit result. It does not by itself create a new theorem. New mathematics begins only when BFA proves a new implication, counterexample, substitution theorem, completion theorem, or domain-independent failure principle.

BFA.1.10 Core-P, Core-DG, and Bridge invocation discipline

Policy 8 (BFA-CORE-01: no forced Core invocation)**.** *Core-P and Core-DG are invoked in BFA only when an exact, theorem-backed Interface supplies the required object, morphism, variance, preservation or reflection property, and consumer strength. If the external Navier–Stokes proof route is analytic and does not require persistence or finite-diagram mathematics, then the invocation fields*

$$\mathsf{Invoke(Core\text{-}P)} = \mathrm{not_invoked}, \qquad \mathsf{Invoke(Core\text{-}DG)} = \mathrm{not_invoked}$$

are valid and preferred readings. A layer or category itself is never identified with a status token.

Proposition BFA.1.25 (No ornamental Core promotion)**.** *Adding persistence, Ext, finite-diagram, descent, or Collapse terminology to an analytic Navier–Stokes route without a theorem proving preservation or reflection of the consumed PDE predicate cannot strengthen the route.*

Proof. The added objects inhabit different source or target types. Without a proved Interface, they do not compose with the analytic route and therefore cannot discharge any missing PDE obligation. The insertion is mathematically ornamental rather than theorem-bearing. □

BFA.1.11 Analytic proof, formal proof, and official target

Definition BFA.1.26 (Triangular interpretation obligation)**.** For a benchmark supplied as an analytic paper and a formal artifact and claimed to establish an official external problem, BFA requires three separately auditable directed relations

$$\mathsf{Paper} \xrightarrow{\rho_{PF}} \mathsf{Formalization}, \qquad \mathsf{Formalization} \xrightarrow{\rho_{FT}} \mathsf{OfficialTarget}, \qquad \mathsf{Paper} \xrightarrow{\rho_{PT}} \mathsf{OfficialTarget}.$$

No ρ is assumed to be an equivalence. Each relation records its direction, theorem statement, parameter conventions, solution class, forcing class, domain, quantifiers, conclusion strength, artifact identities, and prohibited stronger readings. A reverse arrow requires separate evidence.

Policy 9 (BFA-FORMAL-01: formalization nonpromotion)**.** *Successful parsing, compilation, kernel checking, or replay of a formal proof establishes only the formal theorem actually checked in the pinned formal environment. It establishes the intended analytic or Clay-level statement only after the corresponding interpretation relation has passed at the required strength. Conversely, a mismatch of interpretation does not by itself falsify the formal theorem in its own semantics.*

BFA.1.12 Generic extraction and promotion quarantine

Definition BFA.1.27 (Generic extraction candidate)**.** A BFA generic extraction candidate is a proposed abstraction obtained by replacing source-specific notation and assumptions with explicitly declared abstract data and hypotheses. The abstraction step itself proves neither hypothesis weakening nor domain generalization. Until an AK-owned proof establishes the generalized statement, its role is

$$\mathrm{generic_theorem_candidate},$$

not theorem and not stable-Core result.

Definition BFA.1.28 (BFA promotion ladder)**.** The permitted successor-assessment ladder is

$$\mathsf{BFA\text{-}local\ candidate} \longrightarrow \mathsf{TechnicalExtension\ candidate} \longrightarrow \mathsf{SuccessorCore\ candidate}.$$

The arrows are assessment transitions, not theorem implications and not canonical promotions.

Policy 10 (BFA-PROM-01: promotion quarantine)**.** *No BFA-local result enters a stable Core merely because it is useful, general-looking, successful on Navier–Stokes, formally verified, or reusable inside BFA. The following minimum gates apply.*

For a Technical Extension candidate*, BFA requires at least:*

(i) a source-independent statement at a declared abstract scope;

(ii) an AK-owned proof of that abstract statement;

(iii) explicit failure boundaries;

(iv) one independently typed non-Navier–Stokes substitution or an equivalent theorem proving the advertised extension scope.

For a successor-Core candidate*, BFA additionally requires:*

(a) demonstrated nonredundancy with the existing Core and generic Bridge calculus;

(b) at least two independently typed source families beyond the originating benchmark, unless one uniform theorem proves the claimed source-independence over a declared ambient class;

(c) counterexamples or sharp failure/no-go boundaries sufficient to fix the theorem's scope;

(d) complete consumer, migration, ownership, governance, and Claim-Register audit in the successor release.

Until those gates are passed, the candidate remains BFA-local or Technical Extension candidate only.

Remark BFA.1.29 (Residual Improvement Completion)**.** The anticipated Residual Improvement Completion (RIC) pattern is therefore introduced later in BFA, if justified by the source audit, only as a generic candidate. BFA.1 intentionally gives RIC no theorem status, no Technical Extension status, and no Core status.

BFA.1.13 Local identities, frozen-input discipline, and governance

Definition BFA.1.30 (BFA 1.0 local UID namespace)**.** During construction of Appendix BFA 1.0, local theorem-facing records use identities of the form

$$\mathrm{ak\text{-}hpdst{:}rs\text{-}bfa{:}1.0{:}BFA\text{-}}\langle\mathrm{token}\rangle.$$

The suffix 1.0 identifies the appendix construction line, not an independently released sub-version of an individual section. Section edits, repairs, and re-audits performed before the BFA 1.0 freeze do not create artificial local release numbers. They are tracked in the construction log. At the BFA 1.0 freeze, the surviving local identities become immutable for that release. Any post-freeze mathematical change that alters theorem-facing semantics requires a successor release and an explicit predecessor relation.

Such identifiers are local research-supplement identities. They are not v21 Claim UIDs and do not authorize theorem-level consumption outside the exact BFA source and evidence record.

Policy 11 (BFA-INPUT-01: frozen external-input identity)**.** *Every external artifact consumed by a replayable BFA audit is frozen by exact artifact identity, semantic locator, retrieval record, canonicalization, and permitted-use policy. A later fresh version is a new input and therefore a new run identity unless a separately proved source-equivalence theorem licenses substitution. Freshness never silently replaces identity [6].*

Policy 12 (BFA-GOV-01: no backflow into frozen v21)**.** *No BFA result, including a successful reconstruction, mathematical audit, formal verification, route falsification, target interpretation, or generic extraction, is imported backward as a premise of a frozen v21 theorem. A successor release may consume a BFA result only through a fresh migration, ownership, dependency, and registration record.*

BFA.1.14 Explicit non-claims

Declaration BFA.1.31 (BFA.1 explicit non-claims)**.** At the completion of BFA.1, none of the following is asserted:

(i) AK-HPDST independently proves the Navier–Stokes Millennium Prize Problem;

(ii) OpenAI's analytic proof has been completely independently verified by AK;

(iii) OpenAI's Lean artifact has been proved semantically identical to every theorem statement in the analytic paper and official Clay formulation;

(iv) the provisional OpenAI source-family label itself proves semantic equivalence among the announcement, paper, and formal repository;

(v) the existence of one valid forced blow-up construction proves blow-up for the unforced problem;

(vi) failure or nonclosure of one OpenAI proof route proves finite-time blow-up impossible;

(vii) invalidating one proposed witness falsifies an existential Clay target;

(viii) Core-P or Core-DG is necessary for the external blow-up construction;

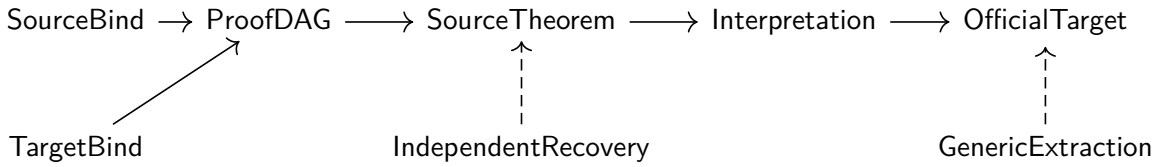
(ix) a BFA-local completion or defect pattern is a new Core theorem;

(x) external institutional recognition or non-recognition is itself a mathematical verifier result;

(xi) Appendix BFA modifies the frozen AK-HPDST v21.0.0 release.

BFA.1.15 BFA dependency and section contract

Definition BFA.1.32 (BFA downstream contract)**.** Every later BFA section shall preserve the following dependency discipline:



The source-faithful track may consume the benchmark proof only through its frozen source bindings. The independent-recovery branch may consume the frozen target identity and precursor-authorized hypotheses but not the benchmark proof or conclusion-derived witnesses. Generic extraction may use only theorem-backed results at their exact strength and may not feed backward into an earlier audit verdict.

Theorem BFA.1.33 (BFA.1 constitutional closure)**.** *Under the records, definitions, policies, and non-claims of BFA.1, Appendix BFA has a well-typed post-v21 research-supplement boundary with the following properties:*

(i) the frozen v21 release is not mutated;

(ii) external source authority, official-target authority, and AK theorem authority remain distinct;

(iii) source-faithful reconstruction and independent recovery remain distinct and benchmark quarantine is enforceable;

(iv) compatibility, reconstruction, interpretation, mathematical audit, formalization, external lifecycle status, extraction, and promotion remain distinct axes;

(v) route nonclosure, route falsification, interpretation failure, and target falsification remain distinct negative-result classes;

(vi) target falsification is quantifier-aware and requires a proof of the exact target negation;

(vii) existing v21 failure-cut and target-cut mathematics is reused rather than duplicated;

(viii) Core-P and Core-DG may remain uninvoked on a purely analytic route;

(ix) paper, formalization, and official-target relations are directional and strength-typed rather than presumed equivalent;

(x) generic extraction is quarantined from Technical Extension or stable-Core promotion until the corresponding independent proof and successor-governance gates are passed;

(xi) every replayable external input is identity-frozen separately from freshness.

Consequently, later BFA sections may conduct an aggressive audit of an external Clay-level proof candidate without prejudging either its correctness or its failure, without confusing missing evidence with falsity, and without inflating a source-specific result into an AK Core theorem.

Proof. Items (i)–(xi) are exactly the ownership, source/target sovereignty, task separation, status, quarantine, negative-result, cut-reuse, Core-invocation, interpretation, promotion, and external-input constraints established above. They are mutually compatible because each acts on a declared typed field or dependency boundary. The final sentence follows because neither positive nor negative benchmark outcomes are imported as undeclared premises of the governing audit semantics. □

Remark BFA.1.34 (BFA.1 final status)**.** BFA.1 is a governance and audit-architecture section. Its mathematical content consists of typed separation results, the route nonclosure/route falsification theorem, and the quantifier-aware target-falsification criterion. It does not yet evaluate the OpenAI Navier–Stokes proof itself. Exact target binding and exact source-artifact decomposition begin in BFA.2, and analytic proof-DAG reconstruction begins in BFA.3.

BFA.2 Exact External Target and Source Binding

BFA.2.1 Purpose and binding discipline

Declaration BFA.2.1 (BFA.2 binding objective)**.** BFA.2 fixes the exact external target semantics and the exact external source statements consumed by the later BFA audit. It does not verify the analytic proof, certify an independent Lean replay, or infer official Millennium-Prize acceptance. Its task is logically prior to all of those questions:

$$\begin{aligned} \text{official target} &\neq \text{analytic source statement},\\ \text{analytic source statement} &\neq \text{formal source statement},\\ \text{formal reference statement} &\neq \text{formal proof object},\\ \text{formal proof object} &\neq \text{institutional announcement},\\ \text{institutional announcement} &\neq \text{external lifecycle status}. \end{aligned}$$

Every later BFA route must name exactly which object it consumes, at which version or commit, and through which directional interpretation edge.

Definition BFA.2.2 (BFA binding mode)**.** Every external BFA record has exactly one primary binding mode:

(i) official_target: an authoritative statement of the external problem to be met;

(ii) analytic_source: a theorem or corollary in the analytic paper;

(iii) formal_reference: an independently stated formal target used for comparison;

(iv) formal_source: a proved declaration in the formal proof repository;

(v) provenance_claim: an institutional announcement or metadata statement about what is claimed, released, or formalized;

(vi) lifecycle_status: an external administrative or community status, such as whether a problem is currently listed as solved or unsolved.

No mode inherits the authority of another merely because the artifacts are co-published, cross-linked, or use the same theorem name.

Policy 13 (BFA-BIND-01: locator, identity, snapshot, and replay are distinct)**.** *A URL, repository branch name such as* main*, theorem title, or displayed filename is a locator, not by itself an immutable artifact identity. BFA therefore separates:*

locator_resolved, identity_frozen, local_snapshot_closed, independent_replay.

A Git commit may resolve immutable repository identity even when no BFA-local mirror or independent build has yet been made. Conversely, a local copy with no source identity does not establish provenance. Final theorem-facing replay requires every consumed identity, snapshot, environment, and semantic locator required by the consumer contract.

Definition BFA.2.3 (Binding-completeness vector)**.** For an external item X, define

$$\mathsf{Bind}(X) := (I_X, L_X, S_X, Q_X, F_X, M_X, R_X),$$

where I_X is source-identity resolution, L_X exact locator resolution, S_X exact statement resolution, Q_X quantifier/type resolution, F_X immutable remote-identity status, M_X BFA-local snapshot/environment closure, and R_X independent replay or verifier execution. Each required coordinate has atomic status

pass, reject, undefined, not_invoked.

A *semantic binding pass* means (L_X, S_X, Q_X) pass at the declared source snapshot. A *theorem-facing binding closure* additionally requires all identity/snapshot coordinates consumed by the downstream verifier. These notions are never serialized into one undifferentiated pass.

BFA.2.2 Official Clay target artifact and exact Fefferman predicates

Fix $\nu > 0$ and the three-dimensional incompressible Navier–Stokes system

$$\partial_t u + (u \cdot \nabla)u - \nu\Delta u + \nabla p = f, \qquad \nabla \cdot u = 0, \tag{BFA.2.1}$$

with $u(x, 0) = u^0(x)$. The target authority is Charles L. Fefferman's official Clay problem description, including its appended erratum [15].

Definition BFA.2.4 (Whole-space admissible datum)**.** A pair (u^0, f) is *Fefferman-admissible on* $\mathbb{R}^3$ when u^0 is smooth and divergence-free and, for every spatial multi-index α and every $K \geq 0$, there is $C_{\alpha K}$ such that

$$|\partial_x^\alpha u^0(x)| \leq C_{\alpha K}(1 + |x|)^{-K},$$

and f is smooth on $\mathbb{R}^3 \times [0, \infty)$ and, for every α, $m \geq 0$, and $K \geq 0$, there is $C_{\alpha mK}$ such that

$$|\partial_x^\alpha \partial_t^m f(x, t)| \leq C_{\alpha mK}(1 + |x| + t)^{-K}.$$

This is the BFA expansion of Fefferman conditions (4)–(5).

Definition BFA.2.5 (Whole-space physically reasonable global solution)**.** For an admissible (u^0, f), let $\mathsf{PhysSol}_{\mathbb{R}^3}(\nu, u^0, f; p, u)$ mean that p, u are smooth on $\mathbb{R}^3 \times [0, \infty)$, satisfy (BFA.2.1), incompressibility, and the initial condition, and obey the Fefferman bounded-energy condition

$$\exists C < \infty \; \forall t \geq 0 : \qquad \int_{\mathbb{R}^3} |u(x,t)|^2 \, dx < C.$$

This expands conditions (6)–(7).

Definition BFA.2.6 (Periodic admissible datum)**.** A pair (u^0, f) is *Fefferman-admissible on* $\mathbb{T}^3 = \mathbb{R}^3/\mathbb{Z}^3$ when u^0 is smooth, divergence-free, and 1-periodic in every coordinate; f is smooth and 1-periodic in every spatial coordinate; and for every α, m, K there is $C_{\alpha m K}$ such that

$$|\partial_x^\alpha \partial_t^m f(x,t)| \leq C_{\alpha m K}(1+t)^{-K} \qquad (t \geq 0).$$

This expands conditions (8)–(9).

Definition BFA.2.7 (Periodic physically reasonable global solution with erratum)**.** For a periodic admissible datum, let $\mathsf{PhysSol}_{\mathbb{T}^3}(\nu, u^0, f; p, u)$ mean that p, u are smooth on $\mathbb{R}^3 \times [0, \infty)$, satisfy (BFA.2.1), incompressibility, and the initial condition, and that both velocity and pressure are 1-periodic in each spatial coordinate for every $t \geq 0$. Periodicity of p is included because the erratum appended to the official Clay PDF states that this condition should be explicit. No whole-space energy condition is added to the periodic target. This is the corrected BFA reading of conditions (10)–(11) together with the erratum.

Definition BFA.2.8 (Fixed-viscosity Clay alternatives $\mathsf{Clay\text{-}C}_\nu$ and $\mathsf{Clay\text{-}D}_\nu$)**.** For one fixed $\nu > 0$, define

$$\mathsf{Clay\text{-}C}_\nu :\Longleftrightarrow \exists (u^0, f) \; \big[\mathsf{Adm}_{\mathbb{R}^3}(u^0, f) \wedge \neg \exists (p,u) \, \mathsf{PhysSol}_{\mathbb{R}^3}(\nu, u^0, f; p, u)\big],$$
$$\mathsf{Clay\text{-}D}_\nu :\Longleftrightarrow \exists (u^0, f) \; \big[\mathsf{Adm}_{\mathbb{T}^3}(u^0, f) \wedge \neg \exists (p,u) \, \mathsf{PhysSol}_{\mathbb{T}^3}(\nu, u^0, f; p, u)\big].$$

Here Adm abbreviates exactly the predicates of Definitions BFA.2.4 and BFA.2.6 and no stronger solution class is silently inserted.

Definition BFA.2.9 (All-viscosity BFA target portfolio)**.** The BFA comparison portfolio is

$$\mathsf{Clay\text{-}C} := \forall \nu > 0 \; \mathsf{Clay\text{-}C}_\nu, \qquad \mathsf{Clay\text{-}D} := \forall \nu > 0 \; \mathsf{Clay\text{-}D}_\nu.$$

This notation records that the OpenAI source statements are uniform over every positive viscosity. The official Fefferman alternatives are read fixed-viscosity-wise under "Take $\nu > 0$"; BFA does not use the all-viscosity wrapper to strengthen any fixed-ν implication. The current local target identities are

$$\text{bfa:1.0:target:clay-C}, \qquad \text{bfa:1.0:target:clay-D}.$$

Proposition BFA.2.10 (Exact quantifier burden for target falsification)**.** *For fixed $\nu > 0$,*

$$\neg \mathsf{Clay\text{-}C}_\nu \Longleftrightarrow \forall (u^0, f) \, \big(\mathsf{Adm}_{\mathbb{R}^3}(u^0, f) \Rightarrow \exists (p,u) \, \mathsf{PhysSol}_{\mathbb{R}^3}(\nu, u^0, f; p, u)\big),$$

and analogously for $\mathsf{Clay\text{-}D}_\nu$*. For the all-viscosity portfolio,*

$$\neg \mathsf{Clay\text{-}C} \Longleftrightarrow \exists \nu > 0 \; \neg \mathsf{Clay\text{-}C}_\nu, \qquad \neg \mathsf{Clay\text{-}D} \Longleftrightarrow \exists \nu > 0 \; \neg \mathsf{Clay\text{-}D}_\nu.$$

Thus rejection of one proposed witness, one viscosity value, or one proof route is not by itself target falsification.

Proof. This is direct classical negation of the quantifier patterns in Definitions BFA.2.8 and BFA.2.9. The statement is recorded explicitly because BFA.1 requires target falsification to match the exact quantifier structure rather than an informal "counterexample" slogan. □

Proposition BFA.2.11 (Zero initial datum is target-admissible)**.** *The choice $u^0 = 0$ lies in both official initial-data classes.*

Proof. The zero vector field is smooth, divergence-free, rapidly decaying, and 1-periodic. Hence it satisfies the initial-data parts of Definitions BFA.2.4 and BFA.2.6. □

BFA.2.3 Analytic-paper source binding: whole-space theorem

Definition BFA.2.12 (OpenAI analytic source OAI-R3)**.** Let

$$\mathsf{OAI\text{-}R3} := \text{bfa:1.0:source:oai-paper-thm1.1}$$

denote Theorem 1.1 of *Finite Time Blowup for Navier–Stokes* [12]. For every $\nu > 0$ it prints the existence of

$$f \in C_c^\infty(\mathbb{R}^3 \times (0,\infty); \mathbb{R}^3),$$

a compact $K \subset \mathbb{R}^3$, and smooth u, p on $\mathbb{R}^3 \times [0,1)$ satisfying (BFA.2.1), $u(\cdot, 0) = 0$, and fixed compact spatial support before time one, together with

$$\sup_{0 \le t < 1} \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty, \qquad \limsup_{t \uparrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty.$$

The theorem then states that there is no global smooth Navier–Stokes solution with the same force and initial datum whose kinetic energy is uniformly bounded. The semantic locator is Theorem 1.1, printed p.1.

The public PDF URL is source-resolved, but BFA has not yet produced a local cryptographic snapshot of that PDF. Accordingly the theorem statement may be reconstructed at the observed source revision, while local artifact closure remains unresolved.

Proposition BFA.2.13 (Whole-space source-to-target implication)**.** *For every $\nu > 0$, if the exact statement* OAI-R3 *is true at the bound paper revision, then* Clay-C$_\nu$ *holds. Hence the all-viscosity source theorem implies the BFA portfolio target* Clay-C.

Proof. The initial datum is zero and hence admissible by Proposition BFA.2.11. Compact spacetime support of the smooth force implies every Fefferman rapid-decay condition in Definition BFA.2.4. The final sentence of Theorem 1.1 excludes precisely a global smooth solution for the same datum and force with uniformly bounded kinetic energy, i.e. the solution class of Definition BFA.2.5. Thus the source witnesses Clay-C$_\nu$ for each $\nu > 0$. □

Remark BFA.2.14 (Statement implication is not proof verification)**.** Proposition BFA.2.13 is a theorem about the relation between two printed statements. It neither verifies the analytic proof of Theorem 1.1 nor promotes any OpenAI lemma to AK-owned mathematics.

BFA.2.4 Analytic-paper source binding: periodic corollary

Definition BFA.2.15 (OpenAI analytic source OAI-T3)**.** Let

$$\mathsf{OAI\text{-}T3} := \text{bfa:1.0:source:oai-paper-cor10.6}$$

denote Corollary 10.6 of [12]. For every $\nu > 0$ it prints the existence of a smooth force on $\mathbb{T}^3 \times [0,\infty)$, compactly supported in time, for which the periodic Navier–Stokes solution with zero initial velocity is smooth on $[0,1)$ and has

$$\limsup_{t \uparrow 1} \|U(t)\|_{L^\infty(\mathbb{T}^3)} = \infty,$$

and it states that no global smooth periodic solution exists for the same datum and force. The semantic locator is Corollary 10.6, printed p.125.

Proposition BFA.2.16 (Periodic source-to-target implication with Clay erratum)**.** *For every $\nu > 0$, if the exact statement* OAI-T3 *is true and "periodic Navier–Stokes solution" is consumed at the paper's stated torus meaning, then the source witnesses* Clay-D$_\nu$*. Hence the all-viscosity source corollary implies* Clay-D.

Proof. The zero datum is admissible. Smooth spatial periodicity and compact support in time imply the Fefferman periodic-force and temporal rapid-decay requirements. A global smooth solution on the torus lifts to smooth 1-periodic velocity and pressure on $\mathbb{R}^3$; the pressure-periodicity clause is exactly the Clay erratum incorporated in Definition BFA.2.7. The source excludes such a global smooth periodic solution. Therefore its data witness Clay-D$_\nu$. □

Audit Obligation BFA.2.17 (Periodic pressure semantics)**.** *BFA.2 records the Clay pressure-periodicity erratum as part of the target. The detailed paper-to-formalization audit must still confirm that every formal periodic solution predicate and every analytic use of P_{per} implements the same requirement. Until that triangular audit is complete, this semantic-equivalence field is* undefined*; the statement-level analytic implication of Proposition BFA.2.16 is not converted into a paper–Lean equivalence claim.*

BFA.2.5 Announcement binding and lifecycle separation

Definition BFA.2.18 (OpenAI announcement record)**.** Let

$$\mathsf{ANN}_{\text{OAI-NS}} := \text{bfa:1.0:source:oai-announcement-2026-09-08}$$

denote OpenAI's public announcement of 8 September 2026 [11]. Its primary mode is provenance_claim. It records that OpenAI publicly claims a finite-time singularity result, an analytic writeup, a Lean formalization, and resolution of official alternatives C and D. The page is a living web artifact and may receive updates; any update is provenance metadata unless separately rebound to a new theorem-facing source record.

Policy 14 (BFA-BIND-02: announcement nonpromotion)**.** *No institutional statement that a problem is "resolved", "proved", or "formalized" changes a BFA mathematical verifier status by itself. Such language is recorded in the provenance field; theorem-facing use must be supported by the corresponding analytic or formal artifact.*

Definition BFA.2.19 (Current external lifecycle record)**.** The mathematical target and source identities above retain their BFA.2 birth version. Lifecycle metadata is freshness-sensitive and is refreshed here for the successor assembly. On 10 September 2026 the Clay Mathematics Institute published an announcement stating that the Navier–Stokes problem had "apparently been settled" and that its evaluation process would be "deliberately unhurried" [16]. As of 16 September 2026 the CMI landing page remains marked *Active* [14]. BFA therefore records the time-indexed lifecycle packet

$$\begin{aligned}\mathsf{external_lifecycle.announcement} &= \text{CMI_apparently_settled_2026-09-10},\\ \mathsf{external_lifecycle.adjudication} &= \text{pending_2026-09-16}.\end{aligned}$$

This refresh changes no mathematical verifier status, no source theorem identity, and no target semantics in Definition BFA.2.8. In particular, an institutional announcement is neither a proof verifier nor a counterexample.

BFA.2.6 Pinned Lean formal-source family

Definition BFA.2.20 (Pinned OpenAI formal snapshot identity)**.** BFA.2 binds the OpenAI formal family to the content-addressed Git commit

$$\mathsf{COMMIT}_{\text{OAI}} = 8937\text{a}8\text{f}4\text{cbc}7\text{abaab}5\text{e}9\text{e}97\text{d}1\text{cc}7\text{f}5\text{d}2319\text{d}9538.$$

The local formal-family identity is

$$\mathsf{LEAN}_{8937} := \text{bfa:1.0:formal:oai-lean-8937a8f4}.$$

At this commit, formalization.yaml declares a full formalization of the main Navier–Stokes results, reports sorry_count: 0 and sorry_in_definitions: 0, identifies the two comparator declarations, and labels the project review status self-assessed [20]. The commit resolves remote repository identity; it does not by itself constitute a BFA-local build or independent semantic audit.

Definition BFA.2.21 (Pinned formal reference statement)**.** At commit $\mathsf{COMMIT}_{\text{OAI}}$, the file ComparatorChallenges/NavierStokes.lean is a standalone comparator reference adapted from Formal Conjectures. It imports only Mathlib, contains the target definitions and alternatives C/D, and intentionally leaves the reference theorem bodies as sorry placeholders. It explicitly includes periodic pressure in the periodic solution structure in accordance with the Clay erratum [21].

This artifact is bound as formal_reference, not as the proof object. Its role is to fix the declaration types against which the proof submission is compared.

Definition BFA.2.22 (Pinned proved comparator declarations)**.** At the same commit, Navier-Stokes/ComparatorSolution.lean exposes the proved declarations

$$\text{NavierStokes.Comparator.navier_stokes_breakdown_R3},$$
$$\text{NavierStokes.Comparator.navier_stokes_breakdown_periodic}.$$

Their exact Lean statement shapes are

$$\forall \nu > 0 : \exists u^0, f,\ \mathsf{InitialVelocityConditionDecay}(u^0) \wedge \mathsf{ForceConditionDecay}(f) \\ \wedge \neg\exists v, p\ \mathsf{NavierStokesExistenceAndSmoothnessRn}(\nu, u^0, f, v, p),$$

$$\forall \nu > 0 : \exists u^0, f,\ \mathsf{InitialVelocityConditionPeriodic}(u^0) \wedge \mathsf{ForceConditionPeriodic}(f) \\ \wedge \neg\exists v, p\ \mathsf{NavierStokesExistenceAndSmoothnessPeriodic}(\nu, u^0, f, v, p).$$

The actual Lean declarations are functions of (ν, h_ν); the displayed universal quantifier is their ordinary logical reading. The solution file proves these names by exact adapters into the underlying proof modules [22].

Definition BFA.2.23 (Comparator and environment binding)**.** At the pinned commit, the comparator configuration binds ComparatorChallenges.NavierStokes to NavierStokes.ComparatorSolution, compares exactly the two declaration names in Definition BFA.2.22, and permits only

$$\text{propext}, \qquad \text{Quot.sound}, \qquad \text{Classical.choice}$$

as listed axioms [23]. The pinned lean-toolchain is leanprover/lean4:v4.34.0-rc2, while the lake manifest pins the relevant Mathlib and Comparator revisions [24, 25]. These records resolve environment identity at source level. BFA has not yet executed an independent build or Comparator replay.

Policy 15 (BFA-BIND-03: formal identity pass is not replay pass)**.** *For* LEAN_{8937}*, the content-addressed commit gives*

$$\mathsf{remote_identity_freeze} = \text{pass}.$$

Until BFA independently checks out or otherwise snapshots that commit, reconstructs the pinned environment, builds the project, and executes the required formal comparator/verifier route, the distinct fields remain

$$\mathsf{local_snapshot_closure} = \text{not_invoked}, \qquad \mathsf{independent_Lean_replay} = \text{not_invoked}.$$

Repository metadata reporting sorry_count: 0 *is source evidence about the repository; it is not silently retyped as an independent BFA replay.*

Remark BFA.2.24 (Content-addressed formal identity)**.** The formal source is not trusted through the mutable main branch. BFA binds a concrete commit, thereby resolving remote formal identity while leaving local snapshot closure and independent replay uninvoked. This is a strengthening of source identity, not a mathematical promotion of the formal theorem.

BFA.2.7 Directional interpretation graph and semantic firewalls

Definition BFA.2.25 (BFA.2 interpretation graph)**.** BFA.2 registers the proved statement-level analytic implications

$$\mathsf{OAI\text{-}R3} \xrightarrow{\rho^{\text{stmt}}_{PC}} \mathsf{Clay\text{-}C}, \qquad \mathsf{OAI\text{-}T3} \xrightarrow{\rho^{\text{stmt}}_{PD}} \mathsf{Clay\text{-}D}.$$

The pinned formal declarations provide candidate formal-to-target edges

$$\mathsf{LEAN}^{8937}_{R3} \xrightarrow{\rho_{FC}} \mathsf{Clay\text{-}C}, \qquad \mathsf{LEAN}^{8937}_{T3} \xrightarrow{\rho_{FD}} \mathsf{Clay\text{-}D},$$

and the repository metadata declares source alignments from analytic Theorem 1.1 and Corollary 10.6 to the corresponding formal declarations. The final truth of ρ_{FC}, ρ_{FD} and the full paper–formalization semantic relation remain obligations of the later triangular audit. No inverse edge and no semantic equivalence is asserted in BFA.2.

Proposition BFA.2.26 (No automatic cross-artifact equivalence)**.** *Co-publication of the analytic paper and formal repository, equality of their claimed final target, matching declaration names, a successful source-side Comparator configuration, or metadata status* proved *does not by itself prove*

$$\mathsf{PaperTheorem} \cong \mathsf{FormalTheorem}$$

at the level of definitions, hypotheses, solution classes, or proof objects.

Proof. The artifacts inhabit different formal languages and carry different primitive definitions, source environments, proof objects, and interpretation maps. Agreement of names or terminal target labels determines none of those semantic equalities. Hence equivalence requires its own audit. □

BFA.2.8 Maximal binding matrix and theorem-facing statuses

Item	Statement	Remote identity	Local snapshot	Independent replay	Current BFA.2 reading
Clay C target	pass	locator only	not invoked	not applicable	exact fixed-ν semantics resolved; local hash pending
Clay D target	pass	locator only	not invoked	not applicable	erratum-corrected pressure periodicity included
OpenAI Thm. 1.1	pass	locator only	not invoked	not invoked	exact analytic source statement resolved
OpenAI Cor. 10.6	pass	locator only	not invoked	not invoked	exact periodic source statement resolved
OpenAI announcement	pass	live web page	not required	not applicable	provenance claim only
Lean commit 8937 . . .	pass	pass	not invoked	not invoked	content-addressed formal identity resolved
Formal reference C/D	pass	pass	not invoked	not applicable	comparator target types identified; theorem bodies intentionally sorry
Formal solution C/D	pass	pass	not invoked	not invoked	proved declarations identified at pinned commit
Formal metadata	pass	pass	not invoked	not invoked	reports zero sorries; review status self-assessed
CMI lifecycle listing	pass	freshness-sensitive	not required	not applicable	administrative status only

Table 1: BFA.2 binding matrix for BFA 1.0. Every pass is field-qualified; none denotes independent analytic or formal verification unless the replay column says so.

Policy 16 (BFA-BIND-04: semantic pass versus theorem-facing closure)**.** *A target or source may have*

$$\mathsf{semantic_binding} = \mathrm{pass}$$

while its theorem-facing replay packet remains incomplete because a required local snapshot, cryptographic hash, environment reconstruction, or independent verifier run has not been supplied. BFA never shortens this distinction to a single favorable status token.

Definition BFA.2.27 (BFA.2 failure routes)**.** BFA.2 supplies the following local failure labels for downstream diagnosis:

$$\begin{aligned}
\mathsf{FR}_{\text{BFA-TGT}} &: \text{official target unresolved, misquoted, or erratum-incomplete,}\\
\mathsf{FR}_{\text{BFA-SRC}} &: \text{analytic source statement/revision/locator unresolved,}\\
\mathsf{FR}_{\text{BFA-FORM}} &: \text{formal reference, declaration, or environment unresolved,}\\
\mathsf{FR}_{\text{BFA-FREEZE}} &: \text{required immutable or local snapshot identity absent,}\\
\mathsf{FR}_{\text{BFA-REPLAY}} &: \text{required independent build/comparator replay absent or failed,}\\
\mathsf{FR}_{\text{BFA-INTERP}} &: \text{required cross-artifact interpretation edge absent or false.}
\end{aligned}$$

These are BFA-local refinements of the frozen v21 missing-evidence, source-binding, verifier, and Return-missing families; they do not modify the v21 failure registry.

BFA.2.9 BFA.2 closure and BFA.3 handoff

Theorem BFA.2.28 (BFA.2 maximal source/target separation closure)**.** *Under BFA.1 and the records above:*

(i) the Fefferman C/D targets are fixed at exact fixed-viscosity quantifier strength, with the periodic pressure erratum included;

(ii) the logical negations required for genuine target falsification are explicitly distinguished from witness or route failure;

(iii) the analytic whole-space and periodic source statements are fixed by theorem/corollary locators and imply the corresponding official targets at statement level, conditional on source truth;

(iv) the OpenAI announcement is quarantined to provenance;

(v) the OpenAI formal source is bound to content-addressed commit 8937a8f4cbc7 *(full hash in Definition BFA.2.20);*

(vi) the formal comparator reference, proved comparator declarations, comparator configuration, and pinned toolchain/environment are kept as distinct source records;

(vii) source-side metadata such as sorry_count: 0 *and* self-assessed *review status is recorded without becoming an independent BFA verifier result;*

(viii) paper–formalization semantic equivalence and independent formal replay remain unproved and are not inferred from shared target labels;

(ix) live or unversioned PDF/web artifacts remain subject to later local snapshot/hash closure before a replayable final BFA package verdict.

Consequently BFA.3 may reconstruct the analytic proof DAG from the exact analytic source statements, while every later theorem-facing conclusion remains bounded by the unresolved proof, snapshot, replay, and interpretation obligations recorded here.

Proof. Items (i)–(ii) are Definitions BFA.2.4, BFA.2.7 and BFA.2.8 and Proposition BFA.2.10. Item (iii) is Propositions BFA.2.13 and BFA.2.16. Item (iv) is Policy 14. Items (v)–(vii) are the pinned formal records Definitions BFA.2.20 to BFA.2.23. Items (viii)–(ix) follow from Proposition BFA.2.26 and Policies 13, 15 and 16. No step invokes the unverified analytic proof itself. □

Remark BFA.2.29 (BFA.2 final local state)**.** The handoff state is deliberately vector-valued:

official_target_semantic_binding = pass,
analytic_statement_binding = pass,
analytic_proof_verification = not_invoked,
formal_remote_identity_freeze = pass,
formal_declaration_identification = pass,
formal_local_snapshot = not_invoked,
independent_Lean_build_and_Comparator = not_invoked,
paper-formalization_semantic_equivalence = undefined,
PDF_local_cryptographic_freeze = undefined,
external_consensus_or_Clay_recognition = CMI_apparently_settled; formal_adjudication_pending.

Thus BFA.2 closes source and target semantics strongly enough for BFA.3 proof architecture reconstruction, but it intentionally does not close analytic truth, formal replay, or institutional recognition.

BFA.3 Analytic Proof Architecture and Dependency DAG

BFA.3.1 Purpose, authority boundary, and non-verification status

Declaration BFA.3.1 (BFA.3 architectural objective)**.** BFA.3 reconstructs the analytic dependency architecture of the external Navier–Stokes blow-up source fixed in BFA.2. Its purpose is to expose the mandatory proof spine, the supporting construction blocks, the edge types, and the theorem-facing bottlenecks that later BFA sections must audit. It does *not* certify the truth of any analytic lemma merely by registering it in the graph. In particular,

$$\text{registered source node} \not\Rightarrow \text{verified mathematical node}.$$

The current section is therefore a source-faithful architecture section with verification status $\mathrm{not_invoked}$ on every nontrivial external proof node unless a later BFA section explicitly upgrades that node.

Policy 17 (BFA-ARCH-01: source-faithful granularity)**.** *The proof graph shall be coarse enough to remain auditable and fine enough that no theorem-facing transition hides more than one logically distinct operation. A source chapter heading is not automatically one graph node, and several source lemmas may be grouped only when they form one inseparable consumer contract. Conversely, BFA does not split a source lemma into invented claims unless the source itself supplies the corresponding intermediate assertions.*

Definition BFA.3.2 (BFA analytic-node record)**.** A BFA.3 analytic node is a record

$$\mathfrak{N} = (\mathsf{UID}, \mathsf{SourceLocus}, \mathsf{Input}, \mathsf{Payload}, \mathsf{Output}, \mathsf{Consumers}, \mathsf{RegistryStatus}, \mathsf{VerifierStatus}).$$

Its $\mathrm{SourceLocus}$ names exact theorem, proposition, lemma, equation, or section locations in the analytic paper [12]. $\mathrm{RegistryStatus}$ is lifecycle metadata and records whether the source node has been entered into the BFA architecture; all nodes introduced in this section have $\mathrm{RegistryStatus{=}registered}$. It is *not* an atomic mathematical verifier status. In accordance with BFA.1, the atomic $\mathrm{VerifierStatus}$ is exactly one of

$$\mathrm{pass}, \quad \mathrm{reject}, \quad \mathrm{undefined}, \quad \mathrm{not_invoked}.$$

Every nontrivial external analytic node introduced here begins with $\mathrm{VerifierStatus{=}not_invoked}$. A theorem-facing consumer may use a node as verified evidence only after an authorized verifier assigns $\mathrm{VerifierStatus{=}pass}$.

Proposition BFA.3.3 (Registry/verifier type separation)**.** *For BFA.3 analytic nodes,*

$$\mathrm{RegistryStatus{=}registered} \not\Rightarrow \mathrm{VerifierStatus{=}pass}.$$

Moreover, a missing or incomplete audit changes the verifier state to $\mathrm{undefined}$ *only when the corresponding verifier has been invoked and cannot close; it does not turn a registered source node into a mathematical rejection.*

Proof. This is the BFA.1 status discipline applied to the analytic registry. Registry membership is provenance/lifecycle metadata, whereas pass, reject, $\mathrm{undefined}$, and $\mathrm{not_invoked}$ are the atomic verifier outcomes. Conflating them would allow source registration to masquerade as evidence. □

BFA.3.2 Edge taxonomy for the source proof

Definition BFA.3.4 (Analytic edge classes)**.** Every mandatory edge in the reconstructed analytic proof DAG is assigned one or more of the following edge classes:

(i) exact: algebraic, differential, or identity-preserving edge;

(ii) $\mathrm{construct}$: existence or construction of a field, profile, potential, amplitude, inverse, or correction;

(iii) $\mathrm{estimate}$: quantitative bound, decay gain, support bound, or derivative-loss control;

(iv) $\mathrm{preserve}$: preservation of divergence, support, moments, exterior data, or previously obtained asymptotics;

(v) iterate: finite-stage correction or induction edge;

(vi) complete: finite-to-infinite summation or completion edge;

(vii) localize: cutoff, compact-support, or endpoint-extension edge;

(viii) compare: energy, uniqueness, or comparison edge;

(ix) transport: viscosity scaling or periodization edge;

(x) return: statement-level interpretation into the official target already bound in BFA.2.

An edge may carry several classes. Edge labels are audit metadata, not new analytic theorems.

Policy 18 (BFA-ARCH-02: no compensation across edge classes)**.** *A successful edge of one class does not compensate for a missing mandatory edge of another class. In particular, arbitrarily strong finite-stage decay cannot replace the* complete *edge, smooth localization cannot replace endpoint force extension, and a singular constructed field cannot replace the* compare *edge needed to exclude a global bounded-energy competitor. This policy is a BFA specialization of the v21 non-compensation discipline.*

BFA.3.3 Source proof spine at viscosity one

The source paper first works at viscosity one, then obtains arbitrary $\nu > 0$ by scaling. Its proof outline identifies the momentum residual

$$\mathcal{R}(u, p) := \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p \tag{BFA.3.1}$$

and requires every Cartesian space–time derivative of this residual to extend smoothly through the singular time. The source then constructs the background, realizes its annular stress with oscillatory pulses, improves the remaining residual by repeated correction, completes the correction sequence, and finally localizes and extends the residual to a compact smooth force [12].

Definition BFA.3.5 (Shared summation/completion engine)**.** Before registering the numbered source spine, BFA exposes one shared analytic node

$$\mathsf{SUM} := \text{BFA-SUM}$$

with source locus Lemma 5.4 of [12]. Its payload is the source's generic locally-finite cutoff summation mechanism: under increment orders $g_j \to \infty$, derivative losses ℓ_m independent of j, fixed finite-order/finite-degree differential-polynomial structure, and the stated support/coefficient hypotheses, it constructs a smooth locally finite sum and transfers arbitrarily high finite residual order into flatness of the completed residual. This node is consumed twice in the source proof:

(i) by Proposition 5.5 when completing the all-order background field;

(ii) by Proposition 9.9 when completing the full finite-correction sequence.

Its initial state is

$$\text{RegistryStatus=registered}, \qquad \text{VerifierStatus=not_invoked}.$$

BFA therefore forbids duplicating Lemma 5.4 independently inside the two consumer nodes: one failure of the shared lemma must be visible to both paths.

Definition BFA.3.6 (Primary analytic proof spine)**.** The primary BFA.3 source spine consists of the following mandatory nodes. Every row has RegistryStatus=registered and VerifierStatus=not_invoked; the table records dependency, not truth certification.

UID	Source locus	Architectural payload	Primary output
BFA-O1	Theorem 4.6; Appendices A–C	Construct the concentrating leading profiles, exact heat exterior, radial moment matching, annular stress support, and admissible stress-cone geometry.	Leading profile/stress package
BFA-O2	Section 5; Proposition 5.5; consumes BFA-SUM	Correct the axisymmetric base flow to all formal orders, then use the shared summation engine to realize the smooth background while preserving incompressibility and the exterior, leaving annular stress divergence plus a flat base remainder.	Smooth base (u_B, p_B) with stress residual
BFA-O3	Section 6; Lemmas 6.1–6.2; Proposition 6.6	Build the auxiliary torus, label geometry, support separation, overlap compatibility, and coefficient classes needed to prevent unintended pulse cross-interactions.	Wave support and infrastructure package
BFA-O4	Section 7; Proposition 7.2, Lemma 7.4, Proposition 7.5, Proposition 7.6, Lemma 7.7	Construct pulse inverses, primary positive covariance amplitudes, signed covariance corrections, and divergence-free physical waves obtained from supported potentials.	Oscillatory stress-realization package
BFA-O5	Section 8; Proposition 8.1; Lemma 8.2; Propositions 8.3–8.4; Corollary 8.5; Lemmas 8.6–8.8; Section 8.7	Derive exact mean momentum equations, radial/auxiliary-time inverses, pressure and tangential reconstructions, the three integral defects, the five-equation moment-preserving correction, nonlinear defect changes, and the final recomputation/compatibility package.	Mean/defect correction package
BFA-O6	Proposition 9.3; Definition 9.4; Proposition 9.5	Recompute the complete physical residual, separate supported correctable sources from already-flat remainders, and initialize a finite correction state satisfying the stage bounds.	Stage-0 finite correction state
BFA-O7	Proposition 9.6	Apply one full correction cycle and improve the residual orders by a fixed positive amount while preserving supports, moments, and the fixed base/inverse architecture.	Monotone finite-stage residual improvement
BFA-O8	Lemmas 9.7–9.8	Put every finite stage on one common domain and prove that the physical derivative loss at each fixed derivative order is independent of the correction stage.	Uniform domain and stage-independent loss
BFA-O9	Proposition 9.9; consumes BFA-SUM	Apply the shared summation/completion engine to the stage-uniform correction sequence, preserve divergence, obtain a locally finite smooth field, and promote arbitrarily high finite residual decay to residual flatness with the local blow-up asymptotic intact.	Completed local field $(u_{\mathrm{loc}}, p_{\mathrm{loc}})$
BFA-O10	Proposition 10.1	Localize the local field in space and time through potentials so that incompressibility and the singular core are preserved and the resulting fields have fixed compact spatial support and zero initial datum.	Compactly supported preterminal field
BFA-O11	Lemmas 10.2–10.3	Prove compatible limits of all force derivatives at $t \uparrow 1$ and extend the residual to a globally smooth force compactly supported in space and time.	$f \in C_c^\infty$
BFA-O12	Lemma 10.4	Establish the uniform kinetic-energy bound and finite total dissipation for the constructed velocity up to the singular time.	Bounded-energy constructed solution
BFA-O13	Lemma 10.5; final proof of Theorem 1.1	Prove uniqueness/comparison on every $[0, T]$, $T < 1$, among smooth solutions with the same force, zero datum, and uniform spatial L^2 bound; transfer the local velocity growth to exclude a global smooth bounded-energy competitor.	Whole-space breakdown at $\nu = 1$
BFA-O14	Equations (10.22)–(10.23); Theorem 1.1	Rescale the viscosity-one construction to every fixed $\nu > 0$, preserving the theorem-facing force, energy, and breakdown properties.	OAI-R3(ν) for all $\nu > 0$
BFA-O15	Corollary 10.6	Rescale into the interior of a fundamental cube and periodize disjoint translates, including periodic pressure, to obtain the periodic breakdown construction.	OAI-T3(ν) for all $\nu > 0$

Remark BFA.3.7 (Why Theorem 3.1 is represented by a composite node)**.** The source paper's Theorem 3.1 is the local-field milestone realized at the end of Section 9. BFA therefore records its proof-producing content in BFA-O1–BFA-O9, together with the shared node BFA-SUM, rather than treating Theorem 3.1 as an opaque primitive. Proposition 9.9 explicitly states that the completed local fields satisfy all conclusions of Theorem 3.1. This choice exposes the finite-stage and completion edges that BFA later intends to audit.

BFA.3.4 Dependency DAG and mandatory paths

Definition BFA.3.8 (BFA.3 source dependency DAG)**.** Let G_{ana} be the directed acyclic graph with numbered nodes $\mathsf{O}_i :=$ BFA-Oi and shared node $\mathsf{SUM} :=$ BFA-SUM. Its normative mandatory predecessor map is

node	Pred(node)
O_1	$\varnothing$
SUM	$\varnothing$
O_2	$\{\mathsf{O}_1, \mathsf{SUM}\}$
O_3	$\varnothing$ (source infrastructure built in Section 6)
O_4	$\{\mathsf{O}_2, \mathsf{O}_3\}$
O_5	$\{\mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4\}$
O_6	$\{\mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4, \mathsf{O}_5\}$
O_7	$\{\mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4, \mathsf{O}_5, \mathsf{O}_6\}$
O_8	$\{\mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4, \mathsf{O}_5, \mathsf{O}_7\}$
O_9	$\{\mathsf{SUM}, \mathsf{O}_1, \mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4, \mathsf{O}_5, \mathsf{O}_6, \mathsf{O}_7, \mathsf{O}_8\}$
O_{10}	$\{\mathsf{O}_9\}$
O_{11}	$\{\mathsf{O}_9, \mathsf{O}_{10}\}$
O_{12}	$\{\mathsf{O}_{10}, \mathsf{O}_{11}\}$
O_{13}	$\{\mathsf{O}_9, \mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{12}\}$
O_{14}	$\{\mathsf{O}_{13}\}$
O_{15}	$\{\mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{14}\}$.

The predecessor sets deliberately expose shared source machinery cited by later propositions; transitive redundancy is retained when the source proposition explicitly reuses fixed base fields, primary amplitudes, supports, inverse operators, or compact-support data. BFA.2 then supplies the separate return edges

$$\mathsf{O}_{14} \longrightarrow \mathsf{Clay\text{-}C}_\nu, \qquad \mathsf{O}_{15} \longrightarrow \mathsf{Clay\text{-}D}_\nu$$

through Propositions BFA.2.13 and BFA.2.16.

$$\mathsf{O}_1 \longrightarrow \mathsf{O}_2, \qquad \mathsf{SUM} \longrightarrow \mathsf{O}_2, \qquad \mathsf{O}_3 \longrightarrow \mathsf{O}_4,$$
$$\mathsf{O}_2 \longrightarrow \mathsf{O}_4 \longrightarrow \mathsf{O}_5 \longrightarrow \mathsf{O}_6 \longrightarrow \mathsf{O}_7 \longrightarrow \mathsf{O}_8 \longrightarrow \mathsf{O}_9,$$
$$\mathsf{SUM} \longrightarrow \mathsf{O}_9, \qquad \mathsf{O}_9 \longrightarrow \mathsf{O}_{10} \longrightarrow \mathsf{O}_{11} \longrightarrow \mathsf{O}_{12} \longrightarrow \mathsf{O}_{13} \longrightarrow \mathsf{O}_{14} \longrightarrow \mathsf{O}_{15}.$$

The display is schematic. The predecessor table is normative and contains mandatory shared dependencies not all drawn as arrows. In particular, BFA-SUM is a single source theorem consumed by both BFA-O2 and BFA-O9; BFA-O3 is reused by the wave, mean, finite-cycle, and derivative-control machinery; and the periodic node reuses compact-support outputs from BFA-O10–BFA-O11 in addition to the arbitrary-viscosity route.

Proposition BFA.3.9 (Architectural reachability)**.** *Assume the shared analytic node* BFA-SUM *and every mandatory numbered node* BFA-O1–BFA-O15 *hold at exactly the source strength registered in Definition BFA.3.6, and assume every mandatory edge in Definition BFA.3.8 is semantically valid. Then the analytic source graph reaches the whole-space and periodic source statements registered in BFA.2, and the statement-level return maps of BFA.2 reach* $\mathsf{Clay\text{-}C}_\nu$ *and* $\mathsf{Clay\text{-}D}_\nu$ *for each fixed* $\nu > 0$.

Proof. This is a graph-composition statement, not an independent proof of the source lemmas. The shared node BFA-SUM and nodes BFA-O1–BFA-O9 produce the completed local field; BFA-O10–BFA-O13 produce the viscosity-one whole-space breakdown; BFA-O14 gives the arbitrary-viscosity whole-space source statement; and BFA-O15 gives the periodic source statement. The final source-to-target implications are exactly the BFA.2 maps cited above. □

BFA.3.5 Leading profile, annular stress, and all-order base correction

Definition BFA.3.10 (Background-construction contract)**.** The background contract is the conjunction of the following source-facing outputs:

(i) a concentrating axisymmetric leading field whose azimuthal velocity is unbounded along the prescribed inner path as $t \uparrow 1$;

(ii) smooth Cartesian regularity at the axis for every $t < 1$;

(iii) an exact exterior heat flow with vanishing local residual;

(iv) a tangential annular stress supported between fixed similarity radii;

(v) membership of that stress direction in the admissible cone needed for positive pulse amplitudes;

(vi) preservation of the cumulative radial moments needed to match the prescribed exterior;

(vii) after higher-order base correction, a residual of the form "negative stress divergence plus flat remainder".

BFA treats each item as mandatory input to the later pulse/correction route. No later oscillatory success repairs failure of this background contract.

The source proof uses the anisotropic similarity geometry $\tau = 1 - t$, with radial scale of order $\tau^{1/2}$ and axial scale of order $\tau^{1/2-h}$, $0 < h < 1/100$, while the dominant azimuthal/axial velocities have scale $\tau^{-1/2-h}$. The source emphasizes that the shrinking core may have unbounded velocity while its kinetic energy remains small because the support volume shrinks sufficiently rapidly [12]. BFA.3 records this only as the geometric input to the proof architecture; quantitative verification is deferred.

BFA.3.6 Oscillatory realization and exact defect bookkeeping

Definition BFA.3.11 (Oscillatory-realization contract)**.** The oscillatory route from BFA-O3 through BFA-O5 must jointly provide:

(i) support separation for distinct labels, including after evaluation of the auxiliary torus variables in physical coordinates;

(ii) an inverse for supported nonzero angular harmonics with the derivative and support estimates required by later cycles;

(iii) primary positive amplitudes whose averaged quadratic covariance realizes the prescribed annular stress to leading order;

(iv) signed amplitude perturbations for later stress increments;

(v) divergence-free physical waves obtained from supported potentials, with all curl and cutoff correction terms retained in the residual;

(vi) exact mean equations, pressure reconstruction, radial-defect tracking, and compactly supported mean corrections preserving the two required moments.

The contract explicitly forbids replacing the full physical velocity by only its principal oscillatory amplitude inside nonlinear products.

Remark BFA.3.12 (Two averages remain distinct)**.** The analytic source uses both angular averaging and auxiliary-torus averaging. BFA registers them as distinct operations. A later audit must reject any argument that silently identifies the two averages or moves an averaging operation across a nonlinear term without source authorization.

BFA.3.7 The finite correction state and residual-improvement engine

Definition BFA.3.13 (Finite correction state, BFA abstraction)**.** A BFA finite correction state $\mathfrak{S}_j$ is the source state of Definition 9.4 viewed only through the data needed downstream:

$$\mathfrak{S}_j = (u^{[j]}, p^{[j]}, G^{[j]}_{\mathrm{wave}}, E^{[j]}_{\theta}, E^{[j]}_{z}, P^{[j]}, J^{[j]}_{\theta}, J^{[j]}_{z}, \sigma_j, \mathsf{Mom}_j, \mathsf{Supp}_j).$$

The source decay parameter satisfies

$$\sigma_j = \frac{1}{5} + \frac{j}{10},$$

and Proposition 9.6 supplies one complete correction cycle

$$\mathfrak{S}_j \longmapsto \mathfrak{S}_{j+1}, \qquad \sigma_{j+1} = \sigma_j + \frac{1}{10}.$$

This definition records the monotone improvement mechanism but does not yet assert that the estimates establishing the map are correct.

Definition BFA.3.14 (One-cycle operation order)**.** BFA registers the source correction cycle in the following order:

(1) remove supported nonzero angular harmonics with the pulse inverse;

(2) correct the averaged stress by signed perturbations of the fixed positive primary amplitudes;

(3) remove the angular mean with zero auxiliary average using the mean inverse machinery;

(4) correct the remaining compact-support integral defects while preserving the two velocity moments;

(5) reconstruct pressure where required and recompute the *full* physical residual, retaining all new linear and quadratic interactions.

Only after the full cycle is recomputed may the source decay index be promoted from σ_j to σ_{j+1}.

Policy 19 (BFA-ARCH-03: no principal-part erasure)**.** *A later audit shall evaluate Proposition 9.6 against the exact viscosity-one residual increment identity. Writing $e_j = \delta u_j$ and $\pi_j = \delta p_j$,*

$$\begin{aligned}&\mathcal{R}(u^{[j]} + e_j, p^{[j]} + \pi_j) - \mathcal{R}(u^{[j]}, p^{[j]})\\ &\quad = \partial_t e_j - \Delta e_j + \nabla \pi_j + (u^{[j]} \cdot \nabla) e_j + (e_j \cdot \nabla) u^{[j]} + (e_j \cdot \nabla) e_j.\end{aligned}$$

The audit must include curl corrections, cutoff terms, interactions with all earlier corrections, pressure reconstruction, and every newly generated quadratic term. A cancellation of only the selected principal source is not a pass for the cycle.

BFA.3.8 Common-domain and derivative-loss gate

Definition BFA.3.15 (Stage-uniform differentiability gate)**.** The finite correction sequence may enter the shared summation engine BFA-SUM and its final consumer BFA-O9 only if the source establishes both:

(i) one common domain $0 < q < q_{\mathrm{big}}$ for all finite stages; and

(ii) for every fixed physical derivative order m, an exponent loss ℓ_m independent of the correction stage j.

Stage-dependent multiplicative constants are permitted by the source summation lemma; stage-dependent deterioration of the exponent loss is not.

Proposition BFA.3.16 (Why stage-independent loss is theorem-facing)**.** *Within the registered source architecture, monotone residual orders $\rho_j \to \infty$ do not by themselves justify the completion edge. The stage-independent derivative-loss condition of Definition BFA.3.15 is a separate mandatory hypothesis of the summation mechanism.*

Proof. The source summation lemma assumes increment estimates of the form $q^{g_j - \ell_m}$ with $g_j \to \infty$ and ℓ_m independent of j. If ℓ_m were allowed to deteriorate with the stage without a compensating source theorem, increasing formal residual order would not imply the uniform fixed-derivative control consumed by the locally finite cutoff sum. Therefore residual improvement and differentiability control are distinct graph inputs. □

BFA.3.9 Finite-to-infinite completion and local-field realization

Definition BFA.3.17 (Completion contract)**.** The final completion edge BFA-O9 consumes the already-registered theorem BFA-SUM together with:

(i) a common domain;

(ii) increments represented through potentials/direct angular components with smooth axis and lateral extensions;

(iii) correction orders tending to infinity;

(iv) stage-independent derivative-loss exponents;

(v) a fixed finite-order, finite-degree differential polynomial describing the residual;

(vi) arbitrarily high finite-stage residual decay, including the separately retained flat remainder.

It must output a locally finite smooth sum, preserve incompressibility, preserve the source's endpoint/exterior/inner-growth properties, and make every fixed Cartesian space–time derivative of the physical residual flat to arbitrary order at the singular point.

Policy 20 (BFA-ARCH-04: finite evidence is not completion)**.** *For every finite J, even a source estimate $|\mathcal{R}(u^{[J]}, p^{[J]})| \lesssim q^{\rho_J}$ with very large ρ_J remains finite-stage evidence. BFA shall not infer the completed local field or an infinitely flat residual without separately discharging the summation/completion contract of Definition BFA.3.17.*

Remark BFA.3.18 (RIC extraction is postponed and novelty is not presumed)**.** Lemma 5.4 is already a source theorem stated generically for a fixed finite-order, finite-degree differential polynomial $\mathcal{F}$. Therefore merely renaming Lemma 5.4 as "Residual-Improvement Completion" would create no new AK mathematics. A future RIC result can be promoted only if AK supplies an independent theorem whose hypotheses/conclusion are genuinely broader or structurally reusable beyond the source's q-geometry, potential/cutoff implementation, and source-specific completion contract, and then validates that theorem on independent source classes. Until then, BFA-SUM remains an external source theorem and RIC remains only a BFA-local extraction candidate.

BFA.3.10 Localization, terminal force extension, and solution-class closure

Definition BFA.3.19 (Terminal analytic closure contract)**.** The transition from the completed local field to the whole-space theorem is not one edge. BFA decomposes it as follows:

(i) **Localization**: obtain compactly supported smooth u, p for $t < 1$ while preserving divergence, zero initial datum, and the inner singular asymptotic;

(ii) **Endpoint force compatibility**: show that every spatial and time derivative of $f = \mathcal{R}(u, p)$ has a compatible uniform limit as $t \uparrow 1$;

(iii) **Smooth compact extension**: extend these jets to a C_c^∞ force on all positive time;

(iv) **Energy**: prove the uniformly bounded kinetic energy required by the whole-space target class;

(v) **Comparison**: prove that every smooth bounded-energy solution with the same force and zero datum coincides with the constructed solution on each $[0, T]$, $T < 1$;

(vi) **Breakdown transfer**: combine comparison with the explicit inner blow-up path to exclude a global smooth bounded-energy solution.

No item may be silently absorbed into another.

Proposition BFA.3.20 (Constructed singularity versus target nonexistence)**.** *Within the BFA source architecture, construction of one smooth preterminal field with unbounded velocity is insufficient by itself to establish the whole-space Clay-C source statement. The comparison edge of* BFA-O13 *is mandatory for converting that constructed field into the nonexistence of a global smooth bounded-energy competitor for the same datum and force.*

Proof. Clay-C is encoded in BFA.2 as existence of admissible data for which no global physically reasonable solution exists. A singular preterminal candidate does not logically exclude another global smooth solution with the same data. The source proof supplies this exclusion through Lemma 10.5 and the subsequent comparison with the explicit blow-up path. Hence the comparison edge is logically distinct and mandatory. □

BFA.3.11 Scaling and periodic transport

Definition BFA.3.21 (Transport closure)**.** The source's theorem-facing transport layer has two distinct maps:

(i) viscosity scaling from the viscosity-one whole-space construction to every fixed $\nu > 0$;

(ii) compact-support rescaling and disjoint periodization from the whole-space construction to the torus construction.

The periodic transport consumes not only the arbitrary-viscosity construction but also the compact-support outputs registered in BFA-O10 and BFA-O11. It must preserve the nonlinear equation, periodic pressure, zero datum, smooth forcing, and nonexistence conclusion. BFA.2 separately checks that the resulting source statements imply the official corrected Clay-C/D targets.

BFA.3.12 Critical audit cut and priority map

Definition BFA.3.22 (BFA.3 theorem-facing critical audit set)**.** The re-audited high-priority set is

$$\mathfrak{C}^{+}_{\mathrm{BFA3}} = \{\mathsf{SUM}, \mathsf{O}_1, \mathsf{O}_2, \mathsf{O}_4, \mathsf{O}_5, \mathsf{O}_6, \mathsf{O}_7, \mathsf{O}_8, \mathsf{O}_9, \mathsf{O}_{11}, \mathsf{O}_{13}, \mathsf{O}_{15}\}.$$

This set is not asserted to be the unique minimal causal blocker antichain of the full source proof. It is an adversarial priority set. The addition of SUM exposes a shared theorem whose rejection simultaneously attacks two distinct completion uses, while the addition of O_2 and O_6 reflects their nontrivial all-order background and full-residual state-construction burdens. Existing v21 blocker/cut mathematics remains authoritative for any later formal minimal-cut claim.

Definition BFA.3.23 (Audit-priority tiers)**.** For subsequent BFA sections, assign the following initial priorities:

P1: Shared/completion/Return bottlenecks: BFA-SUM, BFA-O7, BFA-O8, BFA-O9, BFA-O11, BFA-O13. These are the most direct finite-to-infinite, endpoint-regularity, and theorem-facing gates.

P2: Construction-integrity blocks: BFA-O1, BFA-O2, BFA-O4, BFA-O5, BFA-O6. These create and recompute the background stress, oscillatory realization, mean defects, and finite correction state.

P3: Localization/energy/transport: BFA-O10, BFA-O12, BFA-O14, BFA-O15. Each is mandatory; lower audit priority is not a claim of mathematical routine or truth.

P4: Shared support/chart infrastructure: BFA-O3. This node is reused broadly and must be promoted if any dependent node's verification reaches back into its support or inverse hypotheses.

Priority is an audit-order heuristic and never changes logical necessity.

Proposition BFA.3.24 (Typed failure propagation examples)**.** *Relative to the coarse-grained registered DAG of Definition BFA.3.8:*

(i) if BFA-SUM *receives* VerifierStatus=reject*, then both the registered Proposition 5.5 completion route and the Proposition 9.9 completion route are blocked, hence neither Clay-facing lane closes through this source DAG;*

(ii) if BFA-O13 *is rejected, the registered whole-space route to* BFA-O14 *is blocked and therefore the current coarse-grained periodic route through* BFA-O14 *is also nonclosing; this does not prove either Clay target false and does not exclude a later refined periodic route that discharges comparison independently;*

(iii) if BFA-O15 *is rejected while the whole-space lane passes, the registered periodic lane is blocked but the Clay-C lane need not be;*

(iv) an undefined *verifier state propagates required-route nonclosure as* undefined*; it never propagates as mathematical* reject *without a false well-typed obligation.*

Proof. Each assertion follows from the mandatory predecessor relation and the BFA.1 non-compensation/status discipline. The caveat in (ii) is required because a rejected coarse-grained route is not a target-falsification theorem and a later source-faithful refinement may split a parent node into independent subroutes. □

BFA.3.13 AK ownership map and deliberate Core non-invocation

Definition BFA.3.25 (BFA.3 AK ownership map)**.** The source architecture is provisionally owned as follows:

Source component	**AK/BFA role**	**Current promotion status**
Leading profile, stress cone, pulses, mean inverses	Source-specific analytic construction inside the Problem Interface audit lane	BFA-local only
Residual correction cycle	Source-specific iterative Bridge architecture	BFA-local; generic extraction not yet proved
Common-domain/stage-loss gate	Completion precondition / audit obligation	BFA-local
Shared BFA-SUM (Lemma 5.4) + Proposition 9.9 consumer	External finite-to-infinite completion theorem plus source-specific final application; possible future RIC extraction seed	Source theorem / BFA-local extraction candidate only
Force extension, energy, uniqueness	Terminal analytic Bridge and Return prerequisites	BFA-local / source-facing
Viscosity scaling and periodization	Transport / target-adapter edges	Source-facing
Core-P, Core-DG	No source necessity established by this proof architecture	Invoke(Core-P) = not_invoked, Invoke(Core-DG) = not_invoked

No row is a Core-promotion claim.

Policy 21 (BFA-ARCH-05: architecture similarity is not theorem import)**.** *Similarity between a source node and an AK concept—for example finite-to- infinite completion, defect correction, proof-DAG dependency, or non-compensation—does not make the source theorem an AK theorem and does not make the AK concept a premise of the source proof. BFA records compatibility and possible extraction only after preserving provenance in both directions.*

Policy 22 (BFA-ARCH-06: source-faithful refinement without identity drift)**.** *The graph in this section is intentionally coarse-grained. A later BFA audit may split any registered parent node into child audit nodes when source verification requires finer hypotheses, estimates, or branches. Such a split must preserve the parent source locus, consumer contract, provenance, and logical strength. Refinement alone is neither evidence of source failure nor permission to strengthen the source theorem. Parent UIDs remain stable as registry anchors; child UIDs must be subordinate to the parent.*

BFA.3.14 Architecture closure and handoff

Theorem BFA.3.26 (BFA.3 analytic architecture closure)**.** *Subject only to source-faithful registration, BFA.3 provides a complete architecture at the* declared coarse-grained theorem-facing registry level*, relative to the source loci recorded here, with the following properties:*

(i) the analytic source is decomposed into mandatory construction, correction, completion, localization, comparison, transport, and return layers;

(ii) the shared summation theorem BFA-SUM*, the finite correction engine, and its two distinct completion consumers are separated;*

(iii) stage-independent physical derivative loss is an explicit gate rather than an implicit consequence of residual-order improvement;

(iv) full residual recomputation is mandatory at every correction cycle;

(v) force smoothness at the terminal time is separated from preterminal localization;

(vi) bounded energy and uniqueness/comparison are separate theorem-facing obligations;

(vii) whole-space scaling and periodicization are distinct transport edges;

(viii) source-to-Clay interpretation remains owned by BFA.2 rather than being re-proved by architectural registration;

(ix) the high-priority adversarial audit set is exposed without claiming a new minimal-cut theorem;

(x) no Core-P/Core-DG invocation or successor-Core promotion is inferred;

(xi) later source-faithful node refinement is permitted without treating refinement itself as a rejection or as new theorem strength.

Consequently, later BFA sections may attack individual source nodes without losing the exact downstream consequence of each pass, nonclosure, or reject status.

Proof. Items (i)–(vii) follow from the node and edge records Definitions BFA.3.5, BFA.3.6, BFA.3.8, BFA.3.13, BFA.3.15, BFA.3.17, BFA.3.19 and BFA.3.21. Item (viii) is enforced by the BFA.2 return maps and BFA.1 authority boundary. Item (ix) follows from Definition BFA.3.22, which explicitly declines to assert a new minimal-cut theorem. Item (x) follows from Definition BFA.3.25 and Policy 21. Item (xi) is Policy 22. The final sentence is precisely the purpose of making the mandatory dependency edges explicit at the declared registry resolution. □

Remark BFA.3.27 (BFA.3 final status and next consumer)**.** BFA.3 is complete at its declared coarse-grained resolution as a source-faithful analytic proof-architecture registry. It has not yet independently verified the registered analytic nodes. The next BFA section should consume this DAG and perform the typed AK reconstruction: each analytic node will be mapped to source semantics, Bridge ownership, finite/infinite boundary, target adapter, defect record, and exact Return contract without forcing Core-P or Core-DG where the source proof does not use them. Subsequent adversarial sections can then assign node-level pass/reject/undefined status against this fixed graph.

BFA.4 AK Typed Projection of the External Proof

BFA.4.1 Purpose and projection discipline

Declaration BFA.4.1 (BFA.4 objective)**.** BFA.4 projects the source-faithful analytic DAG of BFA.3 into the typed ownership, obligation, boundary, and Return vocabulary of AK-HPDST v21. The projection is deliberately conservative. It records what each external node *does*, who owns it, which proof boundary it crosses, which downstream contract consumes it, and which defect or audit obligation would block that consumer. It does not convert an external analytic theorem into an AK Core theorem merely by changing notation.

Policy 23 (BFA-PROJ-01: representation is not derivation)**.** *For every registered source node N,*

$$\mathsf{AKTypedProjection}(N) \nRightarrow \mathsf{AKDerivation}(N).$$

The left side is a typed representation of an external theorem or construction. The right side would require an independent AK proof with discharged hypotheses. No later BFA consumer may infer the latter from the former.

Definition BFA.4.2 (BFA typed-projection packet)**.** For a registered BFA.3 node N, its typed-projection packet is

$$\Pi_{\mathrm{BFA}}(N) = (\mathsf{SrcSem}, \mathsf{MathAuthority}, \mathsf{Carrier}, \mathsf{Role}, \mathsf{Boundary}, \mathfrak{D}^{\mathrm{src}}, \mathfrak{O}^{\mathrm{audit}}, \mathsf{Consumer}, \mathsf{CoreUse}, \mathsf{Evidence}).$$

Here:

(i) SrcSem is the exact PDE-side semantic action of N;

(ii) MathAuthority identifies who owns the mathematical statement being consumed (external analytic source, official target authority, or BFA-local meta-theorem);

(iii) Carrier identifies the BFA registry/projection artifact that carries the record and is not a second theorem authority;

(iv) Role is its source theorem, source-specific Bridge, completion, Return-prerequisite, target-adapter, or Return role;

(v) Boundary records any nontrivial proof boundary crossed by the node;

(vi) $\mathfrak{D}^{\mathrm{src}}$ records source-internal mathematical defects explicitly manipulated by the source;

(vii) $\mathfrak{O}^{\mathrm{audit}}$ records BFA audit obligations and must not be silently identified with $\mathfrak{D}^{\mathrm{src}}$;

(viii) Consumer is inherited from the reverse adjacency of the normative BFA.3 predecessor map rather than re-authored locally;

(ix) CoreUse records whether Core-P or Core-DG is actually invoked; and

(x) Evidence records the exact BFA.2 source binding, BFA.3 node registration, and field-qualified verifier state.

For every nontrivial external analytic node in BFA.4, MathAuthority = ExternalAnalytic and the initial analytic verifier state remains $\mathrm{not_invoked}$. This packet is a research-supplement projection record. It does not alter the frozen v21 Problem-Interface schema [3].

Proposition BFA.4.3 (Source defect and audit obligation are different sorts)**.** *For any source node N,*

$$\mathfrak{D}^{\mathrm{src}}(N) \neq \mathfrak{O}^{\mathrm{audit}}(N)$$

as typed records unless a separate theorem or exact definition identifies the fields. In particular, an analytic residual, mean-moment defect, or stress error manipulated inside the source proof is not automatically an AK falsification verdict.

Proof. The source defect is part of the PDE construction. The audit obligation is a meta-level requirement that BFA must verify in order to accept the source edge. Their domains, outputs, and failure semantics differ. Treating them as one sort would conflate "the constructed approximate object has a residual" with "the theorem asserting control of that residual has failed." □

BFA.4.2 Authority, carrier, role, and deliberate Core non-invocation

Definition BFA.4.4 (Authority and role sorts)**.** BFA.4 separates theorem authority from BFA carriage and mathematical role. The authority sort used here is

$$\mathsf{MathAuthority} \in \{\mathsf{ExternalAnalytic}, \mathsf{ClayTarget}, \mathsf{AKv21Generic}, \mathsf{BFALocal}\}.$$

while the BFA role sort is

$$\mathsf{Role} \in \{\mathsf{SourceConstruction}, \mathsf{SourceBridge}, \mathsf{CompletionTheorem}, \mathsf{CompletionConsumer}, \mathsf{ReturnPrereq}, \mathsf{TargetAdapter}, \mathsf{ReturnMap}\}.$$

The Carrier field records only where the typed record is stored (BFA.2 binding, BFA.3 registry, BFA.4 projection, or a later consumer). Consequently an external theorem carried by BFA remains externally owned; BFA carriage never manufactures AK theorem authority.

Policy 24 (BFA-PROJ-01A: authority/role nonidentification)**.** *No role token such as* ReturnPrereq*,* TargetAdapter*, or* CompletionConsumer *is a theorem authority. Conversely, the fact that a theorem is externally authoritative does not determine its BFA role. The two axes must be serialized separately.*

Definition BFA.4.5 (Core invocation record)**.** The present source route carries the invocation record

$$\mathsf{CoreUse}_{\mathrm{BFA}} := (\mathrm{Invoke}(\text{Core-P}) = \mathrm{not_invoked}, \mathrm{Invoke}(\text{Core-DG}) = \mathrm{not_invoked}).$$

BFA shall not fabricate a Core-P predicate, persistent-homology certificate, Ext obstruction, Collapse datum, or Core-DG diagram merely to populate the route with familiar AK objects. Any later Core invocation requires a genuine Interface theorem constructing the required Core datum from the source semantics.

Theorem BFA.4.6 (No synthetic Core insertion)**.** *The source-faithful BFA audit route can be represented as a source-specific typed target slice without invoking Core-P or Core-DG. Therefore no Core datum is required merely to audit or reconstruct this external analytic proof. This statement does* not *assert that the direct BFA lane is an instance of the canonical v21 Problem-Interface record* $\mathfrak{R}_{PI} = (S, R, C, T, \Sigma, \Gamma, \mathfrak{D}, A)$*, whose C field is a Core-facing predicate or diagram [3].*

Proof. The v21 ownership rules permit source-specific NS components and Return statements to remain source-bound and prohibit source-specific mathematics from acquiring generic ownership without an explicit substitution theorem [3]. BFA.3 identifies no source step whose hypotheses or conclusion require a Core-P or Core-DG object. Hence inserting a synthetic C field would add mathematics absent from the source proof. BFA therefore uses a post-v21 direct audit target slice and leaves canonical $\mathfrak{R}_{PI}$ instantiation uninvoked unless a future adapter theorem constructs a legitimate Core-facing C object. □

Policy 25 (BFA-PROJ-01B: canonical-schema fidelity)**.** *BFA shall not call a seven-field record obtained by deleting C from* $\mathfrak{R}_{PI}$ *a v21 Problem-Interface record, and shall not fill C with an atomic status token. Atomic verifier statuses and mathematical data fields are different sorts. A future canonical PI instantiation must provide the full eight-field schema or a proved schema adapter.*

BFA.4.3 Proof-boundary taxonomy

Definition BFA.4.7 (BFA proof boundaries)**.** The projection distinguishes the following noninterchangeable boundaries:

$\partial^{\infty}_{\mathrm{base}}$: all-order formal base correction $\longrightarrow$ smooth realized base field,
$\partial^{\infty}_{\mathrm{corr}}$: finite correction stages $\longrightarrow$ completed smooth local field,
$\partial^{\mathrm{reg}}_{t=1}$: preterminal smooth residual $\longrightarrow$ smooth force across the terminal time,
$\partial^{\mathrm{sol}}_{\mathrm{cmp}}$: constructed singular solution $\longrightarrow$ exclusion of a competing global smooth bounded-energy solution,
$\partial^{\mathrm{tr}}_{\nu}$: $\nu = 1$ construction $\longrightarrow$ arbitrary fixed $\nu > 0$,
$\partial^{\mathrm{tr}}_{\mathrm{per}}$: whole-space compactly supported construction $\longrightarrow$ periodic target construction.

A proof of one boundary does not discharge another.

Policy 26 (BFA-PROJ-02: boundary non-compensation)**.** *If any required boundary in Definition BFA.4.7 is* reject, undefined*, or otherwise nonclosing, success at another boundary cannot be substituted for it. This is the BFA specialization of the v21 typed Bridge non-compensation principle [3].*

BFA.4.4 Node-by-node typed projection

Definition BFA.4.8 (Typed projection of the shared completion theorem)**.** The shared source node SUM = BFA-SUM is projected as follows:

SrcSem :	locally finite cutoff summation/completion under stage-uniform derivative loss,
MathAuthority :	ExternalAnalytic,
Carrier :	BFA4Projection,
Role :	CompletionTheorem,
Boundary :	$\partial^{\infty}_{\mathrm{base}}$ and $\partial^{\infty}_{\mathrm{corr}}$ through separate consumers,
$\mathfrak{D}^{\mathrm{src}}$:	truncation/remainder order and cutoff-localization structure,
$\mathfrak{D}^{\mathrm{audit}}$:	stage-independent $\ell_m,\ g_j \to \infty,$ finite differential order/degree, support control,
Consumer :	BFA-O2, BFA-O9,
CoreUse :	$\mathsf{CoreUse}_{\mathrm{BFA}}$.

The theorem remains external. The fact that it has a generic-looking statement is not an AK generic-Bridge promotion.

Definition BFA.4.9 (Construction-layer projections)**.** The construction nodes BFA-O1–BFA-O6 have the following projection.

Node	Authority / role	Source defect manipulated	Primary BFA audit obligation	Boundary / consumer
O1	external / SourceConstruction	profile mismatch, moments, stress-support geometry	verify exact leading profiles, exterior matching, radial moments, and admissible stress-cone hypotheses	no AK completion boundary; feeds O2/O9
O2	external / SourceBridge	base residual/stress remainder	verify all-order correction hypotheses and the first legitimate use of BFA-SUM; preserve divergence and exterior data	crosses $\partial^{\infty}_{\mathrm{base}}$; feeds O4–O9
O3	external / support-infrastructure component	support overlap and chart/inverse compatibility	verify torus labels, separation, overlap rules, and domains of inverse operators	infrastructure; feeds O4–O9
O4	external / oscillatory stress realization	covariance/stress realization error	verify positivity/signed covariance reconstruction, support, potential realization, and divergence-free wave output	finite construction; feeds O5–O9
O5	external / mean-defect correction	mean, radial, auxiliary-time, and moment defects	verify inverse solvability, defect identities, five-equation correction, and nonlinear defect recomputation	finite construction; feeds O6–O9
O6	external / SourceBridge	full physical residual before iteration	verify complete residual recomputation, correctable/flat separation, and initial stage bounds	finite-stage entry; feeds O7/O9

Definition BFA.4.10 (Correction and completion projections)**.** The correction/completion nodes are projected as follows.

Node	Authority / role	Audit obligation and typed meaning	Boundary / output
O7	external / SourceBridge	verify one complete correction operator improves the registered residual order by the stated positive amount while preserving the fixed support, moment, inverse, and background contracts; later-stage nonlinear terms must be included in the recomputed residual	finite $j \mapsto j+1$ edge; no completion yet
O8	external / ExternalAnalytic	verify a common domain and, for each fixed derivative order, a loss exponent independent of the correction stage; this is a theorem obligation, not a readiness string	gate immediately before $\partial^{\infty}_{\mathrm{corr}}$
O9	external / ExternalAnalytic	verify all hypotheses of BFA-SUM on the actual correction sequence, local finiteness, divergence preservation, flat residual, and retention of the singular inner asymptotic	crosses $\partial^{\infty}_{\mathrm{corr}}$; outputs completed local field

Definition BFA.4.11 (Terminal and target-facing projections)**.** The terminal nodes have the following projection.

Node	Authority / role	Primary BFA audit obligation	Boundary / target-facing output
O10	external / SourceBridge	verify localization through potentials preserves divergence, singular core, zero initial datum, and fixed compact support	preterminal localization
O11	external / ExternalAnalytic	verify all required force jets admit compatible terminal limits and the extension is genuinely C^{∞}_{c}	crosses $\partial^{\mathrm{reg}}_{t=1}$
O12	external / ExternalAnalytic	verify the exact uniform L^2 energy and dissipation bounds in the solution class consumed by comparison and target interpretation	theorem-facing hypothesis witness
O13	external / SourceBridge	verify uniqueness/comparison on every $T < 1$ under the exact force, datum, smoothness, and energy hypotheses; verify that the blow-up asymptotic excludes a global smooth bounded-energy competitor	crosses $\partial^{\mathrm{sol}}_{\mathrm{cmp}}$; whole-space breakdown at $\nu = 1$
O14	external / ExternalAnalytic	verify the scaling preserves all target-facing hypotheses and the nonexistence conclusion for every fixed $\nu > 0$	crosses $\partial^{\mathrm{tr}}_{\nu}$; feeds BFA.2 Clay-C Return map
O15	external / ExternalAnalytic	verify scale placement inside a fundamental cell, noninteraction of translates, periodic velocity/force/pressure, zero datum, energy class, and periodic nonexistence conclusion	crosses $\partial^{\mathrm{tr}}_{\mathrm{per}}$; feeds BFA.2 Clay-D Return map

Definition BFA.4.12 (Evidence-coordinate convention)**.** For every external analytic node $N \in \{\mathsf{SUM}, \mathsf{O}_1, \ldots, \mathsf{O}_{15}\}$, BFA.4 sets

$$\mathsf{Evidence}(N) := (\mathsf{BFA2SourceBinding}, \mathsf{BFA3RegistryEntry}(N), \mathsf{VerifierStatus}(N)).$$

The BFA.3 registry supplies the exact theorem/lemma locator and predecessor map; the BFA.2 analytic source binding supplies source identity; and, until a later analytic audit is actually executed, $\mathsf{VerifierStatus}(N) = \mathrm{not_invoked}$. The Consumer coordinate is the reverse adjacency of the normative BFA.3 predecessor map. BFA.4 is not permitted to strengthen or silently edit either relation.

BFA.4.5 Defect/obligation vector and correction semantics

Definition BFA.4.13 (BFA source/audit defect pair)**.** For the source-faithful route define the two-layer record

$$\mathfrak{D}_{\mathrm{BFA}} = (\mathfrak{D}^{\mathrm{src}}_{\mathrm{BFA}}, \mathfrak{D}^{\mathrm{audit}}_{\mathrm{BFA}}).$$

The first coordinate may include, according to source locus, residual order, stress/covariance mismatch, mean or moment defects, support incompatibility, and endpoint regularity defects. The second coordinate contains theorem-facing obligations such as exact hypotheses, domain uniformity, derivative-loss uniformity, completion admissibility, force-jet compatibility, solution-class comparison, and target transport.

Definition BFA.4.14 (Residual-improvement state)**.** For the finite correction lane, BFA records a source-side state abstractly as

$$\mathsf{X}_j = (u^{[j]}, p^{[j]}, \mathcal{R}_j, \rho_j, \mathcal{M}_j, \mathcal{S}_j),$$

where $\mathcal{R}_j$ is the full physical residual, ρ_j its registered vanishing-order lower bound, $\mathcal{M}_j$ the moment/mean constraints, and $\mathcal{S}_j$ the support/inverse architecture. The source node BFA-O7 claims a typed transition

$$\mathsf{C}_j : \mathsf{X}_j \longmapsto \mathsf{X}_{j+1}$$

with strict order improvement and preservation of the registered invariants. This BFA record does not assert a new AK correction theorem; it exposes the contract that must be verified.

Policy 27 (BFA-PROJ-03: monotone defect language is source-bound here)**.** *Although Definition BFA.4.14 suggests a future monotone-defect calculus, BFA.4 does not promote*

$$\rho_{j+1} > \rho_j$$

to a domain-independent AK principle. Such promotion requires an independent theorem specifying the ordered defect space, correction operators, convergence or termination semantics, stability under substitution, and failure boundary. Until then the transition remains source-specific.

BFA.4.6 Finite-to-infinite and endpoint boundaries as typed obligations

Theorem BFA.4.15 (Two distinct completion uses must remain distinct)**.** *The shared source theorem* BFA-SUM *participates in two logically distinct completion contracts:*

$$\text{BFA-SUM} + \text{BFA-O1} \longrightarrow \text{BFA-O2}$$

and

$$\text{BFA-SUM} + \text{BFA-O7} + \text{BFA-O8} \longrightarrow \text{BFA-O9},$$

with the additional predecessor obligations registered in BFA.3. Passing one consumer does not certify the other.

Proof. The consumers instantiate the shared summation theorem on different input families and are used at different proof stages. The v21 non-compensation principle prohibits transferring a discharged consumer-specific hypothesis to a different consumer without a theorem identifying the records [3, 4]. □

Proposition BFA.4.16 (Endpoint regularization is not finite-to-infinite completion)**.** *The edge* BFA-O11 *is not subsumed by the completion edge* BFA-O9*. Even if the preterminal residual is flat in the source sense, BFA must separately verify the terminal force-jet compatibility and compact smooth extension required at $t = 1$.*

Proof. Completion produces a smooth preterminal/local object with the residual properties asserted by the source theorem. Extension across a terminal time is a different statement about all relevant derivatives and support. The codomain of the completion theorem is therefore not, by typing alone, the codomain of the force-extension theorem. □

BFA.4.7 Route-closure contracts and exact target witness packets

Definition BFA.4.17 (Whole-space route-closure packet)**.** For fixed $\nu > 0$, define the internal whole-space route packet

$$\mathcal{Q}_C(\nu) := (Q_{\text{build}}, Q_{\text{comp}}, Q_{\text{loc}}, Q_f, Q_E, Q_{\text{cmp}}, Q_\nu).$$

These fields record, respectively, the source construction, the two completion consumers and their stage-uniform hypotheses, localization, terminal smooth-force extension, the bounded-energy estimate used by the comparison argument, comparison/nonexistence, and viscosity transport. $\mathcal{Q}_C(\nu)$ is a proof-route packet, not the Clay target predicate.

Definition BFA.4.18 (Whole-space target-witness packet)**.** For fixed $\nu > 0$, the exact Clay-C witness packet is

$$\mathcal{W}_C(\nu) := (W^{\mathbb{R}^3}_{\text{adm}}, W^{\mathbb{R}^3}_{\neg\mathsf{PhysSol}}),$$

where $W^{\mathbb{R}^3}_{\text{adm}}$ is one Fefferman-admissible datum pair (u^0, f) and $W^{\mathbb{R}^3}_{\neg\mathsf{PhysSol}}$ is the statement that no (p, u) belongs to the exact whole-space physically reasonable global solution class fixed in BFA.2. Pressure smoothness and bounded kinetic energy belong to that BFA.2 solution predicate; they are not silently replaced by a different solution class here.

Definition BFA.4.19 (Periodic route-closure packet)**.** For fixed $\nu > 0$, define the periodic route packet

$$\mathcal{Q}_D(\nu) := (Q_{\text{whole}}, Q_{\text{per}}, Q_{p,\text{per}}, Q^{\text{per}}_{\text{no-global}}),$$

where Q_{whole} denotes the whole-space ingredients actually consumed by Corollary 10.6, Q_{per} the scaling/placement and disjoint-translate periodization step, $Q_{p,\text{per}}$ preservation of periodic pressure, and $Q^{\text{per}}_{\text{no-global}}$ the periodic nonexistence conclusion. No periodic kinetic-energy hypothesis is inserted: the official Clay-D solution predicate fixed in BFA.2 has no energy condition.

Definition BFA.4.20 (Periodic target-witness packet)**.** For fixed $\nu > 0$, the exact Clay-D witness packet is

$$\mathcal{W}_D(\nu) := (W^{\mathbb{T}^3}_{\text{adm}}, W^{\mathbb{T}^3}_{\neg\mathsf{PhysSol}}).$$

The pressure-periodicity erratum is internal to the target predicate $\mathsf{PhysSol}_{\mathbb{T}^3}$ fixed in BFA.2. It is therefore part of the exact meaning of $W^{\mathbb{T}^3}_{\neg\mathsf{PhysSol}}$, not an additional target hypothesis layered on top of Clay-D.

Policy 28 (BFA-PROJ-04: route/target nonidentification)**.** *The route packets $\mathcal{Q}_C, \mathcal{Q}_D$ and target witness packets $\mathcal{W}_C, \mathcal{W}_D$ are different sorts. Internal proof ingredients such as a singular local field, a comparison lemma, an energy estimate, a viscosity scaling law, or a periodization construction may be necessary for the selected proof route without being clauses of the official target statement. Conversely, no excess strength in an internal route field may compensate for a missing target-witness clause. BFA.5 must prove a typed export*

$$\mathcal{Q}_C(\nu) \longrightarrow \mathcal{W}_C(\nu), \qquad \mathcal{Q}_D(\nu) \longrightarrow \mathcal{W}_D(\nu),$$

through the exact BFA.2 interpretation maps.

Proposition BFA.4.21 (Periodic-energy nonpromotion)**.** *Even if the periodic construction inherits a finite-energy bound as an auxiliary source fact, BFA shall not add that bound to the Clay-D target predicate or require it as a target-witness field. Doing so would silently replace the official target with a stronger statement.*

BFA.4.8 Direct BFA target slices and relation to the canonical v21 schema

Definition BFA.4.22 (Direct source-specific BFA target slices)**.** Let $G_C(\nu)$ be the BFA.3 ancestor sub-DAG ending at the BFA.2 source-to-$\mathsf{Clay\text{-}C}_\nu$ interpretation edge, and let $G_D(\nu)$ be the corresponding ancestor sub-DAG ending at the source-to-$\mathsf{Clay\text{-}D}_\nu$ edge. BFA records the direct audit slices

$$\mathfrak{L}^{\text{dir}}_C(\nu) = (\mathsf{S}_C, \mathsf{T}_C, \Sigma_C, \Gamma_C, \mathsf{A}_C, G_C(\nu), \mathsf{CoreUse}_{\text{BFA}}, \mathcal{Q}_C(\nu), \mathcal{W}_C(\nu)),$$

$$\mathfrak{L}^{\text{dir}}_D(\nu) = (\mathsf{S}_D, \mathsf{T}_D, \Sigma_D, \Gamma_D, \mathsf{A}_D, G_D(\nu), \mathsf{CoreUse}_{\text{BFA}}, \mathcal{Q}_D(\nu), \mathcal{W}_D(\nu)).$$

Here $\mathsf{S}_{C,D}$ are the exact forced Navier–Stokes source classes, $\mathsf{T}_C = \mathsf{Clay\text{-}C}_\nu$ and $\mathsf{T}_D = \mathsf{Clay\text{-}D}_\nu$, $\Sigma_{C,D}$ record the source-faithful direct reconstruction route class, $\Gamma_{C,D}$ the propagated hypotheses, and $\mathsf{A}_{C,D}$ the bound source artifacts. These are BFA-local target-slice records of the kind permitted by the general v21 proof-DAG/route-portfolio discipline. They are deliberately *not* serialized as the canonical v21 Problem-Interface tuple $\mathfrak{R}_{PI}$ because no Core-facing C datum has been constructed.

Proposition BFA.4.23 (Canonical PI non-instantiation is not a route defect)**.** *For the present direct external-proof audit, failure to instantiate $\mathfrak{R}_{PI}$ is not a missing theorem edge. The selected target slices already live entirely in external analytic semantics plus BFA-local source-specific typing and Return interpretation. A future claim that this lane is a canonical PI route requires an explicit schema-adapter theorem that constructs the missing Core-facing C field at the correct type.*

Definition BFA.4.24 (Whole-space lane image)**.** At the declared BFA.3 resolution, $G_C(\nu)$ contains

$$\{\mathsf{O}_1, \mathsf{SUM}, \mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4, \mathsf{O}_5, \mathsf{O}_6, \mathsf{O}_7, \mathsf{O}_8, \mathsf{O}_9, \mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{12}, \mathsf{O}_{13}, \mathsf{O}_{14}\},$$

followed by the BFA.2 statement-level source-to-$\mathsf{Clay\text{-}C}_\nu$ interpretation edge. Every analytic node remains externally authoritative and source-bound.

Definition BFA.4.25 (Periodic lane image)**.** At the declared BFA.3 resolution, $G_D(\nu)$ consumes the registered whole-space construction required by the source proof, adds O_{15}, and terminates at the BFA.2 source-to-Clay-D_ν interpretation edge. This is the current coarse-grained source route, not a theorem that every possible proof of Clay-D must factor through the whole-space lane.

Proposition BFA.4.26 (Typed projection preserves route failure isolation)**.** *Suppose a required node in $G_C(\nu)$ is rejected. Then that registered lane does not export $\mathcal{W}_C(\nu)$ through BFA, but the statement does not falsify Clay-C itself. Likewise a rejection confined to the additional periodic adapter may block $G_D(\nu)$ without blocking the whole-space lane. An* undefined *prerequisite propagates route nonclosure as* undefined*, not as* reject.

Proof. This is the v21 independent-route and target-slice discipline applied to the BFA projection [3, 6]. Route closure is a property of the selected proof path; target truth is not identified with the health of one path, and atomic status sorts are not collapsed across failure modes. □

BFA.4.9 Node-to-authority/role matrix

Definition BFA.4.27 (Normative BFA.4 authority/role matrix)**.** For later sections, the following authority/role assignment is normative at the current coarse-grained resolution. The BFA carrier for these projection records is BFA4Projection unless explicitly stated otherwise.

Node	Authority	Typed role	Promotion ceiling in BFA.4
SUM	ExternalAnalytic	shared completion theorem	external theorem; RIC seed only
O1,O3	ExternalAnalytic	source construction/infrastructure	BFA-local
O2,O4–O7	ExternalAnalytic	SourceBridge / source-specific analytic component	BFA-local/source-bound
O8	ExternalAnalytic	ReturnPrereq / stage-uniform completion gate	BFA-local/source-bound
O9	ExternalAnalytic	CompletionConsumer / source-specific finite-to-infinite consumer	BFA-local/source-bound
O10	ExternalAnalytic	SourceBridge / localization component	BFA-local/source-bound
O11,O12	ExternalAnalytic	ReturnPrereq / force/energy prerequisites	BFA-local/source-bound
O13	ExternalAnalytic	SourceBridge / comparison/nonexistence component	BFA-local/source-bound
O14,O15	ExternalAnalytic	TargetAdapter / viscosity/periodic transport	BFA-local/source-bound
BFA.2 final maps	BFALocal	ReturnMap / exact source-to-Clay interpretation	BFA-local source-specific Return

The orthogonal invocation record remains

$$\mathsf{Invoke}(\mathsf{Core\text{-}P}) = \mathsf{Invoke}(\mathsf{Core\text{-}DG}) = \mathrm{not_invoked}.$$

It is not an entry of the theorem-authority column.

Theorem BFA.4.28 (Authority and source-domain conservation under typed projection)**.** *The projection of a registered source node into a BFA role does not change its mathematical authority, increase its theorem strength, or widen its source domain. In particular,*

$$\mathsf{source\text{-}specific} \xrightarrow{\Pi_{\mathrm{BFA}}} \mathsf{source\text{-}specific}$$

unless a separately proved substitution/promotion theorem changes the mathematical statement and its authorized owner.

Proof. This is exactly the source-specific ownership ceiling of v21: source class, semantics, hypotheses, realization, and target belong to the theorem statement and cannot be erased by registry placement [3, 6]. The projection packet records those fields rather than replacing them. □

BFA.4.10 What AK contributes at this stage

Proposition BFA.4.29 (AK contribution of BFA.4)**.** *At BFA.4, AK contributes the following without claiming an independent Navier–Stokes proof:*

(i) a typed separation of mathematical authority, BFA carrier, source theorem, source-specific Bridge role, completion consumer, Return prerequisite, target adapter, and Return;

(ii) explicit separation of source-internal defects from audit obligations;

(iii) explicit finite-to-infinite, endpoint-regularization, comparison, and transport boundaries, together with direct target-slice records that do not masquerade as canonical PI tuples;

(iv) separate non-compensating route-closure packets and exact target-witness packets for each official target;

(v) exact authority, role, and promotion ceilings for every BFA.3 source node;

(vi) route-level failure semantics that do not collapse route rejection into target falsification.

These are architectural and audit contributions. The analytic truth of the external source nodes remains a separate verification problem.

Remark BFA.4.30 (Why this is stronger than merely drawing the source DAG)**.** BFA.3 answers "what depends on what?" BFA.4 additionally answers "what kind of mathematical object is this edge, where does theorem authority reside, which boundary does it cross, what exact obligation would block it, and what may it legitimately return?" That added typing is precisely what later compatibility and adversarial sections consume.

BFA.4.11 BFA.4 closure and handoff

Theorem BFA.4.31 (BFA.4 typed-projection closure)**.** *Relative to the source binding of BFA.2 and the coarse-grained source DAG of BFA.3, BFA.4 provides a complete typed projection at the declared registry resolution with the following properties:*

(i) every registered source node has separated mathematical authority, BFA carrier, and typed role;

(ii) source-internal defects and BFA audit obligations are separated;

(iii) the two uses of the shared summation theorem remain distinct consumers;

(iv) the base-completion, correction-completion, terminal-regularization, comparison, viscosity-transport, and periodization boundaries are explicit and non-compensating;

(v) whole-space and periodic route-closure packets are separated from exact target-witness packets;

(vi) the current source route does not invoke Core-P or Core-DG, no synthetic Core payload is introduced, and the direct BFA target slices are not mislabeled as canonical $\mathfrak{R}_{PI}$ records;

(vii) projection does not promote source-specific mathematics to generic AK mathematics;

(viii) route failure remains distinct from target falsification.

No item above assigns $\mathrm{VerifierStatus}=\mathrm{pass}$ *to an unverified external analytic node.*

Proof. Items (i)–(iii) follow from Definitions BFA.4.2, BFA.4.8, BFA.4.12 and BFA.4.27, Policy 24, and Proposition BFA.4.3. Item (iv) is Definition BFA.4.7, Policy 26, Theorem BFA.4.15, and Proposition BFA.4.16. Item (v) is Definitions BFA.4.17 to BFA.4.20, Policy 28, and Proposition BFA.4.21. Item (vi) follows from Definition BFA.4.5, Theorem BFA.4.6, Policy 25, and Proposition BFA.4.23. Item (vii) is Theorem BFA.4.28. Item (viii) is Proposition BFA.4.26 and the BFA.1 falsification discipline. The final sentence follows from Policy 23. □

Remark BFA.4.32 (Next consumer: BFA.5)**.** BFA.4 intentionally stops before asserting that the projected route closes. The next section should prove the *source-faithful reconstruction theorem*: assuming the exact external analytic nodes used by the selected lane are true at their registered strength and every BFA.2 interpretation edge is valid, does the typed composition export $Q_C(v)$ and $Q_D(v)$ to precisely the target-witness packets $\mathcal{W}_C(v)$ and $\mathcal{W}_D(v)$, hence the same Clay-C/Clay-D targets—and no stronger target? That question is BFA.5, not BFA.4.

BFA.5 Source-Faithful Reconstruction Theorem

BFA.5.1 Purpose and theorem mode

Declaration BFA.5.1 (BFA.5 objective)**.** BFA.5 proves a conditional, source-faithful reconstruction theorem for the whole-space and periodic routes fixed in BFA.2–BFA.4. The theorem answers the following precise question: if the external analytic nodes consumed by a selected BFA route are true at exactly their registered strength, and if the registered interpretation edges are valid, does the typed BFA composition export the exact Clay target witnessed by the source—without strengthening that target and without attributing the analytic proof to AK?

Policy 29 (BFA-RECON-01: conditionality is part of the theorem)**.** *Every reconstruction theorem in BFA.5 is conditional on the truth of the external analytic source statements that it consumes. A conditional theorem of the form*

$$\mathsf{SourceAssumptions} \Longrightarrow \mathsf{Clay\text{-}C}_\nu$$

or

$$\mathsf{SourceAssumptions} \Longrightarrow \mathsf{Clay\text{-}D}_\nu$$

shall not be serialized, summarized, or cited as an AK-only proof of the antecedent or as an AK-only solution of the Navier–Stokes problem.

Definition BFA.5.2 (Source-truth predicate)**.** For a registered external analytic node N and a fixed viscosity $\nu > 0$, write

$$\mathsf{SrcTrue}_\nu(N)$$

for the meta-level proposition that the exact external statement bound to N is mathematically true at the source revision and statement strength fixed in BFA.2–BFA.3. This is a proposition used in the hypothesis of BFA.5; it is not a verifier status. In particular,

$$\mathsf{SrcTrue}_\nu(N) \neq \big(\mathsf{VerifierStatus}(N) = \mathrm{pass}\big)$$

as typed objects.

Definition BFA.5.3 (Edge-validity predicate)**.** For a directed edge $e : N \to M$ in the frozen BFA.3 source DAG, let $\mathsf{EdgeValid}_\nu(e)$ mean that the conclusion exported by N, together with the other registered predecessors of M, satisfies the exact hypotheses of the source result bound to M. For a BFA.2 interpretation edge ι, let $\mathsf{InterpValid}_\nu(\iota)$ mean that the source statement implies the exact official target statement at the bound quantifier and solution-class strength. These predicates are theorem-facing propositions, not registry states.

Definition BFA.5.4 (Lane type-fidelity predicate)**.** For a selected source lane G at fixed viscosity $\nu > 0$, write

$$\mathsf{TypeFaithful}_\nu(G)$$

when every consumed source and target field preserves its registered domain, quantifier order, viscosity parameter, datum class, force class, solution class, support requirement, and comparison class. This predicate forbids both hidden strengthening and hidden weakening. It is separate from node truth and from edge validity because a collection of individually true and locally composable statements can still be exported to the wrong external target if its terminal interpretation changes type.

Remark BFA.5.5 (Why edge validity is explicit)**.** Node truth alone does not guarantee composability if two statements have been registered with mismatched hypotheses, domains, solution classes, or quantifiers. BFA therefore requires both truth of the consumed statements and validity of the typed interfaces between them.

BFA.5.2 Reconstruction assumption packets

Definition BFA.5.6 (Whole-space reconstruction assumptions)**.** For fixed $\nu > 0$, let $\mathfrak{A}_C(\nu)$ be the conjunction of:

(i) $\mathsf{SrcTrue}_\nu(N)$ for every external analytic node $N \in G_C(\nu)$ of Definition BFA.4.24;

(ii) $\mathsf{EdgeValid}_\nu(e)$ for every required analytic source-DAG edge e in the selected whole-space lane;

(iii) the two registered uses of the shared completion theorem SUM satisfy their own consumer hypotheses, so that base completion and final completion remain distinct typed occurrences;

(iv) the BFA.2 whole-space interpretation edge of Proposition BFA.2.13 satisfies $\mathsf{InterpValid}_\nu$ at the fixed target meaning of Definition BFA.2.8; and

(v) $\mathsf{TypeFaithful}_\nu(G_C(\nu))$ holds.

The packet is sufficient for the reconstruction theorem below; BFA.5 does not claim that it is a logically minimal set of assumptions.

Definition BFA.5.7 (Periodic reconstruction assumptions)**.** For fixed $\nu > 0$, let $\mathfrak{A}_D(\nu)$ be the conjunction of:

(i) $\mathsf{SrcTrue}_\nu(N)$ for every external analytic node consumed by $G_D(\nu)$, including O_{15};

(ii) $\mathsf{EdgeValid}_\nu(e)$ for every required analytic source-DAG edge in the selected periodic lane, including the edges into the periodization node;

(iii) each use of a shared source theorem, including SUM when inherited through the whole-space construction, satisfies the exact hypotheses of that consumer occurrence;

(iv) the periodic pressure semantics agree with the corrected target predicate of Definition BFA.2.7;

(v) the BFA.2 periodic interpretation edge of Proposition BFA.2.16 satisfies $\mathsf{InterpValid}_\nu$; and

(vi) $\mathsf{TypeFaithful}_\nu(G_D(\nu))$ holds, including the rule that no whole-space energy condition is promoted into the official periodic target, consistently with Proposition BFA.4.21.

As in the whole-space lane, this is a sufficient reconstruction packet and is not asserted to be assumption-minimal.

Policy 30 (BFA-RECON-02: assumptions are lane-relative)**.** *$\mathfrak{A}_C(\nu)$ and $\mathfrak{A}_D(\nu)$ certify only the selected source-faithful lanes $G_C(\nu)$ and $G_D(\nu)$. They do not assert that every proof of Clay-C or Clay-D factors through those lanes, and failure of one lane is not target falsification.*

Policy 31 (BFA-RECON-02A: sufficient is not minimal)**.** *The packets $\mathfrak{A}_C(\nu)$ and $\mathfrak{A}_D(\nu)$ are sufficient certificates for the declared source-faithful reconstruction. BFA.5 does not claim that every conjunct is logically necessary, that the packets are minimal cuts, or that every proof of the same target must realize the same factorization.*

BFA.5.3 Internal route reconstruction

Lemma BFA.5.8 (Whole-space route packet reconstruction)**.** *For fixed $\nu > 0$,*

$$\mathfrak{A}_C(\nu) \Longrightarrow \mathcal{Q}_C(\nu).$$

More precisely, the source-faithful composition reconstructs every field of $\mathcal{Q}_C(\nu)$ at exactly the strength declared in Definition BFA.4.17.

Proof. By Definition BFA.5.6, every analytic node and every required typed edge in $G_C(\nu)$ is available at its registered source strength. The source construction and base-completion chain provides Q_{build}; the shared summation theorem together with its two separately checked consumers and the stage-uniformity gate provides Q_{comp}; localization provides Q_{loc}; the terminal force-extension result provides Q_f; the energy estimate provides Q_E; the comparison/nonexistence result provides Q_{cmp}; and viscosity transport provides Q_ν. Boundary non-compensation from Policy 26 prevents any one of these fields from being replaced by excess strength in another. Hence the entire route packet is reconstructed. □

Lemma BFA.5.9 (Periodic route packet reconstruction)**.** *For fixed $\nu > 0$,*

$$\mathfrak{A}_D(\nu) \Longrightarrow \mathcal{Q}_D(\nu).$$

Proof. The consumed whole-space ingredients provide Q_{whole}. The registered periodization/scaling construction provides Q_{per}; its pressure semantics provide $Q_{p,\mathrm{per}}$; and the periodic nonexistence statement provides $Q^{\mathrm{per}}_{\text{no-global}}$. The definition of $\mathfrak{A}_D(\nu)$ explicitly includes the corrected periodic-pressure semantics and forbids promotion of a whole-space energy requirement into the periodic target. Thus the four fields of Definition BFA.4.19 are obtained without target strengthening. □

Theorem BFA.5.10 (Route reconstruction is source-conservative)**.** *The implications in Lemmas BFA.5.8 and BFA.5.9 preserve mathematical authority and source domain. They do not convert any external analytic node into an AK-owned theorem, and they do not invoke Core-P or Core-DG.*

Proof. Authority conservation is Theorem BFA.4.28; deliberate Core non-invocation is Definition BFA.4.5 and Theorem BFA.4.6. BFA.5 composes already registered source statements and does not alter their owners, hypotheses, or codomains. □

BFA.5.4 Typed export from route packets to exact target witnesses

Theorem BFA.5.11 (Whole-space target-witness export)**.** *For every fixed $\nu > 0$, the exact theorem-facing export is*

$$\mathfrak{A}_C(\nu) \wedge \mathcal{Q}_C(\nu) \Longrightarrow \mathcal{W}_C(\nu).$$

The export is through the exact source-to-target interpretation fixed in BFA.2; it does not identify $\mathcal{Q}_C$ with $\mathcal{W}_C$.

Proof. The route packet reconstructs the bound whole-space source theorem at the registered strength: zero initial data, smooth compactly supported forcing, the constructed singular solution, the bounded-energy comparison class, and nonexistence of a global smooth solution in that class for the same datum and force. By Proposition BFA.2.11, zero initial data are admissible; compact spacetime support implies the required Fefferman decay conditions; and by Proposition BFA.2.13, the source nonexistence conclusion is exactly sufficient for the official whole-space physically reasonable solution predicate of BFA.2. Therefore the route exports $W^{\mathbb{R}^3}_{\mathrm{adm}}$ and $W^{\mathbb{R}^3}_{\neg\mathsf{PhysSol}}$, i.e. $\mathcal{W}_C(\nu)$. □

Theorem BFA.5.12 (Periodic target-witness export)**.** *For every fixed $\nu > 0$, the exact theorem-facing export is*

$$\mathfrak{A}_D(\nu) \wedge \mathcal{Q}_D(\nu) \Longrightarrow \mathcal{W}_D(\nu).$$

Proof. The route packet provides a smooth periodic admissible datum and the source statement that no global smooth periodic solution exists for that datum and force. The pressure-periodicity clause is consumed at the corrected target meaning fixed in Definition BFA.2.7, while Proposition BFA.4.21 prevents any auxiliary energy fact from being promoted into Clay-D. The statement-level implication Proposition BFA.2.16 therefore exports precisely $W^{\mathbb{T}^3}_{\mathrm{adm}}$ and $W^{\mathbb{T}^3}_{\neg\mathsf{PhysSol}}$, i.e. $\mathcal{W}_D(\nu)$. □

Corollary BFA.5.13 (Exact Clay-target export)**.** *For every fixed $\nu > 0$,*

$$\mathcal{W}_C(\nu) \Longrightarrow \mathsf{Clay\text{-}C}_\nu, \qquad \mathcal{W}_D(\nu) \Longrightarrow \mathsf{Clay\text{-}D}_\nu.$$

Proof. This is immediate from the definitions of the target-witness packets in Definitions BFA.4.18 and BFA.4.20 and the exact fixed-viscosity target predicates in Definition BFA.2.8. □

BFA.5.5 Main source-faithful reconstruction theorems

Theorem BFA.5.14 (Whole-space source-faithful reconstruction)**.** *For every fixed $\nu > 0$,*

$$\boxed{\mathfrak{A}_C(\nu) \Longrightarrow \mathsf{Clay\text{-}C}_\nu.}$$

Moreover the implication factors, without changing authority, through the registered internal route and exact witness packet:

$$\mathfrak{A}_C(\nu) \Longrightarrow \mathcal{Q}_C(\nu), \qquad \mathfrak{A}_C(\nu) \wedge \mathcal{Q}_C(\nu) \Longrightarrow \mathcal{W}_C(\nu), \qquad \mathcal{W}_C(\nu) \Longrightarrow \mathsf{Clay\text{-}C}_\nu.$$

Thus, assuming the exact external analytic lane is true and correctly typed at its registered strength, BFA reconstructs the same whole-space Clay-C target reached by the source.

Proof. Use Lemma BFA.5.8 to obtain $\mathcal{Q}_C(\nu)$ from $\mathfrak{A}_C(\nu)$, then Theorem BFA.5.11 for the assumption-qualified witness export, and finally Corollary BFA.5.13. □

Theorem BFA.5.15 (Periodic source-faithful reconstruction)**.** *For every fixed $\nu > 0$,*

$$\boxed{\mathfrak{A}_D(\nu) \Longrightarrow \mathsf{Clay\text{-}D}_\nu.}$$

The implication factors as

$$\mathfrak{A}_D(\nu) \Longrightarrow \mathcal{Q}_D(\nu), \qquad \mathfrak{A}_D(\nu) \wedge \mathcal{Q}_D(\nu) \Longrightarrow \mathcal{W}_D(\nu), \qquad \mathcal{W}_D(\nu) \Longrightarrow \mathsf{Clay\text{-}D}_\nu.$$

Proof. Use Lemma BFA.5.9, Theorem BFA.5.12, and Corollary BFA.5.13 in that order, retaining $\mathfrak{A}_D(\nu)$ through the witness-export step. □

Corollary BFA.5.16 (All-viscosity portfolio reconstruction)**.** *If the whole-space reconstruction assumptions hold pointwise for every positive viscosity, then the all-viscosity portfolio target holds:*

$$\big(\forall \nu > 0\ \mathfrak{A}_C(\nu)\big) \Longrightarrow \mathsf{Clay\text{-}C}.$$

Likewise,

$$\big(\forall \nu > 0\ \mathfrak{A}_D(\nu)\big) \Longrightarrow \mathsf{Clay\text{-}D}.$$

Proof. Apply Theorems BFA.5.14 and BFA.5.15 pointwise in ν and unfold the portfolio notation of Definition BFA.2.9. □

Theorem BFA.5.17 (Same-target theorem)**.** *At the target semantics frozen in BFA.2, the conclusions of Theorems BFA.5.14 and BFA.5.15 are neither weaker proxies nor stronger substitutes for the intended Clay alternatives. They are exactly the fixed-viscosity target predicates* $\mathsf{Clay\text{-}C}_\nu$ *and* $\mathsf{Clay\text{-}D}_\nu$ *registered in Definition BFA.2.8.*

Proof. BFA.4 separates internal proof-route packets from exact target-witness packets. BFA.5 exports only through those witness packets and then unfolds the target definitions. No additional target predicate is introduced between $\mathcal{W}_{C,D}$ and $\mathsf{Clay\text{-}C}_\nu$, $\mathsf{Clay\text{-}D}_\nu$. □

BFA.5.6 Conservativity and theorem ceiling

Definition BFA.5.18 (BFA.5 reconstruction ceiling)**.** Let $\mathsf{Ceil}_{\mathrm{BFA5}}$ be the closure of Theorems BFA.5.14 and BFA.5.15 and Corollary BFA.5.16 under only those logical consequences already contained in the registered source statements and exact target semantics. No theorem outside this closure is licensed by BFA.5 merely because it is compatible with, suggested by, or aesthetically related to the reconstructed route.

Theorem BFA.5.19 (No-strengthening / no-leakage theorem)**.** *BFA.5 does not, by itself, establish any of the following:*

(i) finite-time blow-up for the unforced problem $f = 0$;

(ii) blow-up for arbitrary smooth initial data or arbitrary smooth forcing;

(iii) uniqueness or nonexistence outside the exact comparison/solution classes consumed by the source and official targets;

(iv) a generic residual-improvement/completion theorem owned by AK;

(v) a theorem of Core-P or Core-DG;

(vi) equivalence of the analytic paper and the Lean formalization;

(vii) official institutional adjudication of the Millennium Prize; or

(viii) target falsification from failure of the selected BFA lane.

Proof. Items (i)–(iii) are absent from the exact source/target statements consumed by BFA.5 and therefore violate source-domain conservation if inferred without new theorems. Items (iv)–(v) violate the promotion and Core-noninvocation ceilings of BFA.1/BFA.4. Item (vi) belongs to the later triangular semantic audit. Item (vii) is lifecycle metadata, not a mathematical consequence. Item (viii) is forbidden by the route-failure/target-falsification separation of BFA.1 and Proposition BFA.4.26. □

Corollary BFA.5.20 (Forced blow-up does not imply unforced blow-up)**.** *The source-faithful reconstruction of* $\mathsf{Clay\text{-}C}_\nu$ *and* $\mathsf{Clay\text{-}D}_\nu$ *from the present source family does not imply the existence of a singular solution with* $f = 0$.

Proof. The registered source witness uses specially constructed smooth forcing. No edge in the selected DAG eliminates that forcing while preserving the blow-up conclusion. The claim would therefore require a new theorem outside $\mathsf{Ceil}_{\mathrm{BFA5}}$. □

Theorem BFA.5.21 (No circular target import)**.** *The BFA.5 reconstruction does not use* $\mathsf{Clay\text{-}C}_\nu$ *or* $\mathsf{Clay\text{-}D}_\nu$ *as analytic premises for the source construction. The target predicates appear only after the source route has reconstructed the source theorem and are consumed through BFA.2 interpretation maps.*

Proof. The predecessor DAG $G_C(\nu)$ and $G_D(\nu)$ is source-theorem-facing through BFA.3 and BFA.4. The official target is attached only at the terminal interpretation edge described in Definitions BFA.4.24 and BFA.4.25. Thus the target is not an ancestor of the analytic source nodes in the selected route. □

Theorem BFA.5.22 (No converse from target truth)**.** *Within BFA.5, neither* $\mathsf{Clay\text{-}C}_\nu$ *nor* $\mathsf{Clay\text{-}D}_\nu$ *implies the corresponding source-assumption packet* $\mathfrak{A}_C(\nu)$ *or* $\mathfrak{A}_D(\nu)$. *The source-faithful lane is a sufficient reconstruction route, not a necessary factorization of the target.*

Proof. The target predicates quantify over admissible witnesses and assert nonexistence of a global smooth solution for those witnesses. They contain no identifier for the OpenAI construction, no BFA.3 node identities, and no requirement that a proof factor through the selected analytic DAG. Therefore recovering the selected route from target truth would require a new converse or proof-uniqueness theorem, none of which is registered in BFA.5. □

BFA.5.7 Status discipline after reconstruction

Definition BFA.5.23 (BFA.5 proof-status vector and lifecycle field)**.** After proving the conditional reconstruction theorem, BFA records the proof-facing vector

$$\mathsf{V}^{\mathrm{proof}}_{\mathrm{BFA5}} := \begin{pmatrix} \mathsf{typed_composition},\ \mathsf{same_target}, \\ \mathsf{analytic_verifier},\ \mathsf{independent_AK_verifier}, \\ \mathsf{formal_equivalence} \end{pmatrix}.$$

At the close of BFA.5 the intended statuses are

$$\begin{aligned} \mathsf{typed_composition} &= \mathrm{pass}, \\ \mathsf{same_target} &= \mathrm{pass}, \\ \mathsf{analytic_verifier} &= \mathrm{not_invoked}, \\ \mathsf{independent_AK_verifier} &= \mathrm{not_invoked}, \\ \mathsf{formal_equivalence} &= \mathrm{undefined}. \end{aligned}$$

The external institutional lifecycle is recorded separately as

$$\mathsf{L}^{\mathrm{inst}}_{\mathrm{BFA5}} = \mathrm{pending_formal_adjudication}.$$

No field named "analytic truth" receives a verifier token: mathematical truth appears in BFA.5 only as the conditional proposition $\mathsf{SrcTrue}_\nu(N)$.

Policy 32 (BFA-RECON-03: no pass leakage)**.** *A* pass *in* typed_composition *or* same_target *may not be copied into* analytic_verifier, independent_AK_verifier, *or* formal_equivalence *without the corresponding verifier theorem. Lifecycle metadata likewise may not supply a mathematical verifier edge.*

BFA.5.8 BFA.5 closure and handoff

Theorem BFA.5.24 (BFA.5 source-faithful reconstruction closure)**.** *Relative to the frozen source identities and target semantics of BFA.2, the source DAG of BFA.3, and the typed projection of BFA.4, BFA.5 establishes all of the following at the declared registry resolution:*

(i) exact lane-relative assumption packets $\mathfrak{A}_C(\nu)$ and $\mathfrak{A}_D(\nu)$;

(ii) conditional reconstruction of $\mathcal{Q}_C(\nu)$ and $\mathcal{Q}_D(\nu)$;

(iii) typed export to the exact witness packets $\mathcal{W}_C(\nu)$ and $\mathcal{W}_D(\nu)$;

(iv) exact fixed-viscosity Return to $\mathsf{Clay\text{-}C}_\nu$ and $\mathsf{Clay\text{-}D}_\nu$;

(v) all-viscosity portfolio Return when the source assumptions hold pointwise for every $\nu > 0$;

(vi) source-authority conservation and continued non-invocation of Core-P and Core-DG;

(vii) a theorem ceiling preventing unforced-blow-up, generic-Core, paper–Lean, lifecycle, or target-falsification claims from leaking out of the reconstruction; and

(viii) a vector status discipline that separates conditional composition success from independent mathematical verification.

No external analytic node receives $\mathrm{VerifierStatus}{=}\mathrm{pass}$ *merely from this closure theorem.*

Proof. Items (i)–(v) are Definitions BFA.5.6 and BFA.5.7, Lemmas BFA.5.8 and BFA.5.9, Theorems BFA.5.11, BFA.5.12, BFA.5.14 and BFA.5.15, and Corollary BFA.5.16. Item (vi) is Theorem BFA.5.10. Item (vii) is Definition BFA.5.18, Theorems BFA.5.19, BFA.5.21 and BFA.5.22, and Corollary BFA.5.20. Item (viii) is Definition BFA.5.23 and Policy 32. □

Remark BFA.5.25 (Next consumer: BFA.6)**.** BFA.5 has shown that the external source route, *if true at its exact registered strength*, composes inside BFA to the same Clay-C/Clay-D targets and no stronger target. The next task is different: test whether the reconstructed route is compatible with the existing AK-v21 constitutional constraints and whether any contradiction, forbidden shortcut, ownership violation, finite-to-infinite leak, or hidden converse appears. That is the compatibility/no-contradiction audit of BFA.6; it must not be confused with the conditional reconstruction theorem proved here.

BFA.6 AK–OpenAI Compatibility and No-Contradiction Audit

BFA.6.1 Purpose, scope, and audit mode

Declaration BFA.6.1 (BFA.6 objective)**.** BFA.6 audits the source-faithful reconstruction of BFA.1–BFA.5 against the frozen AK-HPDST v21.0.0 governance and theorem-boundary rules. Its purpose is not to reprove any external analytic lemma. It asks whether the registered BFA route, at the exact strength already declared, violates any v21 rule on source binding, typing, finite-to-infinite passage, local-to-global passage, non-compensation, ownership, Return, benchmark quarantine, converse direction, target strength, formalization, or post-release governance.

Policy 33 (BFA-COMP-01: compatibility is not analytic verification)**.** *A compatibility verdict concerns the logical and governance shape of the BFA route relative to AK-v21. It does not set* $\mathsf{SrcTrue}_\nu(N)$*, does not change any external analytic verifier status, and does not establish the truth of the OpenAI source theorem.*

Definition BFA.6.2 (Basis-relative rule compatibility)**.** Let $\mathfrak{P}$ be a declared finite registry of frozen AK-v21 rules. For a declared BFA lane G, define

$$\mathsf{Compat}_{21}^{\mathfrak{P}}(G)$$

to mean that every consumed rule in $\mathfrak{P}$ is respected by the registered node types, dependency edges, status sorts, assumption contexts, source ownership, target strength, and Return direction of G. BFA.6 uses the particular declared basis $\mathfrak{P}_{\mathrm{BFA6}}$ introduced below. No verdict on rules outside that basis is encoded by the symbol $\mathsf{Compat}_{21}^{\mathfrak{P}_{\mathrm{BFA6}}}$.

The associated clause verdict sort is

$$\mathsf{CompatVerdict} = \{\mathrm{conformant}, \mathrm{conflict}, \mathrm{unresolved}, \mathrm{not_applicable}\}.$$

This is not the atomic mathematical verifier sort $\{\mathrm{pass}, \mathrm{reject}, \mathrm{undefined}, \mathrm{not_invoked}\}$.

Definition BFA.6.3 (Rule-level conflict)**.** A rule-level conflict occurs only when BFA positively asserts a theorem, status transfer, ownership transfer, implication, or target promotion that is forbidden by a consumed frozen v21 rule. Mere absence of an analytic proof is not a rule conflict; it is handled by the relevant mathematical verifier or by an unresolved proof obligation.

Theorem BFA.6.4 (Compatibility is not a formal consistency theorem)**.** *A proof of* $\mathsf{Compat}_{21}(G)$ *is neither a proof of formal consistency of AK-HPDST nor a proof that every theorem used in G is mathematically true.*

Proof. By Definitions BFA.6.2 and BFA.6.3, the compatibility audit compares registered BFA claims with frozen v21 rule boundaries. It does not quantify over all sentences of the AK theory and does not discharge the external analytic premises of BFA.5. Hence neither formal consistency nor analytic truth follows. □

Theorem BFA.6.5 (Declared-basis limitation)**.** *A verdict* $\mathsf{Compat}_{21}^{\mathfrak{P}_{\mathrm{BFA6}}}(G)$ *is a compatibility theorem only relative to the explicitly audited basis* $\mathfrak{P}_{\mathrm{BFA6}}$*. It is not an exhaustive theorem that every sentence, policy, appendix-local rule, migration clause, or future-discovered constraint in the frozen v21 corpus has been checked against G.*

Proof. The definition quantifies only over the declared finite rule registry. An unlisted rule is outside that quantifier domain. Therefore absence of a conflict on the declared basis cannot be promoted into corpus-wide logical consistency or exhaustive compatibility. □

Proposition BFA.6.6 (Basis-extension monotonicity)**.** *If* $\mathfrak{P} \subseteq \mathfrak{P}'$*, then* $\mathsf{Compat}_{21}^{\mathfrak{P}'}(G)$ *implies* $\mathsf{Compat}_{21}^{\mathfrak{P}}(G)$*, while the converse need not hold. A conflict already witnessed in* $\mathfrak{P}$ *remains a conflict after basis extension.*

Proof. The first statement is restriction of a universal clause audit. The converse fails whenever a new rule in $\mathfrak{P}' \setminus \mathfrak{P}$ is violated. A previously witnessed forbidden assertion is not erased by adding further rules. □

BFA.6.2 The frozen v21 compatibility basis

Definition BFA.6.7 (BFA.6 compatibility basis)**.** The declared basis $\mathfrak{P}_{\mathrm{BFA6}}$ contains the following rule families at their frozen v21 strengths:

P1 **Exact source binding and source authority.** A citation, filename, hash, or prose similarity does not replace an exact bound statement and permitted-use record.

P2 **Typing before obstruction.** A mathematical verdict is not read before source identity, scope, type, direction, and authority are established.

P3 **Finite-to-infinite non-shortcut.** Finite observation, numerical convergence, finite prefixes, or stored apex data do not create a completion or reduction theorem.

P4 **Local-to-global non-shortcut.** Local certificates or local constructions do not create a global PDE conclusion without the exact assembly, localization, extension, or Return theorem required by the declared route.

P5 **Non-compensation.** A passing independent component cannot fill a missing or false theorem edge of another component.

P6 **Source-specific ownership and Core sovereignty.** A problem-specific PDE theorem is not promoted into Core-P or Core-DG by representation, reuse, validation, or successful application.

P7 **Realization/Bridge/Return separation.** Forward realization, internal construction, reduction, and external Return are distinct arrows with independent hypotheses and directions.

P8 **Benchmark quarantine for independent recovery.** A route advertised as independent Known-Theorem Recovery may not use the benchmark proof, truth value, conclusion-derived witness, or an equivalent oracle to discharge an upstream theorem-bearing edge.

P9 **No hidden converse.** A proved one-way implication is not inverted without a separately proved reverse theorem.

P10 **No target-strength inflation.** A route exports only the exact registered target strength; stronger source classes, weaker hypotheses, or stronger conclusions require new theorem evidence.

P11 **Formalization and infrastructure are nonproof.** Compilation, proof-assistant replay, hash identity, DAG closure, storage, consensus, or package integrity cannot supply missing informal or external mathematics without the required interpretation theorem.

P12 **Frozen-release non-backflow.** A post-v21 supplement does not retroactively mutate the v21 Claim Register, convert a v21 non-claim into a v21 theorem, or reassign canonical v21 ownership.

P13 **Proof-DAG sovereignty and shared-node identity.** A finite theorem-bearing dependency graph is acyclic at the declared ordinary dependency level; shared source theorems remain single identities with explicit multiple consumers; no cycle, graph closure, or repeated use creates a missing mathematical edge.

P14 **Failure-semantics separation.** Missing evidence, an undefined route, rejection of one proof route, falsity of one source lemma, and falsification of the external target are distinct outcomes. In particular, $\mathrm{undefined}$ is not reject, and route failure does not by itself falsify an existential Clay target.

Remark BFA.6.8 (Primary frozen sources)**.** The governing material is the v21 Main proof-DAG and Return calculus, Problem-Interface source/Return discipline, Search/Platform finite-to-infinite and benchmark-quarantine rules, Technical-Extension conservative closure, Companion nonpromotion rules, and the final Claim Register ownership and noncompensation policies; see [1, 3, 4, 5, 6, 7].

BFA.6.3 Source binding, typing, and ownership audit

Theorem BFA.6.9 (Source-binding compatibility)**.** *The BFA.1–BFA.5 source-faithful route is conformant with P1 and P2 at the architectural level.*

Proof. BFA.2 separates official Clay target authority from the OpenAI analytic paper, announcement, and formal artifacts, and binds the analytic and formal source families separately. BFA.4 further separates mathematical authority, route role, source defect, audit obligation, Core invocation, and evidence coordinates. BFA.5 introduces $\mathsf{SrcTrue}_{\nu}(N)$, $\mathsf{EdgeValid}_{\nu}(e)$, and $\mathsf{TypeFaithful}_{\nu}(G)$ as distinct predicates rather than inferring them from registration. No obstruction verdict is read merely from a stored number, source label, or artifact identity. □

Theorem BFA.6.10 (Ownership conservation and Core nonpromotion)**.** *The BFA route is conformant with P6. In particular,*

$$\mathsf{Invoke}(\mathsf{Core\text{-}P}) = \mathrm{not_invoked}, \qquad \mathsf{Invoke}(\mathsf{Core\text{-}DG}) = \mathrm{not_invoked}$$

remains compatible with a source-specific PDE audit route and does not create a Core theorem.

Proof. BFA.4 explicitly classifies the analytic nodes as external analytic authority or BFA-local projection roles and forbids synthetic Core insertion. The v21 Claim Register assigns generic Bridge/Return authority to Main and source-specific PDE components/Returns to the Problem Interface, while registration itself owns no underlying mathematical proof. Therefore the BFA representation neither rebinds the external analytic theorem to a Core owner nor requires an artificial Core object merely to make the source route readable. □

Proposition BFA.6.11 (Post-release supplement does not rewrite v21)**.** *BFA 1.0, as a post-v21 research supplement, does not retroactively alter the v21 Claim Register statement that the frozen NS appendices themselves prove no Clay-level Navier–Stokes theorem.*

Proof. The frozen v21 NS package states that its completed in-release route is the criterion-level track NS-KTR-SERRIN-L6, not a Clay-level route. BFA binds a later external source theorem and constructs a new source-faithful audit lane outside the frozen v21 source set. A later external theorem cannot be inserted backward as a missing v21 edge without a new release-level migration and registration action. Hence there is no retroactive promotion. □

BFA.6.4 Finite-to-infinite and local-to-global audit

Theorem BFA.6.12 (Finite-to-infinite compatibility)**.** *The registered BFA analytic architecture is conformant with P3 at the architectural level: no finite correction prefix is identified with the completed smooth object merely by stabilization, repetition, or graph adjacency.*

Proof. BFA.3 exposes the shared source theorem SUM separately from its two consumers and distinguishes residual improvement, stage-uniform derivative loss, and the final completion step. BFA.4 further separates the base all-order completion boundary $\partial^{\infty}_{\mathrm{base}}$ from the final correction-completion boundary $\partial^{\infty}_{\mathrm{corr}}$. Therefore the DAG contains explicit theorem-bearing completion edges rather than replacing them by a finite prefix. Whether those source theorems are analytically true remains outside this compatibility verdict. □

Proposition BFA.6.13 (Two completion uses remain noncompensating)**.** *A compatibility or verifier success for one use of* SUM *does not certify its other use.*

Proof. BFA.4 gives the base and final completion uses distinct consumers, assumption contexts, and boundaries. P5 forbids cross-component compensation. Shared theorem identity therefore supports reuse of a theorem statement, not reuse of a consumer-specific applicability verdict. □

Theorem BFA.6.14 (Proof-DAG sovereignty compatibility)**.** *The BFA.3 analytic registry is conformant with P13 at its declared coarse-grained dependency level.*

Proof. BFA.3 declares a finite directed acyclic graph with an explicit predecessor map. The shared summation theorem SUM is represented once and has two separately audited consumers rather than being duplicated into two mathematical authorities. Transitive redundancy in predecessor sets records source reuse but does not create reverse arrows. No graph cycle or DAG closure is used as a substitute for a theorem-bearing edge. □

Definition BFA.6.15 (Analytic localization versus AK categorical globalization)**.** For BFA, *analytic localization/extension* means a source theorem that constructs or extends the PDE fields and forcing on the declared physical domain. *AK categorical globalization* means the distinct v21 mechanism involving covers, overlaps, cocycles, coherence, effectivity, or a declared global-object theorem. The two are not identified by the word "local".

Theorem BFA.6.16 (No local-to-global shortcut)**.** *The BFA route is conformant with P4: its terminal whole-space and periodic conclusions consume explicit analytic localization, endpoint regularization, scaling, and periodization theorems rather than inferring a global PDE target from local certificate agreement.*

Proof. The BFA DAG includes separate nodes for localization, smooth extension of the forcing through the terminal time, comparison/uniqueness, viscosity transport, and periodization. BFA.4 explicitly states that endpoint regularization is not the finite-to-infinite completion edge. No sheaf/descent or overlap language is used to manufacture a PDE conclusion from local passes. Thus the route uses the source's actual globalizing analytic theorems at their declared scope. □

BFA.6.5 Non-compensation, Return, and target-strength audit

Theorem BFA.6.17 (Conjunctive Return compatibility)**.** *The BFA route is conformant with P5 and P7. In particular, the whole-space and periodic route packets are typed conjunctions; no strong coordinate can replace a missing coordinate.*

Proof. BFA.4 separates internal route packets $\mathcal{Q}_C(v), \mathcal{Q}_D(v)$ from target witness packets $\mathcal{W}_C(v), \mathcal{W}_D(v)$. BFA.5 requires the source assumption packet together with the internal route packet before witness export, and only the target witness then Returns to the exact Clay predicate. Thus energy, forcing regularity, comparison, scaling, and periodic transport retain separate obligations and cannot compensate for one another. □

Theorem BFA.6.18 (No hidden converse)**.** *The BFA route is conformant with P9.*

Proof. BFA.5 proves only the forward conditional implications

$$\mathfrak{A}_C(v) \Longrightarrow \mathsf{Clay\text{-}C}_v, \qquad \mathfrak{A}_D(v) \Longrightarrow \mathsf{Clay\text{-}D}_v,$$

and separately records that target truth does not imply the corresponding source-assumption packet. No reverse source reconstruction is inferred from Clay target truth. □

Theorem BFA.6.19 (Target-strength compatibility)**.** *The BFA route is conformant with P10. It exports the exact forced Clay-C/D targets bound in BFA.2 and does not promote them to unforced blow-up, arbitrary data blow-up, arbitrary forcing blow-up, broader uniqueness, or an AK-owned Core theorem.*

Proof. BFA.5 defines an explicit theorem ceiling and a no-strengthening theorem. BFA.4 also removes any periodic-energy condition from the official Clay-D target when that condition belongs only to the stronger source construction. Hence both upward and downward target distortion are blocked: extra source strength is not silently promoted into the target definition, and the target is not strengthened beyond its frozen quantifiers. □

Theorem BFA.6.20 (Failure-semantics compatibility)**.** *The BFA route is conformant with P14. In particular,*

$$\textit{missing evidence} \neq \textit{rejected source lemma} \neq \textit{rejected route} \neq \textit{target falsification}.$$

For the existential Clay C/D targets, invalidating one constructed witness or one proof route does not establish the negation of the target.

Proof. BFA.1 separates $\mathrm{undefined}$, reject, route falsification, and target falsification, while BFA.5 treats the source route only as a sufficient route to the frozen target. Hence a blocked or false route removes that route's entitlement to Return but does not supply the universal negation required to falsify an existential target. □

BFA.6.6 Source-faithful use versus benchmark quarantine

Definition BFA.6.21 (Two noninterchangeable BFA research lanes)**.** The BFA research program distinguishes:

$$\mathsf{SF} := \text{source-faithful audit/reconstruction lane},$$
$$\mathsf{IR} := \text{benchmark-quarantined independent-recovery lane}.$$

In SF, the bound external source proof may be consumed as source mathematics because independent recovery is not claimed. In IR, the benchmark proof and conclusion-derived theorem-bearing information are quarantined from the upstream construction.

Definition BFA.6.22 (Independent-recovery precursor and contamination sets)**.** Fix a benchmark-publication cutoff t_*. An IR experiment declares an immutable precursor corpus $\mathcal{P}_{<t_*}$, together with the official target statement. The prohibited benchmark-derived contamination set $\mathcal{C}_{\mathrm{bench}}$ includes, at minimum:

(i) the benchmark analytic proof, formal proof, and benchmark truth-value or success-announcement information;

(ii) benchmark-derived lemma inventories, proof DAGs, correction schedules, bottleneck lists, or source-faithful theorem summaries;

(iii) conclusion-derived witnesses or constructions selected because the benchmark proof revealed them; and

(iv) prompts, memory, retrieval caches, hidden context, or intermediate artifacts that carry theorem-bearing information from items (i)–(iii).

The pre-benchmark official problem statement is not contamination: it specifies the problem to be recovered without revealing the benchmark's later success direction. A post-benchmark target specification that identifies the successful alternative or imports an equivalent benchmark-strength theorem statement is itself theorem-bearing disclosure and is classified by the target-mode rules of BFA.7. A precursor theorem may be admitted only through an immutable, source-identified record showing that its availability and admissibility do not depend on the benchmark proof.

Definition BFA.6.23 (IR provenance eligibility)**.** The provenance sort for an advertised independent-recovery run is

$$\mathsf{IRProvenance} = \{\mathrm{eligible}, \mathrm{contaminated}, \mathrm{unassessed}\}.$$

An IR run is eligible only if its theorem-bearing execution environment has access to the frozen target statement and admissible precursor corpus but not to C_{bench}, and if the isolation boundary itself is recorded by reproducible provenance artifacts.

Theorem BFA.6.24 (Source-faithful quarantine-scope compatibility)**.** *The use of the OpenAI analytic proof in BFA.1–BFA.5 does not violate P8 because those sections advertise the* SF *lane, not an independent Known-Theorem Recovery. No* SF *success transfers an IR provenance status.*

Proof. Benchmark quarantine is an independence requirement: it prevents a route from encoding the theorem it advertises as independently recovered. BFA.1–BFA.5 instead bind the OpenAI proof precisely as an external source and ask whether its theorem chain can be represented and composed without strength inflation. The provenance is therefore explicit rather than hidden. Relabeling the same construction as independent recovery would change the admissible-input policy and is not licensed by any SF verdict. □

Theorem BFA.6.25 (Current BFA construction context is not IR-certifiable)**.** *The present BFA construction context is* contaminated *for purposes of certifying an* IR *result, because BFA.2–BFA.6 intentionally consume the OpenAI analytic proof, its proof architecture, and source-faithful derived records. Any theorem-bearing derivation produced with access to this context may be studied as a post-hoc recovery candidate but may not be labeled an independent recovery.*

Proof. The current context contains elements of C_{bench}, including benchmark-derived node identities, dependency structure, completion bottlenecks, and Return organization. Therefore the isolation condition in Definition BFA.6.23 fails. This is a provenance verdict, not a mathematical verdict on any candidate derivation. □

Policy 34 (BFA-COMP-02: no lane laundering)**.** *An* SF *success may not be relabeled as* IR *success by removing citations, renaming nodes, replaying the same proof, replacing the source text with a benchmark-derived summary, or running an agent whose prompt, memory, retrieval context, or accessible artifacts contain C_{bench}. Independence is a provenance property of the actual theorem-bearing construction.*

Policy 35 (BFA-COMP-03: IR certification requires isolated execution)**.** *A future certified IR claim requires a separately instantiated execution or research environment satisfying Definition BFA.6.23, an immutable precursor-corpus manifest, a recorded cutoff t_*, and an audit of all imported theorem-bearing inputs. Within the present source-exposed BFA context, BFA.7 may specify the IR protocol and evaluate candidate mathematics, but it cannot certify independence by declaration alone.*

BFA.6.7 Compatibility with the frozen v21 Navier–Stokes package

Proposition BFA.6.26 (Serrin recovery and BFA blow-up route are not contradictory)**.** *The frozen v21 track* NS-KTR-SERRIN-L6 *and the BFA source-faithful blow-up route can coexist without contradiction.*

Proof. The frozen Serrin track is an exact one-way recovery of a conditional strict-subcritical regularity implication on its registered unforced source regime. It is explicitly not an unconditional regularity or Clay-level theorem. The BFA route consumes a different later source construction with a smooth forcing and reaches the Clay C/D existence alternatives. Therefore BFA does not negate the Serrin implication, and the Serrin criterion does not prohibit a forced singular construction outside its registered source/hypothesis packet. □

Theorem BFA.6.27 (No contradiction with the v21 NS non-claim)**.** *The BFA source-faithful Clay-target reconstruction does not contradict the v21 statement that Appendix NS-A–NS-C itself proves no blow-up theorem and no Clay-level theorem.*

Proof. The v21 statement is scoped to the mathematics available in the frozen v21 NS appendices and their registered routes. BFA is a later supplement whose premise includes a newly bound external analytic source. Adding a new external premise can support a new conditional route without making the earlier no-claim false or changing what the earlier documents proved. □

BFA.6.8 Formalization, infrastructure, and status-separation audit

Theorem BFA.6.28 (Formalization nonpromotion compatibility)**.** *The BFA route is conformant with P11.*

Proof. BFA.2 binds the formal repository separately from the analytic paper, and BFA.5 keeps formal semantic equivalence distinct from analytic truth and target interpretation. Neither a pinned commit, $\mathrm{sorry_count}: 0$, successful compilation, package hash, nor DAG closure is used to discharge an external analytic edge. This matches the v21 rule that formal wiring and infrastructure support auditability but do not create missing mathematics. □

Proposition BFA.6.29 (Status-sort compatibility)**.** *BFA.1–BFA.5 do not identify source truth, mathematical verifier status, compatibility verdict, workflow state, lifecycle state, or institutional adjudication.*

Proof. BFA.1 separates lifecycle and mathematical status; BFA.3 separates registry status from verifier status; BFA.5 represents analytic truth as the proposition $\mathsf{SrcTrue}_{\nu}(N)$, not as the token $\mathrm{not_invoked}$; and BFA.6 uses the separate sort $\mathsf{CompatVerdict}$. Hence no lexical equality creates a status conversion. □

BFA.6.9 Compatibility matrix and aggregate verdict

Definition BFA.6.30 (BFA.6 compatibility matrix)**.** The architecture-level audit matrix is:

Rule	BFA discharge location	Verdict
P1	BFA.2 exact source/target binding	conformant
P2	BFA.4 typed projection; BFA.5 type fidelity	conformant
P3	BFA.3 SUM, O8/O9; BFA.4 completion boundaries	conformant
P4	BFA.3 O10/O11/O15; BFA.4 terminal/transport boundaries	conformant
P5	BFA.4 route packets and boundary noncompensation	conformant
P6	BFA.4 authority/Core-use separation	conformant
P7	BFA.4 $Q/\mathcal{W}$ separation; BFA.5 Return factorization	conformant
P8	BFA.6 SF/IR lane separation	conformant for SF; current context contaminated for IR certification
P9	BFA.5 no-converse theorem	conformant
P10	BFA.5 ceiling; BFA.4 periodic-energy nonpromotion	conformant
P11	BFA.2 formal-source separation; BFA.5 no-pass leakage	conformant
P12	BFA.1 post-release scope; BFA.6 no-retroactivity	conformant
P13	BFA.3 acyclic predecessor map and shared-node identity	conformant
P14	BFA.1/BFA.5 route/target failure separation	conformant

The matrix evaluates architecture and governance only. It is not a substitute for BFA.8–BFA.12 mathematical verification.

Theorem BFA.6.31 (Declared-basis AK-v21 architectural compatibility theorem)**.** *For the source-faithful BFA lane frozen through BFA.5,*

$$\boxed{\mathsf{Compat}_{21}^{\mathfrak{B}_{\mathrm{BFA6}}}(G_{\mathsf{SF}})}$$

holds at the rule-governance level represented by P1–P14.

Proof. P1–P14 are discharged by Theorems BFA.6.9, BFA.6.10, BFA.6.12, BFA.6.16, BFA.6.17, BFA.6.18, BFA.6.19, BFA.6.24, BFA.6.25, BFA.6.14, BFA.6.20, BFA.6.28, together with Propositions BFA.6.11 and BFA.6.29. No audited clause requires a prohibited status, ownership, implication, or target promotion. □

Theorem BFA.6.32 (No identified AK-v21 rule contradiction)**.** *Within the declared BFA.1–BFA.5 claim set and the declared compatibility basis* $\mathfrak{B}_{\mathrm{BFA6}}$ *of Definition BFA.6.7, no rule-level conflict in the sense of Definition BFA.6.3 is identified.*

Proof. This is the negative formulation of Theorem BFA.6.31 on the declared basis. The qualifiers "rule-level" and "declared basis" are essential: a future mathematical rejection of a source lemma would invalidate or block the corresponding route, but would not retroactively show that BFA violated a governance rule merely by registering that source lemma as an unverified premise. □

Corollary BFA.6.33 (Compatibility does not certify the OpenAI proof)**.** *Neither Theorem BFA.6.31 nor Theorem BFA.6.32 implies* $\mathsf{SrcTrue}_{\nu}(N)$ *for any external analytic node N, nor does it set the OpenAI proof to mathematical* pass.

BFA.6.10 BFA.6 closure and handoff

Definition BFA.6.34 (BFA.6 audit-status vector)**.** At the close of BFA.6 the intended status vector is

$$\begin{aligned}
\mathsf{source_binding_compatibility} &= \mathrm{conformant},\\
\mathsf{typing_compatibility} &= \mathrm{conformant},\\
\mathsf{finite_to_infinite_architecture} &= \mathrm{conformant},\\
\mathsf{local_to_global_architecture} &= \mathrm{conformant},\\
\mathsf{noncompensation_architecture} &= \mathrm{conformant},\\
\mathsf{ownership_and_Core_boundary} &= \mathrm{conformant},\\
\mathsf{return_and_strength_boundary} &= \mathrm{conformant},\\
\mathsf{source_faithful_quarantine_classification} &= \mathrm{conformant},\\
\mathsf{declared_basis_coverage} &= \mathrm{P1–P14},\\
\mathsf{full_v21_corpus_coverage} &= \mathrm{not_claimed},\\
\mathsf{IR_provenance_current_context} &= \mathrm{contaminated},\\
\mathsf{independent_recovery} &= \mathrm{not_invoked},\\
\mathsf{external_analytic_verification} &= \mathrm{not_invoked},\\
\mathsf{paper_formal_target_equivalence} &= \mathrm{undefined}.
\end{aligned}$$

The provenance, independent-recovery, external-verification, and formal-equivalence coordinates are not promoted by the compatibility result; they belong to different theorem, provenance, and verifier sorts.

Theorem BFA.6.35 (BFA.6 compatibility closure)**.** *BFA.6 establishes, at architecture/governance level only, all of the following:*

(i) the BFA source-faithful lane does not require a forbidden AK-v21 finite-to-infinite, local-to-global, converse, ownership, or status shortcut;

(ii) the OpenAI source proof is used transparently as source mathematics, not laundered into an independent-recovery claim;

(iii) the frozen v21 Serrin recovery and v21 NS non-claims remain intact;

(iv) the exact Clay target strength and BFA.5 theorem ceiling remain intact;

(v) formalization, infrastructure, and institutional lifecycle data remain noncompensating downstream evidence;

(vi) no identified rule-level contradiction between BFA.1–BFA.5 and the consumed frozen AK-v21 rules in $\mathfrak{P}_{\mathrm{BFA6}}$ *remains; and*

(vii) the present source-exposed construction context is not eligible to certify an independent-recovery provenance claim.

No source analytic lemma is independently verified by this closure theorem.

Remark BFA.6.36 (Next consumer: BFA.7)**.** BFA.7 should first construct the *Independent-Recovery Eligibility and Quarantine Protocol*. An actual IR certification requires a separately isolated execution satisfying Policy 35; the present BFA context is already source-exposed. BFA.7 may still study candidate precursor mathematics, but within this context the strongest honest provenance label is a *post-hoc recovery candidate*, not an independent recovery. BFA.6 therefore supplies the compatibility and contamination boundary, not an independence certificate.

BFA.7 Independent-Recovery Eligibility and Quarantine Protocol

BFA.7.1 Purpose and non-claim

Declaration BFA.7.1 (BFA.7 objective)**.** BFA.7 specifies the provenance conditions under which a future derivation may be advertised as an *independent recovery* of the Navier–Stokes target rather than as a source-faithful reconstruction or post-hoc recovery candidate. It does not execute a certified independent recovery in the present BFA construction context. That context has already consumed the OpenAI analytic proof, its proof architecture, and source-derived audit records and is therefore contaminated for strict independent-recovery provenance by Theorem BFA.6.25.

Policy 36 (BFA-IR-01: provenance and mathematical truth are orthogonal)**.** *Independent-recovery provenance is a statement about the origin and admissible information flow of a theorem-bearing construction. It is not a mathematical verifier verdict. A mathematically correct proof may have contaminated provenance, and an impeccably isolated run may still produce a false or incomplete proof.*

Theorem BFA.7.2 (No independence by relabeling)**.** *No theorem-bearing artifact produced in the present source-exposed BFA context becomes an independent recovery merely by deleting citations, renaming lemmas, changing notation, paraphrasing the benchmark proof, or replaying the same construction in a fresh document.*

Proof. Policy 34 makes independence a provenance property of the actual theorem-bearing construction. The present context contains elements of the contamination set of Definition BFA.6.22. Surface rewriting does not change that information history. □

BFA.7.2 The independent-recovery experiment object

Definition BFA.7.3 (Independent-recovery experiment)**.** An independent-recovery experiment is a typed record

$$\mathfrak{E}_{\mathrm{IR}} = \left(T_0, \mathsf{TargetMode}, \mathsf{Cut}, \mathcal{P}_{<t_*}, \mathcal{A}, \mathcal{E}, \mathcal{K}_{\mathrm{obs}}, \mathcal{K}_{\mathrm{prior}}, \mathcal{H}_{\mathrm{op}}, \mathcal{L}, \mathcal{O}, \mathcal{U}\right),$$

with the following fields.

(i) T_0: the frozen target package presented to the recovery run;

(ii) $\mathsf{TargetMode}$: the disclosure class of T_0;

(iii) Cut: the benchmark-cutoff record, including a conservative time bound and the evidence used to bind it;

(iv) $\mathcal{P}_{<t_*}$: the immutable admissible precursor corpus;

(v) $\mathcal{A}$: the precursor-admissibility and selection manifest;

(vi) $\mathcal{E}$: the execution environment and tool-permission record;

(vii) $\mathcal{K}_{\text{obs}}$: observable theorem-bearing inputs such as prompts, memory, retrieval, files, and tool outputs;

(viii) $\mathcal{K}_{\text{prior}}$: theorem-bearing prior state of the reasoner, including checkpoint/model prior information to the extent auditable;

(ix) $\mathcal{H}_{\text{op}}$: theorem-bearing human-operator state and intervention channel;

(x) $\mathcal{L}$: the append-only provenance and action log;

(xi) O: the sealed theorem-bearing output; and

(xii) $\mathcal{U}$: the post-sealing unsealing/comparison record.

The record is a provenance object. It does not assert that O is a valid mathematical proof.

Definition BFA.7.4 (Benchmark-cutoff record)**.** The benchmark cutoff is not a bare date string. It is a record

$$\mathsf{Cut} = \big(\mathsf{EventID}, \mathsf{TimeBound}, \mathsf{Authority}, \mathsf{EvidenceHash}, \mathsf{ConservativeRule}\big).$$

For a strict problem-level experiment, t_* is bound no later than the first public disclosure that conveys theorem-bearing information about a successful benchmark solution. If the exact disclosure time cannot be proved, BFA uses an earlier defensible bound rather than a later permissive one.

Definition BFA.7.5 (Target-disclosure modes)**.** BFA distinguishes three noninterchangeable target modes.

(a) **Problem-level mode.** T_0 is the pre-benchmark official Clay problem statement, with no post-benchmark indication of which alternative is successful. Only this mode can support a future claim of independent recovery of the *problem-level solution direction*.

(b) **Specified-theorem mode.** T_0 contains a theorem statement equivalent in strength to the benchmark result, but not its proof or route. A successful isolated run may at most be described as an independent derivation of the specified theorem; the theorem statement itself discloses the successful direction and therefore cannot certify independent discovery of that direction.

(c) **Route-disclosed mode.** T_0 or the upstream context contains benchmark-derived lemma inventories, bottlenecks, witnesses, proof-DAG structure, or comparable route hints. This mode is source-exposed and is not strict-IR eligible.

Definition BFA.7.6 (Problem-level admissible target package)**.** For strict problem-level IR, the target package contains only a pre-benchmark version of the official problem statement together with the minimum definitions needed to parse it. It excludes success announcements, statements that reveal which alternative is known to hold, benchmark-derived proof summaries, source-specific witnesses, and post-benchmark annotations that materially narrow the search.

Theorem BFA.7.7 (Target-disclosure ceiling)**.** *Providing the benchmark theorem statement while withholding its proof can support, at most, a specified-theorem isolated-derivation claim. It cannot by itself support a claim that the solver independently discovered which Clay alternative or solution direction succeeds.*

Proof. The disclosed theorem statement already transports the truth-direction needed to distinguish one successful branch of the original problem from its alternatives. Proof quarantine can still test independent derivation of that specified statement, but it cannot erase information supplied by the target selection itself. □

Policy 37 (BFA-IR-01A: solver-facing outcome blindness)**.** *In problem-level mode, the theorem-bearing reasoner is not told that the benchmark has been solved, which alternative later succeeded, whether a finite- time blow-up construction exists, or that the experiment is expected to recover a known successful route. Such success information is theorem-bearing because it changes the search prior even when no proof details are supplied.*

Remark BFA.7.8 (Why the original problem statement is admissible)**.** The pre-benchmark Clay problem statement specifies the research task without revealing the benchmark's later success direction. This differs from supplying an exact post-benchmark blow-up theorem as the target. BFA therefore treats target admissibility as mode-dependent rather than declaring every target statement automatically clean.

BFA.7.3 Precursor-corpus admissibility and selection bias

Definition BFA.7.9 (Admissible precursor record)**.** A precursor artifact P is admissible only through a record

$$\mathsf{Prec}(P) = \big(\mathsf{ID}, \mathsf{Date}, \mathsf{Hash}, \mathsf{Statement}, \mathsf{Authority}, \mathsf{SelectionOrigin}, \mathsf{PermittedUse}\big),$$

where the artifact predates t_*, its identity is immutable, and its selection origin can be audited without using benchmark-derived theorem-bearing information.

Policy 38 (BFA-IR-02: pre-benchmark existence is not enough)**.** *A theorem that existed before t_* is not automatically an admissible precursor if it was selected only because the benchmark proof later revealed its importance. Benchmark-informed theorem selection is itself a possible information channel and must be recorded as such.*

Definition BFA.7.10 (Pre-registered corpus and broad frozen corpus)**.** Two precursor-selection modes are admissible in principle.

(a) **Pre-registered mode:** the exact precursor list or search policy was frozen before t_*.

(b) **Broad frozen mode:** a sufficiently broad corpus was archived before t_*, and the recovery agent may search within it without access to post-benchmark ranking, annotations, or route hints.

A post-benchmark hand-picked list is not promoted into either mode merely because every item on the list individually predates t_*.

Policy 39 (BFA-IR-02A: frozen AK-corpus admissibility)**.** *A complete immutable AK artifact set demonstrably frozen before t_* may be admitted wholesale as part of a broad precursor corpus. By contrast, a post-benchmark subset chosen because BFA.2–BFA.6 revealed which AK components appear relevant is not strict-IR admissible merely because the selected files are individually old. The post-benchmark BFA supplement itself is excluded from Phase-I theorem-bearing input.*

Proposition BFA.7.11 (Selection-channel contamination)**.** *If the selection of an upstream precursor materially depends on benchmark knowledge, then the resulting run is not strict-IR certifiable even when the selected theorem itself predates the benchmark.*

Proof. The selected theorem may be old, but the map from the available mathematical universe to the chosen theorem carries post-benchmark information. That map is part of the theorem-bearing construction. Policy 38 therefore classifies the selection channel, not merely the publication date of its output. □

BFA.7.4 Contamination closure and accessible knowledge state

Definition BFA.7.12 (Theorem-bearing contamination closure)**.** Let C_{bench} be the benchmark-derived contamination set of Definition BFA.6.22. Define its theorem-bearing semantic closure

$$\mathrm{Cl}_{\mathrm{th}}(C_{\mathrm{bench}})$$

to include not only literal benchmark artifacts but also theorem-bearing paraphrases, summaries, extracted lemma inventories, proof-DAG encodings, source-derived search priorities, witness templates, benchmark-conditioned selection rules, and intermediate artifacts whose mathematical content materially transports information from C_{bench}.

Definition BFA.7.13 (Observable, prior, and operator knowledge channels)**.** For an execution environment $\mathcal{E}$, BFA separates three theorem-bearing channels.

(i) $\mathcal{K}_{\mathrm{obs}}(\mathcal{E})$: observable prompts, persistent or session memory, retrieval indices, caches, connected sources, local files, imported intermediate artifacts, and theorem-bearing tool output;

(ii) $\mathcal{K}_{\mathrm{prior}}(\mathcal{E})$: theorem-bearing prior state of the reasoner, including model/checkpoint prior information, training or corpus provenance, and frozen library state to the extent auditable;

(iii) $\mathcal{H}_{\mathrm{op}}(\mathcal{E})$: theorem-bearing information held by human operators who may select prompts, rank candidates, intervene, or otherwise steer the recovery run.

The separation prevents a clean observable session from silently absorbing an opaque model prior or a source-exposed human hint.

Definition BFA.7.14 (Channel-wise strict quarantine condition)**.** A run satisfies strict theorem-bearing quarantine only if the recorded evidence supports, at every theorem-bearing stage of Phase I,

$$\mathcal{K}_{\mathrm{obs}}(\mathcal{E}) \cap \mathrm{Cl}_{\mathrm{th}}(C_{\mathrm{bench}}) = \varnothing,$$

$$\mathcal{K}_{\mathrm{prior}}(\mathcal{E}) \cap \mathrm{Cl}_{\mathrm{th}}(C_{\mathrm{bench}}) = \varnothing,$$

and either

$$\mathcal{H}_{\mathrm{op}}(\mathcal{E}) \cap \mathrm{Cl}_{\mathrm{th}}(C_{\mathrm{bench}}) = \varnothing$$

or a mechanically enforced operator firewall prevents $\mathcal{H}_{\mathrm{op}}$ from contributing theorem-bearing information. The semantic intersections are policy-level idealizations; certification must be discharged by channel-specific evidence rather than by claiming an algorithmic decision procedure for semantic equivalence.

Definition BFA.7.15 (Source-exposed operator firewall)**.** A source-exposed human operator may perform administrative actions in a strict IR experiment only when those actions are pre-specified or mechanically constrained so that they cannot transmit theorem-bearing choices. In particular, source-exposed operators may not supply lemmas, rank mathematical routes, choose among candidate witnesses using benchmark knowledge, repair proof steps, or adapt prompts in response to theorem-bearing intermediate output.

Proposition BFA.7.16 (Human-channel contamination)**.** *A theorem-bearing intervention by a source-exposed operator makes the affected run non-certifiable as strict IR, even if the model, files, retrieval, and network channels are otherwise clean.*

Proof. The intervention transports benchmark-derived information through $\mathcal{H}_{\mathrm{op}}$ into the theorem-bearing construction. It therefore violates the channel-wise quarantine requirement unless the intervention is blocked from mathematical content by the operator firewall. □

Remark BFA.7.17 (Opaque prior state)**.** For an AI system, model parameters may encode information whose exact training provenance is not inspectable from the recovery session. Absence of visible retrieval from a benchmark is therefore not, by itself, evidence that the model prior is uncontaminated. BFA.7 treats this as an evidence limitation, not as proof of contamination and not as proof of cleanliness.

Remark BFA.7.18 (A practical strict-prior route)**.** A frozen reasoner whose checkpoint and theorem-bearing training corpus are provably bounded before t_*, or a reasoner trained only from the frozen admissible precursor corpus, can in principle discharge the prior-state channel. A current opaque proprietary prior generally cannot receive that certification from session-level evidence alone.

BFA.7.5 Isolation evidence levels

Definition BFA.7.19 (Isolation-evidence level)**.** The isolation-evidence sort is

$$\mathsf{IsolationLevel} = \{\mathrm{strict}, \mathrm{bounded}, \mathrm{insufficient}\}.$$

It is interpreted as follows.

(i) strict: all theorem-bearing input channels relevant to the independence claim, including the prior state of the theorem-bearing reasoner, are auditable and shown to satisfy the quarantine policy;

(ii) $\mathrm{bounded}$: observable prompts, retrieval, files, tools, memory, and human intervention are quarantined, but at least one theorem-bearing prior channel is opaque or only partially auditable;

(iii) $\mathrm{insufficient}$: a known contamination channel exists or the available evidence does not support even bounded isolation.

Definition BFA.7.20 (Certification strength)**.** BFA distinguishes the following labels:

strict_IR	IRProvenance = eligible and IsolationLevel = strict.
bounded_IR_candidate	IRProvenance = unassessed and IsolationLevel = bounded; observable channels may be clean while at least one theorem-bearing prior channel remains unresolved.
posthoc_candidate	IRProvenance = contaminated.
unassessed	The provenance evidence is incomplete.

Only the first label licenses the phrase *certified independent recovery* in BFA.

Definition BFA.7.21 (Recovery-claim class)**.** The public claim licensed by a successful provenance certificate is bounded by both target mode and isolation evidence:

(a) problem_level_IR: requires problem-level target mode and strict_IR;

(b) specified_theorem_independent_derivation: requires specified-theorem mode and strict_IR, but carries no claim of independent discovery of the successful Clay direction;

(c) bounded_isolated_candidate: bounded evidence only;

(d) posthoc_candidate: contaminated or source-exposed provenance.

No stronger public label may be inferred from mathematical correctness alone.

Theorem BFA.7.22 (Opaque-prior ceiling)**.** *If a theorem-bearing reasoner's prior state is not auditable strongly enough to exclude membership in* $\mathrm{Cl}_{\mathrm{th}}(\mathcal{C}_{\mathrm{bench}})$*, then the run cannot be promoted to* strict_IR *solely from clean observable-session logs.*

Proof. Strict IR requires strict quarantine at every theorem-bearing input channel by Definitions BFA.7.14 and BFA.7.19. Clean observable-session channels establish no fact about an unaudited prior channel. Hence the strict certification premise is not discharged. □

Definition BFA.7.23 (Theorem-bearing prompt and search-policy origin)**.** The initial theorem-bearing prompt, decomposition policy, and adaptive search policy are themselves input artifacts. For strict problem-level IR they must be either (i) frozen before t_*, or (ii) generated after t_* by a source-unexposed operator from a target-neutral generic protocol. A source-exposed designer may configure mechanical isolation infrastructure but may not encode theorem-bearing route preferences into prompts, scoring rules, or stopping criteria.

Proposition BFA.7.24 (Prompt-channel leakage)**.** *A benchmark-informed prompt, route ranking, stopping criterion, or adaptive search intervention can contaminate an otherwise frozen model-and-corpus run.*

Proof. These objects alter which mathematical paths are explored or retained and are therefore theorem-bearing selection channels. They belong to $\mathcal{K}_{\mathrm{obs}}$ or $\mathcal{H}_{\mathrm{op}}$, so channel-wise strict quarantine requires them to be clean as well. □

BFA.7.6 Two-phase sealed recovery protocol

Definition BFA.7.25 (Phase I: sealed recovery)**.** In Phase I, the recovery environment receives only the target package permitted by its declared TargetMode, the admissible precursor corpus, and explicitly allowed generic computation/proof tools. The benchmark proof, benchmark-derived summaries, BFA.2–BFA.6 source-faithful records, success announcements, and post-benchmark route hints are unavailable. The run is network-closed unless a frozen pre-authorized corpus endpoint is part of $\mathcal{P}_{<t_*}$. In problem-level mode, the solver-facing interface also satisfies Policy 37: it does not announce that a successful benchmark solution exists.

Definition BFA.7.26 (Sealing event)**.** A theorem-bearing output is sealed before benchmark unquarantining. The sealing record contains, at minimum,

$$\mathsf{Seal} = \big(\mathsf{OutputHash}, \mathsf{Time}, \mathsf{CorpusRoot}, \mathsf{EnvID}, \mathsf{LogRoot}\big).$$

After sealing, the Phase-I output is immutable for purposes of the independence claim.

Definition BFA.7.27 (Phase II: post-sealing comparison)**.** Only after the sealing event may the benchmark proof, BFA source-faithful DAG, and later literature be exposed for comparison. Phase II may classify overlap, novelty, missing edges, or divergence, but no Phase-II repair is retroactively part of the Phase-I independent-recovery output.

Policy 40 (BFA-IR-03: no retroactive repair)**.** *A proof gap found after unsealing may be repaired as a new post-hoc or source-assisted result, but the repaired artifact does not inherit the original strict-IR certificate unless the repair itself was already present and sealed in Phase I.*

Theorem BFA.7.28 (Seal-before-compare necessity)**.** *A run that compares its developing proof with the benchmark proof before the candidate proof is sealed cannot obtain strict-IR provenance for the compared and subsequently modified artifact.*

Proof. The comparison introduces benchmark-derived information into the theorem-bearing construction before output identity is fixed. The modified artifact therefore has access to an element of $\mathrm{Cl}_{\mathrm{th}}(C_{\mathrm{bench}})$, violating strict quarantine. □

BFA.7.7 Execution-environment requirements

Definition BFA.7.29 (Minimal strict-IR environment manifest)**.** A strict-IR environment manifest records at least:

(i) target-package identity, hash, and TargetMode;

(ii) benchmark-cutoff record Cut;

(iii) precursor-corpus Merkle root or equivalent immutable manifest;

(iv) theorem-bearing reasoner identity and auditable prior-state evidence;

(v) system/user prompts, initial context, and prompt/search-policy origin;

(vi) solver-facing outcome-blindness record when in problem-level mode;

(vii) memory configuration;

(viii) network, retrieval, connector, and file permissions;

(ix) permitted theorem/proof-assistant libraries and their versions;

(x) human-intervention policy and, where needed, an operator-firewall specification;

(xi) random seeds or nondeterminism records where reproducibly available;

(xii) append-only action/provenance-log root; and

(xiii) sealing-event record.

Policy 41 (BFA-IR-04: deterministic tools versus theorem-bearing tools)**.** *Deterministic algebra, numerical computation, compilation, and proof-assistant kernel checking may be permitted if their libraries and inputs satisfy the precursor policy. A search engine, retrieval agent, language model, human expert, or theorem database capable of introducing theorem-bearing information is an admissible input channel only when its accessible knowledge state is covered by the quarantine evidence.*

Remark BFA.7.30 (Proof assistants)**.** A proof assistant may verify a candidate proof without violating independence provided that the imported library is admissible and frozen. The kernel's acceptance is still a formal-verification result and does not by itself prove that the formal statement is the intended Clay statement; the paper–formal–target interpretation remains a separate BFA obligation.

BFA.7.8 Canaries, logs, and limits of negative evidence

Definition BFA.7.31 (Quarantine canary)**.** A quarantine canary is a deliberately distinctive marker placed only inside a prohibited benchmark-side artifact or retrieval namespace. Its appearance in Phase-I prompts, logs, or outputs is positive evidence of leakage.

Proposition BFA.7.32 (Canary asymmetry)**.** *Detection of a quarantine canary is evidence against isolation. Failure to detect a canary is not proof of isolation.*

Proof. A canary is only one possible observable trace of leakage. Semantic leakage may occur without copying the marker, and an opaque prior may contain benchmark content without exposing any literal canary. Hence the diagnostic is one-sided. □

Policy 42 (BFA-IR-05: logs are evidence, not proof)**.** *Clean logs, network denial, hash manifests, and sandbox attestations strengthen an independence case but do not compensate for an unaudited theorem-bearing input channel. Infrastructure evidence is governed by the same non-compensation discipline as other BFA evidence.*

BFA.7.9 Mathematical verification after provenance classification

Definition BFA.7.33 (Orthogonal result axes)**.** Every recovery attempt is classified on at least two independent axes:

$$\mathsf{RecoveryResult} = (\mathsf{ProvenanceStatus}, \mathsf{MathematicalStatus}),$$

where ProvenanceStatus is governed by this section and MathematicalStatus uses the appropriate BFA/AK verifier sort. Neither coordinate determines the other.

Proposition BFA.7.34 (Four logical combinations are possible)**.** *At the protocol level, each of the following combinations is coherent:*

(i) strict independent provenance with a valid proof;

(ii) strict independent provenance with a rejected or undefined proof;

(iii) contaminated provenance with a valid proof; and

(iv) contaminated provenance with a rejected or undefined proof.

Therefore independence neither proves nor disproves the theorem.

Proof. Provenance constrains information flow, whereas mathematical verification checks theorem-bearing correctness. The predicates live on different axes by Definition BFA.7.33. No implication between the axes has been registered. □

Proposition BFA.7.35 (Independence does not require structural novelty)**.** *A strictly isolated derivation may converge to a proof architecture similar to or even mathematically equivalent to the benchmark architecture without losing its provenance status. Conversely, a structurally novel proof obtained through a contaminated channel is not strict IR.*

Proof. Provenance concerns admissible information flow, whereas structural similarity and novelty compare mathematical artifacts after the fact. By Policy 36 and the certificate ceiling below, neither property determines the other. □

BFA.7.10 The BFA independent-recovery certificate

Definition BFA.7.36 (IR certificate record)**.** A claimed strict-IR result must export a certificate record

$$\begin{aligned}\mathsf{IRCert} = (&\mathsf{TargetHash}, \mathsf{TargetMode}, \mathsf{Cutoff}, \mathsf{CorpusRoot}, \mathsf{SelectionManifest}, \mathsf{EnvironmentID},\\ &\mathsf{PriorEvidence}, \mathsf{OperatorEvidence}, \mathsf{AccessPolicy}, \mathsf{LogRoot}, \mathsf{OutputHash}, \mathsf{SealTime},\\ &\mathsf{UnsealTime}, \mathsf{IsolationLevel}, \mathsf{IRProvenance}).\end{aligned}$$

The certificate does not contain a mathematical pass field. The proof artifact identified by OutputHash is verified separately.

Theorem BFA.7.37 (Certificate soundness ceiling)**.** *An* IRCert *can certify at most the provenance property represented by its recorded and auditable isolation evidence. It cannot certify a theorem, a proof-assistant interpretation theorem, novelty, priority, or historical influence unless separate evidence discharges those claims.*

Proof. The certificate fields concern target identity, cutoff, admissible inputs, execution environment, logs, sealing, and isolation. Mathematical validity, semantic equivalence, novelty, priority, and historical influence are not conclusions of those fields. Promotion would therefore be a cross-sort status leak forbidden by the BFA constitutional discipline. □

BFA.7.11 Current BFA status and permitted research within this context

Theorem BFA.7.38 (Current-context provenance verdict)**.** *For the present BFA 1.0 construction context,*

$$\mathsf{IRProvenance} = \mathrm{contaminated}, \qquad \mathsf{strict_IR} = \mathrm{not_certifiable}.$$

This verdict is permanent for theorem-bearing artifacts derived from the already source-exposed context, although those artifacts may remain valuable as post-hoc recovery candidates or as designs for future isolated experiments.

Proof. BFA.2–BFA.6 explicitly consumed the benchmark analytic paper, benchmark proof architecture, source-derived dependency DAG, and source-faithful audit records. These are members of $\mathrm{Cl}_{\mathrm{th}}(C_{\mathrm{bench}})$. The present context therefore fails the strict quarantine condition. □

Policy 43 (BFA-IR-06: permitted work in the contaminated lane)**.** *The present BFA context may still:*

(i) formulate an IR protocol;

(ii) identify admissible pre-benchmark mathematics;

(iii) design a future isolated environment;

(iv) construct post-hoc alternative derivations;

(v) compare candidate derivations with the benchmark after labeling them source-exposed; and

(vi) extract generic mathematical candidates subject to the later BFA promotion gates.

None of these activities retroactively produces strict independent provenance.

BFA.7.12 Protocol closure and handoff

Definition BFA.7.39 (BFA.7 protocol status vector)**.** At closure of this section, the registered status is

$$\begin{aligned}
\mathsf{IR_protocol_definition} &= \mathrm{pass},\\
\mathsf{target_disclosure_semantics} &= \mathrm{pass},\\
\mathsf{current_context_provenance} &= \mathrm{contaminated},\\
\mathsf{strict_IR_execution} &= \mathrm{not_invoked},\\
\mathsf{strict_IR_certificate} &= \mathrm{not_issued},\\
\mathsf{posthoc_recovery_research} &= \mathrm{permitted},\\
\mathsf{external_analytic_verification} &= \mathrm{not_invoked}.
\end{aligned}$$

The strings $\mathrm{not_issued}$ and $\mathrm{permitted}$ are workflow/provenance labels, not mathematical verifier values.

Theorem BFA.7.40 (BFA.7 closure)**.** *BFA.7 establishes the provenance protocol required before any future BFA claim of independent recovery. In particular:*

(i) problem-level, specified-theorem, and route-disclosed targets are separated, so the target itself cannot silently leak the benchmark's successful direction;

(ii) *the precursor corpus, selection channel, prompt/search-policy origin, observable knowledge, prior state, human-operator channel, and execution environment are separately typed;*

(iii) *a complete pre-benchmark AK corpus may be admitted as a broad frozen precursor, while post-hoc BFA-guided AK cherry-picking is excluded;*

(iv) *benchmark contamination is closed under theorem-bearing semantic transport rather than limited to literal source files;*

(v) *strict and bounded isolation claims are separated;*

(vi) *a sealed Phase-I output must exist before benchmark comparison;*

(vii) *provenance and mathematical validity remain orthogonal;*

(viii) *the present BFA context is permanently source-exposed for IR certification; and*

(ix) *a future strict-IR claim requires a separately instantiated and auditable environment rather than a relabeled continuation of the present lane.*

No independent recovery is claimed or certified by this section.

Remark BFA.7.41 (Next consumer: BFA.8)**.** With source-faithful composition, v21 compatibility, and independent-recovery provenance now separated, BFA.8 may define the *Adversarial Falsification Semantics*. That section should classify source failure, route failure, formal-interpretation failure, and target falsification without allowing one category to leak into another.

BFA.8 Adversarial Falsification Semantics

BFA.8.1 Purpose, scope, and non-claim

Declaration BFA.8.1 (BFA.8 objective)**.** BFA.8 defines the negative-result semantics used by the remainder of Appendix BFA. Its purpose is to determine, for every adverse finding, the weakest exact mathematical conclusion licensed by that finding, the route object to which the finding attaches, and the precise downstream portion of the registered route that loses entitlement to Return.

BFA.8 does *not* claim that the OpenAI analytic proof is false, does not claim that the OpenAI theorem statement is false, and does not claim that either Clay alternative is false. It defines what would have to be shown before any of those stronger statements could be issued.

Policy 44 (BFA-FALS-02: criticism is not falsification)**.** *A suspected gap, unexplained step, missing estimate, ambiguous notation, failed informal reconstruction, or reviewer disagreement is recorded first as an audit observation or obligation. None is promoted to a theorem-facing* reject *until a verifier certifies the exact well-typed negative claim consumed by that reject state.*

Remark BFA.8.2 (Refinement rather than replacement)**.** BFA.1 already separates source failure, route nonclosure, route falsification, formal mismatch, interpretation failure, and target falsification (Definition BFA.1.21, Theorem BFA.1.22, and Policy 6). BFA.8 does not replace that constitutional taxonomy. It refines the evidence, subject, propagation, strength, repair, and Navier–Stokes-specific semantics needed to use that taxonomy adversarially.

BFA.8.2 Negative-evidence lifecycle and typed subjects

Definition BFA.8.3 (Negative-evidence lifecycle states)**.** For one theorem-facing obligation, BFA.8 uses the tagged evidence states

$$\mathsf{NegState} = \{\mathsf{observation}, \mathsf{open_obligation}, \mathsf{candidate_witness}, \mathsf{certified_nonclosure}, \mathsf{certified_reject}\}.$$

Their meanings are:

(i) observation: a possible issue has been noticed but no exact theorem-facing burden has yet been isolated;

(ii) open_obligation: the exact required statement, type, or interpretation condition has been isolated but not resolved;

(iii) candidate_witness: a putative counterexample, contradiction, mismatch, or failed estimate has been produced but has not yet passed its governing verifier;

(iv) certified_nonclosure: the selected route is certified not to close at the requested strength because a required object is absent, unresolved, or ill-typed;

(v) certified_reject: a required well-typed mathematical or semantic obligation has been certified false at the strength consumed by the selected route.

These states are *not* a total order. In particular, certified_nonclosure and certified_reject are distinct terminal evidence classes rather than stronger/weaker versions of one another.

Definition BFA.8.4 (Permitted negative-state transitions)**.** The permitted evidence-lifecycle transitions are generated by

$$\begin{aligned} \mathsf{observation} &\longrightarrow \mathsf{open_obligation},\\ \mathsf{open_obligation} &\longrightarrow \mathsf{candidate_witness},\\ \mathsf{open_obligation} &\longrightarrow \mathsf{certified_nonclosure},\\ \mathsf{open_obligation} &\longrightarrow \mathsf{certified_reject}, \end{aligned}$$

and

$$\mathsf{candidate_witness} \longrightarrow \mathsf{certified_nonclosure}, \qquad \mathsf{candidate_witness} \longrightarrow \mathsf{certified_reject}.$$

The direct transition from an open obligation to certified reject covers cases where a governing verifier proves the required statement false without first materializing a separately registered candidate witness. A later repaired artifact or new theorem may open a new audit episode, but it does not rewrite the historical certificate of the earlier episode.

Definition BFA.8.5 (Adversarial subject kind)**.** Every atomic adverse finding is attached to exactly one typed subject kind

$$\begin{aligned} \mathsf{SubjectKind} = \{&\mathsf{source_binding}, \mathsf{proof_derivation}, \mathsf{theorem_statement}, \mathsf{route_node}, \mathsf{route_edge},\\ &\mathsf{route_instance}, \mathsf{formal_artifact}, \mathsf{interpretation_edge}, \mathsf{official_target}\}. \end{aligned}$$

This subject field prevents a verifier result about a proof derivation from being silently reinterpreted as a truth-value verdict about the theorem statement it was intended to establish.

Definition BFA.8.6 (Adversarial failure certificate)**.** An adversarial failure certificate is a typed record

$$\begin{aligned} \mathfrak{F} = (&\mathsf{SubjectKind}, \mathsf{Object}, \mathsf{Claim}, \mathsf{RequiredStrength}, \mathsf{Evidence},\\ &\mathsf{Verifier}, \mathsf{EvidenceState}, \mathsf{VerifierStatus}, \mathsf{AffectedConsumers},\\ &\mathsf{LicensedNegativeConclusions}, \mathsf{ForbiddenPromotions}). \end{aligned}$$

The final two fields are mandatory. BFA records the exact negative propositions licensed by the certificate and the upward promotions that remain forbidden. No assumption is made that all negative propositions lie in one total strength order.

Theorem BFA.8.7 (No negative-state promotion without the governing verifier)**.** *No BFA adverse finding in states* observation*,* open_obligation*, or* candidate_witness *may be reported as* certified_reject *unless the governing verifier certifies the exact negative claim at the consumed strength. Likewise, a certified nonclosure does not become a certified reject without new theorem-bearing evidence.*

Proof. BFA.1 separates missing or unresolved evidence from falsity. The transition graph in Definition BFA.8.4 exposes that separation explicitly. The evidence lifecycle cannot strengthen the verifier conclusion by itself. □

Proposition BFA.8.8 (Negative evidence does not redefine verifier status)**.** *For a theorem-bearing obligation whose governing verifier is invoked,* certified_nonclosure *projects to the v21/BFA verifier semantics* undefined*, whereas* certified_reject *projects to* reject*. The states* observation*,* open_obligation*, and* candidate_witness *do not by themselves determine a verifier verdict. The status* not_invoked *remains orthogonal to this lifecycle.*

Proof. The evidence lifecycle records what has been produced during adversarial inspection; verifier status records the result of the governing mathematical or semantic verifier. BFA.1 already separates undefined, reject, and not_invoked; the proposition is the typed crosswalk between those axes rather than a new status system. □

BFA.8.3 Source identity, proof invalidity, and source-theorem falsity

Definition BFA.8.9 (Source-facing adverse classes)**.** For a bound external source theorem S, BFA.8 distinguishes:

$$\begin{aligned} F_{\text{src-bind}}(S) &: \text{ the artifact identity, locator, version, or quoted statement is incorrectly bound,} \\ F_{\text{src-proof-open}}(S) &: \text{ the supplied derivation has an unresolved theorem-facing gap at the registered strength,} \\ F_{\text{src-proof-reject}}(S) &: \text{ the supplied derivation is certified invalid at a required theorem-bearing step,} \\ F_{\text{src-thm}}(S) &: \text{ the exact registered theorem statement is mathematically false.} \end{aligned}$$

Theorem BFA.8.10 (Source-level noncollapse)**.** *In general,*

$$F_{\text{src-bind}} \nRightarrow F_{\text{src-proof-reject}}, \qquad F_{\text{src-proof-open}} \nRightarrow F_{\text{src-proof-reject}},$$

and

$$F_{\text{src-proof-reject}} \nRightarrow F_{\text{src-thm}}.$$

Likewise, falsity of one advertised construction or witness does not by itself falsify an existential source theorem. Such falsification requires an exhaustion theorem, or an equivalent reduction theorem, that reduces every admissible witness to a class eliminated by the certified negative argument; uniqueness helps only when combined with such an exhaustive reduction.

Proof. A binding error concerns BFA's representation of the source. An unresolved proof gap yields nonclosure rather than proof invalidity. A certified proof error removes entitlement to infer the theorem from that derivation, but the theorem can remain true by another derivation. An existential statement is false only when its exact logical negation is proved. □

Policy 45 (BFA-FALS-03: proof failure is not theorem falsity)**.** *Whenever BFA certifies a problem in a supplied proof, the report remains at* $F_{\text{src-proof-open}}$ *or* $F_{\text{src-proof-reject}}$ *unless an independent certificate proves the exact logical negation of the registered source theorem with all source quantifiers preserved.*

Remark BFA.8.11 (All-viscosity source burden)**.** For a source portfolio of the form $\forall \nu > 0\, P_\nu$, a proof error at one viscosity does not imply $\neg P_\nu$. If, however, BFA separately proves $\neg P_{\nu_0}$ for one $\nu_0 > 0$, then the all-viscosity portfolio is false by its own quantifier structure. The distinction is between failure of a derivation and falsity of the theorem it was intended to establish.

BFA.8.4 Route instances, theorem truth, and strength-indexed propagation

Definition BFA.8.12 (Selected route instance)**.** A selected route instance is a record

$$\mathcal{R} = (G_{\mathcal{R}}, \lambda_{\mathcal{R}}, \mathsf{Target}),$$

where $G_{\mathcal{R}}$ is the finite acyclic dependency sub-DAG consumed by that derivation and $\lambda_{\mathcal{R}}$ assigns to every theorem-bearing node or relation the exact strength consumed by each outgoing dependency. Alternative derivations to the same external theorem have distinct route identities unless they are theorem-bearing identical at every registered node, edge, and consumed strength.

Definition BFA.8.13 (Route verdict versus statement verdict)**.** For a theorem-facing object Y, BFA distinguishes

$$\mathsf{RouteVerdict}_{\mathcal{R}}(Y)$$

from

$$\mathsf{StatementVerdict}(Y).$$

The former records whether the selected route establishes Y at the requested strength. The latter records a truth-value verdict on the exact statement of Y. Rejecting a selected derivation does not by itself reject the theorem statement.

Definition BFA.8.14 (Requested-strength dependency)**.** For a route instance $\mathcal{R}$, write

$$N \preceq_{\mathcal{R},\sigma} Y$$

when the selected derivation of Y at strength σ consumes the registered output of N at the strength prescribed by $\lambda_{\mathcal{R}}$. The relation is route-relative and does not claim that every proof of Y must consume N.

Definition BFA.8.15 (Affected-consumer set)**.** For a certified adverse finding on (N, τ), define

$$\mathrm{Affect}_{\mathcal{R}}(N,\tau) := \{(Y,\sigma) : N \preceq_{\mathcal{R},\sigma} Y \text{ and the finding is certified at the strength consumed by } Y\}.$$

This is a route-consequence set, not a theorem-falsity set.

Theorem BFA.8.16 (Strength-indexed route propagation)**.** *Let $\mathcal{R}$ be the selected route.*

(i) If a required obligation N is unresolved, absent, or ill-typed at the strength consumed by $\mathcal{R}$, then every affected downstream derivation in $\mathrm{Affect}_{\mathcal{R}}(N,\tau)$ *is nonclosing at that route strength.*

(ii) If the exact required obligation consumed by $\mathcal{R}$ is certified false, then every affected downstream route derivation *is rejected at the corresponding strength.*

(iii) Neither clause changes $\mathsf{StatementVerdict}(Y)$ *to* reject *without a separate certificate proving the exact statement of Y false.*

Proof. Part (i) is route nonclosure. Part (ii) removes a necessary premise of the selected derivation. The strength qualifier blocks propagation from failure of an unnecessary stronger statement. Part (iii) is BFA.1's route-versus-target discipline applied recursively to intermediate theorem statements. □

Corollary BFA.8.17 (Undefined does not become reject downstream)**.** *An unresolved or missing prerequisite can make every downstream selected-route derivation nonclosing, but it does not become a false theorem merely by being upstream.*

BFA.8.5 Shared dependencies, alternate routes, and repair identity

Theorem BFA.8.18 (Shared-node fan-out semantics)**.** *If one registered theorem node is consumed at multiple distinct sites, a certified adverse finding on that node propagates independently along every consumer edge at the strength actually consumed there. In particular, a certified failure of the shared BFA.3 node* SUM *blocks both the base-completion consumer and the final-completion consumer whenever both consume the failed strength.*

Proof. BFA.3 registers SUM as one shared theorem identity with multiple consumer edges rather than as duplicated lemmas. Propagation therefore follows each consuming edge through Theorem BFA.8.16. □

Definition BFA.8.19 (Statement, binding, artifact, and proof-route identities)**.** BFA distinguishes four identity layers:

(i) the *mathematical statement identity*, determined by the exact quantified proposition and its typed hypotheses/conclusion, independent of which source happens to state it;

(ii) the *bound theorem-record identity*, determined by a statement identity together with its source authority, locator, and source-facing metadata;

(iii) the *artifact revision identity*, determined by the particular paper/code/repository revision carrying a proof or formalization; and

(iv) the *proof-route identity*, determined by the theorem-bearing dependency structure, consumed strengths, joint-composition obligations, and interpretation edges used by the derivation.

This prevents the same mathematical proposition appearing in two sources from being treated as two different propositions merely because the bindings differ.

Definition BFA.8.20 (Route repair)**.** A route repair changes at least one theorem-bearing node, dependency edge, consumed strength, interpretation theorem, or proof obligation after an adverse finding has been identified. Such a change creates a new proof-route identity

$$\mathcal{R}' \neq \mathcal{R}.$$

A purely typographical or bibliographic erratum may preserve the proof-route identity only if BFA certifies that no theorem statement, hypothesis, dependency, consumed strength, or semantic interpretation has changed. The artifact revision identity nevertheless changes.

Policy 46 (BFA-FALS-04: no silent proof patching)**.** *A rejected or nonclosing route is not retroactively relabeled* pass *after a theorem-bearing patch. The original route verdict is preserved; the repaired route is separately registered, rebound, and re-audited from the first changed object through every affected consumer.*

Theorem BFA.8.21 (Alternative-route nonfalsification)**.** *Failure of one route to a theorem or target does not falsify that theorem or target. This nonimplication does not require BFA to exhibit an alternative proof; theorem falsity still requires a certificate of the exact logical negation.*

Proof. The proposition "route $\mathcal{R}$ establishes T" and the proposition T have different logical content. Rejecting the former removes one sufficient derivation. It does not prove $\neg T$. □

BFA.8.6 Audit-expanded DAG for node, edge, and joint-composition obligations

Definition BFA.8.22 (Theorem-bearing edge and joint-composition obligations)**.** For a selected route $\mathcal{R}$, let $E_{\text{th}}(\mathcal{R})$ be the set of dependency or interpretation edges whose validity is itself a required theorem-facing obligation, including when applicable source-to-target relations such as ρ_{PT}.

For a route node v with a predecessor family $P_v = \{u_1, \ldots, u_k\}$, let $J_{\text{th}}(v)$ denote a *joint-composition obligation* whenever the proof requires more than the separate validity of the incoming edges: for example, common parameter values, compatible domains, common solution class, synchronized cutoffs, or another n-ary hypothesis that only makes sense for the predecessor tuple as a whole.

Definition BFA.8.23 (Audit-expanded route DAG)**.** From $G_{\mathcal{R}} = (V, E)$, construct an audit-expanded finite directed graph $G^{\sharp}_{\mathcal{R}}$ in two steps.

(i) For each theorem-bearing edge $e : u \to v$, insert a typed edge-obligation vertex $[e]$ and replace $u \to v$ by $u \to [e] \to v$.

(ii) Whenever $J_{\text{th}}(v)$ is registered for the predecessor family P_v, insert a typed joint-composition vertex $[J_v]$, connect every required predecessor (or its edge-obligation vertex) to $[J_v]$, and connect $[J_v] \to v$. The direct incoming proof-bearing arcs represented by that joint obligation are then removed from the expanded representation so that the compatibility obligation cannot be bypassed.

Purely structural metadata edges need not be subdivided.

Proposition BFA.8.24 (Expansion preserves acyclicity and represented derivations)**.** *Assume the original route DAG is finite and acyclic and every inserted joint-composition vertex $[J_v]$ is placed strictly between predecessors of v and v. Then $G^{\sharp}_{\mathcal{R}}$ is finite and acyclic. Contracting all inserted obligation vertices recovers the original dependency relation. Conversely, every original selected derivation lifts to an expanded derivation precisely when all inserted edge and joint-composition obligations on that derivation are discharged.*

Proof. Each inserted obligation vertex respects the original topological order: edge vertices lie between the endpoints of one original edge, and a joint vertex lies after all registered predecessors but before their common consumer. Therefore no directed cycle is introduced. Contraction restores the original dependency relation. The final statement records the additional fact that an expanded proof path/derivation is entitled to reach the consumer only through the registered compatibility obligations. □

Corollary BFA.8.25 (Legitimate reuse of the v21 causal-blocker theorem)**.** *When the remaining hypotheses of the frozen v21 causal-blocker theorem hold, BFA may apply that theorem to $G^{\sharp}_{\mathcal{R}}$, treating failed or undefined theorem nodes, theorem-bearing edge obligations, and registered joint-composition obligations uniformly as vertices of the audit-expanded DAG. This is a representation step, not a new minimum-cut theorem.*

BFA.8.7 Formal-artifact and semantic-interpretation failures

Definition BFA.8.26 (Formal-artifact failure classes)**.** For the pinned formal source family, BFA.8 distinguishes:

$$\begin{aligned} F_{\text{replay}} &: \text{ the pinned artifact cannot be replayed under the bound environment,} \\ F_{\text{formal-stmt}} &: \text{ the formal declaration differs from the statement BFA attributes to it,} \\ F_{\text{paper}\leftrightarrow\text{formal}} &: \text{ the claimed paper–formal semantic relation fails at the requested strength,} \\ F_{\text{formal}\rightarrow\text{target}} &: \text{ the formal theorem does not establish the claimed official-target relation.} \end{aligned}$$

Theorem BFA.8.27 (Formal failure isolation)**.** *None of*

$$F_{\text{replay}}, \quad F_{\text{formal-stmt}}, \quad F_{\text{paper}\leftrightarrow\text{formal}}, \quad F_{\text{formal}\rightarrow\text{target}}$$

automatically implies that the analytic theorem in the paper is false. Conversely, correctness of the analytic theorem does not certify replay or semantic equivalence of the formal artifact.

Proof. BFA.1 and BFA.2 bind paper, formal artifact, and official target as separate authorities connected by directional interpretation relations. A failure on one relation or artifact therefore remains attached to that typed subject. □

Policy 47 (BFA-FALS-05: infrastructure failure ceiling)**.** *A compiler error, dependency failure, unavailable repository, or bound-environment replay failure is infrastructure evidence. It may reject or leave undefined the claimed replay record, but it does not by itself reject the analytic theorem or the official target. A hash mismatch is different: when the hash is part of the frozen artifact identity, the mismatch rejects the claim that the observed artifact is that pinned revision (a binding/identity failure), while still not implying falsity of the analytic theorem.*

BFA.8.8 Source-to-target interpretation failure

Definition BFA.8.28 (Interpretation failure)**.** Let

$$\rho_{PT} : \mathsf{Paper} \longrightarrow \mathsf{OfficialTarget}$$

denote the BFA.2 paper-to-target interpretation relation. An interpretation failure F_{interp} is certified when the exact source theorem, even if true, does not entail the target predicate at the frozen target strength because of a mismatch in domain, quantifiers, data class, forcing class, solution class, regularity, periodicity, viscosity convention, or conclusion strength.

Theorem BFA.8.29 (Interpretation failure blocks Return but preserves source truth)**.** *If F_{interp} is certified for one source-to-target edge, the corresponding BFA Return through that edge is rejected. This does not imply $F_{\text{src-thm}}$; the source theorem may remain mathematically correct at its own registered strength.*

Proof. Return consumes both source truth and a valid interpretation edge. Falsity of the latter removes composability with the official target but does not alter the truth value of the source theorem statement. □

Corollary BFA.8.30 (No target-strength repair by rhetoric)**.** *If the source theorem is weaker than the official target, stronger prose, announcement language, compilation, or formal packaging cannot repair the missing mathematical implication. A separate theorem supplying the missing interpretation strength is required.*

BFA.8.9 Quantifier-aware source and target falsification

Definition BFA.8.31 (Target-falsification certificate)**.** A target-falsification certificate for Clay-C$_\nu$ or Clay-D$_\nu$ is a proof object whose conclusion is exactly the logical negation fixed in Proposition BFA.2.10. For the all-viscosity wrappers Clay-C and Clay-D, the certificate must prove the corresponding portfolio negation with the exact viscosity quantifier.

Theorem BFA.8.32 (Witness or route failure is strictly weaker than Clay-target falsification)**.** *For fixed $\nu > 0$, invalidating the OpenAI constructed datum/force/solution, invalidating one analytic lemma, or rejecting the entire OpenAI proof route establishes none of*

$$\neg\text{Clay-C}_\nu, \qquad \neg\text{Clay-D}_\nu$$

without an additional proof of the universal existence statement appearing in the exact target negation.

Proof. By BFA.2, Clay-C$_\nu$ and Clay-D$_\nu$ are existential counterexample targets. Their negations quantify universally over every admissible datum and forcing and assert existence of an admissible global smooth solution. Failure of one proposed witness or route does not discharge that universal burden. □

Corollary BFA.8.33 (Portfolio falsification burden)**.** *To falsify the all-viscosity BFA portfolio* Clay-C *or* Clay-D*, it is necessary and sufficient to prove failure of the corresponding fixed-viscosity target for at least one $\nu > 0$, exactly as specified in Proposition BFA.2.10. A proof gap or rejected derivation at one viscosity is not such a certificate.*

BFA.8.10 Navier–Stokes adversarial priority map

Definition BFA.8.34 (Provisional route-specific adversarial seed sets)**.** BFA.8 supplies semantics, not the final critical-cut ranking. It therefore registers only provisional seed sets for BFA.9:

$$\mathfrak{S}_C = \{\mathsf{SUM}, O_7, O_8, O_9, O_{11}, O_{13}, \rho_{PT}\},$$
$$\mathfrak{S}_D = \{\mathsf{SUM}, O_7, O_8, O_9, O_{11}, O_{13}, O_{15}, \rho_{PT}\}.$$

The whole-space and periodic lanes are separated because O_{15} is periodic-route specific. Construction-integrity nodes such as O_1, O_2, O_4, O_5, O_6 remain eligible audit obligations even when not in a seed set. BFA.9 may refine, reorder, enlarge, or shrink these seeds by the actual blocker/cut calculation. No seed set is asserted here to be a minimal cut, antichain, or optimal audit schedule.

Definition BFA.8.35 (Node-local failure consequences)**.** The default source-faithful consequences at the current registry resolution are:

Object	Certified adverse finding at consumed strength	Maximum immediate BFA conclusion
SUM	shared summation theorem statement false, or selected application nonclosing/rejected	block/reject each registered consumer route at the corresponding strength; no target falsification
O_7	residual-improvement obligation nonclosing/rejected	block/reject the registered correction-improvement route downstream of O_7
O_8	stage-uniform derivative-loss gate unresolved/false	make completion nonclosing / reject the selected completion derivation according to the certified state
O_9	final infinite-completion application nonclosing/rejected	block/reject the registered final-completion route
O_{11}	endpoint force extension nonclosing/rejected	block/reject smooth compact-forcing Return for the affected route
O_{13}	comparison/uniqueness edge nonclosing/rejected	block/reject exclusion of a global smooth competitor through that route; no automatic target falsification
O_{15}	periodization/periodic transport nonclosing/rejected	block/reject the periodic Clay-D route only; whole-space Clay-C is not thereby rejected
ρ_{PT}	source-to-target interpretation edge false	reject the affected official-target Return while preserving source-theorem truth status

Theorem BFA.8.36 (Asymmetric lane isolation)**.** *A certified adverse finding in an obligation unique to the periodic route does not reject the whole-space route. A certified adverse finding in a shared obligation propagates to both only at the strengths and consumer edges actually shared.*

Proof. This follows from the frozen BFA.3 predecessor map, the BFA.4 typed consumer records, and Theorem BFA.8.16. Failure propagation follows registered dependency, not thematic similarity. □

Remark BFA.8.37 (Why O_{13} is terminally high leverage)**.** A singular constructed field alone does not exclude the existence of a second global smooth solution for the same datum and forcing in the source theorem's comparison class. Therefore an adverse finding on the comparison/uniqueness edge has a large downstream effect even if the local blow-up construction remains correct. The correct negative conclusion remains route nonclosure or route rejection, not target falsification.

BFA.8.11 Existing v21 blocker/cut calculus and BFA adversarial semantics

Policy 48 (BFA-FALS-06: consume cuts, do not rename them)**.** *BFA.8 uses the frozen v21 target-cut and causal-blocker machinery only at its registered hypotheses and strength [1, 4]. For edge-bearing BFA routes, the representation in Definition BFA.8.23 is used before applying node-based blocker results. BFA.8 contributes the semantic label of a certified adverse finding and its permitted mathematical conclusion; it does not claim a new generic minimum-cut theorem.*

Definition BFA.8.38 (Adversarial audit set versus causal blocker)**.** An adversarial audit set is chosen prospectively for verification leverage. A causal blocker antichain is computed retrospectively from current reject or undefined statuses on required vertices of the applicable finite acyclic audit DAG. Therefore an audit-priority object may pass and never become a blocker, while a low-priority object may become the earliest actual blocker.

Theorem BFA.8.39 (Audit priority does not alter proof semantics)**.** *Changing the order in which BFA verifies obligations may change the cost or speed of finding an error, but it does not change the truth value, dependency relation, consumed strength, or exact negative conclusion of any theorem-facing obligation.*

Proof. Audit order is search metadata. Mathematical truth and dependency are fixed by the bound statements, types, and route records. Priority cannot discharge or falsify an obligation by itself. □

BFA.8.12 Adversarial verdict record and no-scalar-collapse rule

Definition BFA.8.40 (BFA adversarial verdict record)**.** For one audited subject X, the BFA.8 verdict record is

$$\begin{aligned}\mathfrak{V}_{\mathrm{adv}}(X) = (&\mathsf{SubjectKind}, \mathsf{EvidenceState}, \mathsf{VerifierStatus}, \mathsf{FailureClass},\\ &\mathsf{RequiredStrength}, \mathsf{AffectedRoute}, \mathsf{AffectedConsumers},\\ &\mathsf{RouteEffect}, \mathsf{StatementEffect}, \mathsf{TargetEffect}, \mathsf{RepairIdentity}, \mathsf{EvidenceRefs}).\end{aligned}$$

The effect fields are typed separately. In particular, TargetEffect = target_falsified is permitted only with a certificate satisfying Definition BFA.8.31; route nonclosure or route rejection alone cannot populate that value.

Policy 49 (BFA-FALS-07: no scalar suspicion score)**.** *BFA does not replace the verdict record by a single confidence, suspicion, credibility, or "percent falsified" score. Such a scalar cannot preserve the difference between missing evidence, an invalid derivation, a false theorem, an invalid route, a formal mismatch, an interpretation failure, and exact target falsity.*

Definition BFA.8.41 (Atomic and compound negative reports)**.** An *atomic* negative report certifies one exact negative proposition about one typed subject at one registered strength. The same subject may therefore carry several atomic reports if distinct verifiers certify logically distinct negative propositions. A *compound* adverse finding is represented by the finite set of its independently certified atomic reports; it is not forced into one scalar or one artificially strongest class.

Theorem BFA.8.42 (Weakest-exact atomic negative conclusion)**.** *For every atomic certified adverse finding recorded by BFA.8, the certificate asserts exactly the negative proposition proved by its governing verifier and only the downstream consequences licensed by registered dependency. Every strictly stronger or logically incomparable negative proposition remains unasserted unless separately certified. Compound findings are reported as a set of such atomic conclusions and need not possess a unique strongest member.*

Proof. Atomicity is attached to one certified negative proposition, not merely to one object name. BFA.1's weakest-exact rule forbids upward promotion, while the route dependency record controls downstream consequences. Multiple negative facts about one subject therefore remain separately auditable, and no total ordering among incomparable failure kinds is assumed. □

BFA.8.13 BFA.8 closure and handoff

Definition BFA.8.43 (BFA.8 status vector)**.** At completion of this section,

$$\begin{aligned}
\mathsf{falsification_semantics} &= \mathrm{defined},\\
\mathsf{audit_expanded_DAG_semantics} &= \mathrm{defined},\\
\mathsf{critical_cut_ranking} &= \mathrm{not_yet_computed},\\
\mathsf{source_theorem_falsified} &= \mathrm{not_claimed},\\
\mathsf{OpenAI_route_falsified} &= \mathrm{not_claimed},\\
\mathsf{formal_mismatch_certified} &= \mathrm{not_claimed},\\
\mathsf{Clay_target_falsified} &= \mathrm{not_claimed},\\
\mathsf{adversarial_verification} &= \mathrm{not_yet_executed\ at\ full\ depth}.
\end{aligned}$$

Theorem BFA.8.44 (BFA.8 adversarial-semantics closure)**.** *BFA.8 establishes a source-faithful, strength-indexed, quantifier-aware falsification semantics with the following properties:*

(i) the negative-evidence lifecycle is branching rather than falsely linear, and nonclosure remains distinct from reject;

(ii) binding failure, unresolved proof gap, certified proof invalidity, and source-theorem falsity are separated;

(iii) route verdict and theorem-statement verdict are distinct typed fields;

(iv) route propagation is limited to registered consumers at the consumed strength;

(v) shared-node findings propagate through all actual consumer edges while preserving route specificity;

(vi) theorem identity, artifact revision, and proof-route identity are separated, and theorem-bearing repairs receive new route identity;

(vii) theorem-bearing edges and registered multi-input composition conditions can be represented as typed obligation vertices in an acyclic audit-expanded DAG, permitting legitimate reuse of the frozen v21 node-based blocker theorem;

(viii) formal replay, formal statement, semantic correspondence, and paper-to-target failures remain separate;

(ix) interpretation failure blocks Return without rewriting source truth;

(x) Clay-target falsification requires the exact quantifier-aware target negation fixed in BFA.2;

(xi) route-specific Navier–Stokes audit seed sets are kept provisional so that BFA.9, rather than BFA.8, owns the actual critical-cut ranking; and

(xii) no external analytic theorem, formal theorem, OpenAI route, or Clay target has been declared false merely by defining these semantics.

Proof. The listed properties follow from the typed-subject, route-instance, route/statement-separation, subdivision, quantifier, and weakest-exact results of this section together with the constitutional requirements of BFA.1 and the exact target records of BFA.2. The final clause follows from **??** BFA.8.1 and Definition BFA.8.43: BFA.8 defines the semantics under which later audits may issue negative results but does not itself supply a falsification witness for the external proof or target. □

Remark BFA.8.45 (Next consumer: BFA.9)**.** BFA.9 should instantiate the frozen v21 blocker/target-cut machinery on the BFA.3 source DAG and, where theorem-bearing edges must participate, on the audit-expanded route DAG $G^{\sharp}_{\mathcal{R}}$. It should conduct the *Critical-Cut Audit*: identify the highest-leverage required obligations, compute the currently earliest blockers from verifier states, and distinguish prospective audit priority from actual causal blockage. BFA.9 may use the semantics of BFA.8 but must not turn audit priority into mathematical falsity.

BFA.9 Critical-Cut Audit

BFA.9.1 Purpose, scope, and audit ceiling

Declaration BFA.9.1 (BFA.9 objective)**.** BFA.9 instantiates the adversarial semantics of BFA.8 on the frozen source-faithful whole-space and periodic proof routes. Its purpose is to identify, without changing theorem status, the theorem nodes, theorem-bearing edge obligations, and joint-composition obligations whose verification has the greatest route-discriminating value for the selected Navier–Stokes reconstruction.

The governing question is

which obligations should be audited first to determine whether the selected route can still close?

This is an audit-priority and causal-structure problem over a typed proof DAG. It is not a probability-of-error estimate, not a credibility score, not a substitute for proof, and not a claim that a high-priority object is false.

Policy 50 (BFA-CUT-01: criticality is not suspicion)**.** *A node, theorem-bearing edge, or joint-composition obligation is called* critical *only relative to a declared target slice and audit purpose. High criticality may reflect large downstream coverage, a quantifier-changing boundary, shared use by several consumers, a difficult compatibility burden, or severe Return consequences. It does not mean that the corresponding source statement is likely false, insufficiently justified, unusually novel, or mathematically suspect.*

Accordingly,

$$\text{audit priority} \not\Rightarrow \text{probability of failure}, \qquad \text{cut membership} \not\Rightarrow \text{reject}.$$

Remark BFA.9.2 (BFA.9 does not perform the full analytic re-proof)**.** BFA.9 identifies and orders the principal theorem-facing audit surfaces. BFA.10 performs the dedicated finite-to-infinite/completion stress test, BFA.11 performs the terminal Return stress test, and BFA.12 performs the paper–Lean–problem triangular audit. Therefore BFA.9 does not pre-empt those sections by declaring their mathematical obligations passed or rejected.

BFA.9.2 Normative source graph and two target slices

Definition BFA.9.3 (BFA.9 target-slice convention)**.** Let G_{ana} be the normative source dependency DAG of BFA.3 with vertices

$$\mathsf{SUM}, \qquad \mathsf{O}_1, \ldots, \mathsf{O}_{15},$$

and with the mandatory predecessor map of Definition BFA.3.8. For fixed viscosity $\nu > 0$, let

$$G_C(\nu) \quad \text{and} \quad G_D(\nu)$$

denote respectively the whole-space and periodic necessary target slices, including the BFA.2 source-to-target Return relation at the exact frozen target strength.

Let

$$G_C^\sharp(v), \qquad G_D^\sharp(v)$$

be their BFA.8 audit expansions: each theorem-bearing edge may be subdivided by an edge-obligation vertex, and each registered multi-input compatibility burden may be represented by a joint-composition vertex $[J_v]$.

Proposition BFA.9.4 (Whole-space and periodic slices are not identified)**.** *The critical-audit records for $G_C^\sharp(v)$ and $G_D^\sharp(v)$ are distinct. A whole-space audit result is not silently promoted to the periodic lane, and a periodic-only adverse finding does not retroactively reject the whole-space lane.*

Proof. The terminal source objects and official target records differ. In particular, O_{15} adds periodization, periodic pressure, torus-domain semantics, and periodic target matching. Conversely, the periodic route still consumes shared upstream objects, including O_{14}, by the normative BFA.3 predecessor map. Failure propagation therefore follows the actual shared or route-specific dependencies rather than the thematic label "Navier–Stokes." □

Proposition BFA.9.5 (O_{14} is a shared terminal transport bottleneck)**.** *In the current registered BFA.3 route, O_{14} is mandatory for both target lanes: it is the whole-space arbitrary-viscosity source output and is also a mandatory predecessor of the periodic node O_{15}. Hence a certified failure of the registered O_{14} obligation blocks both current lanes at every downstream strength that consumes it, although it does not by itself falsify either Clay target.*

Proof. BFA.3 registers

$$\mathrm{Pred}(\mathsf{O}_{14}) = \{\mathsf{O}_{13}\}, \qquad \mathrm{Pred}(\mathsf{O}_{15}) = \{\mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{14}\}.$$

BFA.2 then maps O_{14} to the whole-space target and O_{15} to the periodic target. The conclusion is therefore a direct application of registered dependency and BFA.8 strength-indexed failure propagation. □

BFA.9.3 Four notions that must remain distinct

Definition BFA.9.6 (Selected-route blocker)**.** Fix one selected theorem-bearing route instance $\mathcal{R}$ and requested target strength Σ. A required obligation vertex x is a *selected-route blocker* when a theorem-facing reject at x, or an invoked theorem-facing $\mathrm{undefined}$ at x, removes entitlement to derive the selected target through $\mathcal{R}$ at strength Σ.

This is a logical necessity statement for one registered route. It is not by itself a graph-theoretic path cut and it does not exclude an independently complete alternative route.

Definition BFA.9.7 (Graph-theoretic target cut)**.** Let S be a declared family of source-side entry vertices and let z be a target vertex in a finite directed target slice. A set

$$K \subseteq V(G^\sharp) \setminus \{z\}$$

is an S-to-z *target cut* when every admissible directed S-to-z path intersects K. It is inclusion-minimal when no proper subset remains a target cut.

Remark BFA.9.8 (Two different antichain statements)**.** The family of inclusion-minimal cuts is automatically an antichain under *set inclusion*. This must not be confused with the stronger and generally false assertion that the vertices inside one inclusion-minimal cut form an antichain under DAG reachability. The frozen v21 Search/Platform counterexample explicitly separates these notions [5, 6].

Definition BFA.9.9 (Derivational dominator)**.** A theorem-facing obligation x *dominates the selected derivation of z* when every registered complete derivation represented by the selected route consumes x at the requested strength.

This is proof-semantic rather than merely path-theoretic: a multi-input theorem may require several incomparable predecessors simultaneously, so a predecessor can be logically mandatory even when a naive source-to-target path calculation can bypass it through another incoming branch.

Definition BFA.9.10 (Audit frontier)**.** An *audit frontier* is a finite family of theorem-facing obligations selected prospectively because resolving them has high expected value for separating route closure from route nonclosure or rejection. An audit frontier may contain source nodes, theorem-bearing edge vertices, and joint-composition vertices.

An audit frontier need not be a target cut, and a target cut need not be an efficient audit frontier.

Theorem BFA.9.11 (Cut/blocker/dominator/frontier separation)**.** *For the BFA source proof architecture,*

$$\mathsf{selected\text{-}route\ blocker}, \qquad \mathsf{target\ cut}, \qquad \mathsf{derivational\ dominator}, \qquad \mathsf{audit\ frontier}$$

are distinct predicates in general. None may be substituted for another without a theorem establishing the required equivalence for the exact selected route representation.

Proof. A conjunctive proof node may require several predecessors at once. Failure of one required predecessor blocks the theorem even if another directed path from a source entry to the consumer remains visible. A path-hitting cut therefore captures one graph property but not the full conjunction semantics of a multi-input theorem. A derivational dominator instead refers to mandatory use in every registered derivation. An audit frontier additionally incorporates verification leverage and cost. The four predicates therefore answer different questions. □

Policy 51 (BFA-CUT-02: do not overclaim a minimum cut)**.** *BFA.9 may reuse the frozen v21 path-cut theorems only when their path and portfolio hypotheses are actually satisfied. The present section does not rename a BFA structural priority family as a "minimum cut," "unique cut," or "minimum causal blocker" merely because it contains high-leverage nodes. A minimum-cardinality, minimum-weight, or unique graph cut requires a separate complete path model and proof.*

BFA.9.4 BFA-local vector criticality profile

Definition BFA.9.12 (Structural criticality vector)**.** For a theorem-facing obligation x in the target slice for $T \in \{\mathsf{Clay\text{-}C}_\nu, \mathsf{Clay\text{-}D}_\nu\}$, define the BFA-local descriptive vector

$$\mathbf{K}_T(x) = \big(\mathsf{Cov}, \mathsf{Fan}, \mathsf{Qnt}, \mathsf{Assm}, \mathsf{Joint}, \mathsf{IndV}, \mathsf{Repair}, \mathsf{Cost}, \mathsf{Sev}\big)_T(x),$$

with the following meanings.

(i) Cov: downstream dependency coverage inside the frozen target slice;

(ii) Fan: number and identity of distinct direct consumers or target lanes sharing the obligation;

(iii) Qnt: quantifier sensitivity—finite stage, all stages, infinite completion, terminal extension, arbitrary viscosity, or exact official-target quantification;

(iv) Assm: sensitivity to source hypotheses, parameter values, support, domain, regularity, or comparison class;

(v) Joint: dependence on n-ary compatibility rather than isolated unary theorem truth;

(vi) IndV: degree to which the obligation admits independent theorem-local checking without re-establishing most upstream source construction;

(vii) Repair: breadth of the route region that must be rebound and re-audited after a theorem-bearing repair;

(viii) Cost: audit-cost class under the currently frozen source decomposition, recorded qualitatively rather than as invented elapsed time;

(ix) Sev: maximum immediate semantic consequence of a certified adverse finding under BFA.8.

Policy 52 (BFA-CUT-03: no scalar collapse)**.** *The coordinates of $\mathbf{K}_T$ inhabit different ordered or qualitative sorts. BFA.9 therefore does not define a universal real-valued criticality score and does not assert a total ordering of all obligations. Where several objects trade audit cost against causal reach or independent verifiability, the result is retained as a Pareto-style partial ranking rather than forced into one scalar.*

Definition BFA.9.13 (Audit-cost classes)**.** For BFA.9 scheduling only, the qualitative cost coordinate takes values

$$\mathsf{local} \prec \mathsf{multi\text{-}lemma} \prec \mathsf{global\text{-}chain},$$

where the order describes breadth of evidence that must be reconstructed, not mathematical difficulty in any absolute sense. A source theorem may be deep while still being locally auditable at the level of its exact statement.

BFA.9.5 Exact load-bearing regions of the source route

Definition BFA.9.14 (Finite correction and completion region)**.** The principal finite-to-infinite audit region is the mandatory source substructure centered on

$$\mathsf{O}_6 \longrightarrow \mathsf{O}_7 \longrightarrow \mathsf{O}_8 \longrightarrow \mathsf{O}_9, \qquad \mathsf{SUM} \longrightarrow \mathsf{O}_9,$$

together with every additional mandatory predecessor, theorem-bearing edge vertex, and joint-composition vertex registered for O_7, O_8, and O_9.

Theorem BFA.9.15 (Residual improvement, uniformity, and completion remain three obligations)**.** *The BFA registry licenses none of the silent implications*

$$\mathsf{finite\ residual\ improvement} \Longrightarrow \mathsf{stage\text{-}uniform\ derivative/domain\ control},$$

$$\mathsf{stage\text{-}uniform\ derivative/domain\ control} \Longrightarrow \mathsf{infinite\ completion},$$

or

$$\mathsf{finite\ residual\ improvement} \Longrightarrow \mathsf{infinite\ completion}.$$

Thus O_7*,* O_8*,* O_9*, and the intervening edge/joint obligations are separate first-class audit objects.*

Proof. BFA.3 assigns different source loci and outputs to the three nodes. BFA.4 also separates the source residual state from the stage-uniform completion contract. Improving a finite-stage residual order does not itself produce a common domain or a stage-independent derivative-loss estimate. Such uniformity, even if established, does not by itself prove the existence, regularity, compatibility, local finiteness, or residual-flatness properties of the infinite completed object. □

Proposition BFA.9.16 (Shared SUM theorem versus two applications)**.** *The source theorem* SUM *has two distinct registered consumers,*

$$\mathsf{SUM} \longrightarrow \mathsf{O}_2, \qquad \mathsf{SUM} \longrightarrow \mathsf{O}_9.$$

A certified falsity of the shared theorem statement itself propagates to both actual consumers. Failure of one consumer-specific instantiation, hypothesis match, or joint-composition obligation does not automatically reject the other application.

Proof. The theorem identity is shared, but BFA.4 records two logically distinct completion contracts on different input families. BFA.8 separates theorem statement, bound theorem record, application edge, and route identity. Hence statement-level falsity and application-level failure have different propagation scopes. □

Definition BFA.9.17 (Terminal analytic region)**.** The terminal analytic region is the registered source substructure

$$\mathsf{O}_9 \longrightarrow \mathsf{O}_{10} \longrightarrow \mathsf{O}_{11} \longrightarrow \mathsf{O}_{12} \longrightarrow \mathsf{O}_{13} \longrightarrow \mathsf{O}_{14},$$

with the additional mandatory predecessor edges listed in BFA.3 and their BFA.8 audit-expansion vertices. Its load-bearing roles include:

O_{10} : space–time localization preserving the singular core,

O_{11} : smooth compactly supported terminal forcing,

O_{12} : bounded kinetic energy and finite dissipation,

O_{13} : comparison and uniqueness and exclusion of a smooth competitor,

O_{14} : viscosity-one to arbitrary-viscosity transport.

Proposition BFA.9.18 (Completion does not absorb terminal extension)**.** *A pass of the completed-local-field obligation* O_9 *does not discharge* O_{11}*. In particular, preterminal residual flatness or smoothness does not by itself supply the force-jet compatibility, support, and extension statement needed to obtain the registered* C_c^∞ *forcing.*

Proof. The two nodes have distinct source statements and distinct outputs. O_9 constructs the completed local field; O_{11} asserts compatible terminal limits of all force derivatives and a global smooth compactly supported extension. BFA.4 explicitly treats endpoint regularization as a separate theorem-facing obligation. □

Proposition BFA.9.19 (Comparison is an independent terminal blocker)**.** *Even if the constructed singular solution, smooth forcing, and bounded-energy properties are correct, failure of the exact comparison and uniqueness obligation consumed by* O_{13} *blocks the registered inference excluding the relevant global smooth competitor. Such a finding does not by itself falsify the upstream construction.*

Proof. The Clay-facing source conclusion is stronger than existence of one singular preterminal candidate: it must exclude the admissible global smooth competitor in the relevant source class. BFA.3 assigns that burden to O_{13}. Rejecting the comparison statement therefore blocks the corresponding nonexistence route but leaves upstream construction statements with their own independent truth statuses. □

BFA.9.6 Trivial terminal cuts versus informative interior cuts

Definition BFA.9.20 (Terminal predecessor cuts)**.** At the coarse BFA.3 graph resolution, the immediate pre-target source nodes form trivial singleton path cuts:

$$\{\mathsf{O}_{14}\} \quad \text{for the whole-space source-to-target lane,} \qquad \{\mathsf{O}_{15}\} \quad \text{for the periodic source-to-target lane,}$$

provided the target slice has the single registered terminal source edge from those nodes. If the BFA.2 interpretation edge is itself represented as an obligation vertex, that interpretation vertex is likewise a terminal singleton cut for its own selected path representation.

Remark BFA.9.21 (Why terminal singleton cuts are insufficient for BFA.9)**.** The cuts of Definition BFA.9.20 are mathematically correct but audit poor: every serial proof architecture has such immediate-pretarget cuts. They identify where Return can fail, but they do not reveal which analytic interior obligations most efficiently discriminate whether the source construction itself survives. BFA.9 therefore records terminal cuts and separately constructs interior adversarial frontiers.

Definition BFA.9.22 (Proper interior audit domain)**.** For target T, the *proper interior audit domain* $I_T^\sharp$ is obtained from the necessary expanded target slice by removing the official target node and, for the purpose of interior ranking only, marking the final source-to-target interpretation and immediate pre-target transport as terminal boundary objects rather than allowing them to trivialize every cut calculation.

This operation changes only the audit question. It does not remove those objects from the actual proof route or weaken their necessity.

BFA.9.7 Target-specific interpretation instances

Definition BFA.9.23 (Target-specific paper-to-target instances)**.** Let

$$\rho_{PT}^{C,\nu} \quad \text{and} \quad \rho_{PT}^{D,\nu}$$

denote the whole-space and periodic fixed-viscosity instances of the generic BFA triangular relation

$$\rho_{PT} : \mathsf{Paper} \longrightarrow \mathsf{OfficialTarget}.$$

They are target-specific projections of the BFA.1–BFA.2 interpretation record, not new theorem identities. Their domain, forcing, solution, periodicity, viscosity, and conclusion-strength fields are audited separately.

Proposition BFA.9.24 (Interpretation instances are noncompensating)**.** *Failure of $\rho_{PT}^{C,\nu}$ blocks the claimed Clay-C Return through that paper-to-target relation but does not alter the source theorem's truth value. Failure of $\rho_{PT}^{D,\nu}$ analogously blocks the Clay-D Return. A passing whole-space interpretation does not discharge the periodic instance.*

Proof. This is the target-specific form of BFA.8 interpretation-failure isolation. The two targets differ in domain and periodicity conditions, and the periodic lane includes the corrected periodic-pressure requirement. Hence the interpretation records have different consumed fields even when they share upstream source mathematics. □

BFA.9.8 Structural audit profile of the principal candidates

Definition BFA.9.25 (Qualitative structural profile table)**.** The following table records the strongest source-supported BFA.9 structural classification that can be assigned before the dedicated mathematical stress tests. "Both" means that the current registered C and D lanes consume the obligation; "D only" means periodic-specific. Cost is breadth-of-audit metadata from Definition BFA.9.13, not a truth or difficulty score.

Object	Lane reach	Quantifier / boundary	Audit-cost class	Maximum immediate adverse consequence
SUM	Both; two direct consumers	all-order summation/completion theorem	multi-lemma	reject the exact shared theorem use or affected application(s); downstream route closure lost at consumed strength
O_7	Both	one-cycle improvement iterated through the finite correction family	global-chain	correction-improvement route blocked/rejected downstream of the failed instance
O_8	Both	all stages on one domain; stage-independent derivative loss	multi-lemma	completion route cannot be licensed at the registered uniform strength
O_9	Both	finite-to-infinite completion; locally finite smooth limit; residual flatness	global-chain	completed local-field route blocked/rejected
O_{11}	Both	terminal force jets and global smooth compact-support extension	multi-lemma	smooth-forcing source route blocked/rejected
O_{13}	Both	comparison and uniqueness against the admissible global smooth competitor	global-chain	source-side breakdown/nonexistence inference blocked/rejected
O_{14}	Both in the current registry	viscosity-one $\to$ arbitrary fixed $\nu > 0$	multi-lemma	whole-space Return blocked and current periodic route also blocked because O_{15} consumes O_{14}
O_{15}	D only	rescaling into a fundamental cube, disjoint periodization, periodic pressure	multi-lemma	periodic source route blocked/rejected; whole-space lane unchanged
$\rho_{PT}^{C,\nu}$	C only	exact official Clay-C interpretation	local / multi-lemma	Clay-C Return through this interpretation rejected while source truth remains separate
$\rho_{PT}^{D,\nu}$	D only	exact corrected Clay-D interpretation	local / multi-lemma	Clay-D Return through this interpretation rejected while source truth remains separate
$[J_\nu]$	route-dependent	simultaneous parameter, domain, support, and solution-class compatibility	local / global-chain	consumer ν remains undefined or is rejected according to the certified state even when unary predecessors pass

Remark BFA.9.26 (Why O_{14} is promoted relative to the earlier heuristic)**.** BFA.3 initially placed O_{14} in a lower audit-order tier together with localization/energy/transport nodes. BFA.9 refines that heuristic. Because the normative predecessor map makes O_{14} a mandatory ancestor of the periodic node as well as the direct whole-space arbitrary- viscosity output, its downstream coverage is larger than the earlier coarse priority label suggests. This is a scheduling refinement only; no theorem status is changed.

BFA.9.9 Joint-composition obligations are first-class audit objects

Theorem BFA.9.27 (Unary passes do not discharge a joint vertex)**.** *Let a source node v consume predecessors $u_1, \dots, u_k$ through a registered joint-composition obligation $[J_v]$. Even if each predecessor theorem and each unary theorem-bearing edge passes separately, v is not discharged until the exact n-ary compatibility predicate represented by $[J_v]$ passes.*

For the present source route, relevant joint fields may include common:

- *viscosity and scaling convention;*
- *source domain and time interval;*
- *solution and pressure class;*
- *support and cutoff architecture;*
- *stage-independent regularity or derivative-loss convention;*
- *terminal extension data;*
- *forcing and comparison object.*

Proof. BFA.8 inserts $[J_v]$ precisely when the proof requires a predicate on the predecessor tuple that is not implied by separate unary edge validity. The audit expansion removes the represented bypassing proof arcs. Therefore the consumer cannot inherit a pass from its components until the tuple-level condition itself is discharged. □

Corollary BFA.9.28 (Joint vertices may outrank source nodes operationally)**.** *A joint-composition vertex may have higher audit priority than any single predecessor theorem when its failure would invalidate a large downstream subroute and its verification is relatively local. This is an audit-efficiency statement only and is not evidence that the joint predicate is false.*

BFA.9.10 Prospective structural frontiers

Definition BFA.9.29 (Tier A: route-killing transition frontier)**.** The primary BFA.9 structural frontier is

$$\mathfrak{F}_A = \{\mathsf{SUM}, \mathsf{O}_8, \mathsf{O}_9, \mathsf{O}_{11}, \mathsf{O}_{13}, \mathsf{O}_{14}\} \cup \mathfrak{J}_A \cup \mathfrak{E}_A,$$

where $\mathfrak{J}_A$ and $\mathfrak{E}_A$ are the registered joint and edge-obligation vertices on the corresponding transition regions.

For the periodic lane the target-specific augmentation is

$$\mathfrak{F}_A^D = \mathfrak{F}_A \cup \{\mathsf{O}_{15}\} \cup \mathfrak{J}_{15} \cup \mathfrak{E}_{15}.$$

The target-interpretation instances $\rho_{PT}^{C,\nu}$ and $\rho_{PT}^{D,\nu}$ are terminal semantic frontier objects and are scheduled for the BFA.12 audit after their source-side inputs have been frozen.

Definition BFA.9.30 (Tier B: correction and energy frontier)**.** The secondary frontier is

$$\mathfrak{F}_B = \{\mathsf{O}_7, \mathsf{O}_{12}\} \cup \mathfrak{J}_B \cup \mathfrak{E}_B.$$

O_7 carries the finite-stage correction engine; O_{12} carries the bounded-energy/dissipation input consumed by the terminal comparison route. Their Tier-B placement is an audit-order decision, not a weaker logical status.

Definition BFA.9.31 (Tier C: construction-integrity frontier)**.** The construction-integrity frontier consists of

$$\mathfrak{F}_C^{\text{build}} = \{\mathsf{O}_1, \mathsf{O}_2, \mathsf{O}_3, \mathsf{O}_4, \mathsf{O}_5, \mathsf{O}_6, \mathsf{O}_{10}\} \cup \mathfrak{J}_C^{\text{build}} \cup \mathfrak{E}_C^{\text{build}}.$$

Every listed object remains mandatory wherever consumed. This tier records that exhaustive reconstruction of these large source packages is scheduled after the highest-leverage transition checks unless a downstream audit traces a suspected failure back into them.

Policy 53 (BFA-CUT-04: priority classes are workflow metadata)**.** *Tier membership is BFA-local scheduling metadata. It is not document role, evidence status, atomic verifier status, source authority, migration status, or Return strength. A Tier-C theorem can be mathematically indispensable, and a Tier-A theorem can pass completely.*

BFA.9.11 Whole-space and periodic frontier records

Definition BFA.9.32 (Whole-space audit frontier)**.** For fixed $\nu > 0$, the prospective whole-space audit frontier is the ordered family

$$\mathfrak{A}_C(\nu) = \big(\mathfrak{F}_A, \mathfrak{F}_B, \mathfrak{F}_C^{\text{build}}, \rho_{PT}^{C,\nu}\big),$$

with target-specific edge and joint vertices included at their exact registered positions. This is an audit schedule, not a claim that the union is a minimum graph cut.

Definition BFA.9.33 (Periodic audit frontier)**.** For fixed $\nu > 0$, the periodic audit frontier is

$$\mathfrak{A}_D(\nu) = \big(\mathfrak{F}_A^D, \mathfrak{F}_B, \mathfrak{F}_C^{\text{build}}, \rho_{PT}^{D,\nu}\big),$$

again with all target-specific theorem-bearing edge and joint vertices included. The periodic audit does not borrow a whole-space pass for O_{15} or $\rho_{PT}^{D,\nu}$.

Theorem BFA.9.34 (Structural dominance of the Tier-A frontier)**.** *Within the current selected BFA source route, each source node in*

$$\{\mathsf{O}_8, \mathsf{O}_9, \mathsf{O}_{11}, \mathsf{O}_{13}, \mathsf{O}_{14}\}$$

is a mandatory ancestor, at its registered strength, of the whole-space source-to-target conclusion; the same nodes are also mandatory ancestors of the current periodic conclusion because O_{15} *consumes the whole-space arbitrary-viscosity construction. The shared* SUM *statement or its* O_9 *application is likewise mandatory for the registered completion lane.*

Proof. This follows from the normative predecessor map in BFA.3, not from the schematic serial display alone. In particular, O_9 consumes O_8 and the shared summation engine; O_{13} consumes $\mathsf{O}_9, \mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{12}$; O_{14} consumes O_{13}; and O_{15} consumes O_{14} together with localization and compact-support outputs. Thus every registered selected derivation uses the displayed obligations, subject to their exact edge and joint contracts. □

Remark BFA.9.35 (Dominance is more informative here than naive path cutting)**.** For the periodic node, O_{14} is logically mandatory even though the graph also contains incoming paths to O_{15} from O_{10} and O_{11}. A path-hitting calculation can therefore miss conjunctive necessity. This is precisely why BFA.9 records both graph cuts and derivational dominators rather than identifying them.

BFA.9.12 Retrospective causal blockers from verifier states

Definition BFA.9.36 (Certified adverse set)**.** For a frozen target slice $G_T^\sharp$, let

$$\mathfrak{N}_T$$

be the set of required theorem-facing vertices whose BFA.8 atomic state has actually been certified as either reject or invoked $\mathrm{undefined}$. Workflow labels such as $\mathrm{scheduled}$, $\mathrm{not_yet_audited}$, or merely being high priority do not place a vertex in $\mathfrak{N}_T$.

Definition BFA.9.37 (Earliest certified blocker set)**.** Assume $\mathfrak{N}_T \neq \varnothing$. Define

$$\mathrm{ECB}(T) = \{x \in \mathfrak{N}_T : \text{no strict necessary ancestor of } x \text{ lies in } \mathfrak{N}_T\}.$$

The necessary-ancestor relation is taken from the frozen selected target projection, including theorem-bearing edge and joint vertices.

Theorem BFA.9.38 (Earliest certified blockers form a causal antichain)**.** *If the selected expanded target projection is finite and acyclic and* $\mathfrak{N}_T \neq \varnothing$*, then* $\mathrm{ECB}(T)$ *is finite and pairwise incomparable under strict necessary ancestry. Every member of* $\mathfrak{N}_T$ *lies at or downstream of at least one member of* $\mathrm{ECB}(T)$.

This theorem does not assert that $\mathrm{ECB}(T)$ *is a source-to-target graph cut.*

Proof. A finite acyclic necessary-dependency relation is a finite partial order after identifying no aliases. Every nonempty finite subset has minimal elements; these are exactly $\mathrm{ECB}(T)$. Distinct minimal elements are incomparable. Repeatedly descending through adverse necessary ancestors from any nonminimal adverse vertex terminates at a minimal adverse vertex. The final sentence follows because causal minimality among adverse necessary vertices is a different predicate from meeting every admissible directed path. □

Proposition BFA.9.39 (Failure precedence does not cancel later failures)**.** *Using* $\mathrm{ECB}(T)$ *as a preferred repair or re-audit frontier does not remove any later independently certified* reject, undefined*, or mathematical obstruction from the manifest.*

Proof. This is the BFA-local specialization of the v21 failure-precedence discipline: causal ordering controls diagnosis and repair order, whereas the target manifest retains every independently nonpassing required clause [1]. □

Proposition BFA.9.40 (Current BFA.9 blocker state)**.** *At the end of BFA.9, before the dedicated BFA.10–BFA.12 verifier passes, no nonempty* $\mathrm{ECB}(\mathsf{Clay\text{-}C}_\nu)$ *or* $\mathrm{ECB}(\mathsf{Clay\text{-}D}_\nu)$ *is asserted solely from audit priority. The structural frontiers of this section are computed prospectively; the retrospective earliest-blocker sets are instantiated only from certified BFA.8 theorem-facing states produced by later audits.*

Proof. BFA.8 closes with adversarial verification not yet executed at full depth and explicitly forbids converting suspicion or priority into reject. Therefore BFA.9 may determine where to test first, but it cannot manufacture an adverse verifier state from structural centrality alone. □

BFA.9.13 Adverse-result ceiling for critical objects

Theorem BFA.9.41 (Critical-object failure ceiling)**.** *Suppose a BFA.9 critical object receives a certified adverse result. BFA may conclude only the weakest applicable statement below.*

(i) Missing, unresolved, or ill-typed required evidence yields route nonclosure at the consumed strength, not theorem falsity.

(ii) A certified false well-typed required obligation rejects every registered downstream use that actually consumes it at that strength.

(iii) Falsity of an imported source theorem statement propagates to every registered consumer of that exact source theorem; failure of one application edge or joint condition remains application-specific.

(iv) Failure of O_{15} *or* $\rho_{PT}^{D,\nu}$ *can reject the periodic lane without rejecting the whole-space lane.*

(v) Failure of a shared ancestor such as the current registered O_{14} *route propagates to both current C and D lanes.*

(vi) None of these route conclusions alone proves $\neg\mathsf{Clay\text{-}C}_\nu$ *or* $\neg\mathsf{Clay\text{-}D}_\nu$*.*

Proof. Items (i)–(v) are instances of BFA.8 typed negative-evidence, subject, strength-propagation, application, and lane-isolation semantics combined with the BFA.3 predecessor map. Item (vi) is the quantifier-aware target ceiling: rejecting one witness or one proof route is strictly weaker than proving the universal statement that forms the logical negation of the existential Clay counterexample target. □

BFA.9.14 Audit execution order exported to BFA.10–BFA.12

Definition BFA.9.42 (Phase-ordered adversarial schedule)**.** Subject to any newly discovered upstream certified blocker, the BFA.9 schedule is:

Phase 1: freeze the exact theorem identities, source loci, parameter values, and joint hypotheses consumed by the O_7–O_9 correction/completion region;

Phase 2: stress-test O_7, O_8, the $\mathsf{SUM} \to \mathsf{O}_9$ application, and O_9, separating residual improvement, stage-uniform control, and infinite completion;

Phase 3: audit O_{10}–O_{11} terminal localization, force jets, support, and smooth extension;

Phase 4: audit the exact O_{12}–O_{13} bounded-energy/comparison route and its joint hypotheses;

Phase 5: audit O_{14} as the shared arbitrary-viscosity transport, then audit periodic-specific O_{15} separately;

Phase 6: audit the directional triangle

$$\rho_{PF}, \qquad \rho_{FT}, \qquad \rho_{PT}^{C,\nu}, \qquad \rho_{PT}^{D,\nu}$$

at exact paper, formal, and official-target strengths.

Proposition BFA.9.43 (Ownership of the next three sections)**.** *Under Definition BFA.9.42:*

BFA.10 *owns Phases 1–2 at full mathematical depth,*
BFA.11 *owns Phases 3–5 at full mathematical depth,*
BFA.12 *owns Phase 6 and the final triangular consistency audit.*

BFA.9 owns the selection, typing, causal separation, and scheduling architecture; it does not own the final truth verdicts of those later stress tests.

BFA.9.15 BFA.9 critical-cut verdict

Theorem BFA.9.44 (BFA.9 structural critical-cut theorem)**.** *For the currently registered BFA source-faithful reconstruction, the following statements are established at structural-audit level.*

(i) *Whole-space and periodic target slices remain distinct, while their exact shared ancestry is preserved.*

(ii) *Selected-route blockers, graph-theoretic cuts, derivational dominators, and prospective audit frontiers are not conflated.*

(iii) *Criticality is represented by a typed vector profile rather than a universal scalar suspicion score.*

(iv) *The finite correction, stage-uniformity, and infinite-completion burdens at* O_7, O_8, *and* O_9 *remain logically separate.*

(v) *The shared theorem* SUM *is one theorem identity with two separately auditable applications.*

(vi) *Terminal forcing extension* O_{11} *and comparison* O_{13} *remain independent of successful completion.*

(vii) O_{14} *is promoted in audit priority because it is a mandatory shared ancestor of both current target lanes.*

(viii) O_{15} *and the periodic interpretation instance remain D-specific obligations and cannot borrow a whole-space pass.*

(ix) *Joint-composition vertices are first-class theorem-facing audit objects; unary passes cannot discharge them.*

(x) *The prospective Tier-A/B/C frontiers are computed, but no actual adverse blocker antichain is fabricated before later verifier results exist.*

(xi) *No classification in this section proves the external analytic route, rejects the external analytic route, falsifies a source theorem, or decides either Clay target.*

Proof. Items (i)–(ii) follow from Definition BFA.9.3 and Theorem BFA.9.11. Item (iii) is Definition BFA.9.12 and Policy 52. Items (iv)–(vi) follow from Theorem BFA.9.15 and Propositions BFA.9.16, BFA.9.18 and BFA.9.19. Item (vii) is Proposition BFA.9.5 and Remark BFA.9.26. Item (viii) follows from Propositions BFA.9.4 and BFA.9.24. Item (ix) is Theorem BFA.9.27. Item (x) is Proposition BFA.9.40. The final item is the BFA.8 nonpromotion and falsification ceiling inherited throughout this section. □

BFA.9.16 BFA.9 closure and handoff

Definition BFA.9.45 (BFA.9 status vector)**.** At completion of BFA.9,

$$\begin{aligned}
\mathsf{critical_cut_semantics} &= \text{defined},\\
\mathsf{whole_space_target_slice} &= \text{separated},\\
\mathsf{periodic_target_slice} &= \text{separated},\\
\mathsf{structural_criticality_vector} &= \text{defined},\\
\mathsf{prospective_audit_frontiers} &= \text{computed},\\
\mathsf{O14_shared_transport_priority} &= \text{recognized},\\
\mathsf{minimum_graph_cut} &= \text{not_claimed without separate certification},\\
\mathsf{actual_earliest_blocker_set} &= \text{not_yet_instantiated at full depth},\\
\mathsf{source_theorem_falsified} &= \text{not_claimed},\\
\mathsf{OpenAI_route_falsified} &= \text{not_claimed},\\
\mathsf{Clay_target_falsified} &= \text{not_claimed},\\
\mathsf{completion_stress_test} &= \text{delegated to BFA.10},\\
\mathsf{terminal_return_stress_test} &= \text{delegated to BFA.11},\\
\mathsf{triangular_audit} &= \text{delegated to BFA.12}.
\end{aligned}$$

Theorem BFA.9.46 (BFA.9 closure)**.** *BFA.9 converts the provisional BFA.8 adversarial seed sets into a source-faithful critical-audit architecture that:*

(i) preserves the exact BFA.3 predecessor map rather than relying on the schematic serial display alone;

(ii) distinguishes trivial terminal singleton cuts from informative interior audit structure;

(iii) preserves conjunctive proof semantics through derivational dominance and joint-composition vertices;

(iv) refines the earlier heuristic by recognizing O_{14} *as a shared C/D transport bottleneck;*

(v) retains a vector-valued, non-scalar criticality discipline;

(vi) separates prospective priority from retrospective certified causal blockers;

(vii) gives BFA.10–BFA.12 a deterministic order of attack without pre-judging their mathematical outcomes; and

(viii) preserves the BFA.8 ceiling that route criticism, route nonclosure, and route rejection are all weaker than Clay-target falsification.

Proof. The claims are exactly the collected consequences of Definitions BFA.9.3, BFA.9.12, BFA.9.20, BFA.9.22 and BFA.9.42, Theorems BFA.9.11, BFA.9.27 and BFA.9.41, and Propositions BFA.9.5 and BFA.9.40. □

Remark BFA.9.47 (Next consumer: BFA.10)**.** BFA.10 shall perform the dedicated *Finite-to-Infinite / Completion Stress Test*. Its principal source family is

$$\{\mathsf{SUM}, \mathsf{O}_6, \mathsf{O}_7, \mathsf{O}_8, \mathsf{O}_9\}$$

together with every mandatory predecessor, theorem-bearing edge vertex, and joint-composition vertex consumed by the completion route.

The stress test must determine separately:

(i) whether one correction cycle really improves the exact registered physical residual order while preserving every stated invariant;

(ii) whether the common-domain and derivative-loss bounds are genuinely independent of the correction-stage index at each consumed derivative order;

(iii) whether coefficient growth, shrinking supports/cutoffs, local finiteness, nonlinear interactions, and inverse operations remain compatible with one infinite construction;

(iv) whether the SUM theorem applies to the exact O_9 input family rather than only to separately finite stages or to the different O_2 application;

(v) whether the completed object exists in the claimed topology and has the exact smoothness, divergence, support, and local asymptotic properties used downstream;

(vi) whether arbitrarily high finite residual decay really yields the exact residual-flatness statement consumed by the terminal extension; and

(vii) whether any interchange of limit, derivative, summation, localization, nonlinear operation, or inverse construction is used without a theorem at the required uniform strength.

Only after these obligations are resolved may BFA.11 consume the completed local object as theorem-facing input.

BFA.10 Finite-to-Infinite / Completion Stress Test

BFA.10.1 Purpose, audit resolution, and non-promotion ceiling

Declaration BFA.10.1 (BFA.10 objective)**.** BFA.10 executes the first focused adversarial stress test selected by the critical-cut architecture of BFA.9, at an explicitly bounded source-interface audit resolution. Its theorem-facing source family is

$$\boxed{\mathfrak{C}_{\mathrm{comp}} = \{\mathsf{SUM}, \mathsf{O}_6, \mathsf{O}_7, \mathsf{O}_8, \mathsf{O}_9\} \cup \mathfrak{E}_{\mathrm{comp}} \cup \mathfrak{I}_{\mathrm{comp}},}$$

with all mandatory predecessors inherited from the normative BFA.3 dependency map.

The stress test asks whether the source proof actually supplies five logically separate transitions:

$$\text{constructed finite ingredients} \xrightarrow{\mathsf{O}_6} \text{admissible stage-zero correction state},$$

$$\text{finite correction state} \xrightarrow{\mathsf{O}_7} \text{improving finite states},$$

$$\text{improving finite states} \xrightarrow{\mathsf{O}_8} \text{one common domain with stage-independent exponent loss},$$

$$\text{stage-uniform sequence} \xrightarrow{\mathsf{SUM}} \text{smooth locally finite cutoff sum},$$

$$\text{cutoff sum} \xrightarrow{\mathsf{O}_9} \text{completed local Navier–Stokes field with flat residual}.$$

No one arrow is identified with another.

Policy 54 (BFA-COMP-01: source-text audit is not a complete independent re-proof)**.** *BFA.10 checks the exact statements, quantifier order, source interfaces, uniformity requirements, completion mechanism, and the principal estimates needed to connect Proposition 9.3, Definition 9.4, Proposition 9.5, Proposition 9.6, Lemmas 9.7–9.8, Lemma 5.4, and Proposition 9.9 of the external analytic source [12]. It also checks for the principal finite-to-infinite failure modes fixed by BFA.8–BFA.9.*

It does not reproduce line by line every estimate in Sections 6–9 of the external paper and therefore does not change the external theorem's authority to an AK theorem. A favorable BFA.10 disposition means only that no certified finite-to-infinite nonclosure or contradiction has been obtained at the declared audit resolution.

Policy 55 (BFA-COMP-00: construction-stage source-identity firewall)**.** *The source-text comparisons in BFA.10 are made against the analytic PDF served from the canonical OpenAI locator during the present construction audit on 15 September 2026. BFA.2, however, still records the local cryptographic snapshot of that PDF as unresolved. Therefore BFA.10 may close a construction-stage source-interface audit, but it may not be frozen*

as an immutable theorem-facing source audit until BFA.2 records the exact PDF byte identity, hash, and retrieval metadata.

This is a provenance/governance obligation. It is not a mathematical adverse finding against the analytic proof.

Definition BFA.10.2 (BFA.10 completion-audit fields)**.** For each completion obligation X, BFA.10 records the five independent fields

$$\mathfrak{B}_{10}(X) = (\mathsf{SourceExplicit}, \mathsf{TypeMatch}, \mathsf{AdverseFinding}, \mathsf{ReproofCeiling}, \mathsf{SourceIdentityClosure}).$$

Their intended values in this section are descriptive strings, not v21 atomic verifier statuses. In particular, $\mathrm{no_certified_adverse_finding}$ is not synonymous with pass, and $\mathrm{source_explicit}$ is not synonymous with $\mathrm{proved_by_AK}$.

BFA.10.2 Exact source completion contract

Definition BFA.10.3 (Completion contract extracted from the source)**.** The external source's finite-to-infinite completion contract consists of the following obligations.

(i) **Admissible initialization.** Proposition 9.5 must produce an actual stage-zero state satisfying Definition 9.4, including the residual classes, cumulative correction bounds, exact moment constraints, pressure reconstruction, support discipline, and divergence-free representative structure consumed by Proposition 9.6.

(ii) **Finite improvement.** Proposition 9.6 must provide a stage map

$$(u^{[j]}, p^{[j]}) \longmapsto (u^{[j+1]}, p^{[j+1]})$$

preserving the fixed background/support/inverse architecture and improving the registered residual orders by a positive amount.

(iii) **Common domain.** Lemma 9.7 must produce one $q_{\mathrm{big}} > 0$ independent of correction stage and derivative order on which every finite partial state before summation cutoffs is defined.

(iv) **Summation-coordinate geometry.** The actual $q = q(z,t)$ used in Proposition 9.9 must satisfy the smoothness, compact-subset lower-bound, and derivative estimates required in (5.28) of Lemma 5.4 on the common domain.

(v) **Stage-independent exponent loss.** Lemma 9.8 must provide, for each fixed Cartesian space–time derivative order m, a nonnegative nondecreasing exponent-loss sequence ℓ_m independent of stage j.

(vi) **Increasing decay order.** The physical increments must satisfy a bound of the form

$$|Z_j|_m + |\Delta u_j|_m \le C_{j,m} q^{g_j - \ell_m} \Lambda_{\log}(q)^{P_{j,m}}, \qquad 0 < g_1 \le g_2 \le \cdots \to \infty.$$

(vii) **Finite-stage residual decay.** The actual finite partial sums must satisfy the Lemma 5.4 residual hypothesis

$$|\mathcal{R}(u^{[j]}, p^{[j]})|_m \le C_{j,m} q^{\rho_j - K_m} \Lambda_{\log}(q)^{P_{j,m}} + E_{j,m}, \qquad \rho_j \to \infty,$$

with K_m independent of j and $E_{j,m}$ flat at every fixed stage.

(viii) **Controlled base.** The base/finite-initialization block U_0 consumed by the summation theorem must have the derivative bounds required by Lemma 5.4 after any one-time initialization cutoff.

(ix) **Fixed finite differential-polynomial structure and coefficients.** The completed residual must be one fixed finite-order, finite-degree differential polynomial whose coefficient functions, derivative bounds, and comparison regions satisfy (5.31)–(5.32) independently of the truncation index.

(x) **Cutoff summability and local finiteness.** The cutoff scales must be choosable recursively so that every fixed derivative tail is summable and the resulting infinite sum is locally finite for $q > 0$.

(xi) **Structural invariants.** The completion must preserve divergence, smooth Cartesian representatives, support extensions, pressure representatives, and the source-specific overlap/relabeling consistency actually consumed by Proposition 9.9.

(xii) **Residual flatness after completion.** Arbitrarily high finite residual order must pass to

$$\forall m, N\ \exists \delta, C > 0: \quad |\mathcal{R}(u_{\mathrm{loc}}, p_{\mathrm{loc}})|_m \le C q^N \quad (0 < q < \delta).$$

(xiii) **Retention of the source asymptotic and output class.** The cutoff completion must preserve the local inner blow-up asymptotic, exact exterior behavior, one-sided endpoint regularity away from the singular point, and vector-potential structure required by Theorem 3.1 and by the next terminal stage.

Remark BFA.10.4 (The critical finite-to-infinite distinction)**.** The completion contract does *not* require stage-uniform constants $C_{j,m}$ or $P_{j,m}$. What must be stage-independent is the loss in the power of the small parameter for each fixed derivative order. This separation is explicit in Lemma 5.4 and is exactly the burden assigned to O_8 by BFA.3–BFA.4.

Policy 56 (BFA-COMP-01A: viscosity-normalization firewall)**.** *The completion stress test is performed on the normalized viscosity-one local construction used in Sections 3–10 of the analytic source, where*

$$\mathcal{R}(u, p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p.$$

BFA.10 does not silently replace this residual by an arbitrary-ν residual. Transport from the viscosity-one terminal construction to every fixed $\nu > 0$ remains the separate O_{14} obligation audited in BFA.11.

BFA.10.3 Stress Test 0: stage-zero initialization and consumer-domain closure

Audit Obligation BFA.10.5 (BFA10-A0: admissible initialization)**.** *Verify that the finite ingredients constructed before the iteration actually land in the exact state class consumed by Proposition 9.6, rather than merely having qualitatively small residuals.*

Proposition BFA.10.6 (Source-level closure of the $\mathsf{O}_6 \to \mathsf{O}_7$ interface)**.** *Definition 9.4 declares the finite correction-state contract, including the wave and mean residual classes, the cumulative correction bounds (9.9), the two exact moment constraints (9.10), pressure reconstruction, supported representatives, smooth zero extensions, and exact divergence-free structure. Proposition 9.5 then states that the initialized fields form a stage-zero state in precisely this sense, with*

$$B_0 = 0.7, \qquad C_0^* = 1.2.$$

Thus the output type of O_6 is source-explicitly the input type of O_7.

Remark BFA.10.7 (Initialization ceiling)**.** This closes only the source-level consumer-domain match. BFA.10 does not independently re-prove every exponent estimate inside Proposition 9.5 or the upstream wave/mean constructions on which it depends. A certified failure of one of those estimates would remain an upstream adverse witness under BFA.8.

BFA.10.4 Stress Test I: one-cycle residual improvement

Audit Obligation BFA.10.8 (BFA10-A1: exact monotone-improvement burden)**.** *Verify that Proposition 9.6 closes one complete correction cycle on an actual stage state, recomputes the residual from the updated fields, preserves the registered cumulative bounds and moment constraints, and produces a strictly larger residual-decay order for the next stage.*

Proposition BFA.10.9 (Source-level closure of the one-cycle interface)**.** *At the source-statement and proof-interface level, Proposition 9.6 supplies the exact improvement contract consumed by* O_7*:*

$$B_{j+1} = B_j + \frac{1}{10}, \qquad C^*_{j+1} = C^*_j + \frac{1}{10},$$

while preserving the same base, primary amplitudes, supports, inverse operators, cumulative bounds, and exact moment constraints.

Proof. The proposition states the stage-to-stage map and lists the four ordered corrections. Its proof recomputes pressure and the full residual after each update and closes the cycle by inequalities giving a gain strictly larger than 0.1 in all relevant error classes before selecting the common 0.1 increment. It then invokes the preceding source mechanism to recover the hypotheses for the next cycle. This matches the BFA.3 output type "monotone finite-stage residual improvement." The present argument audits the source interface and its quantifier shape; it does not independently re-prove every constituent estimate. □

Corollary BFA.10.10 (No finite-stage stagnation at the registered order level)**.** *Assuming the internal estimates of Proposition 9.6, the registered residual orders increase linearly with the stage index. Hence the route does not rely on a merely bounded or eventually constant sequence of finite-stage orders.*

Remark BFA.10.11 (Primary adversarial attack surface inside O_7)**.** A genuine rejection of O_7 would require locating an exact error term, interaction, support effect, inverse-operator loss, or recomputation term whose exponent fails one of the closing inequalities, or showing that a preserved moment/support condition is actually lost after one of the four updates. BFA.10 found no such certified witness at the declared source-text resolution.

BFA.10.5 Stress Test II: one common domain for all finite stages

Audit Obligation BFA.10.12 (BFA10-A2: stage-independent domain)**.** *Verify that iteration does not require a shrinking sequence* $q_j \downarrow 0$ *whose intersection removes the domain needed by the later summation theorem.*

Proposition BFA.10.13 (Lemma 9.7 supplies the required common domain)**.** *The source explicitly states one*

$$q_{\mathrm{big}} > 0$$

independent of correction stage and derivative order such that every finite partial state before summation cutoffs is defined on

$$0 < q < q_{\mathrm{big}}.$$

At the interface level this discharges the domain component of O_8*.*

Proof. Lemma 9.7 chooses the threshold once using the fixed background, the fixed positive lower bound for the phase gradient, the primary covariance, and the fixed supports of the normalized mean-correction profiles. Later amplitude problems use fixed-coefficient linear equations and fixed inverse maps. The proof expressly allows the constants and the required number of input derivatives to depend on the correction stage while maintaining the same q-threshold. Thus the conclusion is precisely common-domain existence, not uniform boundedness of every auxiliary constant. □

Theorem BFA.10.14 (Domain uniformity does not require derivative-count uniformity)**.** *The fact that the number of input amplitude derivatives required to control a fixed output derivative may grow with the finite correction stage does not by itself contradict the common-domain conclusion or Lemma 5.4.*

Proof. Lemma 9.7 requires finiteness of the derivative dependency graph at each fixed stage and keeps the geometric/inverse domain fixed. Lemma 5.4 allows stage-dependent constants $C_{j,m}$ and does not require one stage-independent finite derivative count. The later summation burden concerns the exponent loss ℓ_m, which is handled separately by Lemma 9.8. □

BFA.10.6 Stress Test III: stage-independent derivative loss

Audit Obligation BFA.10.15 (BFA10-A3: exponent-loss uniformity)**.** *Verify that every fixed physical derivative order costs a power of q bounded independently of correction stage. Stage-dependent constants are permitted; stage-dependent exponent losses are not.*

Proposition BFA.10.16 (Explicit source witness for ℓ_m)**.** *Lemma 9.8 supplies the explicit sufficient choice*

$$\boxed{\ell_m = 2A + (m+1)\left(1 + \frac{3h}{2}\right),}$$

which is independent of the correction stage j, together with

$$g_j = \frac{h}{10} j \longrightarrow \infty.$$

Under the source parameters $A = 1/2 + h > 0$ and $h > 0$, this choice is also nonnegative and strictly increasing in m. Thus the source conclusion has exactly the stage-independence, nonnegativity, and monotonicity profile required by (5.30) of Lemma 5.4.

Proof. The proof of Lemma 9.8 enumerates the fixed physical derivative costs from the chart operators, cylindrical frame, phase, physical rescaling, and recovery of velocity from vector potentials. Harmonic ranges may grow from stage to stage, but at each fixed stage they affect constants rather than the displayed power loss. The resulting ℓ_m depends on m and fixed source parameters, not on j. □

Theorem BFA.10.17 (Constant-growth firewall)**.** *Arbitrary finite stage-dependent growth of the numerical constants and finite logarithmic powers allowed by Lemma 5.4 is not, by itself, a finite-to-infinite obstruction in the source completion mechanism, provided the stage-independent power-loss hypotheses hold and $g_j \to \infty$.*

Proof. Lemma 5.4 recursively chooses the cutoff scale a_j after the stage-j constants are known, imposing only finitely many derivative requirements at that step. Because $g_j > 0$, each such finite requirement can be forced on a sufficiently small q-neighborhood. The cutoff scale therefore absorbs stage-specific constant growth into a smaller support without requiring a uniform bound in j. □

Remark BFA.10.18 (Sharp failure mode)**.** The preceding firewall would fail if a hidden loss entered the *exponent* and grew with j fast enough to cancel the gain g_j, or if a later operation required a stage-dependent domain threshold. This is why Lemmas 9.7 and 9.8 are separate critical nodes rather than cosmetic estimates.

BFA.10.7 Stress Test IV: exact hypothesis crosswalk into the shared summation theorem

Definition BFA.10.19 ($\mathsf{SUM} \to \mathsf{O}_9$ hypothesis crosswalk)**.** The final application of Lemma 5.4 is audited by the following source-to-source crosswalk.

Lemma 5.4 requirement	Proposition 9.9 source witness	BFA.10 disposition
One common domain $0 < q < q_0$	Lemma 9.7; choose $q_* = q_{\text{big}}$	source-explicit; type match
q smooth, bounded below on compacta, and satisfying (5.28)	similarity identities give (5.28); $q \geq 1 - t$ gives the required local lower bound	source-explicit; type match
Smooth tuple representatives $Z_j = (A_j, B_j, p_j)$ with appropriate zero extensions	representatives (9.15); smooth zero extensions; annular representatives vanish near the axis	source-explicit; type match
$0 < g_1 \leq g_2 \leq \cdots \to \infty$	Lemma 9.8: $g_j = hj/10$, $j \geq 1$	exact match
Nonnegative nondecreasing losses ℓ_m independent of j	Lemma 9.8; $\ell_m = 2A + (m+1)(1 + 3h/2)$	exact match under source parameters
Controlled base U_0	fixed slow base plus finite initialization block; one common initial cutoff if needed	source-explicit
Fixed finite-order/finite-degree differential polynomial F	$F(u,p) = \mathcal{R}(u,p) = \partial_t u + (u \cdot \nabla)u - \Delta u + \nabla p$	exact viscosity-one source instance
Coefficient bounds and truncation-independent comparison regions (5.32)	Navier–Stokes residual has fixed constant coefficients; Proposition 9.9 uses one fixed local domain/support region and exact exterior residual zero	source-explicit; no truncation-dependent coefficient introduced
Finite-stage residual exponent $\rho_j \to \infty$	(9.18) with $\rho_j = h\sigma_j$, $\sigma_j = 1/5 + j/10$	exact match
Flat remainder at each fixed stage	$E_{j,m} \leq C_{j,m,N} q^N$ for every N, uniformly over the full local domain at that stage	exact match
Divergence-preserving cutoff representation	cutoffs applied to wave/mean potentials; direct azimuthal term remains divergence-free	source-explicit

Theorem BFA.10.20 (Source-explicit type match at the second summation consumer)**.** *At the declared source-interface resolution, the* SUM *application inside Proposition 9.9 is not merely an analogy with the earlier base-field application. The source explicitly rechecks the hypotheses of Lemma 5.4 on the actual finite-correction representatives used by* O_9.

Proof. Proposition 9.9 explicitly chooses its own representatives Z_j, invokes the common-domain conclusion of Lemma 9.7, invokes the derivative estimate of Lemma 9.8, identifies the Navier–Stokes residual as the fixed differential polynomial in the summation lemma, and identifies (9.18) with the residual hypothesis (5.33). Hence the second use is a distinct source-explicit, type-matched instantiation of the shared theorem rather than a silent transfer from the Proposition 5.5 consumer. This sentence does not promote the instantiation to an independent BFA verifier pass. □

BFA.10.8 Stress Test V: shrinking cutoffs, local finiteness, and summability

Audit Obligation BFA.10.21 (BFA10-A4: cutoff recursion must dominate stage growth)**.** *Verify that the cutoff parameters are selected only after the relevant stage-specific constants are known, that finitely many conditions are imposed at each recursive step, and that the resulting infinite sum is locally finite away from the singular boundary.*

Theorem BFA.10.22 (Diagonal cutoff selection is logically sufficient)**.** *Under the hypotheses of Lemma 5.4, the recursive condition imposed on the* j*th cutoff scale can simultaneously control all derivative orders* $0 \le m \le j$ *required at that stage, despite arbitrary finite stage-dependent constants.*

Proof. At stage j only finitely many inequalities are imposed. Each has a positive power $q^{g_j/2}$, so after the stage-specific constants are fixed there is a sufficiently small neighborhood in which every inequality holds. Choosing a_j large enough shrinks the support into that neighborhood. The recursive constraint $a_{j+1} \ge 2a_j$ can be imposed simultaneously. This is the standard diagonal feature built directly into the source lemma. □

Proposition BFA.10.23 (Local finiteness is genuine, not formal convergence)**.** *For every compact subset of the source domain with* $q \ge \delta > 0$*, only finitely many correction cutoffs are nonzero. Consequently the completed field is defined there by a finite sum of smooth terms; no interchange of an infinite series with differentiation is needed on such a compact set.*

Proof. The cutoff of stage j vanishes once $a_j q \ge 1$, while $a_j \to \infty$. For $q \ge \delta$, all stages with $a_j > \delta^{-1}$ vanish. Therefore local smoothness on $q > 0$ follows termwise. The singular boundary $q = 0$ is treated by asymptotic estimates rather than by claiming smooth extension of the velocity through the blow-up point. □

Remark BFA.10.24 (No convergence of the unmodified formal series is asserted)**.** The source explicitly motivates Lemma 5.4 by allowing the formal coefficient constants to grow too fast for the unmodified infinite series to converge. The realized object is the cutoff-localized, locally finite sum. BFA therefore does not import a nonexistent claim of ordinary convergence of the raw formal series.

BFA.10.9 Stress Test VI: nonlinear residual comparison and quantifier order

Audit Obligation BFA.10.25 (BFA10-A5: no invalid limit through the nonlinearity)**.** *Verify that flatness of the completed nonlinear residual is proved by a quantified estimate rather than by a formal operation such as*

$$\mathcal{R}\left(\lim_j u^{[j]}\right) \overset{?}{=} \lim_j \mathcal{R}(u^{[j]}).$$

Theorem BFA.10.26 (Fixed-finite-stage comparison principle)**.** *The completion proof does not require an uncontrolled interchange of the infinite cutoff sum with the Navier–Stokes nonlinearity. For each requested derivative order* m *and decay order* N*, Lemma 5.4 first chooses one sufficiently high but finite stage* J*, keeps it fixed, and compares the completed field with that finite partial state.*

Proof. The source obtains a tail estimate for $e = U - U^{[J]}$ and then bounds $F(U) - F(U^{[J]})$ by expanding each finite-degree monomial one factor at a time. For fixed (m, N), it chooses J sufficiently large to dominate the finite derivative and polynomial losses, and only afterwards shrinks the q-neighborhood. Thus the nonlinear conclusion is a direct estimate, not a formal continuity assertion without a topology. □

Theorem BFA.10.27 (Correct quantifier order for infinite-order flatness)**.** *The source proof establishes the quantifier pattern*

$$\boxed{\forall m, N\ \exists J\ \exists \delta, C > 0 : \quad |\mathcal{R}(u_{\mathrm{loc}}, p_{\mathrm{loc}})|_m \le C q^N \quad (0 < q < \delta),}$$

which implies the required infinite-order flatness

$$\forall m, N\ \exists \delta, C > 0 : \quad |\mathcal{R}(u_{\mathrm{loc}}, p_{\mathrm{loc}})|_m \le C q^N.$$

No single truncation stage uniform in all (m, N) is required or claimed.

Proof. Lemma 5.4 explicitly chooses the data in the order: first (m, N), then a finite stage J, then the neighborhood and constants. Since J is an auxiliary witness eliminated from the final conclusion, the resulting quantifier statement is exactly infinite-order flatness. Requiring one universal J would be a strictly stronger and unnecessary statement. □

BFA.10.10 Stress Test VII: the flat remainders are not illegally summed

Audit Obligation BFA.10.28 (BFA10-A6: flat-remainder aggregation)**.** *A potentially fatal shortcut would be to infer that an infinite sum of stagewise flat remainders is flat without a summability theorem. Verify that the source does not make this inference.*

Theorem BFA.10.29 (No summation of the stagewise flat remainders)**.** *The source's final completion argument does not sum the family $\{E_{j,m}\}_j$. Instead, for each requested (m, N) it uses the flat remainder from one fixed finite stage selected by the comparison argument.*

Proof. Lemma 5.4 compares the completed field to one fixed $U^{[J]}$ and invokes the flatness of $E_{J,m}$ only after J has been fixed. Proposition 9.9 repeats this point explicitly for the final correction sequence: the finite-stage flat terms are controlled through comparison with a fixed finite state and are not summed. Therefore no unproved interchange between "flat for each stage" and "flat after infinite summation" occurs at this location. □

Remark BFA.10.30 (Why this matters)**.** This is one of the strongest finite-to-infinite safeguards in the completion mechanism. A statement such as

$$E_j = O(q^N) \text{ for every fixed } j, N$$

does not control $\sum_j E_j$ without additional uniformity. The source avoids needing such uniformity altogether.

BFA.10.11 Stress Test VIII: divergence, representatives, support, and overlap

Audit Obligation BFA.10.31 (BFA10-A7: structural invariants under cutoff)**.** *Verify that the cutoff operation used to gain local finiteness does not destroy incompressibility or the representative structure needed by the completed field.*

Proposition BFA.10.32 (Divergence-preserving realization)**.** *The final summation is performed on wave and mean vector potentials before applying curl. The direct azimuthal term remains divergence-free after multiplication by the q-cutoff because its scalar amplitude is independent of θ. Thus the cutoff mechanism preserves the divergence-free output by construction.*

Proposition BFA.10.33 (Support-boundary and chart compatibility are source-explicit)**.** *Lemma 9.8 and Proposition 9.9 explicitly use smooth zero extensions at support boundaries and prescribed torus-coordinate changes/relabeling on overlaps. At the source-interface level these obligations are not silently inferred from pointwise bounds alone.*

Remark BFA.10.34 (Scope of the present check)**.** BFA.10 checks that the completion stage names and consumes these structural obligations. A complete independent reconstruction of every torus chart, label, support, and inverse operator would require reopening the internal proofs represented by O_3–O_5, which is outside the focused completion audit performed here.

BFA.10.12 Stress Test IX: preservation of the local output class

Audit Obligation BFA.10.35 (BFA10-A8: completed output must be strong enough for BFA.11)**.** *Verify that Proposition 9.9 outputs more than a formal residual-flat sequence: it must produce actual smooth fields on the punctured/local domain, preserve the source's Cartesian regularity, exterior formula, one-sided endpoint regularity away from the singular point, and the inner growth used to prove breakdown.*

Proposition BFA.10.36 (The O_9 output is an actual local field)**.** *The source defines*

$$\mathcal{A} = \mathcal{A}_0 + \sum_{j\geq 1} \chi(a_j q) A_j, \qquad Be_\theta = B_0 e_\theta + \sum_{j\geq 1} \chi(a_j q) B_j,$$

$$p_{\mathrm{loc}} = p_0 + \sum_{j\geq 1} \chi(a_j q) p_j, \qquad u_{\mathrm{loc}} = \operatorname{curl} \mathcal{A} + Be_\theta,$$

with locally finite sums for $q > 0$. Proposition 9.9 then establishes the source's Theorem 3.1 conclusions for this actual pair rather than merely for a formal coefficient expansion.

Theorem BFA.10.37 (Completion-output sufficiency for the next BFA stage)**.** *At the declared source-interface resolution, the output type of O_9 is sufficient for BFA.11 to begin the localization and terminal-extension audit. This is a type-level handoff only, conditional on the unresolved independent re-proof obligations and the source-identity firewall of Policy 55. It does not itself discharge O_{10} or O_{11}.*

Proof. BFA.3 assigns O_9 the completed local field and assigns localization and terminal force extension to distinct later nodes. The source Proposition 9.9 supplies the local smooth pair, divergence condition, flat residual, exterior behavior, and local growth properties used by those later steps. BFA.4 and BFA.9 prohibit identifying this with terminal regularization. □

BFA.10.13 Interchange audit

Definition BFA.10.38 (Potentially dangerous interchange classes)**.** The principal finite-to-infinite interchange risks are:

(i) infinite summation with physical differentiation;

(ii) infinite summation with curl/divergence constraints;

(iii) infinite summation with the quadratic convection term;

(iv) infinite summation with pressure reconstruction;

(v) infinite summation of the stagewise flat residual remainders;

(vi) stage limit with changing support/domain or changing inverse operators;

(vii) limit passage with a stage-dependent exponent loss.

Theorem BFA.10.39 (BFA.10 interchange disposition)**.** *At the declared audit resolution, the source route avoids the seven shortcuts of Definition BFA.10.38 as follows:*

(i) local finiteness handles differentiation for every $q > 0$;

(ii) vector-potential cutoff realization handles divergence;

(iii) the finite differential-polynomial difference estimate handles the nonlinear residual;

(iv) pressure is included as a summed representative with stagewise telescoping increments before the completion application;

(v) flat remainders are handled by one fixed finite comparison stage and are not summed;

(vi) Lemma 9.7 fixes one domain and fixed inverse architecture;

(vii) Lemma 9.8 supplies a stage-independent power loss.

No certified adverse witness for these interchange classes is obtained in BFA.10.

Proof. The seven clauses are the combined consequences of Propositions BFA.10.13, BFA.10.16, BFA.10.23 and BFA.10.32 and Theorems BFA.10.26 and BFA.10.29 and the explicit pressure representatives in the source's equations (9.15)–(9.16). The final sentence is an audit disposition at the declared resolution, not an independent proof of every upstream analytic estimate. □

BFA.10.14 Adversarial audit matrix

Definition BFA.10.40 (BFA.10 completion stress matrix)**.** The current BFA.10 source-facing findings are summarized as follows.

ID	Obligation	Source witness	BFA.10 disposition
C0	admissible stage-zero state	Definition 9.4; Proposition 9.5	source-explicit consumer-domain match
C1	finite-stage improvement	Proposition 9.6	source-explicit; no certified adverse witness
C2	common domain	Lemma 9.7	exact type match
C3	derivative loss independent of stage	Lemma 9.8, explicit ℓ_m	exact type match
C4	decay orders tend to infinity	$g_j = hj/10$	exact type match
C5	finite-stage residual order tends to infinity	(9.18), $\rho_j = h\sigma_j$	exact type match
C6	arbitrary stage constants tolerated	Lemma 5.4 recursive cutoff choice	source-explicit firewall
C7	nonlinear completion estimate	Lemma 5.4 finite-stage comparison	source-explicit; no formal limit shortcut
C8	flat remainder handling	fixed-J comparison; Proposition 9.9	no illegal remainder summation identified
C9	divergence and support structure	potential cutoffs, zero extensions, overlap rules	source-explicit at interface level
C10	actual completed local output	Proposition 9.9, (9.20)–(9.21)	sufficient source type for BFA.11
C11	immutable analytic-source byte identity	BFA.2 source-sovereignty contract	open at construction stage; governance blocker for final freeze, not mathematical rejection

Policy 57 (BFA-COMP-02: matrix entries are not atomic verifier passes)**.** *The favorable entries in Definition BFA.10.40 mean that the relevant finite-to-infinite burden is explicitly represented and type-compatible in the source proof at the declared audit resolution. They do not convert the corresponding BFA.8 atomic verifier status to* pass *unless the exact governing mathematical verifier has independently discharged the entire obligation. Likewise, a favorable mathematical field does not close an open source-identity field.*

BFA.10.15 Strongest current completion verdict

Theorem BFA.10.41 (BFA.10 source-faithful completion stress verdict)**.** *At the declared BFA.10 source-interface adversarial resolution, no certified adverse finding has been obtained against the registered*

$$\mathsf{O}_6 \longrightarrow \mathsf{O}_7 \longrightarrow \mathsf{O}_8 \longrightarrow (\mathsf{SUM}\ \textit{at its second consumer}) \longrightarrow \mathsf{O}_9$$

completion interface. Precisely:

(i) the initialization is source-explicitly a stage-zero state in the exact Definition 9.4 class consumed by the correction cycle;

(ii) the source distinguishes finite residual improvement from stage-uniform derivative control and from completion;

(iii) the iteration gains a fixed positive residual order at every finite stage, conditional on Proposition 9.6's internal estimates;

(iv) one common source domain is asserted independently of stage and derivative order;

(v) the physical exponent loss for every fixed derivative order has an explicit bound independent of stage;

(vi) *the completion theorem intentionally permits stage-dependent constants and absorbs them by recursively shrinking cutoff supports;*

(vii) *the cutoff sum is locally finite for every $q > 0$, so local smoothness does not rely on uncontrolled convergence of the raw formal series;*

(viii) *the nonlinear residual is controlled by comparison with one fixed finite stage, not by an unproved interchange of limit and nonlinearity;*

(ix) *the stagewise flat remainders are not summed;*

(x) *the quantifier order used to prove infinite-order flatness is valid at the source-statement level;*

(xi) *divergence and the required representative structure are preserved by the completion construction; and*

(xii) *Proposition 9.9 outputs an actual completed local field of the type required by the next terminal stage.*

The mathematical/source-interface disposition above is independent of the remaining artifact-identity closure. Because BFA.2 has not yet frozen a local cryptographic snapshot of the analytic PDF, final source-bound freeze remains pending under Policy 55.

Accordingly, BFA.10 obtains

$$\boxed{\mathsf{CompletionAdverseFinding}_{\mathrm{BFA10}} = \mathrm{none_certified_at_declared_resolution}}$$

and does not obtain either

$$\mathsf{certified_nonclosure} \quad \textit{or} \quad \mathsf{certified_reject}$$

for the completion interface, while separately recording

$$\boxed{\mathsf{AnalyticSourceIdentityClosure}_{\mathrm{BFA10}} = \mathrm{pending_BFA2_cryptographic_snapshot.}}$$

Proof. Items (i)–(iv) follow from the separated source nodes and Propositions BFA.10.6, BFA.10.9 and BFA.10.13. Item (v) is Proposition BFA.10.16. Item (vi) is Theorems BFA.10.17 and BFA.10.22. Item (vii) is Proposition BFA.10.23. Items (viii)–(x) are Theorems BFA.10.26, BFA.10.27 and BFA.10.29. Item (xi) follows from Propositions BFA.10.32 and BFA.10.33. Item (xii) is Proposition BFA.10.36 and Theorem BFA.10.37. None of these source-interface checks supplies a certified adverse proposition, while Policies 54 and 57 prohibit promoting that absence to a complete independent re-proof. The final source-identity qualification is governed separately by Policy 55. □

BFA.10.16 Residual open obligations and falsification handles

Definition BFA.10.42 (Completion-layer residual audit obligations)**.** The strongest remaining completion-layer obligations after BFA.10 are not missing source interfaces but deeper mathematical re-verification tasks:

(i) line-by-line independent verification of every exponent inequality closing Proposition 9.6;

(ii) independent verification that every inverse operator used at arbitrary finite stage has exactly the fixed-domain and derivative properties consumed by Lemma 9.7;

(iii) independent verification of all chart, harmonic, phase, and rescaling estimates entering the explicit ℓ_m of Lemma 9.8;

(iv) independent verification that the residual estimate (9.18) is uniform over every label/support family exactly as claimed;

(v) independent verification of every source-specific endpoint/exterior statement retained by Proposition 9.9.

Each item is a legitimate future adversarial target, but none is converted into $\mathrm{candidate_witness}$ merely because it remains expensive to re-prove.

Separately, final release freeze requires closure of the analytic-paper byte identity under Policy 55. That requirement is a governance/source-binding obligation and is not counted as an analytic falsification handle.

Theorem BFA.10.43 (Sharp completion falsification handles)**.** *A decisive completion-layer attack would be obtained by certifying any one of the following exact propositions:*

(a) a stage-j correction fails the Proposition 9.6 next-stage bounds;

(b) the required domain threshold necessarily shrinks with j;

(c) the physical exponent loss contains an unavoidable term growing with j that defeats $g_j \to \infty$;

(d) one Lemma 5.4 hypothesis fails for the actual Proposition 9.9 representatives;

(e) the cutoff-tail estimate fails at some fixed derivative order;

(f) the residual difference estimate does not control the Navier–Stokes nonlinearity at the required order;

(g) the final proof implicitly sums a non-summable family of flat remainders; or

(h) the cutoff construction destroys divergence, support compatibility, or the singular asymptotic required downstream.

A certified instance would propagate according to BFA.8–BFA.9. BFA.10 produces no certified instance of (a)–(h).

BFA.10.17 Generic extraction seeds, without promotion

Definition BFA.10.44 (BFA.10 generic extraction seeds)**.** The completion audit exposes four source-independent-looking patterns for later BFA.13 analysis:

(i) *monotone defect improvement*: a finite correction map increases a defect order by a fixed positive amount;

(ii) *uniform-loss / nonuniform-constant separation*: exponent losses are stage-independent while multiplicative constants may depend on stage;

(iii) *diagonal shrinking-support completion*: stage-specific constants are absorbed by recursively shrinking supports so that the infinite realization is locally finite;

(iv) *fixed-stage nonlinear comparison*: the completed nonlinear output is controlled by comparison with one sufficiently accurate finite stage rather than by an abstract interchange of limit and nonlinearity.

Policy 58 (BFA-COMP-03: extraction seed is not RIC theorem)**.** *The patterns in Definition BFA.10.44 are only extraction seeds. BFA.10 does not prove a domain-independent RIC theorem, an AK Core theorem, or a generic completion theorem. Such promotion requires BFA.13 abstraction, independent proof, hypothesis minimization, cross-domain substitution, and the BFA.14 promotion controls.*

BFA.10.18 BFA.10 closure and BFA.11 handoff

Definition BFA.10.45 (BFA.10 status vector)**.** At completion of BFA.10,

$$\begin{aligned}
\text{stage0_initialization_interface} &= \text{source_explicit},\\
\text{finite_stage_improvement_interface} &= \text{source_explicit},\\
\text{common_domain_interface} &= \text{source_explicit},\\
\text{stage_independent_loss_interface} &= \text{source_explicit},\\
\text{summation_q_geometry} &= \text{source_explicit},\\
\text{SUM_O9_consumer_crosswalk} &= \text{type_matched},\\
\text{cutoff_local_finiteness} &= \text{source_explicit},\\
\text{nonlinear_fixed_stage_comparison} &= \text{source_explicit},\\
\text{flat_remainder_illegal_sum} &= \text{not_identified},\\
\text{hidden_limit_nonlinearity_interchange} &= \text{not_identified},\\
\text{completion_certified_nonclosure} &= \text{not_obtained},\\
\text{completion_certified_reject} &= \text{not_obtained},\\
\text{analytic_source_identity_closure} &= \text{pending_BFA2_cryptographic_snapshot},\\
\text{full_independent_analytic_reproof} &= \text{not_claimed},\\
\text{generic_completion_promotion} &= \text{not_claimed},\\
\text{terminal_stress_test} &= \text{delegated_to_BFA.11}.
\end{aligned}$$

Theorem BFA.10.46 (BFA.10 closure)**.** *BFA.10 closes the construction-stage finite-to-infinite audit at its declared source-interface resolution with no certified adverse finding against the registered completion interface. Final immutable source-bound closure remains pending because the analytic-paper byte identity has not yet been frozen under BFA.2. The principal reason is structural rather than rhetorical: the source contains an explicit common-domain theorem, an explicit stage-independent derivative-loss theorem, a generic summation lemma designed to tolerate stage-dependent constants, a locally finite cutoff realization, and a fixed-finite-stage nonlinear residual comparison that avoids summing the stagewise flat remainders.*

This closure is strictly weaker than a complete independent verification of the external analytic proof and strictly stronger than merely observing that finite residual orders tend to infinity.

Proof. The first statement is Theorem BFA.10.41; the source-identity qualification is Policy 55. The structural reasons are proved or source-matched in Propositions BFA.10.13, BFA.10.16 and BFA.10.23 and Theorems BFA.10.17, BFA.10.26 and BFA.10.29. The final sentence follows because BFA.10 has audited and source-matched the exact finite-to-infinite interface architecture and quantifier discipline while Policy 54 retains the independent-reproof ceiling. □

Remark BFA.10.47 (Next consumer: BFA.11)**.** BFA.11 shall perform the *Terminal Return Stress Test*. It must treat the completed local field of O_9 only as an input and independently audit:

(i) O_{10}: localization by potentials, preservation of the singular core, fixed compact spatial support, and zero initial datum;

(ii) O_{11}: existence and compatibility of every terminal force jet and the genuinely smooth compactly supported extension across $t = 1$;

(iii) O_{12}: the exact uniform kinetic-energy and dissipation bounds required by the comparison class;

(iv) O_{13}: uniqueness/comparison on every $[0, T]$, $T < 1$, and the transfer from local velocity growth to exclusion of a global smooth bounded-energy competitor;

(v) O_{14}: viscosity scaling as the shared C/D terminal transport bottleneck; and

(vi) O_{15}: the additional periodization and periodic-pressure obligations for the Clay-D lane.

No result of BFA.10 may be used to pre-discharge these terminal obligations.

BFA.11 Terminal Return Stress Test

BFA.11.1 Purpose, inherited input, and audit ceiling

Declaration BFA.11.1 (BFA.11 objective)**.** BFA.11 audits the terminal analytic and target-facing region of the currently registered source-faithful Navier–Stokes route. It begins only after the completed local field of O_9 has been admitted as the output of BFA.10 at BFA.10's declared source-interface resolution. For navigation it uses the terminal audit spine

$$\mathsf{O}_{10} \longrightarrow \mathsf{O}_{11} \longrightarrow \mathsf{O}_{12} \longrightarrow \mathsf{O}_{13} \longrightarrow \mathsf{O}_{14},$$

and the additional periodic branch

$$\{\mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{14}\} \longrightarrow \mathsf{O}_{15},$$

followed by the statement-level interpretation edges to $\mathsf{Clay\text{-}C}_\nu$ and $\mathsf{Clay\text{-}D}_\nu$. These two displays are schematic audit spines only. The normative mandatory predecessor relation remains the BFA.3 source DAG; in particular O_{13} has the full predecessor set registered there and O_{15} consumes O_{14} together with the compact-support outputs of O_{10}–O_{11}.

The governing question is

> Does the completed local construction survive localization, terminal extension, comparison, scaling, periodization, and exact target interpretation without a certified terminal defect?

Policy 59 (BFA11-RES-01: source-interface resolution)**.** *BFA.11 is an adversarial source-interface audit, not an independent line-by-line reproof of every estimate in Section 10 of the external paper. Its favorable maximum verdict is therefore*

$$\mathrm{none_certified_at_declared_resolution},$$

not proved*,* verified*, or* formally_verified*. A favorable BFA.10 completion verdict does not pre-discharge any BFA.11 obligation.*

Policy 60 (BFA11-STAT-01: audit disposition is not atomic verifier status)**.** *BFA.11 keeps three fields separate:*

$$\mathsf{SourceStatementConformance}, \qquad \mathsf{TerminalAuditDisposition}, \qquad \mathsf{VerifierStatus}.$$

The first asks whether the printed source statement and the consumed interface fields have been read consistently. The second records whether the present adversarial source-interface inspection produced a certified mismatch or blocker. The third is the BFA.8 atomic mathematical verifier status and is restricted to

$$\mathrm{pass}, \quad \mathrm{reject}, \quad \mathrm{undefined}, \quad \mathrm{not_invoked}.$$

Because BFA.11 does not independently reprove the terminal source lemmas, none of the favorable phrases "source-matched", "conformant", or "no certified adverse finding" is serialized as $\mathsf{VerifierStatus} = \mathrm{pass}$*. Unless a separately identified mathematical verifier is actually run, the external analytic nodes retain*

$$\mathsf{VerifierStatus}(\mathsf{O}_i) = \mathrm{not_invoked}, \qquad 10 \le i \le 15.$$

Policy 61 (BFA11-SRC-01: analytic-source identity firewall)**.** *The source locators in this section refer to the currently resolved public analytic PDF [12]. As already recorded in BFA.2 and BFA.10, the canonical BFA source record has not yet frozen a local byte-level cryptographic snapshot of that PDF. Consequently*

$$\mathsf{AnalyticSourceIdentityClosure}_{\mathrm{BFA11}} = \mathrm{pending_BFA2_cryptographic_snapshot}.$$

A successful present-day URL resolution or source reading is not substituted for the missing canonical content-addressed snapshot. This provenance state neither rejects nor verifies the mathematics.

Definition BFA.11.2 (Terminal audit family)**.** The BFA.11 theorem-facing family is

$$\mathfrak{C}_{\mathrm{term}} := \{\mathsf{O}_{10}, \mathsf{O}_{11}, \mathsf{O}_{12}, \mathsf{O}_{13}, \mathsf{O}_{14}, \mathsf{O}_{15}\} \cup \mathfrak{J}_{\mathrm{term}} \cup \mathfrak{E}_{\mathrm{term}} \cup \{\rho_{PT}^{C,\nu}, \rho_{PT}^{D,\nu}\},$$

where $\mathfrak{J}_{\mathrm{term}}$ and $\mathfrak{E}_{\mathrm{term}}$ are the joint-composition and theorem-bearing edge obligations supplied by the BFA.8 audit expansion.

BFA.11.2 Exact source contract for the terminal region

Definition BFA.11.3 (Source-facing terminal contract)**.** At viscosity one, the terminal source contract is the following typed chain.

(i) O_{10}: Proposition 10.1 constructs smooth u, p on $\mathbb{R}^3\times[0,1)$ with fixed compact spatial support, divergence-free velocity, zero initial datum for sufficiently small time, and agreement with the completed local fields near the singular core for late time.

(ii) O_{11}: Lemmas 10.2–10.3 prove compatible one-sided limits of all force derivatives at $t\uparrow 1$ and realize those jets by one force $f\in C_c^\infty(\mathbb{R}^3\times(0,\infty);\mathbb{R}^3)$.

(iii) O_{12}: Lemma 10.4 proves uniform L^2 kinetic-energy control and finite integrated dissipation on $[0,1)$.

(iv) O_{13}: Lemma 10.5 and the final proof of Theorem 1.1 compare every smooth bounded-energy whole-space solution with the constructed solution on every $[0,T]$, $T<1$, and use the retained inner growth to exclude a global smooth bounded-energy competitor.

(v) O_{14}: equations (10.22)–(10.23) transport the viscosity-one construction to every fixed $\nu>0$.

(vi) O_{15}: Corollary 10.6 spatially/time rescales the compactly supported whole-space construction into the interior of a fundamental cube, periodizes disjoint translates, periodizes the pressure, and excludes a global smooth periodic competitor.

Definition BFA.11.4 (Target-critical projections inside the terminal source contract)**.** The complete source outputs are not all equally required by the official Return. BFA.11 therefore records the following target-critical projections.

(i) For Clay-C, the target-critical part of O_{12} is the uniform spatial L^2 bound. Finite total dissipation is a stronger source output and is not separately required by Fefferman condition (7).

(ii) Inside O_{13}, the target-critical branch is the comparison of a hypothetical global smooth bounded-energy competitor with the constructed field on every $[0,T]$, followed by the retained velocity growth contradiction. The auxiliary statement identifying the maximal classical H^3 lifespan is useful source information but is not by itself a necessary Clay-C Return edge.

(iii) For Clay-D, no bounded-energy hypothesis belongs to the official periodic solution class. The periodic nonexistence argument must close intrinsically from smooth periodicity, the common datum/force, and torus uniqueness; whole-space energy is not imported as a target hypothesis.

(iv) Compact support of the whole-space pressure is stronger than Clay-C requires, but it is later useful for the periodic adapter O_{15}. Target-critical strength is therefore recorded relative to each lane rather than by one undifferentiated terminal score.

Audit Obligation BFA.11.5 (BFA11-A1: terminal noncompensation)**.** *Each item of Definition BFA.11.3 must be checked at its own source and target type. In particular,*

$$\mathsf{O}_9 \not\Rightarrow \mathsf{O}_{10}, \qquad \mathsf{O}_{10} \not\Rightarrow \mathsf{O}_{11}, \qquad \mathsf{O}_{11} \not\Rightarrow \mathsf{O}_{12}, \qquad \mathsf{O}_{12} \not\Rightarrow \mathsf{O}_{13}.$$

No smoothness, compact-support, energy, comparison, scaling, or periodicity field is manufactured by a favorable sibling obligation.

BFA.11.3 Stress test of O_{10}: localization without destroying the singular core

Audit Obligation BFA.11.6 (BFA11-A2: localization geometry)**.** *The cutoffs used in Proposition 10.1 must remain inside the local domain on which the completed field is defined; the localized velocity must remain smooth and divergence free; its support must be fixed and compact in space; the temporal cutoff must create zero initial datum; and the singular-core asymptotic used later must remain untouched in a late-time neighborhood of* $(0,1)$.

Proposition BFA.11.7 (Localization margin is source-explicit)**.** *At source-interface resolution, Proposition 10.1 supplies an explicit margin from the outer local-domain boundary rather than merely asserting that a cutoff exists. It proves*

$$q \le C_0\big(\tau + |z|^{1/D}\big)$$

and chooses fixed spatial and temporal cutoffs so that their support lies in $q < q_*/2$, while both cutoffs equal one near the terminal singular core.

Proof. This is the printed content of Proposition 10.1 and equations (10.3)–(10.4) in [12]. The fixed margin is important: it makes zero extension across the cutoff support a smooth local operation rather than an assertion made at the local-domain boundary. □

Proposition BFA.11.8 (Incompressibility survives localization)**.** *The source does not localize the poloidal velocity by naive multiplication. It localizes the vector potential first and defines*

$$u = \operatorname{curl}(cA) + cBe_\theta, \qquad p = cp_{\mathrm{loc}}.$$

With axisymmetric c and B, the displayed formula gives $\operatorname{div} u = 0$*, while the Cartesian representatives used in the source preserve smoothness at the axis.*

Proof. The curl term is divergence-free identically, and the azimuthal term has $r^{-1}\partial_\theta(cB) = 0$. The source explicitly records smooth Cartesian representatives for the axisymmetric potentials before invoking the cutoff construction. Hence the obvious "multiply the velocity by a cutoff and lose incompressibility" failure mode is not the route actually used. □

Proposition BFA.11.9 (Compact pressure and non-solenoidal force create no target mismatch)**.** *The localized pressure is compactly supported because the source sets $p = cp_{\mathrm{loc}}$. The corresponding force is then defined by the full momentum residual and may have nonzero divergence. This is source-explicit: taking divergence yields the pressure Poisson equation with the force-divergence term. Neither Fefferman alternative (C) nor alternative (D) requires the external force itself to be divergence free. Therefore BFA.11 does not impose an artificial solenoidal-force condition and does not treat nonzero* $\operatorname{div} f$ *as a defect.*

Proposition BFA.11.10 (The singular growth path survives the cutoffs)**.** *The localization does not erase the blow-up witness consumed downstream. For the source path $x_\tau \to 0$, the spatial and temporal cutoffs are one for all sufficiently small $\tau > 0$, and therefore the source retains*

$$u_\theta(x_\tau, 1-\tau) = \tau^{-A}\big(e_0 + O(\tau^{2h})\big) \longrightarrow +\infty.$$

Proof. Proposition 10.1 makes the localized fields equal to the completed local fields near the origin at sufficiently late time. The proof of Theorem 1.1 then chooses the explicit path (10.20) and states that both cutoffs equal one there for sufficiently small τ, yielding (10.21). □

Proposition BFA.11.11 (BFA.11 disposition of O_{10})**.** *No certified localization defect is found at BFA.11 source-interface resolution. In particular, the source explicitly addresses the four obvious terminal failure modes: leaving the local domain, losing incompressibility, creating a nonzero initial datum, and cutting away the singular core.*

BFA.11.4 Stress test of O_{11}: terminal force jets and smooth compact extension

Audit Obligation BFA.11.12 (BFA11-A3: all mixed terminal jets)**.** *It is insufficient that the residual itself have a limit as $t \uparrow 1$. For every spatial multi-index α and integer $j \ge 0$, the mixed jet*

$$\partial_x^\alpha \partial_t^j f(\cdot, t)$$

must converge in a topology strong enough to define one compatible smooth extension through $t = 1$, with a support and time-decay profile admissible for the official target.

Proposition BFA.11.13 (The endpoint proof separates origin and exterior)**.** *Lemma 10.2 does not infer terminal smoothness from residual flatness at the origin alone. It combines three different source mechanisms:*

(i) bounded physical derivatives on compact sets with q bounded away from zero and from q_*;

(ii) exact exterior heat-flow estimates and the normalized pressure integral at positive radius; and

(iii) arbitrarily high residual flatness near the origin.

These regions cover the terminal spatial slice and yield uniform convergence on $\mathbb{R}^3$ because all derivatives vanish outside the fixed compact support.

Proof. This is the partition and finite-cover argument of Lemma 10.2. The exterior region uses (10.7)–(10.8) and dominated convergence for the pressure integral; the origin uses the flat bound (10.9); compactness of the remaining support supplies the global uniform conclusion. □

Proposition BFA.11.14 (The terminal jets are mutually compatible)**.** *The source proves more than separate existence of limits. It constructs $F_j \in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$ such that*

$$\lim_{t\uparrow 1} \partial_x^\alpha \partial_t^j f(x,t) = \partial_x^\alpha F_j(x),$$

and uses the fundamental theorem of calculus in time to prove compatibility of consecutive time jets.

Proof. Equation (10.10) expresses a finite-time derivative as its terminal candidate minus the integral of the next derivative. Uniform convergence of the next derivative therefore identifies the derivative relation at the endpoint; spatial compatibility is obtained analogously from coordinate segments. □

Audit Obligation BFA.11.15 (BFA11-A4: Borel-type realization must converge after differentiation)**.** *A formal Taylor series with coefficients F_j is not enough. The post-terminal extension must converge after every fixed mixed derivative and must remain compactly supported in time and space.*

Proposition BFA.11.16 (The source supplies a convergent compact-support realization)**.** *Lemma 10.3 realizes the terminal jets for $\sigma = t - 1 \geq 0$ by*

$$f(x, 1+\sigma) = \sum_{j=0}^{\infty} \chi_0(b_j\sigma)\frac{\sigma^j}{j!}F_j(x),$$

with increasing integers b_j chosen recursively so that every fixed mixed differentiated tail is absolutely and uniformly summable. Each term retains support in K, and every term is zero for $\sigma \geq 1$.

Proof. The source estimate (10.12) bounds a mixed derivative of the j-th term by $C_{j,m}b_j^{m-j}\|F_j\|_{C^{|\alpha|}}$. For each j, only finitely many derivative constraints with total order at most $\lfloor j/2 \rfloor$ are imposed when choosing b_j, and the exponent $m - j$ is negative. Hence an admissible larger integer exists. The resulting bound is at most 2^{-j} on each fixed differentiated tail for all sufficiently large j. This proves the exact convergence property required by **??** BFA.11.15. □

Proposition BFA.11.17 (Official whole-space force decay is automatic after extension)**.** *The force produced by Lemma 10.3 satisfies the Fefferman whole-space force condition (5) for every fixed derivative order and decay power.*

Proof. The force is smooth and supported in $K \times [0,2]$. A smooth function on this compact set has a finite supremum for every mixed derivative, and multiplication by a sufficiently large fixed power of $(1+|x|+t)$ gives the required constant. This is also stated explicitly at the end of Lemma 10.3. □

Proposition BFA.11.18 (BFA.11 disposition of O_{11})**.** *No certified terminal-extension defect is found at BFA.11 source-interface resolution. The source distinguishes endpoint jet existence, compatibility, and convergent compact-support realization, rather than treating any one of them as a synonym for the others.*

BFA.11.5 Stress test of O_{12}: energy and dissipation

Audit Obligation BFA.11.19 (BFA11-A5: exact energy class)**.** *The constructed preterminal velocity must satisfy the uniform spatial L^2 control consumed by the comparison theorem. A hypothetical official global whole-space solution has a bound for all $t \geq 0$; restricting that solution to any $[0,T]$, $T < 1$, must place it in the same local comparison class. This requires a bound uniform in $t < 1$ for the constructed field, not merely finite energy at every fixed time. It does not turn the singular preterminal construction itself into a global physically reasonable solution.*

Proposition BFA.11.20 (The source energy estimate is uniform to the terminal time)**.** *Let*

$$F(t) = \int_0^t \|f(s)\|_{L^2(\mathbb{R}^3)}\, ds.$$

Lemma 10.4 proves

$$\|u(t)\|_2^2 + 2\int_0^t \|\nabla u(s)\|_2^2\, ds \leq F(t)^2, \qquad 0 \leq t < 1,$$

with $F(1) < \infty$. Hence

$$\sup_{0\leq t<1} \|u(t)\|_2 < \infty$$

and the total dissipation before the singular time is finite.

Proof. On each $[0,T]$, $T < 1$, the fields are smooth with fixed compact spatial support, so pairing the equation with u gives the standard energy identity without boundary-at-infinity terms. Dividing the identity by $(\|u\|_2^2 + \delta^2)^{1/2}$, discarding dissipation, integrating from the zero datum, and taking $\delta \downarrow 0$ gives $\|u(t)\|_2 \leq F(t)$. Reinserting this bound gives the displayed estimate; monotone convergence extends the dissipation bound to $[0,1)$. □

Remark BFA.11.21 (Energy is a target condition, not the blow-up mechanism)**.** The finite energy in Proposition BFA.11.20 does not contradict the pointwise velocity growth. The blow-up occurs in a shrinking spatial region; Clay-C requires bounded total kinetic energy of a hypothetical global smooth solution, not a uniform L^∞_x bound as an input condition.

Proposition BFA.11.22 (BFA.11 disposition of O_{12})**.** *No certified energy-class mismatch is found at the declared resolution. The source proves the uniform spatial L^2 bound consumed by the whole-space comparison. The official condition (7) applies to a hypothetical global physically reasonable solution; its restriction to every shorter interval supplies the Lemma 10.5 hypothesis. Finite total dissipation is retained as a stronger source output, not silently promoted to a separate Clay requirement.*

BFA.11.6 Stress test of O_{13}: comparison with an arbitrary smooth bounded-energy competitor

Audit Obligation BFA.11.23 (BFA11-A6: no hidden pressure-growth hypothesis)**.** *The official whole-space target imposes smoothness and bounded kinetic energy on a putative global solution, but no independent spatial decay condition on its pressure. The comparison theorem may therefore not assume a pressure decay condition stronger than the official target.*

Proposition BFA.11.24 (The comparison theorem states the correct competitor class)**.** *Lemma 10.5 assumes, on a fixed $[0,T]$, only a smooth solution of the same viscosity-one equation with the same force and zero initial datum together with*

$$v \in L^\infty([0,T]; L^2(\mathbb{R}^3)).$$

It does not print a spatial decay hypothesis on the competitor pressure.

Proof. This is the statement of Lemma 10.5. It matches the local-in-time restriction of the official whole-space competitor class used in Clay-C. □

Proposition BFA.11.25 (Pressure-gradient recovery removes the obvious infinity loophole)**.** *For the difference $w = v - u$, Lemma 10.5 reconstructs the pressure gradient from the quadratic difference tensor using Riesz transforms and proves*

$$\nabla\pi = \nabla\pi_*$$

in the space–time interior. The argument is distributional and uses the L^2 bound on w and the resulting L^1 bound on the quadratic tensor; it does not impose an additional pointwise pressure-growth condition.

Proof. The source defines $\pi_* = \sum_{i,j} R_i R_j g_{ij}$, observes that $g \in L^\infty_t L^1_x$, and places π_* in negative Sobolev spaces. The difference between the actual pressure gradient and $\nabla\pi_*$, after a time test, is both harmonic and an H^{-3} distribution; its Fourier transform is a weighted L^2 function supported at the origin and is therefore zero. This is the source mechanism specifically designed to avoid adding an uncontrolled pressure-at-infinity hypothesis. □

Proposition BFA.11.26 (Pressure gauge freedom does not obstruct the comparison)**.** *The comparison reconstructs the pressure* gradient*. Thus the actual pressure difference may still differ from the Riesz representative by a function of time only. This gauge freedom is immaterial: the Navier–Stokes momentum equation consumes ∇P, the difference-energy identity contains only the pressure gradient after integration by parts, and the final Clay-C contradiction is carried by the velocity. BFA.11 therefore does not mistake failure to identify an absolute pressure normalization for a failure of the comparison theorem.*

Audit Obligation BFA.11.27 (BFA11-A7: boundary flux must vanish as the comparison ball expands)**.** *A localized difference-energy identity on a growing ball does not prove whole-space uniqueness unless transport and pressure fluxes can be driven to zero as the cutoff radius tends to infinity.*

Proposition BFA.11.28 (The source provides an expanding-ball flux estimate)**.** *The comparison proof introduces weighted energy and dissipation quantities E_R, A_R, controls the pressure commutator through Riesz-transform bounds, and obtains a differential inequality of the form*

$$\frac{1}{2}E_R'(t) + \frac{1}{2}A_R(t)^2 \le \|\nabla u(t)\|_\infty E_R(t) + \frac{C_T}{R}.$$

Since $E_R(0) = 0$ and $\|\nabla u\|_\infty$ is integrable on every fixed $[0,T]$, Gronwall gives $E_R(t) = O(R^{-1})$. Letting $R \to \infty$ therefore gives $v = u$ on $[0,T]$.

Proof. The pressure-flux estimate is equation (10.19), followed by the weighted difference-energy inequality at the end of Lemma 10.5. The powers of the weighted L^6 quantity are strictly below the quadratic dissipation power and are absorbed by Young's inequality. The final R^{-1} remainder is explicit in the source. □

Proposition BFA.11.29 (Comparison converts the retained growth into global nonexistence)**.** *Assume a global smooth whole-space solution with the same force and zero datum and with uniformly bounded kinetic energy exists. Then for every $T < 1$, Lemma 10.5 implies it equals the constructed solution on $[0,T]$, hence on all of $[0,1)$. The retained path from Proposition BFA.11.10 then forces its velocity to become unbounded at points converging to $(0,1)$, contradicting smoothness on a compact space–time neighborhood of that point.*

Proof. The global bounded-energy hypothesis restricts to the hypothesis of Lemma 10.5 on every $[0,T]$. Equality on all such intervals yields equality before time one. A globally smooth field is bounded on each compact neighborhood, while the explicit sequence of source points lies in one such neighborhood and carries velocity magnitude to infinity. □

Proposition BFA.11.30 (BFA.11 disposition of O_{13})**.** *No certified comparison or terminal-nonexistence defect is found at the declared resolution. In particular, the source comparison is not restricted to compactly supported competitors, does not import an unprinted pressure decay hypothesis, and is unaffected by spatially constant pressure gauge freedom. The source's auxiliary H^3-continuation/maximal-lifespan paragraph is not used as a substitute for the target-critical comparison and blow-up contradiction.*

BFA.11.7 Stress test of O_{14}: arbitrary positive viscosity

Audit Obligation BFA.11.31 (BFA11-A8: viscosity transport must preserve every target-facing field)**.** *The viscosity-one result does not itself prove Clay-C for an arbitrary fixed $\nu > 0$. The scaling must preserve the PDE, zero datum, smooth compact force, whole-space decay condition, bounded-energy competitor class, and terminal nonexistence conclusion.*

Proposition BFA.11.32 (Viscosity scaling is exact)**.** *For fixed $\nu > 0$, the source defines*

$$u_\nu(x,t) = \sqrt{\nu}\, u(x/\sqrt{\nu}, t), \qquad p_\nu(x,t) = \nu p(x/\sqrt{\nu}, t), \qquad f_\nu(x,t) = \sqrt{\nu}\, f(x/\sqrt{\nu}, t).$$

The chain rule gives the Navier–Stokes equation with viscosity ν and preserves incompressibility and the zero datum.

Proposition BFA.11.33 (The scaling preserves admissibility and exclusion)**.** *For each fixed $\nu > 0$, the scaled force remains smooth and compactly supported in space–time and therefore satisfies the official whole-space decay condition. Moreover*

$$\|u_\nu(t)\|_2^2 = \nu^{5/2}\|u(t)\|_2^2,$$

and a smooth bounded-energy competitor at viscosity ν rescales back to a smooth bounded-energy competitor at viscosity one.

Proof. The source gives the mixed-derivative scaling explicitly after (10.22) and records the energy and dissipation scaling in (10.23). The inverse competitor scaling is also printed. Since ν is fixed and positive, all resulting constants are finite and the support remains compact. □

Policy 62 (BFA11-NU-01: fixed-viscosity quantifier discipline)**.** *The scaling conclusion is interpreted exactly as*

$$\forall \nu > 0\ \exists\, (u_0^{(\nu)}, f^{(\nu)}) \text{ with the stated breakdown property.}$$

Constants, compact supports, and decay constants may depend on the fixed ν. BFA.11 does not strengthen the source to one datum, one support, or one family of constants uniform over all positive viscosities.

Theorem BFA.11.34 (Whole-space source-to-target match at terminal resolution)**.** *Assuming the truth of the exact external analytic statements already consumed in O_9–O_{14}, the terminal source construction supplies a datum admissible for Fefferman alternative (C) and excludes every global physically reasonable whole-space solution for that datum. Hence the statement-level implication*

$$\mathsf{O}_{14} \Longrightarrow \mathsf{Clay\text{-}C}_\nu$$

has no statement-level mismatch at BFA.11 resolution.

Proof. The initial datum is zero and therefore satisfies every spatial decay condition in Fefferman (4). By Propositions BFA.11.17 and BFA.11.33, the force is smooth and satisfies the official rapid-decay requirement. By Proposition BFA.11.29, no global smooth solution with uniformly bounded kinetic energy exists. These are exactly the relevant data and solution predicates in Fefferman alternative (C), as bound in BFA.2. □

Proposition BFA.11.35 (BFA.11 disposition of O_{14})**.** *No certified arbitrary-viscosity transport defect is found at the declared resolution. The earlier BFA.9 promotion of O_{14} to a shared C/D terminal bottleneck is therefore retained: it is structurally critical, not presently rejected.*

BFA.11.8 Stress test of O_{15}: periodic transport and pressure periodicity

Audit Obligation BFA.11.36 (BFA11-A9: nonlinear periodization requires disjoint support)**.** *Periodizing a nonlinear PDE solution by summing translates is not valid merely because the underlying linear terms periodize. The source must eliminate cross-interactions in $(U \cdot \nabla)U$, and it must periodize the pressure as required by the Clay erratum.*

Proposition BFA.11.37 (The support is rescaled strictly inside one fundamental cube)**.** *For fixed $\nu > 0$, Corollary 10.6 chooses $\lambda > 1$ such that the rescaled common spatial support satisfies*

$$\lambda^{-1} K_\nu \Subset Q_0, \qquad Q_0 = (-1/2, 1/2)^3.$$

A parabolic space–time rescaling is then performed so that the singular time remains exactly $t = 1$, and the rescaled fields vanish on an interval after their new initial time, permitting smooth extension by zero.

Proposition BFA.11.38 (Disjoint periodization preserves the nonlinear equation)**.** *The periodic fields are the integer-translate sums*

$$U = \sum_{k\in\mathbb{Z}^3} \widetilde{u}(\cdot + k), \qquad P_{\mathrm{per}} = \sum_{k\in\mathbb{Z}^3} \widetilde{p}(\cdot + k), \qquad F_{\mathrm{per}} = \sum_{k\in\mathbb{Z}^3} \widetilde{f}(\cdot + k).$$

Because distinct translates have disjoint supports separated by a positive gap, at every point at most one translate is nonzero. Consequently

$$(U \cdot \nabla)U = \sum_{k\in\mathbb{Z}^3} (\widetilde{u}(\cdot + k) \cdot \nabla)\widetilde{u}(\cdot + k),$$

so no quadratic cross-term is created by periodization.

Proposition BFA.11.39 (Periodic pressure and official temporal decay are source-explicit)**.** *The pressure used by the periodic construction is P_{per}, the same integer-translate periodization as the velocity and force; hence it is 1-periodic. The periodic force has compact time support, so for every α, m, N it satisfies*

$$\|\partial_x^\alpha \partial_t^m F_{\mathrm{per}}(t)\|_\infty \le C_{\alpha,m,N}(1 + t)^{-N}.$$

Thus the source construction meets the pressure-periodicity correction in the official Clay statement and the force condition (9).

Proposition BFA.11.40 (The periodic target does not inherit a whole-space energy hypothesis)**.** *Fefferman alternative (D) asks for nonexistence of a global smooth periodic solution satisfying the periodicity and smoothness conditions; it does not add the whole-space bounded-energy condition (7). Corollary 10.6 correspondingly uses the intrinsic torus difference-energy identity only as a uniqueness* proof mechanism. *BFA.11 does not serialize bounded kinetic energy as an additional periodic target hypothesis.*

Audit Obligation BFA.11.41 (BFA11-A10: periodic nonexistence must not borrow whole-space infinity control)**.** *The periodic uniqueness argument must close intrinsically on the torus. It may not rely on a whole-space pressure-at-infinity estimate that has no periodic analogue.*

Proposition BFA.11.42 (Periodic comparison closes intrinsically)**.** *For two smooth periodic solutions on $[0, T]$, the source difference-energy estimate is*

$$\frac{1}{2}\frac{d}{dt}\|w\|^2_{L^2(\mathbb{T}^3)} + \nu\|\nabla w\|^2_{L^2(\mathbb{T}^3)} \le \|\nabla U\|_\infty \|w\|^2_{L^2(\mathbb{T}^3)}.$$

Periodic integration eliminates the transport and pressure terms, and Gronwall gives uniqueness on every shorter interval. The retained growth path then excludes a global smooth periodic competitor.

Theorem BFA.11.43 (Periodic source-to-target match at terminal resolution)**.** *Assuming the truth of the exact external analytic statements consumed through O_{15}, the periodic source datum is admissible for the corrected Fefferman alternative (D), including periodic pressure, and no global smooth periodic solution exists for the same datum and force. Hence*

$$\mathsf{O}_{15} \Longrightarrow \mathsf{Clay\text{-}D}_\nu$$

has no statement-level mismatch at BFA.11 resolution.

Proof. Zero initial datum is periodic and smooth. The periodized force and velocity are smooth and periodic; compact time support gives the required temporal decay; Proposition BFA.11.39 gives periodic pressure; and Proposition BFA.11.42 excludes a global smooth periodic solution without importing condition (7). These are exactly the corrected target fields registered in BFA.2. □

Proposition BFA.11.44 (BFA.11 disposition of O_{15})**.** *No certified periodic-transport defect is found at the declared resolution. The source explicitly neutralizes the two main structural hazards: nonlinear cross-terms between translates and omission of periodic pressure.*

BFA.11.9 Statement-level Clay target audit and lifecycle firewall

Audit Obligation BFA.11.45 (BFA11-A11: statement implication is not institutional adjudication)**.** *A source theorem may match the mathematical quantifiers of Clay-C or Clay-D without the Clay Mathematics Institute having completed its independent adjudication or prize process. BFA must keep these predicates separate.*

Proposition BFA.11.46 (The official target predicates consumed here are exact)**.** *For whole space, the official physically reasonable solution class is smooth (p, u) together with a uniform-in-time kinetic-energy bound. For the periodic alternative, the datum and force are spatially periodic, the solution velocity is periodic and smooth, and the appended Clay erratum makes pressure periodicity explicit. No whole-space energy condition is part of the periodic target. BFA.11 adds no stronger solution hypothesis to either target.*

Proof. This is the official Fefferman problem description [15] as already normalized in BFA.2. The pressure-periodicity correction is part of the appended errata in the official source. □

Policy 63 (BFA11-LIFE-01: announcement and prize status are nonmathematical fields)**.** *The Clay Mathematics Institute announcement of 10 September 2026 states that the Navier–Stokes problem has "apparently been settled" while emphasizing that evaluation and credit assignment remain an unhurried process [16]. BFA.11 therefore records*

$$\mathsf{ClayLifecycle} \neq \mathsf{MathematicalTruth} \neq \mathsf{BFAAuditVerdict}.$$

No institutional acceptance or prize award is claimed in this section.

Proposition BFA.11.47 (Statement-level target edges survive the terminal stress test)**.** *At the declared source-interface resolution, no certified statement-level semantic mismatch is found in either directional edge*

$$\rho_{PT}^{C,\nu} : \mathsf{OAI\text{-}R3}(\nu) \to \mathsf{Clay\text{-}C}_\nu, \qquad \rho_{PT}^{D,\nu} : \mathsf{OAI\text{-}T3}(\nu) \to \mathsf{Clay\text{-}D}_\nu.$$

This is a statement-level compatibility result only. It is not an atomic mathematical verifier pass for the analytic proof.

Proof. The whole-space match is Theorem BFA.11.34; the periodic match is Theorem BFA.11.43. BFA.2 already records the exact target predicates and the directional source-to-target implications. □

Audit Obligation BFA.11.48 (BFA11-A12: paper–formal–target triangle remains open)**.** *BFA.11 does not identify the analytic paper statement, the checked formal repository statement, and the official Clay target as equivalent objects. The relations*

$$\rho_{PF}, \qquad \rho_{FT}, \qquad \rho_{PT}$$

remain separately typed. Their triangular audit is owned by BFA.12.

BFA.11.10 Adversarial terminal audit matrix

Node	Primary stress surface	Source mechanism observed	BFA.11 disposition
O_{10}	cutoff geometry, divergence, zero datum, retained core	fixed margin in q; cutoff on vector potential; late-time identity near origin	no certified adverse finding
O_{11}	all terminal jets and smooth compact extension	Lemma 10.2 uniform mixed-jet limits; Lemma 10.3 differentiated Borel-type series with shrinking supports	no certified adverse finding
O_{12}	uniform whole-space energy class	exact energy identity on compactly supported fields; $F(1) < \infty$	no certified adverse finding
O_{13}	arbitrary smooth bounded-energy competitor; pressure infinity	Riesz pressure reconstruction; expanding-ball flux; Gronwall; retained growth path	no certified adverse finding
O_{14}	fixed arbitrary $\nu > 0$	exact spatial scaling, derivative scaling, energy scaling, inverse competitor scaling	no certified adverse finding
O_{15}	nonlinear periodization and periodic pressure	support strictly inside cube; disjoint translate sums; intrinsic torus energy uniqueness	no certified adverse finding
$\rho_{PT}^{C,\nu}$	exact Clay-C quantifiers	zero datum, compact smooth force, no global smooth bounded-energy solution	no statement-level mismatch found
$\rho_{PT}^{D,\nu}$	exact corrected Clay-D quantifiers	periodic force/velocity/pressure, compact time support, no global smooth periodic solution	no statement-level mismatch found

BFA.11.11 Open obligations retained after BFA.11

Definition BFA.11.49 (Non-discharged obligations)**.** The following remain open after the present stress test:

(i) a complete independent analytic reproof of every estimate used inside Proposition 10.1 and Lemmas 10.2–10.5;

(ii) independent rederivation of every Riesz-transform commutator and expanding-ball exponent appearing in Lemma 10.5;

(iii) byte-level cryptographic freezing of the analytic PDF under the BFA.2 source manifest;

(iv) paper–Lean theorem equivalence and exact formal source identity;

(v) Lean–official-target semantic equivalence, including the periodic pressure clause;

(vi) institutional Clay adjudication and prize lifecycle; and

(vii) any future upgrade of O_{10}–O_{15} from $\mathrm{VerifierStatus{=}not_invoked}$ to pass, which requires a separately identified mathematical verifier or independent reproof.

None of these is silently encoded as a mathematical reject of the analytic terminal route, and none is silently promoted to pass by the favorable source-interface disposition.

Policy 64 (BFA11-NONPROM-01: no generic AK promotion)**.** *The terminal audit does not promote localization, smooth jet extension, whole-space comparison, viscosity scaling, or periodization into a generic AK Core theorem. Any future generic extraction requires the BFA.13 promotion pipeline and a separate theorem identity.*

BFA.11.12 BFA.11 verdict and BFA.12 handoff

Theorem BFA.11.50 (BFA.11 terminal stress-test verdict)**.** *At BFA.11's declared source-interface resolution, the adversarial audit finds no certified defect in the registered terminal chain*

$$\mathsf{O}_{10} \to \mathsf{O}_{11} \to \mathsf{O}_{12} \to \mathsf{O}_{13} \to \mathsf{O}_{14}$$

or in the additional periodic route through O_{15}, and finds no certified statement-level mismatch in the directional mappings to $\mathsf{Clay\text{-}C}_\nu$ and $\mathsf{Clay\text{-}D}_\nu$.

Accordingly,

$$\boxed{\mathsf{TerminalAdverseFinding}_{\mathrm{BFA11}} = \mathrm{none_certified_at_declared_resolution}}$$

while, independently,

$$\boxed{\mathsf{MathematicalVerifierStatus}(\mathsf{O}_{10}, \ldots, \mathsf{O}_{15}) = \mathrm{not_invoked}}$$

and

$$\boxed{\mathsf{FullIndependentTerminalReproof}_{\mathrm{BFA11}} = \mathrm{not_claimed}}$$

and

$$\boxed{\mathsf{AnalyticSourceIdentityClosure}_{\mathrm{BFA11}} = \mathrm{pending_BFA2_cryptographic_snapshot.}}$$

Thus "no certified adverse finding" is explicitly weaker than a verifier pass.

Proof. The localization conclusion is Proposition BFA.11.11; the force extension conclusion is Proposition BFA.11.18; energy is Proposition BFA.11.22; comparison is Proposition BFA.11.30; viscosity transport is Proposition BFA.11.35; periodic transport is Proposition BFA.11.44; and the exact statement-level target edges are Proposition BFA.11.47. The independent-reproof ceiling and source-identity qualification are Policies 59 and 61. □

Definition BFA.11.51 (BFA.11 status vector)**.** At completion of BFA.11,

$$\begin{aligned}
\mathsf{localization_contract} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{terminal_jet_contract} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{smooth_force_extension} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{energy_contract} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{comparison_contract} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{viscosity_transport} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{periodic_transport} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{periodic_pressure_analytic_lane} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{ClayC_statement_edge} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{ClayD_statement_edge} &= \mathrm{no_statement_mismatch_found},\\
\mathsf{terminal_adverse_finding} &= \mathrm{none_certified_at_declared_resolution},\\
\mathsf{terminal_mathematical_verifier_status} &= \mathrm{not_invoked},\\
\mathsf{analytic_source_identity_closure} &= \mathrm{pending_BFA2_cryptographic_snapshot},\\
\mathsf{full_independent_terminal_reproof} &= \mathrm{not_claimed},\\
\mathsf{paper_formal_target_triangle} &= \mathrm{delegated_to_BFA.12},\\
\mathsf{Clay_institutional_adjudication} &= \mathrm{not_a_BFA11_mathematical_field}.
\end{aligned}$$

Theorem BFA.11.52 (BFA.11 closure)**.** *BFA.11 closes the construction-stage terminal Return audit at its declared source-interface resolution. It establishes neither an AK-owned proof of the external theorem nor paper–formal equivalence nor institutional recognition. Its positive content is narrower and audit-oriented: at the present source binding, no statement-level type or quantifier mismatch has been found among the terminal interfaces printed by the analytic source, the current BFA route, and the official whole-space and periodic target predicates; and no certified terminal blocker has been produced by the present stress test. This source conformance statement leaves the mathematical verifier status of the imported analytic nodes at* not_invoked.

Proof. This is exactly the conjunction of Theorem BFA.11.50, Policies 63 and 64, and **??** BFA.11.48. □

Remark BFA.11.53 (Next consumer: BFA.12)**.** BFA.12 shall perform the *Paper–Lean–Problem Triangular Audit*. It must now treat the analytic paper result as only one vertex and independently compare:

(i) the exact paper theorem and corollary statements;

(ii) the pinned Lean reference target declarations;

(iii) the exact proved Lean declarations and assumptions;

(iv) the formalization metadata and pinned commit identity;

(v) the official Clay-C and corrected Clay-D target predicates;

(vi) the directions and strengths of ρ_{PF}, ρ_{FT}, and ρ_{PT}; and

(vii) whether any lexical or structural similarity has been mistaken for theorem equivalence.

BFA.11's favorable terminal verdict supplies no automatic pass on any of these formal or semantic-equivalence obligations.

BFA.12 Paper–Lean–Problem Triangular Audit

BFA.12.1 Purpose, resolution, and nonpromotion ceiling

Declaration BFA.12.1 (BFA.12 objective)**.** BFA.12 audits the three external theorem-facing vertices that were deliberately kept distinct in BFA.2:

$$\mathsf{Paper}, \qquad \mathsf{Formal}, \qquad \mathsf{OfficialTarget}.$$

The purpose is to determine, separately for the whole-space and periodic lanes, whether the currently bound analytic source statement, the pinned Lean Comparator statement, and the corrected Clay target have compatible quantifiers, datum classes, solution classes, and conclusion strength.

The governing question is not

"Do the three artifacts look equivalent?"

but rather

Which directional statement-strength and semantic-interpretation maps are actually licensed?

Policy 65 (BFA-TRI-01: statement audit is not proof replay)**.** *BFA.12 is a source-level and mathematical statement-semantic audit. It does not independently rebuild the pinned Lean repository, execute Comparator, replay every formal proof object, or independently reprove the analytic paper. Therefore the following fields remain distinct:*

$$\mathsf{statement_shape_audit}, \qquad \mathsf{semantic_alignment_audit},$$
$$\mathsf{independent_Lean_replay}, \qquad \mathsf{analytic_proof_verification}.$$

A favorable value of either of the first two fields does not promote either of the final two fields to pass.

Policy 66 (BFA-TRI-02: no cross-language equality by naming)**.** *Equality of theorem names, metadata alignment, common publication date, common target label, zero reported* sorry *count, or a source-side Comparator configuration does not establish*

$$\mathsf{PaperTheorem} = \mathsf{LeanTheorem}$$

or a semantic equivalence between their complete conclusions. A relation between artifacts written in different formal languages is recorded only at the exact directional strength established below.

BFA.12.2 Eight objects in the triangular audit

Definition BFA.12.2 (Analytic paper statements)**.** Let

$$P_C := \mathsf{OAI\text{-}R3}, \qquad P_D := \mathsf{OAI\text{-}T3}$$

denote the analytic statements already bound in BFA.2: respectively Theorem 1.1 and Corollary 10.6 of *Finite Time Blowup for Navier–Stokes* [12].

The whole-space object P_C contains strictly more printed witness data than Clay-C itself: zero initial datum, a compactly supported smooth force, an explicit smooth solution on $[0, 1)$, fixed compact spatial support, uniformly bounded kinetic energy, and an L^∞ blow-up as $t \uparrow 1$, followed by the nonexistence of a global smooth bounded-energy competitor.

The periodic object P_D likewise contains an explicit preterminal periodic solution, L^∞ blow-up, a smooth force compactly supported in time, and the nonexistence of a global smooth periodic competitor.

Definition BFA.12.3 (Pinned formal reference statements)**.** At the pinned commit

$$\mathsf{COMMIT}_{\mathrm{OAI}} = 8937\mathrm{a}8\mathrm{f}4\mathrm{cbc}7\mathrm{abaab}5\mathrm{e}9\mathrm{e}97\mathrm{d}1\mathrm{cc}7\mathrm{f}5\mathrm{d}2319\mathrm{d}9538,$$

let

$$R_C, \qquad R_D$$

denote the two challenge declarations in ComparatorChallenges/NavierStokes.lean [21]. Their bodies are intentionally sorry; their theorem types are the reference signatures against which the submission is compared.

Definition BFA.12.4 (Pinned proved formal statements)**.** At the same commit let

$$F_C, \qquad F_D$$

denote the proved declarations

$$\mathrm{NavierStokes.Comparator.navier_stokes_breakdown_R3},$$
$$\mathrm{NavierStokes.Comparator.navier_stokes_breakdown_periodic}$$

in NavierStokes/ComparatorSolution.lean [22]. The exposed theorem types are

$$F_C(\nu): \quad \exists u^0, f,\ \mathsf{InitialVelocityConditionDecay}(u^0) \wedge \mathsf{ForceConditionDecay}(f) \wedge \neg\exists v, p\ \mathsf{NavierStokesExistenceAndSmoothnessRn}(\nu, u^0, f, v, p),$$

$$F_D(\nu): \quad \exists u^0, f,\ \mathsf{InitialVelocityConditionPeriodic}(u^0) \wedge \mathsf{ForceConditionPeriodic}(f) \wedge \neg\exists v, p\ \mathsf{NavierStokesExistenceAndSmoothnessPeriodic}(\nu, u^0, f, v, p),$$

for an input $\nu > 0$.

Definition BFA.12.5 (Corrected official targets)**.** Let

$$T_C := \mathsf{Clay\text{-}C}, \qquad T_D := \mathsf{Clay\text{-}D}$$

be the corrected official target semantics of BFA.2. Thus T_C is Fefferman alternative (C), including the global smoothness and bounded-energy solution class, and T_D is alternative (D) with the appended Clay erratum requiring periodicity of the pressure as well as the velocity [15].

Remark BFA.12.6 (Why eight objects rather than three)**.** The analytic paper, the formal reference challenge, the formal proved submission, and the official target answer different audit questions. Conflating R_C with F_C, for example, would confuse a challenge theorem type whose body is intentionally unresolved with the proved submission that has the same theorem type. Conflating P_C with F_C would erase the stronger explicit blow-up witness printed by the paper.

BFA.12.3 Reference–submission signature audit

Proposition BFA.12.7 (Source-text Comparator theorem-signature match)**.** *At the pinned commit, the theorem signatures printed for R_C and F_C have the same namespace, theorem name, parameters, and displayed proposition text, up to the fact that the former has an intentional challenge* sorry *body and the latter has a proof term. The same statement holds for R_D and F_D.*

Because the solution import closure uses an independent copied definition module rather than importing the challenge module, this proposition is a source-text statement. Equality of the elaborated cross-environment objects is exactly the kind of claim assigned to Comparator execution and is not inferred here merely from matching prose.

Proof. The reference file declares the two challenge theorems

$$\text{navier_stokes_breakdown_R3}, \qquad \text{navier_stokes_breakdown_periodic}$$

with the existential datum/force predicates and the negated global-solution predicates displayed in Definition BFA.12.4. The solution file exposes declarations with the same namespace, names, parameters (ν, h_ν), and theorem types, and proves them through the underlying bridge modules [21, 22]. □

Proposition BFA.12.8 (Comparator configuration targets exactly the two declarations)**.** *The pinned Comparator configuration names* ComparatorChallenges.NavierStokes *as challenge module,* NavierStokes.ComparatorSolution *as solution module, and lists exactly F_C, F_D as theorem names. Its listed permitted axioms are*

$$\text{propext}, \qquad \text{Quot.sound}, \qquad \text{Classical.choice}.$$

Proof. This is the literal content of the pinned ComparatorChallenges/NavierStokes.json [23]. □

Policy 67 (BFA-TRI-03: signature match is not independent Comparator pass)**.** *Propositions BFA.12.7 and BFA.12.8 license the source-level status*

$$\mathsf{reference_submission_signature_match} = \text{source_text_theorem_signature_match}.$$

They do not license

$$\mathsf{independent_Comparator_replay} = \text{pass},$$

because BFA has not independently reconstructed and executed the pinned formal environment.

BFA.12.4 The paper and the exposed formal theorem are not equivalent statements

Theorem BFA.12.9 (Whole-space paper statement has a strictly richer printed witness schema)**.** *At statement resolution, the printed theorem P_C exposes strictly more witness fields than the proposition exported as F_C. In particular, P_C prints an explicit smooth preterminal solution with compact spatial support, uniformly bounded kinetic energy, and unbounded L^∞ norm as $t \uparrow 1$, whereas F_C exports only the Clay-C-style existential datum/force proposition together with nonexistence of a global smooth bounded-energy solution.*

BFA therefore licenses the forgetful target-consequence projection

$$P_C \rightsquigarrow \pi_C(P_C) \approx_{\text{sem}} F_C,$$

but it does not *claim a strict logical implication order between the two cross-language theorem objects, and it does not license a converse that recovers the complete printed witness content of P_C from the exposed Comparator proposition alone.*

Proof. The additional witness fields are printed directly in Theorem 1.1, and its final consequence has the target-level nonexistence shape consumed by Clay-C. The exported proposition F_C does not itself contain fields asserting a smooth solution on $[0, 1)$, finite-time L^∞ blow-up, compact spatial support of the constructed velocity and pressure, or the particular witness mechanism. This proves strictness of the *printed field schema*. It does not by itself prove a model-theoretic nonimplication $F_C \not\Rightarrow P_C$; accordingly BFA records only the one-way forgetting operation and the absence of an established converse. □

Theorem BFA.12.10 (Periodic paper statement has a strictly richer printed witness schema)**.** *Likewise, P_D exposes a richer printed witness schema than F_D: it includes a smooth periodic preterminal solution, finite-time L^∞ blow-up, and the support-localized periodization construction before concluding global nonexistence. BFA records*

$$P_D \rightsquigarrow \pi_D(P_D) \approx_{\text{sem}} F_D,$$

with no claim that the exported Comparator proposition logically reconstructs the full paper witness.

Proof. This is the periodic analogue of Theorem BFA.12.9. Corollary 10.6 prints the preterminal blow-up witness and then the global periodic nonexistence conclusion. The exposed theorem type F_D records the target-level nonexistence proposition but not the entire witness schema. □

Remark BFA.12.11 (Meaning of repository metadata "formalizes")**.** The pinned formalization.yaml declares that the project formalizes the main Navier–Stokes results, reports zero sorry count for the named main results, and aligns paper Theorem 1.1 and Corollary 10.6 to the two Comparator declarations [20]. BFA records that metadata as source provenance. It does not reinterpret the exposed Comparator theorem type as textually or logically identical to every stronger witness field printed in the paper. The internal Lean development may formalize substantially more structure; BFA.12's theorem-facing formal objects are the pinned exposed Comparator declarations fixed in BFA.2.

BFA.12.5 Cross-language semantic bridge ledger

Definition BFA.12.12 (Semantic bridge obligations)**.** The comparison between the Fefferman statement and the Lean comparator is not purely lexical. BFA.12 therefore isolates the following cross-language interpretation obligations:

(i) $\mathsf{SB}_{\mathrm{der}}$: all-order Fréchet-derivative norm bounds versus coordinate multi-index partial-derivative bounds in finite dimension;

(ii) SB_K: the Lean quantification over real decay exponents K versus the usual "faster than every polynomial" formulation;

(iii) SB_∂: ContDiffOn and derivWithin on the closed half-space versus the intended smooth-up-to-the-boundary and one-sided time-derivative semantics at $t = 0$;

(iv) SB_{L^2}: the explicit MemLp guard plus the totalized Lean integral versus genuine finite kinetic energy; and

(v) $\mathsf{SB}_{\mathrm{per}}$: invariance under each positive unit coordinate translation versus $\mathbb{Z}^3$-periodicity.

These are interpretation obligations, not extra hypotheses on the PDE.

Proposition BFA.12.13 (Standard finite-dimensional derivative and decay bridges)**.** *The obligations* $\mathsf{SB}_{\mathrm{der}}$, SB_K, *and* $\mathsf{SB}_{\mathrm{per}}$ *are standard finite-dimensional identifications at the mathematical level:*

(i) for a smooth map between finite-dimensional Euclidean spaces, the operator norm of the mth Fréchet derivative is equivalent, up to dimension- and order-dependent constants, to the maximum of its coordinate mth partial derivatives;

(ii) decay for every sufficiently large integer exponent implies decay for every real exponent because $1 + \|x\| + t \geq 1$*, while the converse follows by choosing integer exponents among the real ones; and*

(iii) invariance under every $x \mapsto x + e_i$ *implies invariance under negative and arbitrary integer coordinate translates by substitution and iteration.*

Thus no target-critical mismatch is generated by these three encoding choices. BFA.12 does not claim that these equivalences have themselves been replayed as Lean theorems inside the pinned repository.

Policy 68 (BFA-TRI-03A: boundary-smoothness interpretation remains explicit)**.** *The fields* ContDiffOn *and* derivWithin *are intentionally within-set notions on* $\mathbb{R}^3 \times [0, \infty)$*. BFA reads them as the formal implementation of the classical closed-half-space convention used by the Clay statement: smoothness up to the boundary and the right time derivative at* $t = 0$*. This reading is mathematically natural and no adverse example is identified here, but BFA.12 does not elevate it to a separately formalized cross-library equivalence theorem. Therefore* SB_∂ *is recorded as* standard_interpretation_not_independently_formalized*, not as an independent verifier pass.*

Proposition BFA.12.14 (Energy guard is target-preserving)**.** *Under the formal smoothness field, the additional* MemLp *clause in the whole-space comparator prevents totalized-integration junk values and enforces the genuine* L^2 *meaning already intended by Fefferman's bounded-energy condition. It does not introduce a distinct physical regularity requirement. Hence* SB_{L^2} *is target-preserving at the declared semantic resolution.*

BFA.12.6 Formal whole-space target versus Clay-C

Definition BFA.12.15 (Formal whole-space predicate decomposition)**.** The pinned reference decomposes F_C into the following theorem-facing requirements:

(i) u^0 is smooth and divergence-free and has all-order rapid spatial decay;

(ii) f is smooth on the closed future half-space and has all-order rapid decay in space and time;

(iii) a candidate global solution satisfies the Navier–Stokes equation, incompressibility, initial condition, and smoothness of both velocity and pressure;

(iv) velocity is L^2 at each time and the kinetic energy is uniformly bounded.

The first two clauses formalize Fefferman (4)–(5); the latter two formalize (1)–(3), (6)–(7) [21, 15].

Proposition BFA.12.16 (Junk-value guards do not weaken the intended whole-space solution class)**.** *The reference file's separate smoothness and integrability fields are semantically load-bearing guards rather than target inflation. Specifically:*

(i) smoothness prevents the fderiv*/derivative operators from being consumed at their nondifferentiable junk values; and*

(ii) the explicit L^2 membership field prevents Lean's totalized integral from representing a nonintegrable energy profile by a junk value.

At ordinary mathematical semantics these guards implement, rather than strengthen away from, Fefferman's smoothness and bounded-energy requirements.

Proof. The reference source explicitly documents both conventions. Its base solution structure contains smoothness of velocity and pressure in addition to the PDE fields, while the whole-space extension contains both pointwise-in time L^2 membership and a uniform energy integral bound. Fefferman's condition (6) supplies smoothness and condition (7) requires a genuinely finite uniformly bounded kinetic-energy integral. The additional formal guards ensure that the totalized Lean operators represent those intended classical predicates. □

Proposition BFA.12.17 (Whole-space formal-to-official semantic alignment)**.** *At the declared statement-semantic resolution, no target-critical quantifier, datum-class, solution-class, or energy-condition omission is identified between F_C and T_C. The comparison nevertheless uses the explicit semantic bridges of Definition BFA.12.12. BFA therefore records a directional interpretation*

$$\rho^{C}_{FT} : F_C \rightsquigarrow T_C$$

with status

$$\mathsf{formal_to_ClayC_alignment} = \mathrm{supported_modulo_declared_semantic_bridges}.$$

This is neither an independent Lean proof of the interpretation map nor a claim that cross-language predicate equivalence has been machine-checked.

Proof. Both statements have the fixed-viscosity pattern $\forall \nu > 0\, \exists (u^0, f)$ followed by nonexistence of a global physically reasonable solution. The formal datum predicates implement smoothness, divergence-freeness, and rapid decay. The formal solution predicate contains the Navier–Stokes equation, divergence-free constraint, initial condition, smooth pressure and velocity, and the bounded-energy condition. These are precisely the fields used by Fefferman alternative (C), modulo the explicit formal guards discussed in Proposition BFA.12.16. No extra conclusion is needed to reach T_C. □

BFA.12.7 Formal periodic target versus corrected Clay-D

Definition BFA.12.18 (Formal periodic predicate decomposition)**.** For F_D, the pinned reference requires:

(i) smooth divergence-free 1-periodic initial velocity;

(ii) a smooth force, spatially 1-periodic for all $t \geq 0$, whose all-order future derivatives decay faster than any power of time;

(iii) the Navier–Stokes equation, incompressibility, initial condition, and smooth velocity and pressure; and

(iv) spatial 1-periodicity of both velocity and pressure for all $t \geq 0$.

No whole-space bounded-energy condition is inserted into the periodic solution predicate.

Proposition BFA.12.19 (Clay pressure erratum is represented in the pinned formal target)**.** *The periodic formal solution structure contains the explicit field*

$$\mathrm{isOnePeriodic_pressure},$$

requiring 1*-periodicity of the pressure in every spatial coordinate for all* $t \geq 0$*. This matches the erratum appended to the official Clay problem statement, which says that pressure periodicity should be made explicit.*

Proof. The field is printed in the pinned reference source. The final errata page of the official Clay PDF states that the further condition $p(x + e_j, t) = p(x, t)$ should be made explicit in equation (8). The analytic paper's periodic section likewise periodizes the pressure and explicitly cites that erratum. Thus pressure periodicity is not an OpenAI-added strengthening of the corrected target. □

Proposition BFA.12.20 (Periodic formal-to-official semantic alignment)**.** *At the declared statement-semantic resolution, no target-critical quantifier, datum-class, solution-class, periodicity, or pressure-erratum omission is identified between* F_D *and the corrected target* T_D*. The comparison uses the declared semantic bridges, so BFA records*

$$\rho^{D}_{FT} : F_D \rightsquigarrow T_D$$

with

$$\mathsf{formal_to_ClayD_alignment} = \mathrm{supported_modulo_declared_semantic_bridges}.$$

Again this is a source-semantic audit result, not an independently replayed formal proof or a machine-checked cross-language equivalence theorem.

Proof. The fixed-ν existential quantifiers agree. The formal periodic datum and force predicates implement Fefferman (8)–(9). The solution structure implements (1)–(3), (10)–(11) and explicitly adds the pressure periodicity required by the Clay erratum. It does not add the whole-space energy clause. Thus no target inflation or missing corrected periodic-pressure condition is identified. □

BFA.12.8 Paper-to-formal relation: projection, not equivalence

Definition BFA.12.21 (Target-consequence projection)**.** For each lane $X \in \{C, D\}$, define

$$\pi_X(P_X)$$

to be the weakest target-level existential nonexistence proposition obtained from the stronger explicit witness statement P_X by forgetting witness fields not required by the corresponding Clay alternative.

Concretely, π_C forgets the displayed finite-time blow-up profile, preterminal support data, and the particular singular witness after retaining the datum/force and global bounded-energy nonexistence conclusion; π_D forgets the corresponding preterminal periodic blow-up witness after retaining the global periodic nonexistence conclusion.

Proposition BFA.12.22 (Paper-to-formal target alignment)**.** *At statement resolution,*

$$\pi_C(P_C) \quad \textit{and} \quad F_C$$

co-target the same alternative (C) after the declared source-language semantic bridges are applied to the admissibility and solution predicates. Likewise,

$$\pi_D(P_D) \quad \textit{and} \quad F_D$$

co-target the corrected alternative (D) under the same declared bridge discipline.

Accordingly BFA registers

$$\rho^C_{PF} : P_C \rightsquigarrow F_C, \qquad \rho^D_{PF} : P_D \rightsquigarrow F_D$$

as one-way target-consequence projections*, not as theorem isomorphisms.*

Proof. For C, the paper explicitly concludes nonexistence of a global smooth solution with the same force and zero initial datum whose kinetic energy is uniformly bounded. The formal bridge theorem in the pinned repository uses the compact candidate and chooses zero initial data before exposing the exact Comparator-C proposition [27]. For D, the pinned bridge uses the constructed periodic candidate, zero initial data, the rescaled force, and pressure periodicity to expose the exact Comparator-D proposition [28]. The additional preterminal blow-up fields printed in the paper are not present in the exposed Comparator theorem types, so the relation is directional. □

Corollary BFA.12.23 (No paper–formal equivalence claim)**.** *No converse translation recovering the complete printed paper witness schema from the exposed Comparator declarations has been established. In particular, BFA does not register either*

$$F_C \rightsquigarrow P_C \qquad \textit{or} \qquad F_D \rightsquigarrow P_D$$

as a proved cross-language reconstruction map. Hence

$$\boxed{\mathsf{PaperFormalEquivalence} = \mathrm{not_claimed}}.$$

BFA.12.9 Target-consequence commutativity of the triangle

Definition BFA.12.24 (Triangular paths)**.** For $X \in \{C, D\}$, BFA compares two statement-level paths from the analytic paper to the official target:

$$\mathsf{direct}_X : P_X \xrightarrow{\rho^X_{PT}} T_X,$$
$$\mathsf{formal}_X : P_X \xrightarrow{\rho^X_{PF}} F_X \xrightarrow{\rho^X_{FT}} T_X.$$

The symbol $\rightsquigarrow$ denotes a BFA semantic-interpretation or statement-strength map across source languages; it is not a morphism in a single formal category unless separately realized.

Theorem BFA.12.25 (Target-consequence co-targeting of the two audit routes)**.** *At the current source-bound statement resolution, the direct paper route and the paper–formal route are assigned the same terminal target labels after the explicit witness-forgetting projection and declared semantic bridges:*

$$\boxed{\mathsf{Target}_{\text{audit}}(\rho^C_{PT}) = \mathsf{Target}_{\text{audit}}(\rho^C_{FT} \circ \rho^C_{PF}) = T_C,}$$

$$\boxed{\mathsf{Target}_{\text{audit}}(\rho^D_{PT}) = \mathsf{Target}_{\text{audit}}(\rho^D_{FT} \circ \rho^D_{PF}) = T_D.}$$

This is a statement about the BFA target registry. It is deliberately weaker than categorical commutativity, equality of proof objects, or a machine-checked cross-language equivalence.

Proof. BFA.2 and BFA.11 bind the analytic whole-space and periodic consequences to T_C and corrected T_D. Proposition BFA.12.22 records the forgetful target-consequence translations into the exposed Comparator propositions, while Propositions BFA.12.17 and BFA.12.20 records their source-semantic interpretation into the same target records, modulo the declared bridge ledger. Hence the two audit routes have the same terminal target labels. No equality of the intermediate theorem objects is used. □

Remark BFA.12.26 (Why BFA no longer calls this a commuting triangle)**.** Even after π_C, π_D, BFA has not constructed all three arrows as morphisms in one formal category and has not independently machine-checked the semantic bridge maps. The correct conclusion is therefore *co-targeting at the declared audit resolution*, not literal diagram commutativity. This terminology prevents "same Clay alternative" from being misread as "same theorem" or "same proof".

BFA.12.10 Formal bridge implementation and witness specificity

Proposition BFA.12.27 (Whole-space formal bridge uses zero initial data and compact-candidate decay)**.** *The pinned formal bridge proving F_C does not obtain Clay-C merely from an abstract existential principle. Its printed adapter takes a selected compact candidate, chooses zero initial velocity, proves the Comparator force-decay condition from compact spatial/time support and smoothness, and excludes a global Comparator solution through the finite-energy comparison theorem [27].*

Proposition BFA.12.28 (Periodic formal bridge discharges periodic force and pressure requirements)**.** *The pinned periodic bridge chooses zero initial data, rescales the candidate force, proves smoothness, spatial periodicity, and future-time decay, and excludes a global solution only after transporting both velocity and pressure periodicity into the normalized candidate comparison [28].*

Remark BFA.12.29 (Witness specificity is proof-internal, not exposed theorem strength)**.** Propositions BFA.12.27 and BFA.12.28 explain why the formal proof is consistent with the zero-datum, compactly supported analytic construction even though the exported Comparator theorem types quantify over an arbitrary existential u^0, f. Existential generality in the theorem type does not mean that the proof uses a nonzero or noncompact witness.

BFA.12.11 Source identity, metadata, and replay separation

Proposition BFA.12.30 (Pinned remote formal identity remains resolved)**.** *The raw sources at commit* 8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538 *resolve the reference statement, solution statement, Comparator configuration, and formalization metadata used in this audit. The repository metadata reports* sorry_count: 0 *for the two named Navier–Stokes main results and labels review status* self-assessed *[20].*

Policy 69 (BFA-TRI-04: remote resolution is not local snapshot closure)**.** *The current audit obtained fresh remote source resolution for the pinned formal files, but did not create and cryptographically close an independent BFA-local checkout of the entire repository or independently execute its Lean build. Therefore*

$$\begin{aligned} \mathsf{formal_remote_identity} &= \mathrm{resolved_at_pinned_commit}, \\ \mathsf{formal_local_snapshot_closure} &= \mathrm{not_invoked}, \\ \mathsf{independent_Lean_build} &= \mathrm{not_invoked}, \\ \mathsf{independent_Comparator_replay} &= \mathrm{not_invoked}. \end{aligned}$$

Likewise, the analytic PDF is source-resolved and freshly re-read, but the BFA-local cryptographic byte snapshot requested in BFA.10–BFA.11 remains a separate downstream provenance task.

Policy 70 (BFA-TRI-05: mutable main branch noninterference)**.** *Any later change on the repository's mutable default branch has no automatic effect on this triangular audit. A new revision enters BFA only through a new source-binding record or a proved identity relation to the pinned commit.*

BFA.12.12 Independent third-party formal-audit evidence

Definition BFA.12.31 (External independent formal-audit record)**.** As of 16 September 2026, BFA records an independent public audit repository that targets the same pinned OpenAI commit 8937a8f4cbc7abaab5e9e97d1cc7f5d2319d9538. The repository reports a successful reconstruction of the OpenAI proof chain, official Comparator execution, Lean-kernel and nanoda acceptance, a global repository build, and an independently written formal bridge from the Comparator alternatives to a separate Lean transcription of the relevant Clay alternatives [26]. BFA classifies this material as external_third_party_formal_audit_evidence. It is not a BFA-local replay record and is not imported as a BFA verifier token.

Policy 71 (BFA-TRI-05A: external replay evidence does not rewrite BFA-local status)**.** *Third-party reports of a successful build, Comparator run, kernel check, nanoda check, or semantic bridge may strengthen the external evidence landscape but do not change*

$$\mathsf{independent_Lean_build}_{\mathrm{BFA}} = \mathrm{not_invoked},$$
$$\mathsf{independent_Comparator_replay}_{\mathrm{BFA}} = \mathrm{not_invoked}.$$

A BFA-local replay status can change only after BFA itself binds the external audit artifacts or independently reproduces the corresponding execution under a separately recorded evidence identity. Likewise, the final human interpretation boundary from the Fefferman/Clay prose to any independent Lean specification remains a semantic-review boundary rather than a consequence of kernel acceptance alone.

Proposition BFA.12.32 (Effect of the external formal audit)**.** *The third-party audit in Definition BFA.12.31 is positive corroborating evidence for the formal-artifact and Comparator-to-target lanes, but it neither proves the analytic paper inside BFA nor changes any* not_invoked *BFA-local verifier field. Hence its safe BFA status is*

$$\mathsf{third_party_formal_audit} = \mathrm{observed_external_not_reproduced}.$$

BFA.12.13 Adversarial mismatch catalogue

Definition BFA.12.33 (Triangular mismatch classes)**.** BFA.12 distinguishes the following failure classes:

(i) PF-strength-inflation: treating a target-level formal theorem as equivalent to the stronger analytic blow-up witness theorem;

(ii) RF-signature-mismatch: different challenge and submission theorem types;

(iii) FT-quantifier-mismatch: a change in the order or scope of $\forall \nu > 0$ and existential witness quantifiers;

(iv) FT-datum-mismatch: formal initial-data or force predicates fail to implement the official admissible class in the required direction;

(v) FT-solution-mismatch: the formal global solution class omits or adds a target-critical condition;

(vi) periodic-pressure-omission: pressure periodicity required by the erratum is absent;

(vii) semantic-bridge-gap: a cross-language encoding is treated as definitionally identical without discharging the corresponding bridge obligation;

(viii) formal-replay-leakage: source metadata is retyped as an independent formal verification pass;

(ix) lifecycle-leakage: institutional or prize-adjudication status is used as a mathematical premise.

Theorem BFA.12.34 (BFA.12 adversarial mismatch disposition)**.** *At the current declared resolution:*

(i) PF-strength-inflation *is explicitly blocked rather than accepted;*

(ii) no RF-signature-mismatch *is identified in the pinned printed sources;*

(iii) *no target-critical* FT-quantifier-mismatch, FT-datum-mismatch, *or* FT-solution-mismatch *is identified;*

(iv) periodic-pressure-omission *is not present in the pinned reference or the analytic periodic source;*

(v) semantic-bridge-gap *is prevented by the explicit ledger Definition BFA.12.12, Propositions BFA.12.13 and BFA.12.14, and Policy 68; no unrecorded definitional equality is used;*

(vi) formal-replay-leakage *is blocked by Policies 67, 69 and 71; and*

(vii) lifecycle-leakage *remains prohibited by the BFA.2 lifecycle separation.*

Proof. The first item is Theorems BFA.12.9 and BFA.12.10. The second is Proposition BFA.12.7. The third is Propositions BFA.12.17 and BFA.12.20. The fourth is Proposition BFA.12.19. The fifth follows from the explicit semantic-bridge ledger. The final two follow from the cited policies and the BFA.2 authority firewall. □

BFA.12.14 Triangular audit matrix

Relation / object	Source-level observation	BFA.12 interpretation	Status
$P_C \to T_C$	Paper Theorem 1.1 explicitly states the Clay-C consequence	Direct statement-level Return	no_mismatch_found
$P_D \to T_D$	Corollary 10.6 states no global smooth periodic solution; pressure is periodized	Direct corrected Clay-D Return	no_mismatch_found
$R_C \leftrightarrow F_C$	Same printed theorem name and type at pinned commit	Reference/submission signature match only	source-text theorem-signature match
$R_D \leftrightarrow F_D$	Same printed theorem name and type at pinned commit	Reference/submission signature match only	source-text theorem-signature match
$P_C \rightsquigarrow F_C$	Paper has stronger explicit blow-up witness; formal export is target-level nonexistence	Forgetful target-consequence translation	one-way supported; converse not established
$P_D \rightsquigarrow F_D$	Same strength asymmetry in periodic lane	Forgetful target-consequence translation	one-way supported; converse not established
$F_C \rightsquigarrow T_C$	Quantifiers, decay, smoothness, PDE, and bounded-energy solution class align	Semantic target interpretation modulo bridge ledger	supported; no target-critical omission identified
$F_D \rightsquigarrow T_D$	Periodicity and Clay pressure erratum are represented; no energy inflation	Semantic target interpretation modulo bridge ledger	supported; no target-critical omission identified
Paper ≅ Formal	Stronger paper witness is not recoverable from exported Comparator proposition	Equivalence prohibited	not claimed
Third-party formal audit	Independent public audit reports pinned build, Comparator, kernel/nanoda checks, and an independent Comparator-to-Clay bridge	Corroborating external evidence only; not a BFA replay token	observed / not reproduced by BFA
Independent Lean replay	Source says zero sorry count and supplies Comparator config	Requires local pinned environment and execution	not invoked

BFA.12.15 BFA.12 verdict and exact claim ceiling

Theorem BFA.12.35 (BFA.12 triangular audit verdict)**.** *Relative to the exact sources bound in BFA.2 and freshly re-read in this section, BFA.12 obtains no certified target-critical statement mismatch in the paper–formal–official-target comparison* after making every cross-language semantic bridge explicit*. More precisely:*

(i) the pinned reference and submission theorem signatures agree in the inspected source text;

(ii) the exposed formal whole-space and periodic propositions carry the same fixed-viscosity existential breakdown targets as corrected Clay-C and Clay-D at the declared semantic resolution;

(iii) the periodic pressure erratum is represented in the formal solution predicate and the analytic periodic construction;

(iv) the analytic paper prints additional witness data not present in the exposed Comparator conclusions, so paper–formal equivalence is not asserted;

(v) the direct paper route and the paper–formal route co-target the same corrected official alternative only after witness-forgetting projection and the declared semantic bridge ledger; and

(vi) an independent third-party audit reports successful pinned formal replay and an independently formalized Comparator-to-Clay bridge, but BFA has not reproduced or adopted that execution as a BFA-local verifier token; and

(vii) no independent Lean build, Comparator replay, or full cross-language proof-object equivalence has been performed by BFA itself.

Therefore the strongest safe BFA.12 result is

$$\boxed{\mathsf{TriangleAdverseFinding}_{\mathrm{BFA12}} = \mathrm{none_certified_at_declared_source_semantic_resolution.}}$$

Policy 72 (BFA-TRI-06: exact nonclaims)**.** *The BFA.12 verdict does not assert any of the following:*

(i) that the paper proof is correct merely because a Lean repository exists;

(ii) that the Lean development has been independently rebuilt or replayed by BFA;

(iii) that $P_C \cong F_C$ or $P_D \cong F_D$ as theorem objects;

(iv) that printed witness-schema richness proves a strict logical nonimplication in the reverse direction;

(v) that the paper–formal–target comparison is a commuting diagram in one formal category;

(vi) that repository metadata is an independent proof certificate;

(vii) that a reported third-party formal-audit pass is identical to a BFA-local replay or discharges BFA's analytic-verification obligations;

(viii) that Clay has completed prize adjudication;

(ix) that an institutional announcement determines mathematical truth; or

(x) that the BFA statement audit replaces specialist line-by-line review of the analytic or formal proof.

Definition BFA.12.36 (BFA.12 status vector)**.** At completion of BFA.12,

$$\begin{aligned}
\text{reference_submission_signature_match} &= \text{source_text_theorem_signature_match},\\
\text{paper_to_formal_C} &= \text{supported_one_way_target_projection},\\
\text{paper_to_formal_D} &= \text{supported_one_way_target_projection},\\
\text{formal_to_ClayC} &= \text{supported_modulo_declared_semantic_bridges},\\
\text{formal_to_ClayD} &= \text{supported_modulo_declared_semantic_bridges},\\
\text{periodic_pressure_triangle} &= \text{source_semantics_aligned},\\
\text{paper_formal_equivalence} &= \text{not_claimed},\\
\text{target_consequence_triangle} &= \text{co_targets_at_declared_audit_resolution},\\
\text{formal_remote_identity} &= \text{resolved_at_pinned_commit},\\
\text{formal_local_snapshot_closure} &= \text{not_invoked},\\
\text{independent_Lean_build} &= \text{not_invoked},\\
\text{independent_Comparator_replay} &= \text{not_invoked},\\
\text{third_party_formal_audit} &= \text{observed_external_not_reproduced},\\
\text{analytic_local_cryptographic_snapshot} &= \text{pending_provenance_task},\\
\text{semantic_bridge_ledger} &= \text{explicit_and_nonpromotional},\\
\text{triangle_adverse_finding} &= \text{none_certified_at_declared_source_semantic_resolution}.
\end{aligned}$$

Theorem BFA.12.37 (BFA.12 closure)**.** *BFA.12 closes the* source-semantic audit record *for the paper–formal–official-target comparison at its declared resolution. It does not close an independent formal-verification task. The principal ambiguity left open in BFA.2 is resolved as follows: the paper and the exposed Lean Comparator theorem are not identified as equivalent theorem objects; the paper prints a richer witness schema and admits a one-way forgetting translation to the target-level proposition; and the formal target propositions co-target corrected Clay-C and Clay-D modulo an explicit ledger of standard semantic bridges. No target-critical omission is identified. A third-party audit reporting pinned Comparator/kernel replay and an independent formal semantic bridge is recorded as corroborating external evidence, but independent Comparator execution, local repository rebuild, and proof verification by BFA itself remain separate, uninvoked tasks.*

Remark BFA.12.38 (Next consumer: BFA.13)**.** BFA.13 shall perform *Generic Mathematical Extraction*. It may consume only structures that survive BFA.8–BFA.12 without depending on the truth of a Navier–Stokes-specific external theorem. In particular, it shall examine whether the following can be abstracted into reusable AK mathematics:

(i) monotone residual-improvement contracts with stage-independent loss;

(ii) completion under stage-dependent constants and recursively shrinking supports;

(iii) fixed-stage comparison as a substitute for summing infinitely many flat residuals;

(iv) terminal jet-to-smooth-extension contracts;

(v) critical-cut / joint-obligation audit semantics; and

(vi) asymmetric cross-artifact target-consequence projections such as the triangle audited here.

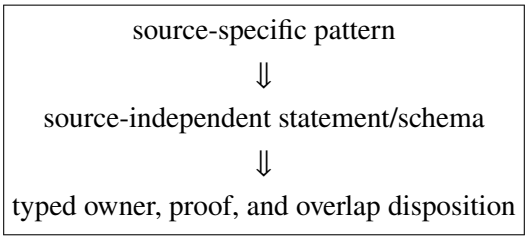
No item is promoted merely because it appears in this list; BFA.13 must separate source-specific structure from genuinely substitution-stable generic mathematics.

BFA.13 Generic Mathematical Extraction

BFA.13.1 Purpose, extraction ceiling, and reaudited classification

Declaration BFA.13.1 (BFA.13 objective)**.** BFA.13 asks which structures surviving the source-faithful reconstruction and adversarial audits of BFA.1–BFA.12 admit a genuinely source-independent formulation. It does

not ask whether a useful source pattern can be renamed in AK notation. The admissible extraction chain is



with explicit failure boundaries and no backflow into the earlier benchmark audit.

Definition BFA.13.2 (Two-axis extraction type)**.** Every BFA.13 object receives two independent labels.

Ownership class is exactly one of:

(i) source_retained: essentially Navier–Stokes/source-specific;

(ii) already_owned_generic: already represented by frozen AK-v21;

(iii) bfa_local: a source-independent local statement or schema not assigned a competing frozen-v21 owner;

(iv) successor_program: a research direction only, with no current theorem promotion.

Maturity class is exactly one of:

(a) proved_lemma;

(b) definition_or_normal_form;

(c) specialization_of_existing_theory;

(d) contract_only;

(e) research_program_only.

No value on one axis promotes the other. In particular, bfa_local does not imply novelty, Technical-Extension eligibility, or successor-Core status.

Definition BFA.13.3 (Generic extraction record)**.** A replayable extraction record is

$$\mathfrak{E}_{\text{gen}} = (\mathsf{id}, \mathsf{seed}, \mathsf{owner_class}, \mathsf{maturity}, \mathsf{abstract_data}, \mathsf{retained_hyp}, \mathsf{discarded_source_data}, \mathsf{statement}, \mathsf{proof_owner}, \mathsf{overlap_owner}, \mathsf{classical_overlap}, \mathsf{substitutions}, \mathsf{failure}, \mathsf{status}, \mathsf{nonclaims}).$$

The fields discarded_source_data and classical_overlap are mandatory. Renaming variables or removing source nouns is not generic extraction.

Policy 73 (BFA-EXT-01: novelty, ownership, and nonbackflow firewall)**.** *BFA.13 obeys the following rules.*

(i) An imported source theorem is not an AK theorem after renaming.

(ii) An elementary consequence proved here is not advertised as a new mathematical discovery merely because it receives a BFA label.

(iii) An already-owned v21 proof-DAG, semantic-field-map, target-cut, causal-frontier, noncompensation, tail, or promotion rule is reused through its frozen owner rather than republished.

(iv) A source-independent statement is not automatically promotion-worthy; mathematical nontriviality, nonredundancy, consumer value, and external overlap remain separate gates.

(v) No BFA.13 item may be fed backward into BFA.1–BFA.12 unless a new explicitly versioned audit route is constructed.

Theorem BFA.13.4 (Extraction nonbackflow)**.** *No theorem, schema, normal form, or candidate first introduced in BFA.13 may upgrade, repair, or reinterpret an earlier BFA.1–BFA.12 verdict inside the same audit identity.*

Proof. BFA.1 places generic extraction downstream of source binding, reconstruction, compatibility, and adversarial audit. Reusing a post-hoc abstraction as an undeclared upstream premise changes the proof ancestry of the benchmark run. A later route may consume the extracted object, but that route has a new identity and dependency record. □

BFA.13.2 Frozen-v21 and classical-overlap screen

Definition BFA.13.5 (Mandatory frozen-v21 overlap screen)**.** Before an object may receive a BFA-local owner, it is screened against at least:

(i) Main Chapter 5 / Appendix D finite-to-infinite and tail reduction;

(ii) Main Chapter 8 proof-DAG node/edge semantics, typed conjunction, assumption transport, failure propagation, and noncompensation;

(iii) Main Chapter 9 and Appendices HT/UB typed Bridge/Return composition;

(iv) Technical Extension Appendix M ordered ledgers, residual reserve, zero-reflection, and causal frontiers;

(v) Search/Platform U–Z semantic field maps, consumer projections, target cuts, substitution, promotion certificates, replay, and repair;

(vi) Problem-Interface MS/NS source-specific interfaces and Returns.

Proposition BFA.13.6 (No generic gain from architectural resemblance)**.** *The correspondences*

$$\begin{aligned} \textit{finite-to-infinite source step} &\rightsquigarrow \textit{AK tail/reduction},\\ \textit{source bottleneck} &\rightsquigarrow \textit{AK target cut},\\ \textit{failed obligation} &\rightsquigarrow \textit{AK failure route},\\ \textit{source implication} &\rightsquigarrow \textit{AK typed Return} \end{aligned}$$

do not create new BFA mathematics without an additional theorem or schema not already owned by the right-hand architecture.

Proof. Each right-hand object already has a frozen-v21 semantic owner. Similarity is not theorem inequality, strengthening, or a new proof object. Conservative evolution therefore permits only specialization, exact extension, or a new independently proved result. □

Policy 74 (BFA-EXT-02: classical-iteration overlap quarantine)**.** *Any future theorem combining iterative correction, loss of derivatives, smoothing or localization, rapidly improving residuals, and convergence must be compared explicitly with classical Nash–Moser/Hamilton-type schemes and modern abstract Nash–Moser variants before any AK novelty or promotion claim is considered. Such theories already treat smoothing operators, derivative loss, nonuniform/tame estimates, and rapidly convergent correction schemes [8, 9, 10].*

Consequently, the phrase "adaptive RIC" denotes only a BFA research program unless a successor theorem proves a mathematically precise distinction, nonredundancy, and independently useful generalization.

BFA.13.3 Extraction I: separated-filter residual limit closure

Definition BFA.13.7 (Separated residual-limit datum)**.** A *separated residual-limit datum* consists of:

(i) a topological space X;

(ii) a topological space Y with distinguished point 0_Y and closed subsets

$$Y = F^0Y \supseteq F^1Y \supseteq F^2Y \supseteq \cdots, \qquad \bigcap_{N\geq 0} F^NY = \{0_Y\};$$

(iii) a continuous map $\Phi : X \to Y$;

(iv) a sequence $x_j \to x_\infty$ in X;

(v) integers $r_j \to \infty$ such that

$$\Phi(x_j) \in F^{r_j} Y.$$

Lemma BFA.13.8 (BFA-local separated-filter residual-limit lemma)**.** *For every datum of Definition BFA.13.7,*

$$\Phi(x_\infty) = 0_Y.$$

Proof. Fix $N \geq 0$. Since $r_j \to \infty$, for all sufficiently large j one has $\Phi(x_j) \in F^N Y$. Continuity gives $\Phi(x_j) \to \Phi(x_\infty)$, and closedness of $F^N Y$ yields $\Phi(x_\infty) \in F^N Y$. This holds for every N, hence separatedness gives $\Phi(x_\infty) = 0_Y$. □

Remark BFA.13.9 (Why this is not a completion theorem)**.** Lemma BFA.13.8 assumes that the state limit x_∞ already exists. It proves only closure of residual vanishing under a separated closed filtration. It does not construct a completion, prove Cauchy convergence, control stage-dependent constants, repair derivative loss, or choose localizers. Calling it a general "completion theorem" would therefore inflate its content.

Definition BFA.13.10 (Local RIC-0 alias and ceiling)**.** For continuity with the BFA research vocabulary, the implication of Lemma BFA.13.8 may be referred to locally as the *post-completion residual-closure kernel* RIC_0. The alias does not change the theorem's exact content and carries no promotion or novelty claim.

BFA.13.4 Extraction II: fixed-stage filtered residual closure

Definition BFA.13.11 (Filtered additive residual target)**.** Let $(Y, +)$ be an abelian group equipped with subgroups

$$Y = F^0 Y \supseteq F^1 Y \supseteq \cdots, \qquad \bigcap_{N \geq 0} F^N Y = \{0\}.$$

Let $\Phi : X \to Y$ be any map. No topology or continuity is required.

Lemma BFA.13.12 (Fixed-stage filtered residual-closure lemma)**.** *Assume that for every $N \geq 0$ there exists a finite stage $J = J(N)$ such that*

$$\Phi(x_J) \in F^N Y, \qquad \Phi(x_\infty) - \Phi(x_J) \in F^N Y.$$

Then $\Phi(x_\infty) = 0$.

Proof. For fixed N, subgroup closure gives

$$\Phi(x_\infty) = \big(\Phi(x_\infty) - \Phi(x_J)\big) + \Phi(x_J) \in F^N Y.$$

This holds for every N, so separatedness gives zero. □

Corollary BFA.13.13 (Fixed-stage comparison can replace an infinite remainder sum)**.** *Under Lemma BFA.13.12, one need not define an infinite sum of stagewise residual remainders. For each requested filtration depth it is enough to exhibit one finite comparison stage satisfying the two membership conditions.*

Remark BFA.13.14 (Exact mathematical weight)**.** The lemma isolates the elementary filtered-group logic used in the BFA.10 fixed-stage safeguard. The hard work in an application is proving the second membership and constructing the completed object. Accordingly, this lemma is a BFA-local proof locator, not a current Technical-Extension recommendation.

BFA.13.5 Extraction III: monotone defect improvement as a research schema

Definition BFA.13.15 (Ranked defect-improvement system)**.** A ranked defect-improvement system consists of

$$x_0 \xrightarrow{C_0} x_1 \xrightarrow{C_1} x_2 \to \cdots,$$

defects $D_j = D(x_j) \in \mathcal{D}$, an ordered set $(\Lambda, \preceq)$, and $\rho : \mathcal{D} \to \Lambda$ such that

$$\rho(D_j) \preceq \rho(D_{j+1})$$

for every defined correction step. Strict inequality records a strict improvement step. A separate *cofinal improvement* condition is required when one needs the ranks to escape every fixed defect level.

Proposition BFA.13.16 (MDI alone has neither convergence nor zero force)**.** *Monotone, even strictly cofinal, defect-rank improvement alone implies neither convergence of the state sequence nor vanishing of a residual at a state limit.*

Proof. For failure of state convergence, take $X = \mathbb{R}$, $x_j = j$, distinct defects D_j, and define $\rho(D_j) = j$. The rank improves strictly while (x_j) diverges.

For failure of residual vanishing even when a state limit exists, take $x_j = 1/(j+1) \to 0$, again set $\rho(D_j) = j$, and define an unrelated residual map $\Phi(x) = 1$. Then the state converges and the defect rank is cofinal, but $\Phi(0) = 1$. A theorem connecting defect rank to a zero-reflecting residual filtration is therefore indispensable. □

Corollary BFA.13.17 (MDI factors through residual closure only after an explicit rank-to-filter theorem)**.** *Suppose an MDI system has a separately proved map from cofinal defect rank to a descending residual filtration,*

$$\rho(D_j) \text{ cofinal} \quad \Longrightarrow \quad \Phi(x_j) \in F^{r_j} Y, \qquad r_j \to \infty,$$

and suppose the remaining hypotheses of Lemma BFA.13.8 hold. Then $\Phi(x_\infty) = 0_Y$.

Proof. The rank-to-filter theorem supplies the only nontrivial bridge from the defect schema to the hypotheses of Lemma BFA.13.8; the conclusion then follows from that lemma. □

Remark BFA.13.18 (Reaudited status of MDI)**.** MDI is not presently a BFA theorem family. AK-v21 already owns ordered defect ledgers, zero-reflection guardrails, causal frontiers, and several finite-to-infinite mechanisms. BFA contributes only a possible dynamic research syntax coupling correction steps to improvement ranks. Until a nonredundant positive theorem and a cross-domain application exist, MDI is classified successor_program/research_program_only.

BFA.13.6 Extraction IV: adaptive-localizer completion remains quarantined

Definition BFA.13.19 (Adaptive-localizer research pattern)**.** The BFA source audit suggests the following research pattern:

(i) stage increments with stage-dependent finite constants;

(ii) improvement orders tending to infinity;

(iii) derivative-loss exponents controlled independently of the stage;

(iv) smoothing/localizing operators chosen adaptively after local constants are known;

(v) only finitely many new inequalities imposed at each stage;

(vi) a separate theorem producing convergence or local finiteness;

(vii) a finite-order residual-transfer theorem.

Policy 75 (BFA-EXT-03: no adaptive-RIC theorem is extracted)**.** *BFA.13 registers no general adaptive-localizer completion theorem. The external source already supplies a generic analytic summation lemma, while classical Nash–Moser and related abstract iteration theories already occupy a large part of the conceptual territory involving smoothing, derivative loss, and rapidly improving correction schemes [8, 9, 10].*

A future AK theorem must therefore do more than abstract the notation of the source. It must state a precise ambient class, prove an independent theorem, identify a mathematically meaningful difference from existing iteration frameworks, and provide a failure theorem or counterexample demonstrating the necessity of a genuinely new hypothesis.

Remark BFA.13.20 (Promotion burden for any future RIC theorem)**.** At minimum, a successor theorem would need independent proofs of:

(i) adaptive localizer/smoother selection under nonuniform stage constants;

(ii) a stage-independent loss criterion;

(iii) state convergence or local finiteness;

(iv) nonlinear residual transfer;

(v) exact zero reflection at the limit; and

(vi) nonredundancy relative to established Nash–Moser/Hamilton-type or other complete-filter iterative frameworks.

The current BFA does not discharge these obligations.

BFA.13.7 Extraction V: terminal jet realization is a contract, not a new theorem

Definition BFA.13.21 (Jet-realization consumer contract)**.** A terminal-extension audit separates

$$\text{jet existence} \longrightarrow \text{jet compatibility} \longrightarrow \text{smooth realization} \longrightarrow \text{structural preservation}.$$

Support, symmetry, divergence, boundary, equivariance, or other downstream constraints belong to the last arrow unless the realization theorem already contains them.

Policy 76 (BFA-EXT-04: jet contract ceiling)**.** *BFA.13 extracts only the typed consumer contract of Definition BFA.13.21. Classical Borel/Whitney-type realization mathematics is not republished as BFA mathematics. Any stronger constrained realization theorem requires its own exact source or independent proof.*

BFA.13.8 Extraction VI: joint-obligation vertices are a v21 normal form

Definition BFA.13.22 (BFA joint-obligation normal form)**.** When a consumer requires several predecessors to satisfy one simultaneous compatibility predicate, BFA may represent that predicate by an explicit vertex $[J_v]$ rather than leaving it implicit inside the consumer's statement or assumption context.

Proposition BFA.13.23 (No new theorem authority from joint-vertex expansion)**.** *The $[J_v]$ representation is a normalization of the frozen-v21 Chapter 8 proof-DAG semantics, not a new generic theorem family. A valid expansion must preserve the exact consumer statement, assumptions, node/edge types, and necessary target sub-DAG.*

Proof. Main Chapter 8 already requires every node to contain its exact statement, assumption context, incoming necessary edges, and verifier policy, and already provides coherence nodes and noncompensating typed conjunction semantics. An explicit $[J_v]$ merely materializes one simultaneous conjunct as a node. If the transformation changes the consumer predicate or deletes another necessary dependency, it is not semantics-preserving. □

Remark BFA.13.24 (Reaudited ownership)**.** BFA.9's explicit joint vertices remain useful audit instrumentation, but their generic semantics are classified already_owned_generic/specialization_of_existing_theory under Main Chapter 8. No new Governance or Core owner is proposed.

BFA.13.9 Extraction VII: cross-artifact semantic projection is a Search/Platform specialization

Definition BFA.13.25 (BFA cross-artifact semantic projection specialization)**.** For theorem-bearing artifacts A and B written in different formal or mathematical languages, BFA.12 uses the packet-style record

$$\mathfrak{S}_{A\to B} = (A, B, \text{direction}, \text{quantifiers}, \text{objects}, \text{hypotheses}, \text{strength}, \text{bridge ledger}, \text{loss}, \text{converse status}, \text{artifacts}).$$

This is a BFA specialization of the frozen Search/Platform semantic-field-map and consumer-contract architecture of SP-v21-DEF-06.

Policy 77 (BFA-EXT-05: semantic-projection composition remains v21-owned)**.** *BFA.13 creates no new semantic-projection composition theorem. Composition is licensed by the existing typed composition of preservation certificates in SP-v21-DER-03: the middle semantics must lie in the domain of the next semantic field map, and only fields with complete witnessed lineage survive the composite [5]. BFA's Paper–Lean–Clay ledger is an application of that rule at theorem-statement level.*

Remark BFA.13.26 (No converse from common target)**.** Co-targeting one external proposition does not establish equivalence of the source artifacts or reverse reconstruction of their witness data. In BFA.12 this remains an explicit directional-loss field, not a new generic theorem.

BFA.13.10 Extraction VIII: negative-evidence subject typing is a Chapter 8 specialization

Definition BFA.13.27 (BFA negative-evidence subject view)**.** For external-proof falsification BFA projects the frozen node/edge records to

$$(\text{subject}, \text{strength}, \text{evidence kind}, \text{scope}, \text{propagation}, \text{repair}),$$

where subject distinguishes theorem statement, theorem application, edge instantiation, joint obligation, interpretation relation, source binding, and target statement.

Proposition BFA.13.28 (Subject identity is preserved while consequences propagate)**.** *A downstream consumer may lose entitlement because an ancestor is rejected or undefined, but this propagation does not change the proposition or artifact that was the subject of the original negative evidence.*

Proof. Frozen Chapter 8 node and edge semantic records contain immutable identities, exact statements/relations, scopes, assumptions, and statuses. Dependency propagation changes consumer status; it does not rewrite the ancestor's semantic record. BFA's subject field is therefore a projection of existing typed identity rather than a new failure calculus. □

Remark BFA.13.29 (Reaudited ownership)**.** The subject view is useful for BFA.8 falsification reports, but its generic content is classified $\text{already_owned_generic/specialization_of_existing_theory}$ under Main Chapter 8. No additional Governance promotion is recommended.

BFA.13.11 Source-specific material that does not survive extraction

Definition BFA.13.30 (Nonextractable source kernel)**.** The following material remains source-retained at the present resolution:

(i) the similarity coordinates, pulse geometry, stress cone, and oscillatory realization of the analytic construction;

(ii) source-specific wave/mean inverses, moment constraints, pressure reconstruction, and exponent arithmetic inside one correction cycle;

(iii) the exact Navier–Stokes residual and source-specific derivative estimates;

(iv) compact-support localization through vector potentials;

(v) terminal energy/comparison, viscosity scaling, and periodic disjoint-support construction as PDE-specific theorems;

(vi) the Clay-C/Clay-D Returns themselves.

Policy 78 (BFA-EXT-06: no abstraction by erasure)**.** *Deleting source nouns from Definition BFA.13.30 does not create a generic theorem. Removed source data must be replaced by abstract hypotheses that are sufficient for an independent proof and meaningful in another source family.*

BFA.13.12 Reaudited extraction matrix

Definition BFA.13.31 (BFA.13 reaudited extraction matrix)**.** The strongest safe disposition after the frozen-v21 and external-overlap audit is:

Object	Generic content	Reaudited status	Promotion ceiling after BFA.13
RIC_0 alias	residual zero closes under an already-existing state limit and separated closed filtration	BFA-local elementary proved lemma	remain BFA-local; not a completion theorem and no current TE recommendation
Fixed-stage closure	one finite comparison stage per requested filtration depth	BFA-local elementary proved lemma	implementation/proof locator only unless an independent consumer justifies migration
MDI	correction steps carry monotone/cofinal improvement ranks	successor research schema only	no promotion; positive theorem, nonredundancy, and cross-domain application required
Adaptive localizers	nonuniform stage constants plus stage-independent loss and adaptive smoothing/localization	source-derived research pattern under classical-overlap quarantine	no theorem promotion; Nash–Moser/related distinction must be proved first
Jet realization	jet existence, realization, and structural preservation kept separate	BFA audit contract only	no mathematical promotion; classical realization theory remains external
Joint obligations	explicit vertex for one simultaneous consumer predicate	specialization of Main Ch. 8 proof-DAG semantics	no new owner; BFA instrumentation only
Semantic projection	directional cross-artifact field/statement preservation	specialization of SP-v21 semantic field maps	no new theorem owner; BFA.12 application only
Negative subject view	preserve exact failed subject while consequences propagate	specialization of Main Ch. 8 node/edge records	no new Governance owner; BFA reporting schema only
NS analytic kernel	geometry, correction estimates, PDE terminal route	source-retained	BFA/Problem Interface only

Theorem BFA.13.32 (BFA.13 reaudited extraction theorem)**.** *At the present construction stage, the strongest source-independent mathematical content newly owned locally by BFA.13 consists only of the two elementary filtered-residual lemmas Lemmas BFA.13.8 and BFA.13.12 and their immediate corollaries. The remaining apparent generic structures are either:*

(i) research programs not yet represented by an independent general theorem (MDI and adaptive-localizer RIC);

(ii) audit contracts without new mathematical theorem authority (terminal jet realization);

(iii) specializations or normal forms of frozen-v21 generic semantics (joint obligations, semantic field maps, and negative-evidence subject views); or

(iv) source-retained Navier–Stokes mathematics.

No item currently warrants Technical-Extension or successor-Core promotion on the evidence of BFA.13 alone.

Proof. The two local lemmas are proved directly above. MDI has explicit counterexamples to both convergence and zero force without additional bridges. The adaptive-localizer pattern has no AK-owned general theorem and lies in a classically populated iteration-theory region. The jet item is explicitly a consumer contract. Main Chapter 8 already owns exact node/edge statements, coherence nodes, typed conjunctions, assumption transport, and failure propagation, while Search/Platform already owns typed semantic field maps and their witnessed composition [1, 5]. The remaining PDE objects retain source-specific hypotheses. Therefore no stronger promotion is licensed. □

BFA.13.13 Research-program consequences after re-audit

Definition BFA.13.33 (Three surviving successor questions)**.** BFA.13 leaves three research questions rather than three promoted theories:

(i) **Nontrivial RIC question.** Is there a source-independent correction/completion theorem that handles a class not already subsumed by standard complete-filter or Nash–Moser-type iteration machinery and that is genuinely useful in multiple domains?

(ii) **Dynamic defect question.** Can AK's existing static defect and ledger semantics be extended by correction operators and improvement ranks in a way that yields a nontrivial convergence/zero theorem?

(iii) **Formal audit-normalization question.** Can BFA's joint-node and cross-artifact views be implemented as lossless normal forms of the existing Chapter 8/Search-Platform schemas, with machine-checkable preservation witnesses?

Policy 79 (BFA-EXT-07: successor challenge requirement)**.** *A successor mathematical project may not proceed by definitions alone. For items (i) and (ii) of Definition BFA.13.33, promotion requires at least:*

(a) one nontrivial positive theorem;

(b) one failure/no-go theorem or sharp counterexample;

(c) one independently typed non-Navier–Stokes application;

(d) an explicit nonredundancy comparison with frozen v21; and

(e) an external-literature overlap analysis appropriate to the theorem class.

Item (iii) is primarily an implementation/governance task and does not become Core mathematics merely by successful formalization.

BFA.13.14 Status vector, closure, and handoff

Definition BFA.13.34 (BFA.13 reaudited status vector)**.** At completion of the BFA.13 re-audit,

$$
\begin{aligned}
\mathsf{extraction_typing} &= \mathrm{two_axis_defined},\\
\mathsf{RIC0_postcompletion_residual_closure} &= \mathrm{bfa_local_elementary_proved},\\
\mathsf{RIC0_is_completion_theorem} &= \mathrm{false},\\
\mathsf{fixed_stage_residual_closure} &= \mathrm{bfa_local_elementary_proved},\\
\mathsf{monotone_defect_improvement} &= \mathrm{successor_research_program_only},\\
\mathsf{adaptive_localizer_RIC} &= \mathrm{classical_overlap_quarantined_program},\\
\mathsf{terminal_jet_realization} &= \mathrm{audit_contract_only},\\
\mathsf{joint_obligation_vertices} &= \mathrm{v21_chapter8_normal_form},\\
\mathsf{cross_artifact_semantic_projection} &= \mathrm{v21_searchplatform_specialization},\\
\mathsf{negative_subject_typing} &= \mathrm{v21_chapter8_specialization},\\
\mathsf{NS_analytic_kernel} &= \mathrm{source_retained},\\
\mathsf{TechnicalExtension_promotion_recommended} &= \mathrm{none},\\
\mathsf{SuccessorCore_promotion} &= \mathrm{not_licensed},\\
\mathsf{v21_backflow} &= \mathrm{forbidden}.
\end{aligned}
$$

Theorem BFA.13.35 (BFA.13 reaudited closure)**.** *BFA.13 closes without converting the external Navier–Stokes proof into AK-owned general mathematics. Two small filtered-residual lemmas survive as BFA-local proved statements. The previously proposed adaptive RIC and dynamic defect structures remain research programs; joint-obligation, semantic projection, and negative-subject structures are reclassified as specializations of frozen-v21 semantics rather than new theorem owners. No Technical Extension or successor-Core promotion is recommended from BFA.13 alone.*

Remark BFA.13.36 (Next consumer: BFA.14)**.** BFA.14 shall build the *Promotion / Non-Promotion Matrix* from the reaudited rows of Definition BFA.13.31. In particular, it must be allowed to return retain/no-promotion for every row. Promotion is not a required outcome of generic extraction.

BFA.14 Promotion / Non-Promotion Matrix

BFA.14.1 Purpose, constitutional boundary, and decision semantics

Declaration BFA.14.1 (BFA.14 objective)**.** BFA.14 converts the reaudited extraction results of BFA.13 into explicit owner-preserving promotion and non-promotion dispositions. Its task is not to manufacture mathematical strength, novelty, or publication priority by score. The admissible decision chain is

$$\boxed{\text{BFA.13 extracted object} \longrightarrow \text{typed owner-readiness assessment} \longrightarrow \text{owner-preserving disposition}}$$

with mathematical proof, internal AK ownership, external-literature novelty, consumer value, migration cost, and provenance kept as separate axes.

Policy 80 (BFA-PROM-01: promotion is optional)**.** *No mathematical, architectural, or publication requirement forces BFA.14 to promote any extracted object. The disposition* retain_bfa_local *is a positive governance verdict when it is the strongest useful current reading, even if no theorem-theoretic impossibility would prevent a future successor artifact from migrating the statement.*

Policy 81 (BFA-PROM-02: frozen-v21 nonbackflow)**.** *BFA.14 cannot insert a theorem, schema, owner, or migration record into the already frozen AK-HPDST v21.0.0 source set. Any future owner transition occurs only in a separately versioned successor artifact with a new identity, complete proof DAG, exact source/proof artifacts, and explicit migration record.*

Definition BFA.14.2 (Promotion disposition sort)**.** Every BFA.13 row receives exactly one primary disposition from

(i) retain_bfa_local: retain a proved local statement or useful BFA-specific record without opening a new global owner;

(ii) bind_existing_owner: treat the object as a specialization, normal form, or consumer view of an already frozen v21 owner;

(iii) research_program: retain as a successor research question without present theorem authority;

(iv) implementation_candidate: nominate a future implementation under existing mathematical ownership, without creating new mathematics;

(v) technical_extension_candidate: nominate an item for future Technical-Extension owner review after a complete successor packet is supplied;

(vi) successor_core_candidate: nominate an item for future Core-level review after all successor-Core gates are supplied;

(vii) source_retained: remain source-specific in BFA or a Problem Interface; and

(viii) promotion_rejected: reject a proposed stronger owner reading because it is duplicate, inflated, ill-typed, or otherwise unsupported.

Candidate status is a governance nomination, not proof of eligibility, novelty, or eventual promotion.

Definition BFA.14.3 (Independent promotion-readiness sort)**.** For a proposed owner transition, BFA records one readiness value

$$\text{blocked}, \quad \text{incomplete}, \quad \text{ready_for_governance_review}, \quad \text{not_applicable}.$$

This sort is independent of the primary disposition. In particular, retain_bfa_local does not mean blocked; a proved local lemma may simply have no opened promotion packet or no current architectural need for a new owner.

Definition BFA.14.4 (Independent novelty-status sort)**.** External-literature novelty is recorded separately as one of

$$\text{not_claimed}, \quad \text{not_assessed}, \quad \text{overlap_unresolved}, \quad \text{comparison_completed}.$$

The final value means only that a declared literature comparison was completed; any positive priority or originality claim still requires its own precise statement and evidence. BFA.14 makes no positive priority claim.

Definition BFA.14.5 (Promotion assessment profile)**.** For an extracted object E, define the non-scalar profile

$$\mathfrak{P}(E) = (T, P, G, O, L, C, F, M, Q),$$

where:

(i) T records exact theorem/contract typing and strength;

(ii) P records proof maturity;

(iii) G records source independence and domain generality;

(iv) O records frozen-v21 owner overlap or conflict;

(v) L records external-literature comparison status;

(vi) C records actual consumer breadth and architectural value;

(vii) F records failure boundaries and prohibited stronger readings;

(viii) M records migration and maintenance cost; and

(ix) Q records provenance separation between motivating sources and genuinely AK-owned proof content.

No coordinate is inferred from another.

Policy 82 (BFA-PROM-03: no weighted promotion score)**.** *BFA.14 defines no weighted scalar*

$$s(E) = \sum_i w_i \mathfrak{P}_i(E)$$

whose magnitude licenses promotion. High consumer value, implementation convenience, literature novelty, or low migration cost cannot compensate for a missing proof, ill-typed statement, occupied owner, or absent migration record.

BFA.14.2 Three gate families: owner admission, novelty, and adoption

Definition BFA.14.6 (Internal AK owner-admission gates)**.** A future theorem-facing owner transition inside AK requires the applicable conjunction below.

A1: Exact typed statement. Complete hypotheses, source and target types, conclusion strength, and prohibited stronger readings are fixed.

A2: Proof or authoritative theorem source. The intended owner has an immutable proof artifact or an explicitly attributed external theorem source at exactly the consumed strength; a research pattern is insufficient.

A3: Generic-scope discipline. If the target owner is generic, source-specific hypotheses are removed or remain explicitly typed as Interface hypotheses; no noun-erasure generalization is allowed.

A4: Frozen-owner comparison. The exact content is compared with existing AK owners. Duplicate content is rebound to the existing owner; a claimed strengthening receives a distinct statement and proof.

A5: Failure boundary. Counterexamples, excluded regimes, converse failures, and scope ceilings are preserved.

A6: Conservative migration. A successor identity, dependency DAG, artifact bindings, migration relation, and nonbackflow record are available.

A7: Provenance honesty. Imported mathematics, source inspiration, adapted proof components, and AK-local derivations are distinguished explicitly.

For Core-level promotion the stronger successor constitution additionally requires the applicable conservative-extension, Core-UID, proof-locator, and register obligations of the frozen v21 policy [1, 7].

Definition BFA.14.7 (External novelty/publication gate)**.** A claim that an extracted theorem is mathematically new, AK-original in a literature-priority sense, or a distinct general iteration theory requires a separate theorem-level literature comparison against the declared relevant corpus. This gate is mandatory for a positive novelty/priority claim, but it is *not* a constitutional prerequisite for honest internal registration of known or externally sourced mathematics under correct attribution.

Remark BFA.14.8 (Why novelty is not an internal owner gate)**.** Frozen v21 selective promotion requires domain-independent scope, complete proof/hypotheses, source-assumption separation, conservative extension, identity/proof locators, failure boundaries, and migration governance. It does not make external literature novelty a theorem-admission axiom [1]. BFA therefore refuses both errors:

(i) unresolved literature overlap cannot support a novelty claim; but

(ii) unresolved literature overlap alone does not make an otherwise correctly attributed theorem mathematically false or constitutionally impossible to register in a future internal owner.

Definition BFA.14.9 (Adoption and portfolio signals)**.** Consumer breadth, cross-domain reuse, implementation value, maintenance cost, and package complexity are adoption signals. They can justify or discourage opening a new owner, but they are not theorem-truth predicates. Core-level consideration may reasonably demand a substantially broader portfolio than a local or Technical-Extension placement, yet that is a governance/adoption criterion rather than a proof of mathematical truth.

Theorem BFA.14.10 (Typed noncompensation across gate families)**.** *The three gate families have distinct failure consequences.*

(i) Failure of a required owner-admission gate blocks the corresponding owner transition.

(ii) Failure or noncompletion of the novelty gate blocks a positive novelty/priority claim, but does not by itself falsify the theorem or prohibit honest attributed internal use.

(iii) Weak adoption signals can justify local retention, but cannot turn a false or unproved statement into a theorem.

(iv) Strong adoption signals cannot override owner duplication, missing proof, ill typing, or migration failure.

Proof. Owner admission, novelty, and adoption answer different predicates. Frozen v21 already separates theorem truth, ownership, workflow, migration, registration, and publication status [1, 5, 7]. Therefore a value in one sort cannot be substituted for a missing value in another without a separately proved or governed conversion. □

Definition BFA.14.11 (Immediate vetoes at the current identity)**.** The following current-identity vetoes are distinguished by target:

(a) `already_owned_generic` vetoes a competing new generic owner;

(b) `research_program_only` vetoes theorem promotion until a theorem exists;

(c) unresolved Nash–Moser or comparable classical overlap vetoes a positive novelty claim for adaptive-RIC, but not merely the possibility of a future correctly attributed theorem;

(d) `source_retained` vetoes generic-owner promotion absent a new source-independent theorem or exact attributed import contract;

(e) an audit contract or machine normal form cannot be promoted merely because it is operationally useful; and

(f) the frozen-v21 boundary vetoes direct backflow under every favorable assessment.

BFA.14.3 Mathematical-candidate decisions

Definition BFA.14.12 (Disposition of the residual-limit lemma)**.** For the BFA-local lemma of Lemma BFA.13.8,

$$\mathsf{Disposition}(\mathsf{RIC}_0) = \mathrm{retain_bfa_local}.$$

Its current record is:

$$\begin{aligned}\mathsf{ProofMaturity} &= \mathrm{proved_lemma},\\ \mathsf{OwnerReadiness} &= \mathrm{incomplete},\\ \mathsf{NoveltyStatus} &= \mathrm{not_claimed}.\end{aligned}$$

The readiness value means that no successor owner packet has been opened; it does not mean that a mathematical contradiction or permanent constitutional barrier has been found.

Proposition BFA.14.13 (Current local-retention recommendation for RIC_0)**.** *The strongest current governance recommendation is to retain* RIC_0 *locally rather than nominate it for Technical-Extension review.*

Proof. The lemma is correct but elementary: it assumes the state limit and then uses continuity, closedness, and separatedness to reflect residual zero. No current BFA consumer requires a distinct generic owner, and nearby frozen-v21 tail/zero-reflection machinery already occupies the architectural territory. These facts support local retention. They do not prove that a future successor artifact could never migrate the lemma if a concrete owner need, precise nonduplication analysis, and migration packet were later supplied. □

Definition BFA.14.14 (Disposition of fixed-stage residual closure)**.** For Lemma BFA.13.12,

$$\mathsf{Disposition}(\mathsf{FixedStageClosure}) = \mathrm{retain_bfa_local},$$

with

$$\mathsf{OwnerReadiness} = \mathrm{incomplete}, \qquad \mathsf{NoveltyStatus} = \mathrm{not_claimed}.$$

The result is retained as a proof locator for the useful quantifier pattern

$$\forall N\ \exists J(N),$$

not as a new completion theory.

Proposition BFA.14.15 (Useful quantifier structure is not an automatic owner transition)**.** *The fact that fixed-stage comparison avoids summing stagewise residuals is a useful proof pattern but does not itself create a need for a new generic owner.*

Proof. The abstract lemma is elementary once the two $F^N Y$-memberships are supplied. The difficult mathematical content in applications lies in constructing the completed object and proving those memberships. A future owner transition is possible only through a new successor packet that gives the logical shell substantive independent content or a concrete architectural need; neither is presently supplied. □

Definition BFA.14.16 (Disposition of monotone defect improvement)**.** For the MDI schema of BFA.13,

$$\mathsf{Disposition}(\mathsf{MDI}) = \mathrm{research_program}, \qquad \mathsf{OwnerReadiness} = \mathrm{blocked}.$$

The blocking reason is mathematical rather than bibliographic: monotone or cofinal defect-rank improvement alone proves neither state convergence nor residual zero, and no independent positive general theorem has yet been supplied.

Definition BFA.14.17 (Disposition of adaptive-localizer RIC)**.** For the proposed adaptive-localizer completion calculus,

$$\mathsf{Disposition}(\mathsf{AdaptiveRIC}) = \mathrm{research_program}, \qquad \mathsf{OwnerReadiness} = \mathrm{blocked}.$$

The blocking owner-admission defect is the absence of an AK-owned source-independent abstract completion theorem. Separately,

$$\mathsf{NoveltyStatus}(\mathsf{AdaptiveRIC}) = \mathrm{overlap_unresolved}$$

because Nash–Moser/Hamilton-type iteration and modern variants already cover substantial territory involving smoothing, derivative loss, nonuniform estimates, and rapidly improving corrections [8, 9, 10].

Theorem BFA.14.18 (No current successor-Core theorem candidate from dynamic-defect data)**.** *Neither MDI nor adaptive-localizer RIC is a theorem-level* $\mathrm{successor_core_candidate}$ *at the present BFA.14 state.*

Proof. Both fail the theorem/proof owner-admission gate: MDI is a schema with explicit nonimplication boundaries, and adaptive-localizer RIC has no independent source-general completion theorem. This suffices to block theorem-level Core nomination irrespective of consumer value. Adaptive RIC additionally has an unresolved novelty comparison, but that is a separate ceiling on originality claims rather than the primary theorem-admission defect. □

BFA.14.4 Audit, proof-semantic, and infrastructure decisions

Definition BFA.14.19 (Terminal jet realization disposition)**.** The terminal jet object extracted in BFA.13 receives

$$\mathsf{Disposition}(\mathsf{TerminalJetContract}) = \mathrm{retain_bfa_local}.$$

It remains an audit contract separating jet existence, compatibility, smooth realization, and structural preservation. Classical Borel/Whitney-type realization mathematics is not reissued under a BFA owner, and no new generic constrained-realization theorem is presently proved.

Definition BFA.14.20 (Joint-obligation disposition)**.** The BFA.8–BFA.9 explicit joint-obligation vertex receives

$$\mathsf{Disposition}([J_v]) = \mathrm{bind_existing_owner}.$$

Its generic semantics are owned by Main Chapter 8's typed proof-DAG, assumption, coherence, and necessary-conjunction machinery. BFA retains $[J_v]$ only as an explicit adversarial-audit normal form.

Definition BFA.14.21 (Cross-artifact semantic projection disposition)**.** The Paper–formal–official-target semantic projection record receives

$$\mathsf{Disposition}(\mathfrak{S}_{A\to B}) = \mathrm{bind_existing_owner}.$$

Search/Platform already owns partial semantic field maps and witnessed typed composition [5]; BFA.12 is a source-specific application rather than a competing generic theorem family.

Definition BFA.14.22 (Negative-evidence subject disposition)**.** The negative-evidence subject view receives

$$\mathsf{Disposition}(\mathsf{NegativeSubjectView}) = \mathrm{bind_existing_owner}.$$

Its generic identity and propagation semantics remain with Main Chapter 8. BFA retains only the reporting projection needed to distinguish theorem, application, edge, joint obligation, interpretation, binding, and target.

Definition BFA.14.23 (Machine-normalization implementation candidate)**.** There is one operational successor candidate:

$$\mathsf{BFAAuditNormalForm} := \big([J_v], \mathfrak{S}_{A\to B}, \mathsf{NegativeSubjectView}\big),$$

with

$$\mathsf{Disposition}(\mathsf{BFAAuditNormalForm}) = \mathrm{implementation_candidate}.$$

The candidate is a proposal for a machine normalization under the existing Chapter 8 and Search/Platform owners. Its required target property is *lossless preservation of every consumed semantic field*; BFA.14 does not claim that such losslessness has already been proved or implemented.

Proposition BFA.14.24 (Implementation-candidate ceiling)**.** *At the present BFA.14 state,*

$$\mathsf{Losslessness}(\mathsf{BFAAuditNormalForm}) = \mathrm{open_obligation},$$

and even a later complete implementation with proved losslessness, serialization, replay, and proof-store integration would not by itself create a new mathematical theorem owner.

Proof. Search/Platform already separates semantic preservation from infrastructure success and proves that storage, replay, route multiplicity, and other platform predicates cannot create missing mathematics [5]. Therefore losslessness must be established by explicit field-preservation witnesses, and successful implementation remains operational evidence rather than theorem-strength promotion. □

BFA.14.5 Source-specific and document-level decisions

Definition BFA.14.25 (Navier–Stokes analytic-kernel disposition)**.** For the source-specific objects listed in Definition BFA.13.30,

$$\mathsf{Disposition}(\mathsf{NSAnalyticKernel}) = \mathrm{source_retained}.$$

They remain owned by the external analytic source at their exact statements and may be referenced through BFA or a future source-specific Problem Interface. BFA audit success or non-adversity does not convert them into AK generic mathematics.

Definition BFA.14.26 (BFA document-level disposition)**.** The combined BFA.1–BFA.14 architecture itself is classified as

$$\boxed{\mathsf{BFA} = \text{post-v21 source-specific research/audit supplement}}$$

with generic proof-semantic components bound to their frozen-v21 owners, source-specific analytic mathematics retained at external-source authority, and the two BFA-local residual lemmas retained without owner escalation. Document-level usefulness is independent of theorem promotion.

Policy 83 (BFA-PROM-04: successful external audit is not a promotion event)**.** *A successful or non-adverse audit of a difficult external proof may increase confidence in a route, expose reusable research questions, and validate the utility of AK's typed architecture. It does not, by itself, transfer theorem ownership from the external source to AK, establish literature novelty, or strengthen the frozen Core.*

BFA.14.6 Canonical Promotion / Non-Promotion Matrix

Definition BFA.14.27 (BFA.14 canonical matrix)**.** The current disposition matrix is:

Object	Maturity	Disposition	Owner readiness / novelty	Reason / next gate
Post-completion residual closure RIC_0	elementary proved BFA lemma	retain local	incomplete / not claimed	no opened successor packet; local retention is architectural, not a permanent mathematical veto
Fixed-stage residual closure	elementary proved BFA lemma	retain local	incomplete / not claimed	useful proof locator; difficult content remains in source-specific membership/construction estimates
MDI	schema plus nonimplication theorems	research program	blocked / not claimed	no positive convergence/zero theorem; needs rank-to-residual and completion theory
Adaptive-localizer RIC	source-derived research pattern	research program	blocked / overlap unresolved	no AK-owned abstract theorem; novelty additionally requires theorem-level Nash–Moser comparison
Terminal jet contract	audit contract only	retain local	not applicable / not claimed	separates obligations; no new generic constrained-realization theorem
Joint obligation $[J_v]$	explicit BFA normal form	bind existing	blocked for competing owner / not claimed	generic conjunction/coherence semantics already owned by Main Ch. 8
Cross-artifact semantic projection	BFA.12 specialization	bind existing	blocked for competing owner / not claimed	semantic field-map composition already owned by Search/Platform
Negative-subject view	BFA.8 reporting specialization	bind existing	blocked for competing owner / not claimed	generic identity/status propagation already owned by Main Ch. 8
BFA audit normal form	proposed machine bundle	impl. candidate	incomplete / not claimed	lossless mapping is an open implementation obligation; no new theorem owner
NS analytic kernel	source-specific mathematics	source retained	blocked for generic owner / not claimed	exact external ownership retained; genericization requires a new source-independent theorem

Proposition BFA.14.28 (Current owner-escalation disposition)**.** *At the end of the present BFA.14 assessment, no row of Definition BFA.14.27 is* nominated *as either a* technical_extension_candidate *or a* successor_core_candidate.

Proof. The two proved local lemmas are deliberately retained because no successor owner packet or current architectural need has been opened; this is a retention choice rather than a theorem of permanent ineligibility. MDI and adaptive-localizer RIC are blocked by the absence of a positive abstract theorem. The jet item is a contract. Joint obligations, semantic projection, and negative-subject typing are specializations of existing owners. The audit-normal-form package is operational with losslessness still open, and the remaining analytic mathematics is source-specific. Hence no present row is assigned an owner-escalation candidate disposition. □

Corollary BFA.14.29 (No-promotion disposition is not theorem rejection)**.** *Proposition BFA.14.28 neither rejects the BFA-local proofs nor asserts that future migration is mathematically impossible. It records only the strongest current owner assignment under the present construction identity and evidence packet.*

Proposition BFA.14.30 (Current hard-block register with blocker types)**.** *The present stronger readings are blocked for different reasons and must not be conflated:*

(i) ***mathematical counterexample block:*** *MDI alone cannot support the implications "defect improvement ⇒ state convergence" or "defect improvement ⇒ residual zero", because BFA.13 supplies explicit counterexamples;*

(ii) ***evidence/readiness block:*** *adaptive-localizer RIC cannot presently be registered as an AK-owned general completion theorem because no such independent abstract theorem has been proved in BFA; this records missing evidence, not a proof that no such theorem can exist; and*

(iii) ***owner-collision block:*** *the joint-obligation, semantic-projection, and negative-subject structures cannot receive competing generic owners at their current content because frozen-v21 owners already supply the governing generic semantics.*

Proof. Item (i) is Proposition BFA.13.16. Item (ii) is the explicit BFA.13 adaptive-RIC ceiling and absence of an extracted theorem. Item (iii) follows from the owner comparisons in BFA.13 against Main Chapter 8 and Search/Platform semantic-field-map composition [1, 5]. □

BFA.14.7 Promotion debt and successor-entry conditions

Definition BFA.14.31 (Typed promotion-debt ledger)**.** A future successor project reopens a disposition only by identifying the debt class and supplying the corresponding new evidence.

Object	Debt class	Required new evidence	Earliest plausible owner
RIC_0 family	adoption / owner packet	concrete independent consumer, exact owner comparison, migration packet, and rationale for a distinct owner	TE if later justified
Fixed-stage closure	adoption / owner packet	substantive theorem family or concrete consumer showing value beyond a local proof locator	TE if later justified
MDI	mathematical theorem debt	positive convergence/zero theorem, sharp no-go boundary, rank-to-residual bridge, independent application, owner comparison	TE; Core only after later portfolio evidence
Adaptive RIC	theorem + novelty debt	abstract completion theorem, failure theorem, source-independent applications; separately, theorem-level Nash–Moser overlap analysis for any novelty claim	TE first; Core deferred
Jet contract	mathematical theorem debt	generic structural-preservation theorem beyond classical realization plus source checks	TE if proved/needed
Audit normal form	implementation debt	exact schema mapping, field-preservation witnesses proving losslessness, replay tests, versioning and migration behavior	SP/Companion implementation
NS kernel	abstraction and provenance debt	source-independent theorem with exact proof and failure boundary, not mere removal of PDE nouns; external source ownership preserved	PI first; generic owner only after separate proof/import contract

Policy 84 (BFA-PROM-05: no automatic escalation)**.** *Discharging one debt does not automatically move an object upward through Technical Extension and Core. Every owner transition requires a new complete assessment. A completed literature comparison may change novelty status without changing theorem truth; a new consumer may change adoption value without repairing a missing proof.*

Definition BFA.14.32 (Successor re-entry packet)**.** A successor project seeking to reopen a disposition supplies

$$\mathfrak{P}^{+}(E) = (E, \text{requested owner}, \text{new statement}, \text{proof/source}, \text{old owner comparison}, \\ \text{literature status}, \text{consumers}, \text{failures}, \text{migration}, \text{provenance}, \text{artifacts}).$$

Fields irrelevant to the requested disposition may be marked not_applicable, but they may not be silently omitted. A prose claim that an object is "more general" or "more useful" is not a valid re-entry packet.

BFA.14.8 BFA.14 status vector and final handoff

Definition BFA.14.33 (BFA.14 status vector)**.** At completion of the strongest BFA.14 re-audit,

$$
\begin{aligned}
\text{assessment_profile} &= \text{non_scalar_defined},\\
\text{gate_families} &= \text{owner_novelty_adoption_separated},\\
\text{weighted_promotion_score} &= \text{forbidden},\\
\text{external_novelty_as_owner_hard_gate} &= \text{rejected},\\
\text{RIC0_disposition} &= \text{retain_bfa_local},\\
\text{RIC0_owner_readiness} &= \text{incomplete},\\
\text{RIC0_permanent_hard_veto} &= \text{not_claimed},\\
\text{fixed_stage_disposition} &= \text{retain_bfa_local},\\
\text{MDI_disposition} &= \text{research_program},\\
\text{MDI_owner_readiness} &= \text{blocked},\\
\text{AdaptiveRIC_disposition} &= \text{research_program},\\
\text{AdaptiveRIC_owner_readiness} &= \text{blocked},\\
\text{AdaptiveRIC_novelty} &= \text{overlap_unresolved},\\
\text{jet_contract_disposition} &= \text{retain_bfa_local},\\
\text{joint_obligation_disposition} &= \text{bind_existing_owner},\\
\text{semantic_projection_disposition} &= \text{bind_existing_owner},\\
\text{negative_subject_disposition} &= \text{bind_existing_owner},\\
\text{audit_normal_form_disposition} &= \text{implementation_candidate},\\
\text{audit_normal_form_losslessness} &= \text{open_obligation},\\
\text{NS_analytic_kernel_disposition} &= \text{source_retained},\\
\text{TechnicalExtension_candidate_nominated_now} &= \text{none},\\
\text{SuccessorCore_candidate_nominated_now} &= \text{none},\\
\text{hard_block_types} &= \text{mathematical_evidence_owner_separated},\\
\text{positive_literature_novelty_claim} &= \text{none},\\
\text{v21_backflow} &= \text{forbidden}.
\end{aligned}
$$

Theorem BFA.14.34 (BFA.14 reaudited closure)**.** *BFA.14 closes the current owner-disposition question without conflating mathematical admissibility, literature novelty, and architectural adoption. The two proved residual lemmas remain local by current governance choice, not by a claimed permanent theorem veto. MDI and adaptive-localizer RIC remain research programs because theorem-level owner-admission obligations are missing; adaptive RIC additionally has unresolved novelty overlap. Generic proof-semantic structures are rebound to frozen-v21 owners. The audit normal form remains only an implementation candidate with losslessness unproved, and Navier–Stokes mathematics remains source-retained. No Technical-Extension or successor-Core candidate is presently nominated.*

Proof. The owner/novelty/adoption separation is established in Definitions BFA.14.6, BFA.14.7 and BFA.14.9. The objectwise dispositions are established in Definitions BFA.14.12, BFA.14.14, BFA.14.16, BFA.14.17, BFA.14.19 to BFA.14.23 and BFA.14.25. Proposition BFA.14.28 records the present governance assignment, while Proposition BFA.14.30 separates mathematical, evidence/readiness, and owner-collision blocks. Frozen-v21 backflow remains prohibited by Policy 81. □

Remark BFA.14.35 (Next consumer: BFA.15)**.** BFA.15 shall issue the final BFA construction verdict. It must combine, but not conflate:

(i) source-faithful reconstruction and compatibility status;

(ii) adversarial findings from BFA.8–BFA.12;

(iii) open verification, replay, and source-identity obligations;

(iv) BFA.13 generic-extraction limits;

(v) the present owner dispositions; and

(vi) the distinction among theorem validity, owner readiness, novelty status, implementation readiness, and adoption value.

The final verdict must separately state what BFA has shown, what remains conditional or not invoked, what is retained without owner escalation, and what is explicitly not claimed.

BFA.15 Final Verdict, Open Obligations, and Explicit Non-Claims

BFA.15.1 Purpose and closure discipline

Declaration BFA.15.1 (BFA.15 objective)**.** BFA.15 closes the BFA 1.0 construction draft without introducing a new analytic theorem, a new formal theorem, or a new AK owner. Its sole task is to compress BFA.1–BFA.14 into a final typed verdict while preserving every status boundary established earlier. The closure map is

$$\begin{array}{|l|}\hline \text{binding / reconstruction / compatibility} \\ \quad + \text{ adversarial audit} \\ \quad + \text{ formal–target semantic audit} \quad \longrightarrow \text{ typed final verdict} \\ \quad + \text{ provenance / verifier status} \\ + \text{ generic extraction / promotion status} \\ \hline \end{array}$$

and is not a map from "many positive-looking facts" to a single scalar notion of correctness.

Policy 85 (BFA-FINAL-01: no scalar final score)**.** *BFA.15 shall not assign a numerical probability, confidence percentage, weighted score, or rank to the mathematical truth of the external proof or the official target. The final record is a product of independent typed coordinates.*

Policy 86 (BFA-FINAL-02: closure is conservative)**.** *No BFA.15 sentence may upgrade* not_invoked*,* not_claimed*,* none_certified_at_declared_resolution*,* supported_modulo_declared_semantic_bridges*, or* retain_bfa_local *into a stronger mathematical, formal, provenance, ownership, or institutional status merely because the construction has reached its final section.*

Policy 87 (BFA-FINAL-03: no v21 backflow)**.** *The final closure of BFA 1.0 does not alter frozen AK-HPDST v21.0.0, its Claim Register, its Core-P/Core-DG status, or any frozen theorem owner. BFA remains a post-v21 Problem-Interface research supplement unless a separately versioned successor artifact performs an explicit migration.*

BFA.15.2 Final six-axis verdict

Definition BFA.15.2 (BFA.15 final verdict product)**.** The final BFA verdict is the ordered product

$$\mathfrak{V}_{\mathrm{BFA15}} := (\mathsf{B}, \mathsf{R}, \mathsf{A}, \mathsf{F}, \mathsf{P}, \mathsf{O}),$$

where:

(i) B is source/target binding status;

(ii) R is source-faithful reconstruction and AK-compatibility status;

(iii) A is adversarial analytic-route audit status;

(iv) F is formal-artifact / official-target semantic status;

(v) P is provenance and independent-verification status; and

(vi) O is AK ownership, extraction, and promotion status.

No coordinate is defined to dominate or compensate another.

Definition BFA.15.3 (Binding coordinate)**.** The strongest safe final binding coordinate is

$$\mathsf{B} = \mathrm{target_semantics_and_declared_source_locators_bound},$$

with the following retained qualifications:

(a) Clay whole-space and periodic targets remain distinct;

(b) the periodic-pressure erratum is part of the periodic target binding;

(c) the analytic paper, formal repository/commit, and official target are distinct authorities;

(d) the pinned formal commit identity is resolved at source level; and

(e) a BFA-local cryptographic snapshot of the analytic PDF remains an open provenance task as recorded in BFA.10–BFA.12; fresh remote access to the source URL does not substitute for byte-level local freeze.

Definition BFA.15.4 (Reconstruction/compatibility coordinate)**.** The strongest safe reconstruction coordinate is

$$\mathsf{R} = \mathrm{source_faithful_route_reconstructed_and_v21_compatible}.$$

This means that the selected analytic route was decomposed into a typed DAG, its same-target Return composition was made explicit, and the route was found architecturally conformant with the declared v21 basis. It does *not* mean that every analytic node is mathematically verified, that the route is necessary for the target, or that v21 independently derives the source result.

Definition BFA.15.5 (Adversarial audit coordinate)**.** Let the declared BFA adversarial resolution be the union of the BFA.8–BFA.12 source-semantic tests actually executed. Then

$$\mathsf{A} = \mathrm{no_certified_route_reject_found_at_declared_BFA_resolution}.$$

More explicitly:

(a) BFA.10 obtained no certified nonclosure or reject in the audited finite-to-infinite completion interfaces;

(b) BFA.11 obtained no certified adverse finding in the audited terminal localization, force extension, energy, comparison, viscosity, and periodic transport interfaces; and

(c) BFA.12 obtained no certified target-critical mismatch in the audited paper/formal/official-target triangle at its declared source-semantic resolution.

The value of A is deliberately not named proved, $\mathrm{verified}$, or true.

Definition BFA.15.6 (Formal/target semantic coordinate)**.** The strongest safe formal/target coordinate is

$$\mathsf{F} = \mathrm{target_co_alignment_supported_modulo_declared_semantic_bridges}.$$

It includes the source-text theorem-signature match at the pinned formal commit and the target-level one-way paper-to-formal projections audited in BFA.12. It excludes literal Paper=Lean theorem identity, a proved converse from Comparator target propositions to the full printed witness schema, and formal categorical commutativity of the three authorities. BFA.12 also records a third-party independent formal audit as corroborating evidence; that external record is not retyped as a BFA-local formal-verifier pass.

Definition BFA.15.7 (Provenance/verifier coordinate)**.** The final provenance and independent-verification coordinate is

$$\mathsf{P} = \text{source_exposed_and_independent_verification_not_completed}.$$

Its exact components are:

$$\begin{aligned}
\mathsf{strict_independent_recovery} &= \text{not_certifiable_in_current_context},\\
\mathsf{full_independent_analytic_reproof} &= \text{not_claimed},\\
\mathsf{independent_Lean_build} &= \text{not_invoked},\\
\mathsf{independent_Comparator_replay} &= \text{not_invoked},\\
\mathsf{third_party_formal_audit} &= \text{observed_external_not_reproduced},\\
\mathsf{analytic_local_cryptographic_snapshot} &= \text{pending}.
\end{aligned}$$

These fields are not defects in the source theorem; they are explicit limits on what BFA has independently certified.

Definition BFA.15.8 (Ownership/promotion coordinate)**.** The final AK ownership/promotion coordinate is

$$\mathsf{O} = \text{no_new_global_mathematical_owner_nominated_now}.$$

Specifically:

(a) the two elementary filtered-residual lemmas of BFA.13 remain BFA-local;

(b) MDI and adaptive-localizer RIC remain research programs;

(c) joint-obligation, semantic-projection, and negative-subject views are rebound to existing frozen-v21 owners;

(d) the audit normal form remains an implementation candidate with losslessness unproved;

(e) the Navier–Stokes analytic kernel remains source-retained; and

(f) no Technical-Extension or successor-Core candidate is presently nominated.

Theorem BFA.15.9 (BFA.15 strongest justified final verdict)**.** *For the exact source bindings, selected route, and audit resolution declared in BFA.1–BFA.14, the strongest justified BFA conclusion is:*

> *The selected OpenAI Navier–Stokes route is source-faithfully reconstructed, is compatible with the declared AK-HPDST v21 audit architecture, and has not been assigned a certified BFA route-reject by the completion, terminal, or paper–formal–target stress tests executed at the declared resolution. Nevertheless, BFA has not independently re-proved the full analytic route, has not itself independently rebuilt/replayed the formal artifacts, and only records a reported independent third-party formal audit as external evidence; it cannot certify strict independent recovery in the present source-exposed context, does not identify Paper and Lean as equivalent theorems, does not decide official institutional adjudication, and does not promote the source mathematics into AK Core or Technical Extension.*

Equivalently,

$$\boxed{\textit{survives the present BFA audit} \quad \not\Rightarrow \quad \textit{independently verified mathematical truth}}$$

and

$$\boxed{\textit{route survives} \quad \not\Rightarrow \quad \textit{target truth is proved by AK}}.$$

Proof. The binding and authority separation is established in BFA.1–BFA.2. The source DAG and typed reconstruction are established in BFA.3–BFA.5, while BFA.6 records compatibility without verifier leakage and BFA.7 fixes the provenance boundary. BFA.8 defines falsification semantics, BFA.9 identifies the critical audit frontiers, and BFA.10–BFA.12 record no certified adverse finding at their declared resolutions while explicitly leaving independent analytic/formal verification uninvoked or unclaimed. BFA.13 extracts only two modest local lemmas and reclassifies the remaining generic-looking structures, and BFA.14 nominates no new Technical-Extension or successor-Core owner. By Policy 86, none of these coordinates may be silently upgraded during final closure. The displayed verdict is therefore the conjunction of the strongest already-established coordinate-wise results and no stronger one. □

BFA.15.3 What BFA has actually established

Definition BFA.15.10 (Positive closure ledger)**.** The positive content of BFA 1.0 consists of the following finite ledger:

(i) a governance-safe separation of source theorem, proof route, formal artifact, interpretation edge, official target, and institutional lifecycle;

(ii) exact target/source binding sufficient for a typed audit;

(iii) a source-faithful analytic dependency DAG with explicit shared, terminal, and target-specific dependencies;

(iv) same-target conditional reconstruction without converse inflation;

(v) a declared-basis AK-v21 compatibility result with Core-P/Core-DG remaining uninvoked on the analytic lane;

(vi) a strict provenance protocol distinguishing independent recovery from post-hoc source-exposed work;

(vii) typed adversarial falsification semantics distinguishing route nonclosure, route falsification, formal mismatch, interpretation failure, and exact target falsification;

(viii) critical-frontier and joint-obligation audit instrumentation;

(ix) executed completion and terminal source-interface stress tests with no certified adverse witness found at their declared resolution;

(x) a paper/formal/target semantic audit with one-way target projections, pinned formal-source identity, periodic-pressure alignment, an explicit semantic-bridge ledger, and a separately typed record of corroborating third-party formal-audit evidence;

(xi) two modest BFA-local filtered-residual lemmas proved independently of the source-specific PDE construction; and

(xii) an owner-preserving non-promotion decision separating internal owner admission, external novelty, and adoption value.

Remark BFA.15.11 (Meaning of the positive ledger)**.** The ledger is intentionally heterogeneous. Items (i)–(ix) and (xii) are primarily audit/governance architecture; item (x) is a source-semantic audit; item (xi) is local mathematics. Their conjunction must not be relabeled "an AK proof of Navier–Stokes."

BFA.15.4 Open obligations after closure

Definition BFA.15.12 (Open-obligation classes)**.** The final open-obligation set is partitioned as

$$\mathfrak{O}_{\mathrm{open}} = \mathfrak{O}_{\mathrm{src}} \sqcup \mathfrak{O}_{\mathrm{an}} \sqcup \mathfrak{O}_{\mathrm{form}} \sqcup \mathfrak{O}_{\mathrm{sem}} \sqcup \mathfrak{O}_{\mathrm{prov}} \sqcup \mathfrak{O}_{\mathrm{succ}}.$$

The six classes are source identity, analytic verification, formal replay, semantic bridges, provenance/independent recovery, and successor research.

Definition BFA.15.13 (Source-identity obligations)**.** $\mathfrak{O}_{\mathrm{src}}$ contains at least:

(a) freeze a BFA-local byte-identical copy of the analytic paper consumed by BFA.10–BFA.12;

(b) record its cryptographic hash, retrieval metadata, and immutable artifact identifier; and

(c) if the external paper or repository changes, bind the new artifact to a new source identity rather than silently mutating the current route.

Definition BFA.15.14 (Analytic verification obligations)**.** $\mathfrak{O}_{\mathrm{an}}$ contains any future claim stronger than the present source-interface audit, including:

(a) line-/lemma-level independent checking of the source-specific inequalities, exponent arithmetic, inverse estimates, compatibility conditions, and nonlinear recomputations underlying the audited interfaces;

(b) an independently written proof of the finite-to-infinite completion step if a full analytic verifier token is sought;

(c) independent verification of the terminal comparison and transport arguments at the exact claimed solution class; and

(d) explicit certification of any route reject should a well-typed false obligation later be found.

No such obligation is discharged merely by the absence of a current adverse finding.

Definition BFA.15.15 (Formal replay obligations)**.** $\mathfrak{O}_{\mathrm{form}}$ contains the following tasks if a BFA-local replay token is desired, notwithstanding the independently reported third-party audit recorded in BFA.12:

(a) a local snapshot of the pinned Lean repository and dependency lock state;

(b) an independent build under the pinned toolchain;

(c) an independent Comparator replay of the declared challenge/solution theorem declarations; and

(d) preservation of build logs, exact commit, toolchain, dependency hashes, and theorem identifiers as replay evidence.

Successful replay would upgrade a formal-artifact verifier coordinate only; it would not by itself independently prove every informal analytic statement in the paper.

Definition BFA.15.16 (Semantic-bridge obligations)**.** $\mathfrak{O}_{\mathrm{sem}}$ contains any desired BFA-local strengthening of BFA.12's cross-language correspondence. A third-party audit now reports an independently written Lean Clay specification and formal Comparator-to-Clay bridge [26]; BFA has not reproduced or adopted those artifacts as its own replay. A BFA-local upgrade therefore requires importing and pinning that bridge with an explicit verification record, or independently formalizing the derivative, boundary-smoothness, rapid-decay, L^2, and periodicity correspondences. Until then, the BFA.12 status remains semantic support at the declared audit resolution rather than literal theorem equality.

Definition BFA.15.17 (Provenance obligations)**.** $\mathfrak{O}_{\mathrm{prov}}$ contains the requirements for any future strict independent-recovery claim. Because the present BFA construction is source-exposed, no later editing of this artifact can retroactively make it a strict independent-recovery run. A strict claim requires a new isolated execution satisfying the BFA.7 precursor freeze, disclosure control, sealing, and compare-after-seal protocol.

Definition BFA.15.18 (Successor research obligations)**.** $\mathfrak{O}_{\mathrm{succ}}$ contains:

(a) for MDI: a nontrivial positive convergence/zero theorem plus a sharp failure theorem or counterexample and a genuinely independent non-Navier–Stokes consumer;

(b) for adaptive-localizer RIC: an independent abstract theorem, explicit hypotheses replacing source nouns, and a careful comparison with Nash–Moser and related iteration theories before any positive novelty claim;

(c) for the audit normal form: a machine implementation together with semantic-field preservation witnesses sufficient to prove the claimed normalization lossless; and

(d) for any future owner migration: a separately versioned successor packet satisfying the owner, proof, failure-boundary, and migration rules recorded in BFA.14.

Proposition BFA.15.19 (Open obligations are noncompensable)**.** *No completed obligation from one class in Definition BFA.15.12 discharges a missing obligation in another class unless an explicit theorem provides that implication.*

Proof. The classes have different subjects and verifier sorts. For example, Lean replay concerns a pinned formal artifact, strict independent recovery concerns provenance, source hashing concerns identity, and analytic reproof concerns the mathematical route. BFA.1, BFA.5–BFA.8, and BFA.12 repeatedly prohibit cross-sort pass leakage. The claim follows. □

BFA.15.5 Explicit non-claims

Policy 88 (BFA-FINAL-04: explicit non-claim register)**.** *The closed BFA 1.0 construction draft explicitly does* not *claim any of the following:*

(i) that AK-HPDST independently proves the Navier–Stokes Millennium Prize Problem;

(ii) that the OpenAI analytic proof has been independently re-proved in full by BFA;

(iii) that the OpenAI analytic proof is false;

(iv) that absence of a certified BFA adverse finding is itself a proof of correctness;

(v) that survival of the selected route implies uniqueness of that proof route or necessity of its factorization;

(vi) that Paper and Lean are identical or logically equivalent theorem packages;

(vii) that the Lean artifact has been independently rebuilt or Comparator- replayed by BFA;

(viii) that the independently reported third-party formal audit is a BFA-local replay token, an analytic reproof, or institutional adjudication;

(ix) that a source-text theorem-signature match is identical to successful elaboration/replay;

(x) that the current source-exposed BFA work has strict independent- recovery provenance;

(xi) that a route failure, should one later be found, would by itself falsify Clay-C or Clay-D;

(xii) that source-specific Navier–Stokes mathematics becomes AK-owned by being represented in the BFA DAG;

(xiii) that RIC_0*, the fixed-stage lemma, MDI, or adaptive RIC constitutes a new successor-Core theory at present;*

(xiv) that adaptive RIC has established external-literature novelty;

(xv) that the proposed audit normal form is already proved lossless;

(xvi) that BFA changes the frozen v21 Claim Register or retroactively modifies v21 theorem ownership;

(xvii) that institutional recognition, prize adjudication, publication status, or public announcements supply a mathematical verifier token; or

(xviii) that this BFA 1.0 closure automatically covers future revisions of the analytic paper, Lean repository, Clay errata, or external target semantics.

Theorem BFA.15.20 (Non-claim coverage over the registered BFA status vocabulary)**.** *Within the status distinctions explicitly registered by BFA.1–BFA.14, the principal identified routes by which the final verdict could be overread as proof, disproof, independent recovery, formal replay, target falsification, AK ownership, novelty, migration, or institutional adjudication are blocked by Policy 88. No claim is made that this list enumerates every conceivable future misreading outside the registered BFA vocabulary.*

Proof. The listed non-claims correspond respectively to the theorem/route/target separation of BFA.1, conditional reconstruction and no-pass-leakage of BFA.5, compatibility boundary of BFA.6, provenance boundary of BFA.7, falsification semantics of BFA.8–BFA.9, incomplete BFA-local independent verification in BFA.10–BFA.12, generic-extraction limits of BFA.13, and promotion/nonbackflow rules of BFA.14. They cover the major typed status classes actually used by the present construction without asserting logical completeness over unknown future vocabulary. □

BFA.15.6 Reopen triggers and immutable-history rule

Definition BFA.15.21 (BFA.15 reopen triggers)**.** The closed BFA 1.0 construction may be superseded by a newly identified audit artifact if at least one of the following occurs:

(i) a theorem-bearing external source changes identity or statement;

(ii) Clay target prose or errata relevant to the Return edge changes;

(iii) the formal repository/commit consumed by the audit changes;

(iv) an independent analytic verifier certifies a pass or a well-typed reject for a currently unverified source obligation;

(v) an independent Lean build/Comparator replay produces new formal evidence;

(vi) a strict independent-recovery experiment is executed under the BFA.7 protocol;

(vii) a semantic bridge is formally proved or refuted; or

(viii) a successor AK promotion packet is created for an extracted object.

Policy 89 (BFA-FINAL-05: immutable audit history)**.** *A reopen trigger does not rewrite the historical status of BFA 1.0. New evidence shall create a new artifact/revision identity, preserve the old verdict and source bindings, and record the explicit delta. In particular, a future certified pass does not make an earlier* $\mathrm{not_invoked}$ *field retroactively* pass*, and a future certified failure does not make an earlier "no certified adverse finding" statement historically false.*

Proposition BFA.15.22 (Monotone audit-history principle)**.** *Under Policy 89, BFA audit history is monotone in recorded evidence but non-retroactive in verdict identity: new records may refine the evidence order while old typed verdicts remain statements about the evidence state under which they were issued.*

Proof. The source, verifier, and status identities in BFA are versioned records. The truth of the historical proposition "status s was issued under evidence state E" is preserved when a larger evidence state E' is later constructed. Only a new verdict over E' is licensed. □

BFA.15.7 Global cross-section re-audit closure

Definition BFA.15.23 (Global re-audit invariants)**.** The post-construction global re-audit checks the assembled BFA.1–BFA.15 artifact against the following invariants:

(i) source theorem, formal artifact, official target, lifecycle record, and BFA verifier token remain distinct sorts;

(ii) the BFA.3 normative predecessor relation and the shared role of SUM and O_{14} are unchanged downstream;

(iii) no $\mathrm{not_invoked}$, $\mathrm{not_claimed}$, or $\mathrm{none_certified}$ field is upgraded by later compatibility, external evidence, extraction, or closure language;

(iv) BFA.13–BFA.14 ownership assignments agree with frozen-v21 owners and do not create an unlicensed competing owner;

(v) source-locator/semantic binding is distinguished from byte-identical local source freeze; and

(vi) newly observed third-party formal-audit evidence is typed as external corroboration and does not rewrite the historical BFA-local replay fields.

Theorem BFA.15.24 (Global re-audit consistency result)**.** *After correcting the assembly-level stale summary metadata and applying the invariants of Definition BFA.15.23, the assembled BFA.1–BFA.15 construction contains no identified cross-section status contradiction and no new certified analytic-route reject at the declared BFA resolution. The following limits remain explicit:*

$$\begin{aligned}
\mathsf{analytic_byte_freeze} &= \mathrm{pending},\\
\mathsf{full_independent_analytic_verification} &= \mathrm{not_completed},\\
\mathsf{BFA_local_Lean_replay} &= \mathrm{not_invoked},\\
\mathsf{BFA_local_Comparator_replay} &= \mathrm{not_invoked},\\
\mathsf{third_party_formal_audit} &= \mathrm{observed_external_not_reproduced}.
\end{aligned}$$

Thus the global re-audit strengthens source/status hygiene and the external evidence ledger without converting absence of a detected defect into proof.

Remark BFA.15.25 (Byte-freeze execution note)**.** During the global re-audit the canonical analytic PDF URL was rechecked as a live theorem-facing locator, but the present execution environment did not successfully obtain raw PDF bytes through the local download path. BFA therefore deliberately leaves the cryptographic byte-snapshot obligation open rather than manufacturing a hash from a non-identical representation.

BFA.15.8 Final release record

Definition BFA.15.26 (BFA 1.0 construction closure record)**.** At the end of BFA.15, the construction record is

$$\begin{aligned}
\mathsf{BFA1_0_construction_draft} &= \mathrm{closed},\\
\mathsf{selected_route_reconstruction} &= \mathrm{closed_at_declared_source_resolution},\\
\mathsf{v21_compatibility} &= \mathrm{conformant_at_declared_basis},\\
\mathsf{certified_BFA_route_reject} &= \mathrm{none_obtained_at_declared_audit_resolution},\\
\mathsf{full_independent_analytic_verification} &= \mathrm{not_completed},\\
\mathsf{independent_formal_replay} &= \mathrm{not_invoked},\\
\mathsf{third_party_formal_audit} &= \mathrm{observed_external_not_reproduced},\\
\mathsf{analytic_local_cryptographic_snapshot} &= \mathrm{pending},\\
\mathsf{strict_independent_recovery} &= \mathrm{not_certifiable_in_current_context},\\
\mathsf{paper_formal_equivalence} &= \mathrm{not_claimed},\\
\mathsf{AK_independent_Clay_solution} &= \mathrm{not_claimed},\\
\mathsf{TechnicalExtension_candidate_nominated_now} &= \mathrm{none},\\
\mathsf{SuccessorCore_candidate_nominated_now} &= \mathrm{none},\\
\mathsf{v21_backflow} &= \mathrm{forbidden},\\
\mathsf{open_obligation_ledger} &= \mathrm{explicit},\\
\mathsf{explicit_nonclaim_register} &= \mathrm{explicit},\\
\mathsf{reopen_policy} &= \mathrm{new_identity_required}.
\end{aligned}$$

Theorem BFA.15.27 (BFA 1.0 final closure theorem)**.** *Subject to the registered target semantics, source locators/identities, the explicitly pending analytic byte-freeze qualification, and the audit scope in this appendix, BFA 1.0 is internally closed as a post-v21 audit/research supplement: every downstream task registered in BFA.1–BFA.14 has either a typed final status, an explicit open obligation, an existing-owner binding, a source-retained classification, or an explicit non-claim. No registered unresolved item requires an implicit status upgrade in order for the construction draft to close.*

Proof. BFA.1–BFA.7 close governance, binding, route reconstruction, compatibility, and provenance setup. BFA.8–BFA.12 close the declared adversarial audit rounds by issuing typed outcomes and preserving uninvoked BFA-local verifiers. BFA.13 classifies extractable mathematics and BFA.14 assigns owner dispositions. The unresolved work registered by BFA.1–BFA.14 is routed to the open-obligation classes of Definition BFA.15.12,

while prohibited readings are listed in Policy 88. Reopening is governed by Definition BFA.15.21 and Policy 89. Hence no undeclared consumer or silent upgrade remains necessary for closure. □

Remark BFA.15.28 (Final architectural reading)**.** The principal architectural result of BFA 1.0 is therefore not a new Navier–Stokes theorem. It is a disciplined way to place a high-stakes external proof candidate inside AK without either laundering external claims into AK truth or discarding mathematically useful structure merely because it originates outside AK. The final state preserves both skepticism and reusability:

$$\boxed{\begin{array}{c}\text{bind exactly} \;\rightarrow\; \text{reconstruct faithfully} \;\rightarrow\; \text{attack typed obligations} \\ \rightarrow\; \text{separate what survives} \;\rightarrow\; \text{promote nothing without a new proof/owner case}\end{array}}$$

This is the terminal state of the BFA 1.0 construction draft.

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